Answer Booklet

simplilearn

1. A team decided to use personas for writing user stories. What is the primary benefit they can expect to get from this?
   a) Better alignment between the story outcomes with the user's needs and goals
   b) Reduces the need to write long stories
   c) Gives the team a go-to person in case they have queries and want to clarify a requirement
   d) Ensures that the story can be completed in a time frame expected by the user

   Explanation: A persona is an imaginary representation of a user role. Personas help reduce waste. It ensures that all solutions are aligned with real end-users' needs, goals, capabilities, attitudes, and behavior.

2. An HR team is writing the job description for a product owner. Which of the following is NOT the responsibility of a product owner?
   a) Ensuring one subset of a team does not receive more focus than the others
   b) Describing and documenting the user roles
   c) Identifying the user roles and personas
   d) Being accountable for the delivery of the product as predicted

   Explanation: The Product Owner prioritizes the backlog and represents the customer's voice, but the entire team, including developers and testers, shares accountability for delivering the product.

3. If the team finds a problem in the story that they implemented as part of the current iteration. What should be done about this?
   a) They should write a story to fix the bug and ask the PO to prioritize it.
   b) They should raise a defect and fix it within the current iteration.
   c) They should flag a risk that the product may fail in some situations.
   d) The team should ignore the defect until it is reported by the customer.

   Explanation: Fixing defects within the current iteration is part of the normal work of a team in Agile development. It ensures that the product meets quality standards and maintains the integrity of the current iteration's work.

4. Which of the following projects will be most suitable for agile development?
   a) Low technical complexity and low certainty of requirements
   b) High technical complexity and high uncertainty of requirement
   c) Establish priorities so that only the highest-valued functionality is being worked on at any time.
   d) Depends on the willingness of the development team.

Explanation: The project most suitable for agile development is where the priorities are established, and feedback loops are established.

5. At the retrospective meeting, the team members highlighted few potential improvement actions. What should be done about those issues?
   a) The PO should convert these into user stories and prioritize for the next iteration.
   b) The team should start working on all the improvement actions in a self-organized manner.
   c) The scrum master should log the improvement actions and help the team prioritize them for action.
   d) The scrum master should assign responsibility to individual team members and follow up until they are resolved.

   Explanation: It is the responsibility of the team to find the solutions to the issues raised. The scrum master should help them come up with an improvement backlog and work through it.

6. Which of the following is NOT true about the daily standup meeting (also known as the daily SCRUM?
   a) Only active participants in the sprint may speak during the daily scrum meeting.
   b) A sprint planning meeting is held midway of each sprint.
   c) The product owner may be present in the meeting but only as an observer.
   d) The total time spent on the daily standup should not exceed 15 minutes per day.

   Explanation: A sprint planning meeting is held at the start of each sprint and not midway.

7. Which of the following is NOT a reason for writing a spike story in the product backlog?
   a) To perform an urgent task, for which the entire team may need to work harder than usual for a limited period
   b) To gain more understanding about a feature or technology
   c) To perform a proof of concept on an uncertain requirement
   d) To get more reliable effort estimates

   Explanation: It is not recommended to write a spike story to perform an urgent task, for which the entire team may need to work harder than usual for a limited period of time.

8. Which of the following techniques is most valuable for quickly estimating a large number of stories during a release planning session?
   a) Work breakdown structure
   b) Function points
   c) Planning poker
   d) Affinity estimations

   Explanation: Affinity estimation helps generate fairly reasonable estimates for a large number of stories in a short span of time.

9. How should the risks on an agile project be managed?
   a) By creating a risk register at the beginning of the project
   b) By asking the team to avoid working on risky items
   c) By identifying specific actions to respond to risks and prioritize these actions through the product backlog
   d) By assigning an owner to each risk and holding them accountable for it

   Explanation: The risks on an agile project be managed by identifying specific actions to respond to risks and prioritize these actions through the product backlog.

10. The primary difference between timeboxed projects and feature-driven projects is that:
   a) Time-boxed projects are never late, they incorporate a "feature buffer" for any schedule overruns.
   b) Time-boxed projects need a larger schedule buffer and should be managed critically
   c) The team working on time-boxed projects needs to be highly mature and able to guarantee timely completion.
   d) Time-boxed projects will only work in case the technical complexity is low.

   Explanation: Time-boxed projects fix the time limit up-front and allow features to vary.

11. Which of the following best expresses the reason for having a customer representative embedded in a team (a practice known as "onsite customer" in Extreme Programming)?
   a) Ensure that all team members are not wasting time while they are working
   b) Establish priorities so that only the highest-valued functionality is being worked on at any time.
   c) Guarantee that the product will be delivered at the time it is needed in the field.
   d) Participate in the discussions and provide timely feedback.

Explanation: The role of the customer is to be a representative of the business and provide timely feedback.

12. An agile coach was asked by the team about the causes of bad quality and how to eliminate it. How should the coach respond?
    a) Provide anecdotes and ideas that have worked on other projects.
    b) Encourage the team to dive into the causes of the quality issues and figure out how to solve them.
    c) Discuss with the functional manager responsible for quality and come back to the team with solutions.
    d) Add more testers to the team.

    Explanation: The coach should ideally put the question back to the team and encourage them to analyze and come up with the appropriate solutions.

13. A team is planning on a group of iterations to predict the outcomes and milestones over the next few months. The product owner will use this information to set expectations with the customers. What is the type of planning mentioned in this scenario?
    a) Daily planning
    b) Release planning
    c) Iteration planning
    d) Sprint planning

    Explanation: A release plan is a plan for a packaged group of iteration outcomes, designed to be delivered to end users and customers.

14. When you coach an agile team, there are different shades and styles of coaching. Which level is more appropriate at the beginning of the sprint?
    a) Individual-level coaching
    b) Whole-team coaching
    c) End user coaching
    d) Product owner coaching

    Explanation: Whole-team coaching is the most appropriate type of coaching at the beginning of the sprint.

15. As the agile coach, you may be asked to help select a product owner for a team. Which of the following is the least important qualities to look for in a product owner?
    a) Technical expertise
    b) Business savvy
    c) Domain knowledge

d) Availability to the team

Explanation: Technical expertise may not be necessary for the PO as the team has technical expertise.

16. A team is considering various methods like scrum, XP, Lean, etc., to enable adaptive development. What would be the best advice for the team?
    a) Choose a starter methodology and blend practices from other methods based on experience.
    b) Use scrum as it is the most popular method and includes all the other methods.
    c) Choose a method based on the advice given by the agile coach.
    d) It is better that the choice is suggested by the customer as they are the end users of the product.

    Explanation: Agile uses a philosophy from martial arts called Shu Ha Ri. Shu means 'Follow the rule,' Ha means 'Break the rule,' Ri means 'Be the rule.' To begin with, a prescription is needed but as the team starts working, they can customize their ways of working by blending diverse set of practices.

17. An agile coach has a habit of providing opinions to the team with regard to different questions and issues raised during the agile events. What is a potential issue with this?
    a) The team may not always appreciate a frank and open expression of ideas by the coach.
    b) It is better for the team to hear about the ideas from their own management for it to be more effective.
    c) The coach may become attached to their opinion and lose objectivity to evaluate other ideas.
    d) Unless the coach has personal experience and can guarantee that an approach will work, they should not give it to the team.

    Explanation: The coach should try not to inject their own ideas, but encourage the team to come up with their own.

18. Who is the intended audience of an information radiator?
    a) The team
    b) The product owner and customers
    c) The scrum master or project manager
    d) All the relevant stakeholders

Explanation: The information radiator uses visible means to display information and increase transparency with the relevant group of stakeholders.

19. During a retrospective, a team found that many of their issues were due to the lack of a clear product vision. Which is the best way to resolve this?
    a) Ask the product owner to write detailed requirements that provide maximum clarity to the team.
    b) Conduct an agile chartering exercise to clarify the vision and mission
    c) Prioritize the requirements using the MoSCoW method
    d) Provide coaching to the team about business issues

    Explanation: Agile chartering will help provide a clear statement of purpose to the team.

20. One of the recommended settings for a collocated team is a WAR room space. Which of the following is NOT a factor for the ideal team space for a team?
    a) Minimize distractions.
    b) Make available necessary spaces.
    c) Arrange seatings by job description.
    d) Improve the hygiene factors.

    Explanation: Seating arrangement by job description is a sign of bad team space and not of ideal agile team space.

21. Which of the following is NOT one of the advantages of the planning poker technique for estimation?
    a) It is fun and gets the whole team involved.
    b) It gives more accurate estimates and ensures timely completion of stories.
    c) It triggers discussion and clarifies assumptions.
    d) It even contributes to clarifying design.

    Explanation: Planning poker is a collaborative technique that fosters discussion, involves the whole team, and helps clarify assumptions and design. However, it doesn't guarantee more accurate estimates or ensure timely completion of stories independently. Other factors such as team experience and task complexity also influence estimate accuracy and completion time.

22. When an agile team is distributed across multiple locations, which of the following usually happens?
    a) They tend to use more formal channels of communication.
    b) They tend to skip the daily standup meetings.

c) They are unlikely to continue using agile as it doesn't work very well with distributed teams.
d) They will not be able to deliver high quality as it is harder to integrate everybody's work.

Explanation: Agile teams which are distributed will have to be more formal in exchanging information.

23. What is the value of osmotic communication in an agile team?
    a) It helps management keep tabs on what is going on within a team.
    b) It ensures complete trust and bonding between the team.
    c) It helps reduce the time for receiving answers.
    d) It keeps the customer informed about the technical solution.

    Explanation: Osmotic communication is important because it reduces the time between needing information and getting it.

24. What is the future value of $300,000 five years down the line at an interest rate of 10%?
    a) 330000
    b) 483100
    c) 410500
    d) 405100

    Explanation: We can calculate the value at the end of the third year using this formula - 300000(1+0.1) ^5.

25. Which of the following best expresses the criteria to be checked before a work item is marked as complete?
    a) Acceptance tests
    b) Integration tests
    c) Non-functional tests
    d) Regression tests

    Explanation: Acceptance tests validate that the functionality to be delivered meets customer expectations.

26. Which of the following projects will you choose if you have a set of candidate projects?
    a) A project with the highest payback period
    b) A project with the lowest IRR
    c) A project with the highest NPV
    d) A project with the highest future value

Explanation: The project with the highest NPV would be chosen as it provides the highest economic value. NPV stands for net present value and is an economic analysis to compare projects.

27. The value of a smartphone increases in proportion with the internal storage space provided. In the context of the Kano Model, storage space provided will be termed as a:
    a) Threshold
    b) Linear
    c) Exciters
    d) Delighters

    Explanation: Linear Features are those where customer satisfaction is correlated linearly with the quantity of the feature.

28. Your prioritized feature list consists of 10 features. According to your customer requirements, you need to deliver a minimum of five features to launch the product. What are these features known as?
    a) Prioritized feature list
    b) Minimally viable product
    c) Product backlog
    d) Sprint backlog

    Explanation: A minimally viable product or MVP is the smallest deliverable that can add value to users.

29. Which of the following is probably the least important factor used in the prioritization of backlog items?
    a) Value of the work to customers
    b) Cost of adding the features
    c) Amount of risk removed by developing the features.
    d) Cost of removing the new features

    Explanation: The cost of removing the new features is not a factor used in prioritization.

30. Which of the following information radiators is most useful for a team trying to track the progress of work in an iteration?
    a) Task board
    b) Burndown bar chart
    c) Impediment log
    d) Test case completion

Explanation: A story or task board is actively used by the team to indicate overall status.

31. A quantitative approach to managing projects is important even in agile projects. Which of the following would be the least appropriate use of metrics on a project?

    a) To focus attention on a problem and encourage solutioning
    b) To determine which team member is contributing the most to the deliverables.
    c) To quantify the extent of improvements made
    d) To prioritize which issues, need to be addressed on a higher priority

    Explanation: It is not a good idea to use metrics to compare between team members as it will set up unhealthy competition and reduce collaboration within a team.

32. If the product backlog is 200 points, velocity is 20 points per iteration, and cost per sprint is $1,500; what is the expected cost of the project?
    a) $15,000
    b) $1,500
    c) $20,000
    d) $30,000

    Explanation: The number of iterations is Product Backlog / Velocity = 200/20 = 10. At 1500 per sprint the cost is 1500 * 10 = $15,000.

33. In a manufacturing process, an average of eight finished products are delivered in an eight-hour shift. This means that one hour is the:
    a) Cycle Time
    b) Velocity
    c) Lead time
    d) Processing Time

    Explanation: Cycle time is the average time between the delivery of successive work items.

34. What is the cost variance(CV) when EV = $20,000 and AC = $15,000?
    a) 5,000
    b) -5,000
    c) 2,000
    d) -2,000

Explanation: Cost Variance (CV) is calculated as the Earned Value (EV) minus the Actual Cost (AC). Therefore, in this case:

$CV = EV - AC$ = 20,000 − 15,000 = 5,000
So, the Cost Variance (CV) is $5,000.

35. In agile project planning, why is it not recommended to develop a detailed plan up-front?
    a) Because it is better to spend the time on execution
    b) Because the actual may not follow the plan
    c) Because agile methods believe in evolving or rolling wave plans
    d) Because agile methods only develop plans at the iteration level

    Explanation: Planning too much detail up-front is not helpful because all the insights will not be available in the beginning and the plan will evolve as details emerge.

36. In a large complex system with multiple interfaces, which type of testing should form the majority of the testing activity?
    a) Integration tests
    b) Unit tests
    c) Non-functional tests
    d) Acceptance tests

    Explanation: Unit testing should form the bulk of the tests as per the testing pyramid.

37. You have identified a technical risk with a 20% probability of a 12-day delay on the project. What is the risk exposure?
    a) 2.4
    b) 0.2
    c) 1.2
    d) 0.6

    Explanation: Risk exposure is the product of probability and size of loss; Risk Exposure = 20% x 12 = 2.4

38. Which of the following best exemplifies "servant leadership"?
    a) The product owner answers the questions from the team to clarify the requirements
    b) The team facilitator drops the task she was working on, in order to help a team member resolve a blocking issue.

c) The technical expert in the team conducts a knowledge sharing session to educate the other team members.
d) The team manager conducts performance appraisals rewarding the top performers in the team.

Explanation: The team facilitator helps the team by unblocking their issues before conducting their tasks, thereby putting the team's interest ahead of their own tasks.

39. Which of the following metrics can be used to quantify customer value?

a) The customer achieves a 10% reduction in operating expense as a result of automation of a business process.
b) The team productivity increases by 20% as a result of knowledge increasing over some time.
c) The number of defects reported in internal testing came down from 15 per sprint to 10 per sprint.
d) The developer's estimate for stories was closer to the actuals as compared to previous iterations.

Explanation: Customer value is achieved only when the customer perceives an improvement or benefit.

40. Find the budget for the following project.

The total release backlog is 100 points, velocity is 20 points per iteration, and cost per unit is $1,500?"

a) $150,000
b) $100
c) $2,000
d) $30,000

Explanation: BAC = Cost per unit x Release Backlog = $1,500 x 100 = $150,000.

41. Warranty costs are rising for a particular product. Which of the following metrics should the team track to address this issue?

a) Escaped defects found
b) Regression errors
c) Cycle time increases
d) Story defects

Explanation: Escaped defects are defects that are reported by the customer after the delivery by the team. These are direct causes of warranty costs.

42. A team wants to see which of the identified risks poses maximum challenges to the team. Which chart should the team look at?
    a) Risk burn-up chart
    b) Risk census
    c) Risk burndown chart
    d) Sprint burndown chart

    Explanation: The risk burndown chart plots the sum of the risk exposure values per the iterations.

43. Which of the following is one of the advantages of pair programming?
    a) It reduces the manpower requirement for programming.
    b) It increases the testing effort.
    c) It is a good mechanism to enable knowledge sharing.
    d) It ensures a better understanding of customer requirements.

    Explanation: Pair programming ensures two pairs of eyes work on a problem and gives rise to knowledge sharing.

44. A team is using the velocity data from past iterations to forecast the amount of work that can be done in a forthcoming sprint. In terms of responses to the estimation or planning risk, what form of risk response action is this?
    a) Avoid
    b) Mitigate
    c) Transfer
    d) Accept

    Explanation: Use of velocity reduces the probability of planning or estimation risk. This is known as mitigation of the risk.

45. Agile teams generally produce more reliable plans. Which of the following is the most important reason for this?
    a) Better testing practices
    b) Use of velocity for planning
    c) Using NPV and IRR to assess business value
    d) Use of good technical practices

    Explanation: Velocity gives empirical measure of work done in a sprint, and understanding it gives rise to more predictable plans.

46. Many agile teams make use of an evolutionary development approach to programming where developers must write a test that fails before they write new functional code. Which of the following describes this approach?

a) **Test first development**
b) Aile frequent verification
c) Agile frequent validation
d) Continuous refactoring

Explanation: Test first development is an agile evolutionary development approach to programming where developers must write a test that fails before they write new functional code.

47. Continuous integration helps in keeping the system integrated and in working condition and all times. Which of the following is NOT a practice of continuous integration?
    a) Automate the build.
    b) Keep the build fast.
    c) **Deploy a build for testing at least once in every iteration.**
    d) Make your build self-testing.

    Explanation: Continuous integration is where all the code is integrated at once and the build is automated, so it is fast and self-testing. Therefore, the deployment of the build is a continuous exercise, not once in an iteration.

48. The use of velocity is an important aspect of agile planning. Which of the following statements is false about velocity?
    a) Only completed work is counted towards velocity.
    b) **Once established, the velocity does not fluctuate from its mean.**
    c) Velocity is not directly related to customer satisfaction.
    d) Velocity tracks early and continuous delivery.

    Explanation: Velocity can always fluctuate from its mean, though it gives more confidence about the planning.

49. In a release burndown bar chart, the top of the bar went down from 100 to 80 points at the end of one iteration. What does this indicate?
    a) **Work completed**
    b) Work to be completed
    c) Work to be estimated
    d) New work added

    Explanation: A release burndown bar chart uses bars to indicate the amount of work in the release. The top of the bar is lowered any time the work is completed.

50. The budget of a project is $50,000 project. The earned value analysis shows that the PV is $25,000, the EV is $20,000, and the AC is $15,000. What can be determined from these figures?
    a) The project is ahead of schedule and over budget.
    b) The project is ahead of schedule and under budget.
    c) The project is behind schedule and over budget.
    d) The project is behind schedule and under budget.

    Explanation: The project is behind schedule and under budget. The data provided in this question is BAC=$50K, EV=$20K, PV=$25K, and AC=$15K. SV = (EV-PV) = $20K-$25K=$-5K. CV = (EV-AC) = $20K-$15K=$5K. Looking at the data, it is evident that the project is behind schedule but is under budget.

51. A team is making use of the release burndown chart to track and forecast project progress. When should they update this chart?
    a) After the sprint review
    b) During the retrospective
    c) During the daily standup
    d) During the release planning

    Explanation: The release burndown chart is based on the following factors: the remaining effort in the product backlog and time. The chart is best created and updated in the sprint review meeting when the sprint outcome is known.

52. To keep track of the risks in an agile project, a project manager plotted a chart showing the risk severity scores in an additive way to give a cumulative severity profile of the project. What type of chart is this?
    a) Risk graph
    b) Risk board
    c) Risk issue board
    d) Risk profile graph

    Explanation: The risk Profile Graph is a graph that shows the risk severity scores plotted in an additive way to give a cumulative severity profile of the project.

53. Two sprints into an agile project, the present value was assessed at $13500 and earned value at $15000. What is the schedule performance index at this stage?
    a) 1.11
    b) 0.9
    c) 9
    d) 11.1

Explanation: The Schedule Performance Index (SPI) is calculated as the ratio of Earned Value (EV) to Planned Value (PV).
SPI = EV / PV = 15000 / 13500 = 1.11

54. A team member wants to bring up a technical issue that she is facing while completing a user story. Which is an appropriate forum to bring this up?
    a) Release planning
    b) Sprint review
    c) Sprint retrospective
    d) Daily stand-up meeting

    Explanation: A daily stand-up meeting is a good forum to bring up blocking issues.

55. User stories express the requirements in an agile project. Which of the following is NOT a clear story?
    a) A recruiter can manage the ads placed.
    b) A recruiter can review resumes from applicants for the ads.
    c) A recruiter can change the expiry date of the ads.
    d) A recruiter can delete an application that is not a good match for a job.

    Explanation: This is not a clear story because managing the ads placed does not indicate which exact operation is being carried out.

56. A business analyst and a developer are working on understanding the usage scenario for a transaction so that it can be developed properly. For this, they make use of a ubiquitous language that can later generate automated tests. Which practice is being followed here?
    a) Behavior driven development.
    b) Test-driven development
    c) Test first development
    d) Verification and validation

    Explanation: Behavior driven development involves creating tests before code, and these tests represent expectations of the software behavior.

57. A 6-month project using agile methods has been completed. Which of the following activities should be performed?
    a) Cleaning up open items
    b) Finalizing documentation
    c) Producing or manufacturing support material
    d) All the above

Explanation: Project closing involves cleaning up open items, finalizing documentation, producing, or manufacturing support material, and preparing required end-of-project administrative reports, release notes, and financial reports.

58. During which stage of the agile project management framework does the team start delivering completed features for a project?
    a) Explore
    b) Execute
    c) Adapt
    d) Speculate

    Explanation: In the "Explore" stage of agile, the team delivers completed features, focusing on understanding requirements, identifying stakeholders' needs, and defining project scope. They iteratively deliver features, ensuring continuous feedback and adaptation to evolving requirements.

59. An agile team is following extreme programming practices. When should the team refactor the code?
    a) At any time that the need is felt
    b) During hardening sprints
    c) During the testing cycle
    d) Before starting any new development

    Explanation: Continuous refactoring is one of the extreme programming practices.

60. How many user personas can a typical system have?
    a) One
    b) One per development team member
    c) Two personas - a customer and an end-user
    d) As many as the number of primary users of the system

    Explanation: User personas represent fictional characters created to embody different user types or roles who might interact with the system. A system can have multiple user personas, typically reflecting the diverse range of primary users or stakeholders involved in using or interacting with the system.

61. Triple constraints in the traditional project management framework represent the boundaries within which a project manager should conduct their projects. What is the equivalent concept in agile projects?
    a) Value, quality, and constraints
    b) Schedule, cost, and value

c) Value, quality, and cost
d) Cost, quality, and constraints

Explanation: The dimensions in the agile triangle are value, quality, and constraints.

62. A team working in a traditional, linear fashion wants to start introducing few agile practices in its work. Which of the following practices can they start with
    a) Weekly status meetings
    b) Daily stand-up meetings
    c) Iteration planning meetings
    d) Iteration review meetings

    Explanation: The daily stand-up meeting provides a forum for team members to coordinate activities daily, monitor the status of work, focus on the work to be done, and raise problems and issues.

63. A team delivered 30 story points in an iteration. How should the burndown bar chart reflect this change?
    a) The top will move up 30 points.
    b) The top will move down 30 points.
    c) The top will remain stable.
    d) The bottom of the bar moves down by 30 points.

    Explanation: The top will move down 30 points in the release burndown chart.

64. Whether a project can apply agile methods should be determined at the beginning of any project. Which of the following elements is least important in making this determination?
    a) Complexity and uncertainty in a project
    b) Compliance factors
    c) Cultural factors
    d) Project duration

    Explanation: Complexity and uncertainty, compliance factors, and cultural factors are important to determine whether a project will be an agile project.

65. A team estimated story points during the sprint planning and reckoned that they would not be able to complete a high-priority story in its entirety. What should be done next?
    a) Split the story into smaller stories.
    b) Estimate the story.
    c) Remove the story from the plan.

d) Estimate the story in ideal days.

Explanation: The solution is to split the stories into smaller stories to provide finer estimates.

66. A distributed team is approaching a critical phase of the project which requires intense collaboration between the team members. Which of the following techniques will best help with increasing the collaboration?
    a) Team meetings
    b) Daily stand-up meetings
    c) Team bonding activities
    d) Team collocation

    Explanation: Team collocation promotes effective communication and collaboration among team members.

67. Approximately how much time of a sprint should be dedicated to sprint planning activities?
    a) 5-10%
    b) 10-20%
    c) Up to 5%
    d) 20-30%

    Explanation: Up to 5% of sprint duration must be dedicated to spring planning activities.

68. How should an agile team decide the allocation of tasks among the team members?
    a) Team members should self-assign the tasks based on availability, interest, expertise, and other parameters.
    b) The scrum master should assign the tasks to the team.
    c) The product owner should approach the functional managers for resources for each sprint.
    d) The team members randomly assign tasks based on the draw of lots.

    Explanation: The objective of workload management is to have team members themselves manage the day-to-day tasks required to deliver stories at the end of each iteration. To the greatest extent possible, teams should manage their workload.

69. Which of the following is the most important attribute while selecting a team facilitator (e.g., scrum master)?
    a) Decision making abilities.
    b) Emotional intelligence

c) Achievement orientation
d) Technical expertise

Explanation: Emotional intelligence is important for the team facilitator as they act as servant leaders to the team.

70. Which metric can be used to determine the business value added through the adoption of sprints?
    a) Reducing the number of errors
    b) Improving the team morale
    c) Delivering completed features earlier
    d) Leading the scrum adoption in the industry

    Explanation: Delivering completed features earlier in the life cycle is a business benefit of using agile methods.

71. Among collaboration methods, agile teams should favor synchronous methods as far as possible. Which of the following is an example of synchronous collaboration?
    a) Centralized site for all released project documentation
    b) Drawing, viewing, and editing website.
    c) Web-based meetings
    d) Electronic exchange of drawings and model

    Explanation: Synchronous collaboration is when participants view information and provide feedback at different points in time, therefore, the web-based meeting is a synchronous collaboration.

72. In the five levels of conflict, the intensity of conflict increases from Level 1 to Level 5. At what level of conflict do team members develop an aim to win whereby the prior conflicts and problems remain unresolved?
    a) Level 1 - Problem to solve
    b) Level 2 - Disagreement
    c) Level 3 - Contest
    d) Level 4 - Crusade

    Explanation: At level 3, during conflict, the aim is to win and team members choose one side or the other to solve the conflict situation.

73. A servant leader leads by serving. Which of the following is NOT a characteristic of a servant leader?
    a) Listening
    b) Foresight

c) Healing
d) Decisiveness

Explanation: Decisiveness is not a characteristic of servant leadership while the other three listed are characteristics of servant leadership.

74. An agile team is self-organizing. Which of the following will enable self-organization and self-management within teams?
a) Empowerment
b) Clear directions
c) Prioritization
d) Technical Trainings

Explanation: Empowerment defines operating by oneself and with enthusiasm, taking responsibility and ownership. An empowered team is more likely to embrace self-organization.

75. Which of the following is NOT an objective of an agile retrospective meeting?
a) Assessing the contribution of individual team members
b) Identifying areas of improvement
c) Prioritizing the improvement actions
d) Assigning the prioritized improvement actions to the team members

Explanation: Performance assessment of team members is not an objective of the agile retrospective meetings.

76. Which response strategy can be successful for applying at Conflict Level 1: Problem to Solve?
a) Strive for consensus.
b) Negotiate for middle ground.
c) Accommodate to one another.
d) De-escalate the conflict.

Explanation: Consensus is the strategy that includes learning where every team member is focusing on the issue and time, arriving at a decision everyone can back.

77. When team members are not in the same time zone and find it hard to exchange information, they rely more on formal communication methods like the exchange of documents. Among the various documentation options, which type of communication is most effective?
a) Audiotape
b) Documentation
c) Videotape

d) Face-to-Face

Explanation: Videotape is the most effective communication under the documentation options, while face-to-face communication is the most effective under the modelling options.

78. Feedback is a continuous improvement process where the output of the process is fed back into the input to be improved. Which of the following feedback mechanisms are available in scrum that can lead to process or product improvements?
    a) Daily feedback on progress through standup meetings
    b) Product feedback after each Sprint
    c) Process feedback after each Sprint
    d) All the above

    Explanation: The feedback specific to SCRUM includes all options that is a, b, and c.

79. A scrum master often makes use of different techniques to demonstrate their listening skills. What method confirms understanding by repeating what the other person said?
    a) Active Listening
    b) Reflective Listening
    c) Listening
    d) Injected Listening

    Explanation: Reflective listening confirms understanding by repeating what the other person said.

80. Which of the following development methods can have variable iteration lengths and may not use iterations at all?
    a) FDD
    b) XP
    c) Kanban
    d) Scrum

    Explanation: Kanban can be applied to any iteration length and may not even use iterations.

81. What is the right time for a product owner to give feedback on the work results?
    a) During the sprint review
    b) At any time during a sprint
    c) During daily standup
    d) During lessons learnt meetings (retrospectives)

Explanation: The product owner does not have to wait until the sprint review meeting to provide feedback on work results. It is often helpful to carry out just-in-time reviews as the results emerge in the sprint. This gives the team an opportunity during the sprint to adjust the results, if necessary.

82. During an iteration, the burndown chart for the iteration moved upwards from one day to the next. Which of the following is NOT a reason for this?
    a) Defects were found in the work completed previously.
    b) Few tasks were completed by team members.
    c) A task was underestimated during planning.
    d) Team member discovered a hidden complexity in a story.

    Explanation: When tasks are completed, the remaining work should decrease, resulting in the burndown chart moving downwards. If few tasks are completed or if new tasks are added, the burndown chart may move upwards. Therefore, option b) is NOT a reason for the burndown chart moving upwards.

83. The sprint backlog comprises all activities necessary to reach the sprint goal. During which scrum event is the sprint backlog created?
    a) Sprint planning meeting
    b) Sprint review meeting
    c) Sprint retrospective meeting
    d) During backlog refinement

    Explanation: The Sprint Backlog is created during the Sprint Planning meeting. In this meeting, the Scrum Team collaboratively selects items from the Product Backlog and defines the work needed to deliver the selected items, creating the Sprint Backlog.

84. During a four-week iteration, the product vision changed drastically and many work items in the iteration backlog were made obsolete. What is the best course of action?
    a) Brainstorm within the team about how to add the changed vision in the current iteration.
    b) Complete the current iteration before starting the new work.
    c) Cancel the iteration and start a new iteration with the changed priorities.
    d) Raise defects for obsolete work items and correct them.

    Explanation: If the product vision changes, the team should cancel the current iteration and start a new one in alignment with the new vision.

85. Which phase in the agile project management life cycle delivers a release plan?

a) Envision
b) Speculate
c) Explore
d) Close

Explanation: The release plan is produced during the speculate phase of a project.

86. The development team should assess the project's progress, technical risks, the requirements evolution, and an ongoing competitive market analysis. Who is primarily responsible for coming up with the product roadmap in alignment with this assessment?
a) Product owner
b) Development team
c) Customer
d) Team facilitator

Explanation: The product owner should produce the roadmap after assessing the following areas: product value, product quality, product status, and team performance. If the product vision changes, the team should cancel the current iteration and start a new one in alignment with the new vision.

87. Agile team members should usually be assigned full-time to a team. Which of the following is NOT a reason why fractional assignment is detrimental to team productivity?
a) It results in task switching.
b) It minimizes requirement for scarce resources.
c) Team members cannot focus on a particular task.
d) Other team members may be stuck when shared resources are not available.

Explanation: Fractional assignment is assigning people to multiple projects simultaneously. This may reduce resource requirements, but it negatively impacts productivity.

88. Which measure gives an idea about the time taken to deliver a single item of work by a team?
a) Earned value
b) Velocity
c) Cycle time
d) Lead time

Explanation: Lead time is the average time taken to deliver a completed work item.

89. A large product contains several work items and it is hard for the product owner to convey the product priorities and explain the big picture to the team. Which of the following classification schemes will help the product owner in this task?

a) Theme
b) Epic
c) Features
d) Business value

Explanation: Theme is the term for a group of related user stories that help achieve a common goal for the customer.

90. Brainstorming is an effective technique used in many different areas of an agile project. In which of the following tasks is it least likely to be effective?

a) To generate ideas
b) To solve complex problems
c) To identify risks
d) To analyze and narrow down choices

Explanation: Brainstorming is not used to narrow down ideas - rather to generate maximum ideas.

91. In comparing story points to ideal days, which of the following is NOT true?

a) Story points are a pure measure of size.
b) Story point express the complexity involved in the requirement.
c) Story points help drive cross-functional behavior.
d) Story point estimates have a longer shelf-life than ideal day estimates.

Explanation: An estimate expressed in story points has a longer shelf life than an estimate in ideal days. An estimate in ideal days can change based on the team experience with the technology, the domain, and themselves, among other factors.

92. A team is struggling to arrive at a common understanding about a story and the most effective technical solution. Which of the following is the most effective way to solve this problem?

a) Document existing solutions and encourage offline analysis to shortlist ideas
b) Get the team into a meeting room to discuss the problem.
c) Provide training to the team in decision making.
d) Make use of collaborative technology like instant messaging and video conferencing.

Explanation: The most effective medium would be to have a face-to-face meeting to exchange ideas and come to a conclusion.

93. Timeboxing is a central principle of agile planning. It means limiting the time available for an activity. Which of the following is NOT true about timeboxing?
   a) It implies setting a fixed time limit to overall development efforts and letting other characteristics such as scope vary.
   b) It creates focus and urgency within the team.
   c) It sets the length of the iteration and decides how much functionality can be delivered in the fixed length of time.
   d) It ensures that teams will stretch themselves and find a way to always complete the task within the timebox.

   Explanation: If you are running behind the schedule, move unfinished work into next the timebox rather than stretch yourself, leading to unsustainable pace of work.

94. Calculate the velocity for the project team that completes five stories in one iteration with the following story points: Story A - 5, Story B - 8, Story C - 3, Story D - 2, Story E - 10. Story F-8 (50% complete)
   a) 26
   b) 28
   c) 18
   d) 25

   Explanation: Velocity is the addition of the number of story points assigned to each user story completed during the iteration, i.e., Velocity = 5+8+3+2+10 = 28

95. Which of the following tools indicate the different stages of work items at any point in an agile project?
   a) Cumulative flow diagrams
   b) Burnup charts
   c) Burndown charts
   d) Product backlog

   Explanation: A cumulative flow diagram is a tool for tracking and forecasting agile project

96. During an agile retrospective, the team facilitator decided to use a different technique. No team member is allowed to speak. All participants are invited to stand up and slide cards representing improvement items on the whiteboard. This technique is known as:
   a) Result-based management
   b) Mute mapping
   c) Prime directive

d) None of the above

Explanation: Mute Mapping is a variant of affinity mapping where no one speaks and all participants are invited to stand-up and slide cards on the whiteboard.

97. During backlog refinement, it was discovered that some stories were too large to be completed in an iteration. Which of the following should be done?
    a) Establish an analogy.
    b) Disaggregation to break down the stories into smaller units.
    c) Defer the stories which are too large.
    d) Prioritize the stories based on business value.

    Explanation: Disaggregation is breaking down epics or large stories into smaller user stories. If the stories are too large and cannot be completed in an iteration, they must be disaggregated or broken down into smaller units.

98. In your project, the sum of all stories is 100 story points. Based on past experience, it is known that the team velocity is 10 story points per one-week iteration. How many iterations does the project need to complete the user stories?
    a) 10
    b) 1
    c) 0.1
    d) 20

    Explanation: The number of iterations needed to complete the user stories is calculated by dividing the sum of all stories (100 story points) by the team velocity (10 story points per one-week iteration).

99. Which of the following charts shows the total work in a project and changes in the total work?
    a) Burnup charts
    b) Burndown charts
    c) Task board
    d) Iteration graph

    Explanation: The burnup chart, in addition to showing how much work is completed, also shows the scope of the project.

100. One of the first requirements to follow a Kanban system is to visualize the work. Which of the following tools is helpful in achieving this step?
    a) Task board
    b) Burnup charts

c) Burndown charts
d) Risk profile graphs

Explanation: The task board is a form of Kanban board that shows the current status of the work-in-progress (WIP) and limits the amount of work in-progress the team allows.

101. Planning poker is an important estimation technique commonly used in agile teams. Which of the following best expresses a principle of estimation utilized in planning poker?
   a) Analogous estimation
   b) Detailed estimation
   c) Work breakdown structure
   d) Three point estimation

   Explanation: Planning poker combines the estimation techniques of expert opinion, disaggregation, and analogy.

102. In order to prioritize the work items targeted in a given timebox, the team has ranked them based on value and risk. Which of these features should be done first?
   a) High risk high value
   b) High risk low value
   c) Low risk high value
   d) Low risk low value

   Explanation: High risk high value features are risky to implement, but are desirable to the customer. In order to drive down the risk early, the team should address them first.

103. The daily standup meeting is an important event in a sprint. Which of the following represents a primary intention of the daily stand-up meeting?
   a) Inspecting and adapting work plans
   b) Discussing architectural issues
   c) Prioritizing the feature requests from the project manager
   d) Reporting progress to the team facilitator

   Explanation: Stand-up meetings allow the team members to share information among themselves and evaluate the progress of work to make changes as needed.

104. Risks need to be tracked on any project including agile projects. Who is the most appropriate person to monitor all the risks in an agile project?

a) Quality tester
b) Customer support
c) Developer
d) Project manager

Explanation: Anybody on the team can be chosen to monitor risks, but the project manager is the most suited person for monitoring risks. Stand-up meetings allow the team members to share information among themselves and evaluate the progress of work to make changes as needed.

105. An agile retrospective needs to be facilitated by an experienced team member who is perceived to be neutral and easily approachable to the team. Which of the following is NOT the responsibility of the retrospective facilitator?
a) Create the agenda
b) Begin planning the next project
c) Observe the team
d) Ask questions of the team members

Explanation: The goal of the Retrospective facilitator is to look back at the project to assess its successes and challenges. It is his role to create an agenda, solicit feedback, and encourage open dialogue between the team members. Stand-up meetings allow the team members to share information among themselves and evaluate the progress of work to make changes as needed.

106. Which of the following represents the correct sequence of steps to be followed typically in an agile retrospective meeting?
a) Set the stage, gather data, generate insights, decide what to do, close
b) Gather data, decide what to do, discuss, close
c) Set the stage, generate insights, gather data, develop, close
d) Gather data, generate insights, decide what to do, develop, close

Explanation: The facilitator should follow the following steps when conducting a retrospective: set the stage, gather data, generate insights, decide what to do, close

107. During an agile retrospective, the team members generated a number of improvement suggestions to be worked on. The team facilitator noted the suggestions in an improvement backlog. What should be the next step?
a) Start working on the improvement items.
b) Plan the next sprint with all the improvement items considered.
c) Discuss the suggestions with senior management and see which are appropriate for implementation.
d) Prioritize the improvement backlog and decide who will work on the highest priority items.

Explanation: The team should decide on the priority order for the suggestions and assign ownership of the actions.

108. An escaped defect is one which was found by the customer instead of the team. Which of the following is NOT a potential effect of an undetected or escaped defect in product design?
    a) Loss of comfort
    b) Loss of life
    c) Increase in the velocity of the team
    d) Loss of discretionary or irreplaceable money

    Explanation: An escaped defect will not result in an increase in team velocity.

109. INVEST is an acronym for a set of attributes used to create well-written user stories. What does 'V' mean in INVEST?
    a) Valuable
    b) Verifiable
    c) Validated
    d) Virtual

    Explanation: The 'INVEST' stands for Independent, Negotiable, Valuable, Estimable, Small, and Testable.

110. How do teams utilize process tailoring and frameworks to deliver final products?

    a) Teams begin by building their own framework and tailor it to the needs of the project and stakeholders.
    b) Teams start with a well-known method and tailor it to their needs based on their experience.
    c) Teams begin by appointing an external coach to advise them about a suitable method.
    d) Teams construct frameworks at the end of the projects to use in future projects, which they can then tailor to their needs as it fits.

    Explanation: Process tailoring involves tailoring the agile processes to cater to a situation.

111. Large products require multiple agile teams working on the same product. How can the work of such teams be coordinated in terms of resources and dependencies?
    a) Frequent all-hands meetings
    b) Management meetings
    c) Meta scrums
    d) Product visioning sessions

Explanation: Meta scrum is a communication mechanism used in large projects with multiple agile teams for the purpose of coordinating resources and dependencies.

112. The core agile values state that responding to change is more important than _____.
    a) Reacting to change
    b) Conforming to a plan
    c) Customer collaboration
    d) Working product

    Explanation: Agile values state that it prefers responding to change over performing to plan.

113. The five stages of agile project management framework are:
    a) Envision, plan, explore, adapt, close.
    b) Initiate, speculate, explore, adapt, close.
    c) Envision, speculate, explore, adapt, close.
    d) Envision, speculate, execute, adapt, close.

    Explanation: The five stages of the agile project management framework are envision, speculate, explore, adapt, and close.

114. Which of the following best expresses the term refactoring that is commonly used by teams following extreme programming practices?
    a) Reorganizing the working code to make it better without changing the functionality
    b) Regression testing to make sure existing functions are still working.
    c) Assessing the relative importance of different features and adjusting priorities
    d) Elaborating requirements to a more granular level

    Explanation: Refactoring is the process of reorganizing the working code to make it more organized and more maintainable without changing the functionality.

115. Agile teams often make use of a fixed set of values from a given series to limit the choices while estimating stories. Which of the following is the best estimation scale for estimating large units of work?
    a) 10,11,13,15
    b) 1,2,3,5,8
    c) 1, 3, 5, 7, 9
    d) 2, 4, 6, 8, 10

Explanation: Using non-linear sequences is better for estimating because these sequences give a better-estimating result than linear sequences.

116. In the NPV formula, the term "r" is taken to equal the rate of discounting. What is meant by "discounting"?
    a) Providing discounts on the actual cost (AC)
    b) Reducing the effort estimates of features
    c) Working on the earned value (EV)
    d) Mapping future amounts back to their present value

    Explanation: Discounting is the process of mapping the future amounts back to their present values.

117. A programming practice where developers work in pairs with one programmer driving the keyboard and the other programmer helping from another perspective is called _____.
    a) Pair programming
    b) Refactoring
    c) Test-driven development
    d) Continuous integration

    Explanation: Pair Programming is a programming practice where developers work in pairs with one programmer driving the keyboard and another programmer helping from another perspective.

118. It is good to use a standard set of metrics in different areas to track progress. Which of the following metrics cannot be standardized across different teams?
    a) Response time
    b) Cycle time
    c) Velocity
    d) Process cycle efficiency

    Explanation: Velocity is based on benchmarks that are unique for each team. So it cannot be standardized across multiple teams

119. Teams are naturally hesitant to come to the retrospective meetings when it comes to agile retrospectives. Which of the following is the least likely cause of this hesitation?
    a) Fear of wasting time
    b) Fear of blaming
    c) Fear of speaking up
    d) Fear of failure

Explanation: Retrospective challenges are fear of wasting time, blaming, and speaking up.

120. Which of the following practices best describes an iterative methodology?
    a) Stopping to fix defects found during development
    b) Rapid prototyping to arrive at the desired goal or solution quickly
    c) Identifying the value-adding and non-value-adding steps to reduce waste
    d) Delivering a product in multiple stages

    Explanation: Iterative approaches focus on the correctness of the solution by experimenting and prototyping

121. A team discussed potential improvements for the next iteration. The team facilitator noted 15 action items for improvement that the team wanted to work on. What should they do next?
    a) The team should ask the manager for feedback.
    b) The team should ask the product owner to list backlog items for the next iteration.
    c) The team should start fixing the bugs that they found in the previous iteration.
    d) The team should identify the top priority action items and focus on improving those in the next iteration

    Explanation: The next step that the team must follow is to identify the top priority action items and focus on improving those in the next iteration.

122. Which of the following best describes the purpose of a sprint review meeting?
    a) Determining how to improve the performance of the team
    b) Calculating the velocity of the team
    c) Celebrating the achievements of the team
    d) Assessing the readiness of the product for potential release

    Explanation: The sprint review is primarily meant for gathering feedback and deciding whether the product is ready for release.

123. Agile requirements are often expressed in terms of user stories. Which of the following best describes a user story?
    a) A story that captures the profile of a typical end user.
    b) A day in the life of a user of the system.
    c) A small, granular unit of work that results in added value to the customer.
    d) The collection of all the requirements that the customer wants in a project in order of priority.

Explanation: A user story represents a customer requirement - something that adds value to the customer. It is small and granular and not a big narrative about the users. The collection of such stories makes up the backlog.

124. A user story has to be completed in the upcoming iteration. The work requires multiple people with different skill sets and activities to come together. How should the user story be divided among team members?
    a) Child stories or tasks
    b) Epics
    c) Modules
    d) Sprints

    Explanation: Epics (option B) are collections of stories woven around a theme. Modules (option C) are units within the system, not related to a story. Sprints (option D) are timeboxes within which the work required for accomplishing the sprint backlog is carried out. Stories can be further subdivided into child stories (if the parent story is too large) or into tasks that fulfill the requirements stories.

125. Often, the priority of the work in an agile project (expressed as user stories) is governed by the interplay of different factors. Which of the following best describes the factors governing the priority of a story?
    a) Stakeholder wishes
    b) ROI and risk
    c) Number of customers requesting it
    d) Size estimate given by the team

    Explanation: Higher the ROI from a story, higher is its priority. Similarly, higher risk items should have a higher priority. Stakeholder perceptions may play a role, but it is always better to have a clear view based on these parameters (ROI and risk). The size estimate given by the team may play a role in determining the ROI, but it is not sufficient to determine the priority. The product owner typically determines priorities - it is not a majority wins exercise (number of requesters).

126. The product backlog is typically produced by the product owner. Which of the following is correct regarding backlogs?
    a) A backlog represents the things a team was supposed to complete but did not.
    b) Backlog is a collection of items that a team might do to add value for the customer.
    c) Backlog represents the extent to which a team is behind schedule.
    d) Release backlog is a subset of the iteration backlog.

Explanation: Backlog represents all the different things that a team might do for the customer to add value (which are not yet done). It does not indicate whether or not a team is lagging behind schedule. An iteration backlog is a subset of the release backlog (not the other way around).

127. Before the end of the sprint, a few team members along with the product owner came together to look at possible stories for the upcoming sprint. They clarified doubts about the story, provided rough effort estimates, and discussed the potential sequence of work. What activity is the team performing?
    a) Prioritization
    b) Story mapping
    c) Backlog refinement
    d) Release planning

    Explanation: Backlog refinement is an activity in which the team and the product owner pay attention to the requirements (stories). Requirements and approaches are discussed, estimations are arrived at, and the team gets more clarity about requirements.

128. During an iteration planning session, three team members could not agree on the estimate for a story even after several rounds. Whose estimate should be considered during sprint planning?
    a) Team member assigned to work on the story
    b) Development lead
    c) Product owner
    d) Average of their estimates should be used

    Explanation: A product owner should not ideally participate in the estimation - the estimates have to come from the team. An expert (lead) opinion is valuable and must be considered, but in the event of a disagreement, the estimate of the person who is going to do the actual work should be respected.

129. Ideal time measure is often used as a unit of measure for size estimates. When one uses the term "ideal days" it means:
    a) The lowest possible time it would take to implement something - assuming everything goes well.
    b) The amount of time it would take for the best developer to complete a story.
    c) The goal for a story or a project that is set by the management.
    d) The amount of time it would take to complete a story, assuming there are no distractions.

    Explanation: Ideal days capture the amount of effort required to complete a story after assuming that there will be no distractions. The conversion from ideal

to actual days is determined by factoring in the amount of distractions, the level of experience of the developer, etc.

130. Which of the following information is normally NOT used in agile?
    a) Burndown charts
    b) Burnup charts
    c) Scrum boards
    d) Gantt charts

    Explanation: Agile methodologies typically do not use Gantt charts. Gantt charts are more commonly associated with traditional project management approaches, whereas Agile methodologies emphasize iterative and adaptive planning, which is often visualized through tools such as burndown charts, burnup charts, and Scrum boards.

131. In a test-driven-development approach:
    a) Testers advise the developers on how to develop a story.
    b) Test cases are written along with the story.
    c) Development stops until all bugs found by testers are fixed.
    d) Tests are written before development, code is written to pass the tests.

    Explanation: TDD refers to a way of working where tests are written even before a single line of code is written. The developers then work to write code that pass the tests. Testers and developers have to work hand-in-hand to develop a common understanding of the code and tests that validate the code.

132. In a caves and commons arrangement for an agile work space, the commons area refers to:
    a) The area where the junior team members work.
    b) The stories that are common to multiple teams that will be worked on first.
    c) The common design patterns that are identified and implemented during refactoring.
    d) The space where team members are located when they are working on the project.

    Explanation: Commons is the common team space where all team members normally work. Caves are private spaces used for any private work.

133. A team facilitator in an agile team observes that during the daily standup meeting, two team members start discussing technical details about a potential problem. What should the facilitator do?
    a) Let the discussion continue as long as it adds value.

b) **Timebox the conversation and suggest that the team members continue it offline, after the meeting.**
c) Invite participation from others.
d) Provide own opinion on the matter.

Explanation: The time of the entire team cannot be spent on discussing the technical issue, even if it is very important. For the daily standup to be effective, the facilitator must focus attention on the core agenda and timebox any side conversations. The side conversations can always be conducted later (outside the daily standup).

134. How do you select the metrics that should be tracked in an agile project?
a) Select metrics that showcase the achievements of the team.

b) Select metrics that highlight areas of improvement.
c) Select metrics that customers want to see.
d) **Select metrics that drive the intended behavior within a team.**

Explanation: The metrics must aim to achieve the "right behavior" within the team. Therefore, the performance metrics you select need to be customized to the particular situation that the team is in.

135. The 12 agile principles along with the agile manifesto form the basis on which agile teams manage their day-to-day work. Which statement below is NOT a valid agile principle?
a) Deliver software frequently - every couple of weeks or couple of months.
b) **Simplicity is essential - the art of maximizing the work you do.**
c) Be open to changing requirements, even late in the cycle.
d) Self-managing teams for best requirements, architecture, and design.

Explanation: The agile principle about simplicity says 'Simplicity - the art of maximizing the work NOT done - is essential'. It recommends keeping solutions simple, not trying to over-engineer or deliver more than what the customer requires. Therefore, b is not a correct statement about agile principles.

136. A team found that the automated tests that are produced are taking too much time to execute and slowing down the continuous integration and build process. What should be done?
a) Remove the slow tests from the build.
b) Separate the test and build phases.
c) Retain only the regression tests and remove the unit tests from the build.
d) **Find a way to speed up the testing.**

Explanation: Testing an integrated build is the core purpose of continuous integration. A solution must be found to do it without slowing down the development process.

137. When should the team refine the product backlog while working on iterations?
    a) Frequently during the project
    b) During the iteration review
    c) During the release planning
    d) As soon as the story is added to the backlog

    Explanation: The backlog needs to be refined frequently so that the team can pick up relevant items for subsequent iterations. This needs to be done frequently throughout the release.

138. Which of the following represents the ideal workspace for an agile team?
    a) Open workspaces with no partitions
    b) Desks with cluster seating to separate team members working on the same part of the product
    c) Workspaces with provision for privacy and individual thinking
    d) Distributed teams that can follow the sun

    Explanation: An agile workspace should be designed to facilitate open communication between the team members. Ideally, team members should be within earshot and in sight of each other as far as possible.

139. You are working on a large project that will be executed using agile methodologies. Ten scrum teams will be working on the same product. How many product backlogs are to be created?
    a) As many as the number of teams
    b) Depending upon the functional areas of a product
    c) One backlog per product
    d) Approximately one PO per ten team members

    Explanation: In Agile methodologies, including Scrum, there is typically one Product Backlog per product. This single backlog is managed and prioritized by the Product Owner in collaboration with stakeholders. Having multiple Product Backlogs for the same product could lead to fragmentation and difficulty in managing priorities effectively across teams.

140. You are working on a large project that will be executed using agile methodologies. Ten scrum teams will be working on the same product. How many product owners should be appointed?
    a) One PO per product

b) As many as the number of scrum masters
c) One PO per team
d) One PO for each feature

Explanation: Agile recommends having one PO per team if multiple teams are working on the same product.

141. Which of the following is most important to be an effective leader of an agile team?
a) Empowering the team
b) Monitoring the progress of the team
c) Ensuring that every team member has just enough work and not too much work
d) Ensuring that the deliverables committed to the team are being met

Explanation: A good agile leader should let the team self-manage, within a certain set of core principles. Hence, it is the team which manages the project.

142. As a team facilitator, you are concerned that the team has committed to too many work items in a given iteration. How should you respond to this?
a) Advice the team to set more realistic goals.
b) Ask the team questions about their proposed approach and assess how confident they are about meeting the goals.
c) Ask senior team members to monitor progress closely.
d) Tell the customer that although the team has committed, the goal is unlikely to be met.

Explanation: The team facilitator should tactfully manage the team and not impose their own ideas instead. If necessary, they can ask probing questions that may give clues to the team.

143. One of the core practices of extreme programming is refactoring. Which of the following practices greatly encourages continuous refactoring?
a) Pair programming
b) Continuous integration
c) Coding standards
d) Test-driven-development

Explanation: TDD assures that there is a robust set of tests available for the team to detect problems. It encourages the team to refactor more frequently.

144. During a release planning discussion, the product owner and the team are wondering which epic to prioritize. Epic A has the highest NPV. Epic B has the highest IRR. Epic C has the highest ROI. Epic D has the highest payback period.
   a) A
   b) B
   c) C
   d) D

   Explanation: IRR is a superior measure and should be used for making decisions in the benefit-cost analysis.

145. Which of the following best describes an incremental development method?
   a) Incremental delivery provides early value to the customer by delivering step-by-step and frequently during a release.
   b) Incremental delivery relies on delivering the entire product increment together.
   c) Incremental delivery relies on low rate of change to plan the increments.
   d) Incremental delivery is used when the team is not sure about the solution to a problem.

   Explanation: The incremental delivery method delivers step-by-step to increase the value of visibility for customers.

146. User story describes the requirements for an agile project from a user perspective. All of the following can be candidates for user stories except:
   a) New features
   b) Testing for a bunch of features
   c) Deletion of features
   d) Technical activities

   Explanation: Testing is part of the work required to complete a user story - not a story in itself.

147. How do agile methods help in estimation?
   a) By improving correctness of estimates
   b) By providing quicker feedback about estimates
   c) By making estimation a fun exercise
   d) By assigning financial value to each estimate
   Explanation: Agile projects provide faster feedback about estimates (literally every iteration).

148. A team is using a Kanban system to manage its work. If the speed of working is reduced at a particular step, what would be the result?

a) **Increase in the amount of work-in-progress**
b) Decrease in the amount of work-in-progress
c) Frequent changes in work-in-progress limits
d) Inability to predict from the given data

Explanation: If productivity is reduced at a particular step, then the amount of work-in-progress will increase.

149. In agile estimation techniques, the size of a user story is usually measured by story points. The story is then broken into several tasks which contribute to the completion of the story. What would be the measure used for the size of a task?
a) Task points
b) Function points
c) **Hours**
d) None of the above

Explanation: Tasks are usually estimated in terms of hours taken to complete. There is no such thing as task points. Function points are calculated for entire projects.

150. Which practice is most useful for deciding the minimally marketable feature for a product?
a) **Story mapping**
b) Backlog refinement
c) Brainstorming
d) Value stream mapping

Explanation: Story mapping helps ascertain the MMF by tracking the bare minimum set of stories needed to enable an end-to-end feature.

151. In the 3 Cs of good user stories, one C stands for card. What is meant by card?
a) **Story is written (usually by hand) on a physical card.**
b) Story should lead to a conversation within the team.
c) The acceptance criteria of a story that is written down.
d) A story that the customer pays for instead of the paying for the entire backlog.

Explanation: In many collocated teams, stories are written by hand on physical cards.

152. Which is the best forum to involve external stakeholders?
a) Iteration planning
b) Daily stand-up

c) Iteration review
d) Iteration retrospective

Explanation: The team demonstrates the new functionality during the iteration review. This is the most appropriate forum for the team to invite stakeholders to give them information and get their feedback. The other forums are internal and may not be appropriate for external stakeholders.

153. Lean methods originated in the Toyota production system. Which of the following is NOT a fundamental tenet of lean?
    a) Eliminate waste
    b) Amplify learning
    c) Decide as early as possible
    d) Deliver as fast as possible

    Explanation: Lean actually advocates deciding as late as possible (i.e., taking a lot of time to weigh the possible alternatives and detailed planning). Hence, option C is an incorrect statement.

154. Though it is self-managed, an agile team needs to keep in mind some basic principles to guide its behavior. Which of the following is most important for the team?
    a) Conforming to a plan
    b) Negotiating contracts
    c) Enforcing team discipline
    d) Adapting to change

    Explanation: In keeping with the agile manifesto guidelines, an agile team should be adept at adapting to change. Conforming to a plan is not a fundamental value in agile, nor is contract negotiation or team discipline.

155. A team uses 2 weeks as a common sprint duration. How long should a sprint planning meeting take for this team?
    a) 1 hour
    b) Half an hour
    c) Four hours
    d) Fifteen minutes or less

    Explanation: The ideal amount of time for sprint planning is 1-2 hours per week of iteration. For a two-week sprint, this translates to 2-4 hours. Option c is the only option available within this range.

156. A product manager needs to convey some important information related to a task during an iteration. Where can the PM find out who is working on that task?
   a) The daily stand-up meeting
   b) In the iteration backlog
   c) In the product backlog
   d) By asking the scrum master

   Explanation: During the iteration planning session, the iteration backlog is created. It contains information such as the breakup of stories into tasks, assignment of tasks to team members, etc.

157. What can you usually say about the velocity of a seasoned and functioning agile team?
   a) Their velocity should be greater than the industry average.
   b) Their velocity should improve by 10% after each iteration.
   c) There is no need to track the velocity as the team is self-managed.
   d) Their velocity should be consistent and predictable.

   Explanation: The velocity of a team is an important metric because it indicates the normal rate at which the team completes stories. It is certainly an important piece of information for a team. It is not advisable to compare the velocity across teams, because each team will have a different definition for the story points and the circumstances of each team are different. It is also not reasonable to expect a continuous improvement in velocity. Being predictable (consistent) about the rate of deliveries is probably the best goal to have.

158. A team member wants to implement a new tool to improve code quality. Which is the best forum to talk about this?
   a) Iteration retrospective
   b) Ad-hoc meeting
   c) Iteration review
   d) Daily stand-up

   Explanation: The iteration retrospective is a forum that provides an opportunity to adapt procedures. Hence, it is the most appropriate forum to discuss this.

159. How does the customer communicate with the team in extreme programming?
   a) By going on-site (with the team)
   b) By appointing a single point of contact with the team
   c) By participating in code reviews at random
   d) By participating in the daily stand-up meetings

Explanation: The on-site customer is a fundamental element of extreme programming or XP. XP does not have a product owner role. The customer is not really expected to attend all the meetings or review codes. Rather, they are supposed to work on requirements in conjunction with others in the customer organization, come up with different user personas, answer questions from the team, provide instant feedback if required on UI designs, etc.

160. In a scrum team, who is responsible for ensuring that the method is followed in letter and in spirit?
    a) The team product owner
    b) The scrum master
    c) The manager
    d) The agile coach

    Explanation: Scrum masters are supposed to be team gurus. They keep their teams honest in terms of following the practices in the right spirit.

161. During backlog refinement, a team member felt that a story was too complex to provide an estimate for it. What can be done?
    a) Make suitable assumptions
    b) Defer the story
    c) Separate the story into a timeboxed research story and an implementation story
    d) Ask the expert to provide an estimate

    Explanation: If a story is too complex, it makes sense to first conduct time-boxed research on it and then proceed with implementation.

162. During a daily stand-up meeting, a team member brought up a suggestion to improve the utility of a feature. The scrum master should suggest that it should be discussed instead during:
    a) Sprint planning meeting
    b) Backlog refinement
    c) Sprint retrospective
    d) Release retrospective

    Explanation: Possible product enhancements can be discussed during backlog refinement.

163. Why is diversity a desirable attribute for an agile team?
    a) Diverse teams are less likely to disagree with the PO
    b) Diverse teams guarantee timely delivery
    c) Diverse teams comply with regulations

d) **Diverse teams give rise to different points of view and innovation**

Explanation: The main benefit of diversity in a team is the diverse perspectives it brings to the table, which can lead to innovation. It may even be a mandate, but option D better articulates the justification.

164. How are agile projects usually planned?
    a) Scope and time are fixed, cost and size of the team are flexible.
    b) **Time and resources are fixed, scope is flexible.**
    c) Cost and scope are fixed, time is flexible.
    d) Triple constraint is a waterfall concept - not applicable to agile projects.

    Explanation: Agile projects usually operate within specific timeboxes and pre-assigned teams. The scope, broken down into small increments of epics and stories, is left flexible. You work on the scope in the order of priority.

165. Extreme programming gives excellent guidance on development practices. What does XP recommend about code check-ins?
    a) Developers check-in code every day at a fixed time.
    b) **Developers can check-in at any time but are responsible to make sure their changes compile and pass unit tests.**
    c) Developers cannot check-in code till the end of the iteration or when all features are complete.
    d) Code is first submitted to the release manager, who approves check-ins.

    Explanation: Continuous integration is one of the practices of agile. This means that the code is checked-in regularly. As soon as a developer is finished, the code should be checked in and validated.

166. A product manager was newly appointed to an agile team and is looking for a high-level description of the scope of the project. Which document can the manager find it in?
    a) **The project charter**
    b) The work breakdown structure
    c) The user stories
    d) The epics

    Explanation: The key words in the question are high-level description. This can be found in the charter. The user stories and epics contain some of the requirements, not the entire scope.

167. A team is embarking on a release planning exercise. Which of the following is NOT a required output of a release planning exercise?

a) A release backlog with estimates
b) Stories tentatively allocated to iterations
c) An initial list of risks and open issues
d) A detailed project schedule

Explanation: Release backlog, stories, and risks are all outputs of the release planning process.

168. A project manager is trying to establish a forecast for the end date of an agile project based on the rate at which stories are getting completed. Which of the following charts will show this information?
a) Iteration burndown chart
b) A progress chart
c) Cumulative flow diagram
d) A Kanban board

Explanation: The cumulative flow diagram shows the progress of stories planned for a release through initial definition to completion. It gives a complete picture of progress in a release. The iteration burndown chart pertains to a specific iteration and deals with the task level. A progress chart is also typically for a given iteration, whereas a Kanban board gives the picture as on a given point in time of the stories being worked on.

169. Why is it that the product owner of a team should ideally not be the line manager for any of the team members?
a) The team will not be able to push back if the demands made by the PO are not reasonable.
b) The PO is meant to act like a servant leader.
c) The PO will be too busy with customers to do performance appraisals for the team.
d) The PO will not be able to solve blocking issues.

Explanation: The PO is naturally expected to be somewhat demanding and ask for more work in a given timebox. If they are also line managing the team, the team may not be able to push back, resulting in loss of self-organization.

170. For the last three iterations, a team completed the development tasks for the planned stories, but the testing was lagging behind. What is the best thing you can do?
a) Add more testers to the team
b) Encourage the team to be more cross-functional
c) Reduce the developers on the team
d) Plan for testing to lag development by one iteration

Explanation: We must make sure that the team is doing everything it can to help remove the bottlenecks and help each other out. The team should collectively deliver the near releasable product at the end of the release. Options A, C, and D are all trying to mask or work around the problem, whereas the issue needs to be addressed directly.

171. A team is unable to complete their committed work in iterations due to blocking issues. This indicates that:
    a) The team is not cross-functional enough.
    b) The team facilitator is not effective.
    c) The team needs to be more reasonable in estimating.
    d) The team needs to modify their definition of done.

    Explanation: The team facilitator is expected to remove blocking issues. If they are not effective, stories are likely to remain incomplete.

172. Which of the following best explains the advantage of story points over ideal days?
    a) Story points are easier to explain outside the team.
    b) Story points use separate estimates from each team member contributing to the story.
    c) Story points are based on the Fibonacci series.
    d) Eventually, story point estimation becomes faster.

    Explanation: Since story points are analogous there is better chances of estimates being arrived at faster through comparisons.

173. A team frequently finds that they get stuck within an iteration as some stories have dependencies that are not met. What is the best way to address this?
    a) Establish and follow a definition of ready that clarifies when a story can be accepted in a given iteration.
    b) Drop those stories and focus on those which can be completed.
    c) Escalate the unmet dependencies and seek a timely resolution.
    d) Product owner should ensure that external dependencies are met.

    Explanation: Agile projects deal with the basic triangle of value, quality, and constraints. Scope, cost, and time represent the constraints.

174. A team is encouraged to do value-based prioritization. All of the following represents value derived from a piece of work except:
    a) Additional team members inducted
    b) Additional revenue
    c) Operational cost reduction
    d) Knowledge gained

Explanation: Inducting additional team members as such has nothing to do with value to the customer.

175. The team is working on an agile project and the contract has a fixed price and fixed scope. Midway through the project, the customer requests for a change. What should you do?
    a) Inform the customer that no changes are allowed on a fixed price contract.
    b) Agile is about customer collaboration over contract negotiation - hence we accept the change.
    c) Negotiate with the customer to drop some low priority feature of same size as that of the change requested.
    d) Defer the decision to the management of the organization.

    Explanation: If the customer requires a change, it must be accommodated. The team should try to work out a way in which it can be delivered. Deferring to management may be a tempting option, but before doing so, it seems that the team would have to do some homework in compiling the relevant information. There are many ways a change can be accommodated by an agile team. For example, the team can always negotiate with the customer to drop a low priority feature of same size as that of the change requested.

176. In an agile team, the developers are complaining that testers are not completing their work on time, leading to the team not fulfilling its commitments in the iteration. The testers complain that developers are delaying deliverables to test, which is causing this problem. What is the level of conflict in the team?
    a) Level-1
    b) Level-2
    c) Level-3
    d) Level-4

    Explanation: In the hierarchy of conflicts, when the team discovers a problem it's level-1, when a disagreement comes to the fore it's level-2, when a disagreement degenerates to a contest (generalizations, formation of groups) it's level-3, crusade (open warfare between groups) is level-4, world war (all hell breaks loose) is level-5. It is clear from the description that the team is at level-3.

177. One of the important attributes of agile teams is team self-organization. What is NOT implied in self-organized teams?
    a) Teams self-assign tasks
    b) Technical decisions are made by the team
    c) The team decides what commitments they want to make
    d) The team dictates the priority decisions

Explanation: In self-organized teams, the team manages their work and makes technical decisions but prioritizes tasks based on guidance from the Product Owner, who ensures alignment with stakeholder needs.

178. In agile projects, "technical debt" is a term that refers to:
    a) Loans taken by the organization to undertake the project
    b) Technical team members offered on a temporary basis to the team
    c) Deferring technical decisions till the last responsible moment
    d) Technical tasks that are pending and will add to the backlog of the product in the future

    Explanation: Pending technical tasks are often called technical debt as over the long term, they will come back and haunt the team.

179. A team found that it has a serious shortage of testing team members. As a result the testing tasks remain open for many of the committed stories. What is the best way to address this?
    a) Break each story into development and testing tasks.
    b) Perform testing in a subsequent iteration.
    c) Find a way to test the story with the available resources.
    d) Modify the definition of done for the stories.

    Explanation: Testing is required and the team must find a way to manage it with the existing resources.

180. Which of the following is NOT one of the advantages of pair programming?
    a) Team members get more freedom to think independently
    b) Collective code ownership
    c) Online review
    d) Reduction in overall development time

    Explanation: Pair programming is a practice in XP that brings two programmers together to work on a task. It helps collective code ownership because each line of code is known to at least two programmers. It helps get an online review and results in cleaner code that will more likely work the first time. This leads to reduction in the overall time. It may or may not lead to independent thinking. Indeed, one of the risks in pair programming is that one of the members of the pair may end up listening to and merely following the instructions of the other.

181. Lean and cycle time are two important metrics for any team practicing Lean and Kanban techniques. Which of the following best explains the terms lead time and cycle time?

a) **Lead time is the amount of time it takes for a unit of work to proceed from inception to production delivery. Cycle time is the time it takes to complete a requirement after it is picked up.**
b) Lead time is the waiting time between processes. Cycle time is the total time spent in processing.
c) Lead time is the amount of wait time before a process can start. Cycle time is the total time spent in processing.
d) Lead time and cycle time mean the same thing.

Explanation: Lead time is the amount of time it takes for a unit of work to proceed from inception to production delivery. Cycle time is from when a requirement is picked up for development to when it is completed.

182. Which of the following represents the normal agenda for a daily stand-up meeting?
a) What did you do yesterday? What are you planning to do today? When will you finish your tasks for the sprint?
b) **What did you do yesterday? What are you planning to do today? What issues are blocking you?**
c) What is the status of your tasks? When will you complete them? What are your risks and issues?
d) What is the status of each task? What percentage of work is completed? What is the projected end date?

Explanation: The three questions you answer during a daily stand-up meeting are: What did you do since the last meeting? What do you plan to do before the next meeting? What is in your way (blocking issues)?

183. If you are an agile project manager, what is the best measure of success of your project?
a) **The project delivered maximum value to the customer.**
b) The project was on time and on budget.
c) The customer received all the promised features.
d) The project was delivered with the documented scope.

Explanation: Agile aims to deliver the maximum value possible to the customer. It is not about following a plan (scope, cost, time). It is about being flexible and adaptable in order to achieve this goal.

184. What happens during iteration planning meetings?
a) Team members brainstorm the design for the iteration.
b) **Activities and tasks are identified and story points are estimated.**
c) The team is introduced to the iteration stories for the first time.

d) The product owner shares the goal of the iteration and lists the stories that must be completed.

Explanation: Iteration planning does result in some clarity about the approach or design, but it is not a design meeting. Some of the stories may be new, but in general the team may already have been exposed to the stories before the planning meeting (for example, during release planning). Indeed, it may be necessary to have some exposure prior to the meeting in order to timebox it. The product owner does set the goal for the iteration, but does not decide which stories to complete - that requires buy-in from the team. In the iteration planning session, stories are broken down into tasks, assigned to team members, and story points are estimated.

185. If a team is looking for quick time-to-market, what should be the basis for prioritizing features?
    a) Cost of features
    b) Risk to complete the features
    c) Number and completeness of features
    d) Creating complex stories

    Explanation: Time-to-market is often a critical parameter. In order to reach the market quicker, number of features may have to be compromised, along with the extent to which they are implemented.

186. Project stakeholders have low confidence in the development team. Which of the following gives the maximum benefit in terms of building stakeholder confidence?
    a) More meetings with the stakeholders to boost communication
    b) Reduction in number of defects
    c) Induction of new team members
    d) Delivering working software increments at the end of each iteration

    Explanation: Stakeholders might value all options (A, B, and C) but what they value the most is to see functionality delivered in the form of working software at a good frequency.

187. If the burndown chart showed an upward trend for two days in a row in the middle of the iteration, what might be the reason?
    a) The team is not completing the stories fast enough.
    b) Few team members were absent.
    c) Tasks got added to the iteration backlog.
    d) The team is working overtime.

Explanation: The burndown chart plots the number of hours of work remaining in a given iteration. Slow completion of tasks or absenteeism may see this number stagnate or move down slowly, but it would only move upward if new tasks got added.

188. A scrum team wants the scrum master role to be re-defined so that it is more useful for the team. What should be done in this situation?
    a) It should not be done.
    b) The topic should be discussed at the next scrum meeting.
    c) Discuss the topic at the next sprint retrospective.
    d) The project management office should be consulted for guidance.

    Explanation: The philosophy of agile is flexibility and adaptability. It is legitimate to re-define roles if the team agrees to it. The best forum to discuss bringing in such changes is the sprint retrospective and not the daily scrum, which is more of a sync-up meeting. It is also not appropriate to delegate judgment to a PMO because a team should have the freedom to do what it needs in order to be effective.

189. A team is practicing Kanban and has reached the WIP limit on its constraining step with regard to the number of work items in progress. Another work item came up, which is extremely high priority. What should the team do?
    a) Refuse to accept the work item until the number of work items in progress goes below the WIP limit.
    b) Add team members to the team to address the urgent issue.
    c) Request the removal of one of the existing items from the queue and add the new high priority item in.
    d) Discuss the topic at the next daily huddle.

    Explanation: Kanban allows re-prioritization of items but doesn't permit exceeding the WIP limit. So a reasonable approach is to add the urgent task in exchange for another lower priority item.

190. Which of the following is usually not considered in the prioritization of stories?
    a) The business value of the functionality added.
    b) The effort needed to build the functionality added.
    c) The risk in adding the new functionality.
    d) The number of team members needed to work on the story.

    Explanation: Value is a function of the business value (or return), the cost, and it also needs to be viewed in conjunction with the risk (or variability). Number of team members would be included in the cost so it is not really an additional consideration.

191. A distributed team is facing challenges in collaboration, which is impacting productivity. Which of the following is NOT likely to be helpful in increasing collaboration?
   a) Collocate at least those team members who are in the same location.
   b) Add team members with the specific goal of coordinating across team members.
   c) Make use of tools for high bandwidth, direct communication.
   d) Broadcast important project communication to everybody in the team.

   Explanation: Adding team members will likely introduce more overheads in communication.

192. Agile techniques are based on the well-known principles of empirical process control and incorporate many concepts from process control theory. What is the best representation of these principles?
   a) Inspect and adapt
   b) Deliver frequently
   c) Plan in detail and execute rapidly
   d) Introspect and retrospect

   Explanation: The agile principle of inspect and adapt adheres to the process control principles of feedback loops.

193. Process tailoring and process instantiation are both very common in agile methodologies, which advocate being flexible about the process being followed. Which of the following best explains the difference between the two?
   a) Tailoring introduces new processes and instantiation implements a new process.
   b) Tailoring is adapting the processes, instantiation is implementing the adapted processes.
   c) They are the same.
   d) Tailoring scraps existing processes, instantiation maintains the status quo.

   Explanation: Process tailoring is an important aspect of agile. It means that you start on a process and tailor it in a way to best fit your team and its particular situation. Process instantiation, on the other hand, takes a (often heavy and prescriptive) process and implements it with little or no scope for adaptation.

194. Which of the following is least useful for the project management team in an organization with widely adopted agile methodologies?
   a) Collating data and establishing organizational baselines for average productivity of scrum teams.

b) Reporting on the compliance requirements being met through agile methods.
c) Providing guidelines for making decisions in uncertain environments.
d) Establishing job descriptions and career paths for typical agile team roles.

Explanation: Comparing productivity data across teams is not advisable in agile because each team may have a different baseline.

195. A team is trying to come up with a charter for its agile project. What is least important for the team at this stage?
    a) Visualizing the customer need to think of the best solution
    b) Building a solid business case to justify the project
    c) Maintain the integrity of the agile process to be followed
    d) Get maximum all possible details about the product backlog to add clarity to the vision

    Explanation: Options A, B, and C are all reasonable principles, activities, and deliverables in an agile chartering process. Option D runs contrary to the principle of rolling-wave-planning, or 'inspect and adapt' framework, while trying to add too many details about the product backlog. The backlog will be in a state of flux and agile is all about being flexible to handle changing priorities. Hence, option D is the right answer.

196. A sprint review is an important meeting in the scrum methodology. What is the purpose of the sprint review?
    a) The scrum master reviews progress and gives guidance about corrective actions.
    b) The team provides a working demonstration of the software and solicits feedback.
    c) The customer and other stakeholders carry out a thorough review of the performance of the team members and provide the feedback to the manager of the team.
    d) The team members introspect on the sprint that went by and decide on corrective actions.

    Explanation: The sprint review is a meeting where the team demonstrates progress (in the form of working software) to the product owner and other stakeholders in order to get feedback.

197. A team has recently transitioned to scrum. The product owner wants to know the level of detail to put in to the stories so that the team can work better. What guidance would you give to this end?

a) Stories should be as detailed as possible, so that there is no room for ambiguity.
b) There cannot be more than one index card and the details can be given in a separate document.
c) The level of detail should be worked out between the team and the product owner.
d) Stories can contain less details because agile relies on developing working software more than comprehensive documentation.

Explanation: Level of detail in the requirements should be 'just enough' to ensure clarity. Beyond this, finer level of guidance needs to be worked out by the team along with the product owner. The product owner should not get 'perspective' on this and let them work it out.

198. A team finds that the product owner frequently reverses old decisions and many features are started but never finished. What is a way to resolve this?
a) The team and PO should work closer together to understand the priorities and make informed decisions.
b) The PO has final authority over the backlog so it is okay if they want to keep changing the backlog.
c) The team should go for longer sprints to accommodate the mid-sprint changes.
d) The executives should step and override the PO decisions where appropriate.

Explanation: This situation calls for greater collaboration and communication between the team and the PO.

199. Which of the following best explains internal rate of return?
a) It is the discount rate at which NPV is zero and represents the rate at which the project adds value to the organization.
b) It is the average savings or profit per year divided by the initial investment.
c) It does not take into consideration the time value of money.
d) It calculates how many features or stories were rejected (returned) by the customer.

Explanation: NPV is the summation of the cash flows of each time period, reduced to their present value.

200. In the KANO model of prioritization, what are the priorities that you can assign to stories?
a) Must, Should, Could, Would
b) High, Medium, Low
c) Threshold, Linear, Exciter

d) See if the stated quality level is really needed

Explanation: In the Kano model, Threshold features are must haves without which a product is not acceptable. Linear features are good to have and the more of these features you have, the better it is. Exciters are features that act as clear differentiators from competition and help attract a price premium.

201. Which of the following is a good way to use the MoSCoW prioritization?
    a) In a given time box, the team must only work on the MUST items.
    b) Appropriate rewards and recognition policies must be in place for teams that consistently deliver not just the M, but also the S, C and W.
    c) A team must ensure that at least 50% of the work is classified as W (Won't) in a given timebox.
    d) While planning a timebox, a team must never fill an entire timebox with only the MUSTs.

    Explanation: In the MoSCoW model, M stands for must. As must items should never be compromised, it makes sense that you plan to not fill up the entire timebox with must items.

202. The project management office, in an organization that has adopted agile, regularly issues best practices and records performance data. They just published a comparison of velocity data (story points completed per iteration) for different teams. As a scrum master for one of the teams, what would be your next steps?

    a) Share this data with the team and discuss with them how to improve their numbers in comparison with the rest of the organization.
    b) Write back to the PMO stating that velocities should not be compared across teams.
    c) Write to the management justifying the difference between the organizational benchmark and the figures for your team.
    d) As the team is self-organizing, simply share the data with them and let them figure out what to do.

    Explanation: Comparison of productivity data across teams is useless because each team works under different circumstances and captures story points on different basis. Comparing velocity data is not healthy. Hence, it must be pushed back to the PMO.

203. You are starting a new project with a new agile team. What would NOT be a good approach to develop an initial estimation of the team velocity?
    a) Take a guess.
    b) Run a couple of initial iterations and extrapolate the data.

c) **Use the desired velocity of the management as a starting point and go from there.**
d) Use historical data.

Explanation: Options A, B, and D are all legitimate ways of determining the initial velocity of the team. The notion of management about the desired velocity of the team is a poor starting point because it is completely uncorrelated with the inherent capabilities of the team.

204. Teams following feature-driven development often use a parking lot chart. What does a project parking lot chart indicate?
    a) The features that you have decided to park for now for want of prioritization.
    b) The overall backlog of features in the project.
    c) The amount of time a feature is allowed to stay in the backlog before it is automatically removed.
    d) **The progress made in each feature that the team is working on.**

    Explanation: The project parking lot diagram indicates the progress made in each feature planned in the release.

205. Many organizations already have a business analyst's role separate from the PO. What could be some tasks that the PO can delegate to the BA?
    a) **Writing detailed stories and communicating with the team on a daily basis.**
    b) Developing business justifications for product releases and creating the roadmap.
    c) Communicating with the business users and assigning priorities.
    d) Being a servant leader and helping the team with development activity.

    Explanation: In general, the product manager focuses more on the external stakeholders and tries to reflect their point of view in the roadmap. A product owner has both an inward and outward focus. The outward focus is like that of the product manager, and the inward focus calls upon the product owner to work with the team in order to plan iterations and releases and build a roadmap with team input.

206. A company is trying to transition to agile and is looking for a suitable project to pilot the transition. Which project would be the most suitable for the pilot?
    a) Project A: Estimated duration is 1 year. Team size is 10.
    b) **Project B: Estimated duration is 3 months. Team size is 10.**
    c) Project C: Estimated duration is 2 years. Team size is 50.
    d) Project D: Estimated duration is 3 years. Team size is 55.

Explanation: A medium duration project with a reasonable team is best suited for the pilot. Within the first couple of iterations, you can figure out whether the methodology will work for you.

207. Although agile methodologies give a lot of flexibility to the team members to tailor their processes, teams are also required to comply with standards or regulations at the same time. Which of the following may be a good way to bring the compliance angle into agile projects?
    a) Let the external support teams like the PMO or SEPG handle the compliance needs.
    b) Make use of existing quality frameworks like ISO, CMMi.
    c) Ask the customer to recommend a process.
    d) Treat compliance also as a requirement and include in the product backlog for prioritization and working.

    Explanation: Compliance requirements may come in whenever you need to comply with a certain process, standard, or a framework. It may be due to government regulations, customer-mandated processes, or due to internal requirements to comply with well-known quality frameworks. Compliance also requires work, which needs to be prioritized like any other activity on the project.

208. The primary output of a sprint planning session is:
    a) Product backlog
    b) Release backlog
    c) Sprint backlog
    d) Project schedule

    Explanation: Sprint planning sessions result in a sprint backlog. It may impact the release backlog or the product backlog, but the primary output of the session is the sprint backlog.

209. What is the primary purpose of the agile manifesto?
    a) To outline the basic values and principles that are common across all agile methods
    b) To explain to people what agile can do for their organizations.
    c) To make a detailed business case for adopting agile.
    d) To enable organizations to scale an agile implementation.

    Explanation: The agile manifesto was created by a few leaders in the software industry, who came together to form an agile alliance.

210. To make a portfolio decision, senior management wants to know the approximate cost of a project. At what stage can we estimate the cost of an agile project?

a) You cannot estimate it until the last iteration because the backlog is changeable.
b) It can never be known for sure because agile projects are flexible by design.
c) During release planning, the cost can be estimated by the team.
d) During the business case, the product owner can estimate the cost.

Explanation: Here we are talking about a budgetary estimate, which might be known during the release planning stage based on the broad contours of the release and the estimates assigned by the team. It is true that the release may change up to the last sprint, but an approximate figure for budgeting purposes can still be given. The business case is usually prior to the planning stages. Hence, the estimates in the business case may not be suitable for budgeting purposes since they are not likely given by the team.

211. You are the manager of an agile project. You find that a particular stakeholder is proving to be a major distraction to the team. He frequently calls the team to ask for status updates, get information, provide suggestions, and sometimes to request changes. What should you do about this?
a) Escalate the issue and keep the stakeholder off-limits.
b) Invite the stakeholder to the appropriate planning or review meetings and request him to bring up his views at those meetings.
c) Listen to the stakeholder, but ignore his suggestions.
d) Tell the stakeholder directly not to disturb the team during the iteration.

Explanation: Although interest and involvement of the stakeholders is not a bad thing, it must not be discouraged. Agile provides forums and platforms to listen to and accept stakeholder input. The skill of the manager lies in utilizing these forums to get positive involvement from the stakeholders while protecting the team from distractions.

212. Which statement best describes the term self-organized in agile?
a) A team that has no external dependencies.
b) A team that can prioritize the backlog independently.
c) A team that is responsible for making and delivering on its commitments.
d) A team that does not need a manager to get along.

Explanation: A team being self-organized does not mean there are no external dependencies (option A). It still requires the product owner to prioritize the backlog (option B). It also does not eliminate the need for a manager (option D). What it means is that the team is able to make commitments and accept responsibility and accountability for delivering their commitments.

213. How does pair programming work in extreme programming?
    a) A developer and a tester team up to work on a story.
    b) A customer representative and a developer jointly work on a requirement.
    c) A tester writes the test while a developer writes the code.
    d) **Two developers work together on a task.**

    Explanation: In pair programming, two programmers work in parallel on any piece of code, literally sharing a terminal. One of them works on the code and the other does an online review, provides comments, suggestions, and also, reviews alternative implementation methods.

214. During the estimation process, a team first ordered the stories from the smallest to the largest and then slotted them in the appropriate buckets of estimates (small, medium, large, extra-large). Which technique might be used here?
    a) Story pointing
    b) Affinity estimating
    c) Parametric estimation
    d) Expert estimation

    Explanation: In affinity estimating, stories are estimated by virtue of their affinity to one or the other categories. It forces you to make fairly coarse-grained judgments about the size, which is all you need for planning.

215. A team is experiencing a lot of conflicts. What is the best statement you can make about conflict management in agile?
    a) Conflicts are dangerous in agile and must be prevented at the source.
    b) You must try your best to reach a consensus in all circumstances.
    c) Conflicts may be healthy as long as they are resolved in a way that enhances trust and collaboration.
    d) The manager must act as the arbitrator in such conflicts to ensure a satisfactory resolution.

    Explanation: Conflicts may be good as it indicates that team members bring up different ideas and feel free to disagree. The manager needs to make sure that the conflicts are managed proactively and not allowed to degenerate into something detrimental to the team cause.

216. What is the primary role of a product owner on an agile team?
    a) To staff up the team and ensure that the morale of the team is good
    b) To manage the delivery of the software in a timely manner
    c) To estimate the size of the requirements
    d) To coordinate with the team to ensure they are working on items that are valuable to the customer

Explanation: The product owner is not a line management role (it is the job of the line manager to staff up and ensure morale is good). It is not a delivery management role (the team is responsible for the delivery). The product owner relies on the team estimates and uses them to plan the releases. PO is responsible for making sure that the team has a good set of requirements, and that they are ranked in the right order of priority. The PO tries to deliver maximum possible value to the customer in the shortest possible time.

217. A team finds it difficult to comprehend story points and continues to use ideal days to estimate tasks. Which of the following statements would you make to convince the team about the value of story points as a measure of size?
    - Story points drive cross-functionality in the team
    - Story points estimate do not change with time
    - Estimating tasks in story points can be faster
    - Story points are a pure measure"

    a) I and II only
    b) I and III only
    c) I, II, and IV only
    d) I, II, III, and IV

    Explanation: All the statements (I, II, III, and IV) are true. While deriving the story points, one has to take a holistic view of the work required to complete it (not just development or testing). Hence, it encourages team members to think in more cross-functional ways. Story point estimates are relative estimates; the numbers remain constant over a period of time. It is usually quicker to estimate in story points as one has good reference points to make comparisons against (as opposed to agreeing upon the number of man days to complete work). Story points help measure size purely because it is unit-less (and therefore less susceptible to individual biases while deriving man day estimates).

218. The objective of lean methodologies for work-in-progress is to:
    a) Maximize WIP
    b) Minimize WIP
    c) Have higher WIP for larger teams, and lower WIP for smaller teams
    d) Be neutral to WIP considerations

    Explanation: Work-in-progress means work that is lingering in the system, which requires you to maintain inventories. Lean would emphasize a small WIP (only to the extent that is required to ensure continuity).

219. What is a risk-burndown-chart?
    a) It plots the risk exposure against time for the duration of the project.
    b) It explains how to address the riskiest requirements first.

c) It indicates whether all the risks have been successfully identified or not.
d) It helps prioritize which risks to ignore and which to plan for.

Explanation: Risk exposure is calculated as the product of the probability of the event and the impact of the event (in terms of money or time or whatever unit you choose to track). The risk breakdown chart plots the sum of the risk exposures on all the identified risks on a project. It basically is a measure of the risk level on your project, which you should try to reduce over a period of time.

220. Your organization has outsourced work to a different organization, which follows agile methodologies. As the leader of the customer organization, which of the following is NOT your responsibility?
a) Ensuring that the right requirements are given
b) Ensuring that the requirements are drawn from a wide spectrum of users
c) Ensuring that clarifications about requirements are given in a timely manner
d) Ensuring that the vendor follows the agile methodologies correctly

Explanation: The performing organization and the team performing the work are probably best positioned to come up with the right interpretation of the process to be most effective. This is in the spirit of having a self-organized, empowered team. What the customer is responsible for is to give the right set of requirements, provide clarifications and guidance as needed about the prioritization of requirements.

221. What is the best way to facilitate communication between a pair of developers who are pair programming?
a) Put them in the same cubicle.
b) Put them in adjacent cubicles.
c) Buy video cameras to facilitate video chats.
d) Send them for communication skills training.

Explanation: Pair programming involves two programmers sharing and working on the same terminal, so adjacent cubicles will not work. It is not about being in sight of each other (so video chat is immaterial). Communication training is interesting but not directly related to pair programming, which requires being in the same office or cubicle, working on the same terminal.

222. You are reading a business case for a project and are planning to release a new version of a software product. You find that one of the figures mentioned in the document is incremental revenue. What is your understanding of incremental revenue?
a) Additional revenue from expansion in sales
b) Getting additional revenue by up-selling to existing customers

c) Revenue from annual maintenance contracts
d) A concept that emphasizes constantly growing revenues

Explanation: Incremental revenue means finding ways to get more revenue out of the existing customer base (by up-selling, upgrading, or offering more value).

223. Which of the following is NOT a characteristic of teams practicing extreme programming?
    a) On-site customer
    b) 40-hour work weeks
    c) Test-driven development
    d) Ability to scale to large teams

    Explanation: Extreme programming does not aim to scale to large teams. All other options are features of XP teams.

224. At the end of the iteration, a team which committed to delivering 68 story points ended up not delivering anything. What could most likely be the cause of this?
    a) Most of the team members went on vacation.
    b) The team was not self-organized.
    c) The scrum master did not manage the team well.
    d) The definition of done defined by the team was too stringent.

    Explanation: If the team was going on a vacation, it would likely have known beforehand, hence they would not have made the commitment. The scrum master, in any case, does not manage or drive the team, so that is also unlikely to be the cause. Self-organization is a desirable feature of an agile team, but the lack of it does not seem likely to cause a drastic failure to deliver even a single story point. What is more likely is that the definition of done (the criteria to decide whether to accept a story or not) was so stringent that the team under-estimated what was required to be done.

225. DSDM uses a variant of the Pareto principle to guide the development process. Which of the following best expresses how the principle might be used in this context?
    a) 20% of the features can deliver 80% of the value
    b) 20% of the value comes from 80% of the features
    c) 20% of the team delivers 80% of the work
    d) 20% of the features can deliver 100% of the value
    Explanation: The underlying principle behind rapid application development and DSDM is that 80% of the value that the customer is seeking really comes from 20% of the features. The art of delivering applications rapidly comes from

understanding the 20% features and then relentlessly focusing only on those features.

226. A product owner along with the team has categorized the items in the product backlog in terms of value and risk. They decide to work on high value and high risk items first, high value and low risk next, followed by low value and low risk. What should be done about the low value and high risk?
    a) Do them last, if at all.
    b) Find ways to increase the value or reduce the risk.
    c) Do not work on them.
    d) Run some pilots to validate the assumptions about value and risk.

    Explanation: Do them last. Before implementing such features, we need to make informed decisions as it may actually derail the areas of your application that are working well.

227. A team is wondering whether they should use off-the-shelf components, shareware, or free-ware components instead of developing them from scratch. What technique in agile project management will be useful to make that decision?

    a) De Bono's 6 thinking hats
    b) Planning poker
    c) Make-or-buy analysis
    d) Wide-band Delphi technique

    Explanation: Make-or-buy analysis is used to determine what components should be developed in-house and what should be externally procured.

228. A team is working on a product where the user experience is important. How will agile methods help while working on this project?
    a) They will help to maintain a consistent velocity and meet the user expectations.
    b) They will help deliver more with less due to the shorter iterations.
    c) They will help reduce the cost of delivery.
    d) They will help ensure the use of good design practices and frequent involvement of the users in development.

    Explanation: Agile facilitates frequent involvement of the users and the implementation of good design practices. These will help to improve user experience.

229. A team finds that the flow of work is often impeded by interruptions or delays. What would be a good way to address this?

a) Discuss these issues in the sprint reviews and try to find a solution.
b) Allocate a higher buffer for estimates to make more reasonable commitments.
c) Add more team members to deal with the interruptions.
d) Experiment with Kanban systems to make work visible and improve the flow of work.

Explanation: Kanban systems allow you to visualize the workflow and improve it.

230. Which of the following is characteristic of meetings in agile projects as opposed to conventional projects?
a) Agile recommends allocating sufficient time for frequent meetings
b) Agile meetings are timeboxed and have a fixed agenda.
c) Side discussions are not allowed in any agile meeting.
d) Agile recommends making notes in each meeting and then diligently following up on them.

Explanation: All agile meetings are timeboxed (have defined time limits) with a clear agenda and purpose.

231. Which of the following is most true about agile methodologies?
a) They focus on increasing process speed by eliminating waste.
b) They include an incremental and iterative life cycle for software development.
c) They rely on pull-based systems to deliver the product sooner.
d) They require people to be co-located.

Explanation: Agile methodologies emphasize an incremental and iterative approach to software development. It takes inspiration from lean, but it is not necessarily the same thing. The methodology relying on pull-based system is Kanban, which is related to agile but not quite the same thing. It helps if the team is co-located, but that is not really a central tenet of agile.

232. Agile teams are generally small. This is because smaller teams are:
a) Usually more productive and have less management overheads
b) More likely to deliver as per commitments
c) Unlikely to question the decisions of the product owner
d) Able to make suitable assumptions when the PO is not available

Explanation: Smaller teams are generally more productive as there is less overhead to manage and less social "loafing".

233. Typical roles in an agile team are:
   a) Team, scrum master, and customer
   b) Team, manager, and customer
   c) Team, leads, and architect
   d) Cross-functional team, team facilitator, and product owner

   Explanation: The team, team facilitator, and product owner are typical roles that play an important part in agile methods (agile practice guide).

234. During iteration planning, a team committed to implementing 10 stories. Coding has been completed and basic regression tests are passing. There are 12 bugs pending. Some of them are blocking few tests that execute some of the paths in the story. The fixes for these bugs were checked in on the last day of the iteration, but they have not been tested. In such a scenario, the iteration is:
   a) Successful because the basic functionality is implemented and is working
   b) Unsuccessful because all tests are not passed and few tests are still blocked
   c) Successful or unsuccessful depending upon the done criterion decided by the team
   d) Successful if the customer accepts it

   Explanation: The code at the end of each iteration should be of 'near releasable' quality. This has not happened or has not been validated yet, so the iteration cannot be deemed successful.

235. The manager of a scrum team is also playing the role of a scrum master. Which of the following things should the manager stop doing?
   a) Protect the team from disturbances
   b) Do performance appraisals for the team
   c) Make sure that all the required tasks are assigned and being worked on
   d) Think about growing the company business

   Explanation: Ideally, a manager should not play scrum master's role. But if the manager does, the following recommendation must be followed: A scrum master does not (should not) assign tasks. A self-managing team should do that on its own. All other functions are OK for a manager to do.

236. Which of the following ways would be useful to address poor quality of product increments?
   a) Cancel the iteration demo and continue working till the quality improves.
   b) Make use of test-driven practices and other mistake-proofing methods.
   c) Make use of shorter sprints.
   d) None of the above

Explanation: Agile practice guide recommends making use of test-driven-development and other mistake-proofing methods.

237. In planning poker estimation, apart from the estimates, what is the other desirable outcome from the estimation process?
    a) A guarantee that the sprint will be delivered as committed
    b) The ability to estimate a large number of stories in a short span of time
    c) Expertise in the business or domain
    d) Team buys into the estimates after discussions

    Explanation: The planning poker technique involves the entire team and provides clarity on assumptions and the selected approach for project management.

238. Which of the following is NOT an essential attribute of an agile team?
    a) It is self-managing.
    b) It is cross-functional.
    c) It is co-located.
    d) It is capable of delivering a potentially shippable product.

    Explanation: A scrum team need not be co-located. Scrum works with virtual teams as well. All the other options are characteristics of a scrum team.

239. In an agile team, who usually does NOT provide inputs for creating a backlog?
    a) Product owner
    b) Team
    c) Customer
    d) Competitor

    Explanation: The product owner is ultimately accountable for the product backlog. However, the PO may take inputs from several different sources, which may include the team members, customers, subject matter experts, senior management, etc. Competitive pressures may come into the definition of the backlog but usually competitors are not consulted while making a product backlog.

240. An agile team invited an external expert to provide inputs on what items they should work on. What is wrong with this scenario?
    a) Usually, it is the product owner who provides inputs to the team.

    b) The team facilitator should have been involved in the decision-making process.

c) Decisions should be made based on business value - not on technical merit alone.
d) The team should be making its own decisions as a self-organized team.

Explanation: Product owner is the only person authorized to direct the team in matters of the backlog (items to work on). All others, who want to influence the activities of the team, need to first convince the product owner.

241. In a weak matrix setting, you're the project manager steering a cross-functional team through a challenging project. Your team members don't functionally report to you, and you lack the authority to directly acknowledge their efforts. The project comes with tight deadlines and demanding quality benchmarks. In this situation, which type of project management power is likely to prove most effective?
a) Expert
b) Penalty
c) Formal
d) Referent

Explanation: In a context where the project manager lacks formal authority, their expertise and knowledge in the project domain serve as a powerful means of influence. This expert power, rooted in the manager's respected skills and experience, enables them to provide guidance and direction to the team, proving especially effective in navigating complex projects with tight deadlines and high quality standards.

242. As the project manager overseeing the creation of a novel power plant, your project is advancing well and approaching the product acceptance milestone. Which technique will play a crucial role in ensuring product acceptance?
a) Quality audit
b) Impact analysis
c) Team review
d) Inspection

Explanation: Inspection includes activities such as measuring, examining, and validating to determine whether work and deliverables meet requirements and product acceptance criteria. Inspections are sometimes called reviews, product reviews, and walkthroughs. In some application areas, these different terms have unique and specific meanings.

243. After progressing from project management to portfolio management in your current organization, you now oversee multiple project managers. Your company has secured a prestigious project similar to the one you executed years ago. You request the current project manager to consult the lessons learned from the previous project—where are these stored?
a) Projects record management system.

b) Organizational process assets.
c) Project performance reports.
d) Project status reports.

Explanation: The organizational process assets include historical information and lessons learned repository (e.g., project records and documents, information about the results of previous project selection decisions, and information about previous project performance).

244. In aiming to manage a project with a focus on identifying particular tasks requiring closer monitoring and management, which method should the project manager employ?
a) The GERT method
b) The PCDM method
c) The PERT method
d) The critical path method

Explanation: The critical path method is used to estimate the minimum project duration and determine the amount of schedule flexibility on the logical network paths within the schedule model.

245. Alice has engaged with Six Sigma in her workplace and observes similarities between Project Management and Six Sigma. She strongly believes in reviewing each project phase to assess whether it should proceed to the next stage. The review and decision to continue with the project is:
a) A milestone
b) Based upon stakeholder expectations.
c) Time-consuming and unnecessary for small projects.
d) A kill point

Explanation: In the context of Six Sigma and project management, a kill point refers to a decision point or review gate where the project team evaluates the progress, outcomes, and viability of the project before proceeding to the next phase. At this point, the team decides whether to continue with the project, make adjustments, or terminate it if it does not meet predefined criteria or objectives. Kill points help ensure that resources are allocated effectively and that projects remain aligned with organizational goals.

246. While reaching out to sellers for bids and proposals, you realize that ensuring a clear and shared understanding of the procurement process and project needs might be a time-consuming task. Which technique could assist in saving time in this regard?
a) Advertising
b) Qualified sellers list
c) Four-eyes meetings
d) Bidder conferences

Explanation: Bidder conferences (also called contractor conferences, vendor conferences, and pre-bid conferences) are meetings between the buyer and prospective sellers prior to proposal submittal. They are used to ensure that all prospective bidders have a clear and common understanding of the procurement and no bidders receive preferential treatment.

247. The customer is requesting a minor scope change and anticipates a prompt resolution without incurring additional costs. While you believe you have sufficient authorization to make the decision independently, there is some uncertainty. What should be your next course of action?
    a) Before making a decision you should have a look at the customer's parking lot. If you find there many expensive, new models, it is likely that you can use the requested change to increase the profit from the contract. Otherwise, you should reject the request.
    b) Handle the request according to the integrated change control processes described in your management plans. Then make a decision together with the appropriate change control body, whether the increased customer satisfaction will be worth the extra costs, work, risks, etc.
    c) A requested change is always an opportunity to get more money paid by the customer and to secretly solve schedule and quality problems. You should make some reasonable estimates on time, costs, risks, etc., and then add a nice margin on top of that to calculate the new price.
    d) Customer satisfaction is your top priority. The customer gives you an opportunity to increase their satisfaction, which you should use to the maximum benefit. Most project managers have contingencies to cover risks; these can be used to pay the additional costs.

    Explanation: Since the client is requesting a change in scope, the integrated change control process should be initiated to evaluate the impact of the change.

248. As the deadline for a project, which has been accepted, draws near, the project manager recognizes that only 75% of the work has been completed. In response, the project manager initiates a change request. What should the change request be authorized to do?
    a) Escalation approval to use contingency funding.
    b) Team overtime to meet schedule.
    c) Corrective action based on causes.
    d) Additional resources using the contingency fund.

    Explanation: In this case, the change request should lead to corrective action since the project is behind schedule.

249. In your role as the project manager, you've chosen to organize a team meeting to identify and analyze lessons learned from quality control with stakeholders. What actions should you take regarding these lessons?

a) Discuss them with management and make sure that they remain otherwise confidential.
b) Publish them in the corporate newsletter.
c) Follow your strategic decisions, independent from lessons learned. These decisions have been made and should be implemented whatever the outcomes are.
d) Document them and make them part of the historical database for the project and the performing organization.

Explanation: It's important information that will be used for upcoming projects and to be considered in risk management.

250. Collaborating with your team, you utilized three-point estimation on a critical path comprising two activities. The calculated duration uncertainties, based on a 3 sigma confidence interval, reveal that the first activity has a duration uncertainty of 18 days, while the second has an uncertainty of 24 days. When applying the PERT formula for paths, what is the overall duration uncertainty for the entire path?
a) 30 days
b) 42 days
c) No statement is possible from the information given
d) 21 days

Explanation: To calculate the overall duration uncertainty for the entire path using the Program Evaluation and Review Technique (PERT) formula, we need to apply the square root of the sum of the squares of the individual activity uncertainties.

Activity 1 uncertainty (σ_1) = 18 days

Activity 2 uncertainty (σ_2) = 24 days

σ path = $\sqrt{(\sigma_1^2 + \sigma_2^2)}$

σ path = $\sqrt{(18^2 + 24^2)}$

$= \sqrt{(324 + 576)}$

$= \sqrt{900}$

= 30 days

So, the overall duration uncertainty for the entire path is 30 days.

251. A few colleagues informed you that they are planning, executing, monitoring and controlling a project solely based on milestones, each with durations ranging from 1 to 4 weeks. What are your thoughts on this approach?
a) It is a good approach as long as the milestones reflect fixed or imposed dates during the project lifecycle only.

b) It is a good approach if the milestones are used for reviews between consecutive project phases only.
c) It is a good approach because it saves progress measurement on activities and work packages.
d) The approach is erroneous. A milestone is a significant point with zero duration to highlight achievements in monitoring and controlling a project.

Explanation: A project cannot be managed a monitored effectively by using only the milestones.

252. After creating a schedule network diagram with your team, determining estimates, defining durations, and identifying the Critical Path, you choose to perform a 'Forward Pass' through the Schedule Network Diagram. What specific calculations would you be conducting in this process?
a) The latest start and latest finish for each activity.
b) The duration of the other paths.
c) The duration of the critical path.
d) The Early Start and Early Finish for each activity.

Explanation: The Early Start and Early Finish for each activity. Option c and d are incorrect as the Forward Pass is not used to calculate the duration. Based on duration the critical path is calculated. Option b talks about Late Start and Late Finish - which are calculated using Backward Pass. Using 'Forward Pass' - we calculate the Early Start and Early Finish of all the activities in the network diagram.

253. Due to the diverse industries encompassed by the profession, many organizations have varied definitions for Project Management. As a result, the Project Management Institute (PMI) characterizes Project Management as:
a) Industry-specific because the definition changes based upon whatever industry the company is involved with
b) Initiation, planning, execution, control, and closeout of a project
c) The processes and tools to complete a project
d) The use of knowledge, tools, and techniques to meet requirements

Explanation: Project management is the use of specific knowledge, skills, tools, and techniques to deliver something of value to people.

254. Our team has successfully secured a software development project, bringing great satisfaction to the executive management due to its prestigious nature. However, you harbor concerns given the project's complexity. To address this, you initiate efforts to identify potential risks by gathering relevant stakeholders for extensive discussions. Despite generating a list of risks collaboratively, your experience leads you to believe that not all risks have been accounted for. In preparation for

unidentified but anticipated risks, you opt to allocate a budget. The question arises: where does this budget allocation originate?

a) Contingency Reserves
b) Management Reserve
c) Cost Management Plan
d) Cost Baseline

Explanation: Estimates may also be produced for the amount of management reserve of schedule for the project. Management reserves are a specified amount of the project budget withheld for management control purposes and are reserved for unforeseen work that is within the scope of the project. Management reserves are intended to address the unknown unknowns that can affect a project. Management reserve is not included in the schedule baseline, but it is part of the overall project duration requirements. Depending on contract terms, the use of management reserves may require a change to the schedule baseline.

255. Materials management plays a pivotal role in the project's success, particularly when resources must be available at a designated time and location to maintain continuous progress. When assessing materials costs, it's essential to consider not just the purchase cost but also _____.

a) Transportation, storage, and shortage.
b) Transportation, storage, and loss.
c) Transportation, deterioration, and storage.
d) Storage, damage, and deterioration.

Explanation: Transportation, storage, shortage and the cost of purchase should be considered when evaluating the cost of materials.

256. The triple constraints of Project Management are often a topic of discussion in various contexts, including marketing classes and other subjects. The interplay among _____, _____, and _____ can be visualized as a triangle, where the three sides influence one another.

a) Scope, quality, planning.
b) Time, scope, cost.
c) Quality, resources, time.
d) Money, resources, quality.

Explanation: The triple constraints of project management are time, scope, and cost. These three elements form a triangle where each side represents one of the constraints, and they interact with and influence each other throughout the project lifecycle.

257. In the project planning phase, you decide to maintain a strong hold on project risks. Collaborating with your team, you compile a list of potential risks and now

intend to prioritize them based on their probability and impact. In which process does this activity occur?

a) Risk Identification.
b) Risk Register.
c) Perform Qualitative Risk Analysis.
d) Perform Quantitative Risk Analysis.

Explanation: During the Perform Qualitative Risk Analysis process, the project team identifies and evaluates potential risks based on their probability and impact. The team assesses the likelihood and potential impact of each risk event and assigns a qualitative rating to prioritize them. This process helps identify high-priority risks that require further attention and mitigation efforts.

258. For projects to be efficiently and successfully planned and executed, they should encompass fundamental elements. Insufficient planning often leads to project failures, resulting in the loss of money and resources. Among the documents listed below, one specifically authorizes and initiates the project, empowering the project manager to commence work. This document is commonly referred to as a:

a) Project plan
b) Work authorization form
c) Project approval requisition
d) Project charter

Explanation: The project charter authorizes the project manager to start working on a project.

259. Your management is implementing the 10 percent solution: it has asked you to trim the budget for the project you are currently initiating by 10%. What implications is this likely to have for your project?

a) If the project budget is an honest one, then a reduction in budget must be accompanied by a trade-off in either time or performance.
b) A 10% reduction in the budget will be accompanied by a loss of performance much less than the target reduction in cost.
c) Most projects have "padded" estimates and the budgetary reduction will force out the padding from the project.
d) If the budget is cut by 10%, then the project will have an accompanying degradation of quality by 10%.

Explanation: If the project budget is an honest one, then a reduction in budget must be accompanied by a trade-off in either time or performance.

260. In managing a project, a project manager is handling a scenario where the initial scope baseline was set at a budget of $100,000. Since the project commenced, seventeen authorized and approved changes have been introduced, amounting to $17,000. Additionally, the cost incurred for investigating these changes before approval was $2,500. What is the current budget for the project?

a) 114500
b) 117000
c) 119500
d) 100000

Explanation: To find the current budget for the project, we sum the initial budget with the cost of authorized changes and the cost incurred for investigating these changes.

Initial Budget = $100,000

Cost of Authorized Changes = $17,000

Cost of Investigation = $2,500

Total Budget = Initial Budget + Cost of Authorized Changes + Cost of Investigation

Total Budget = $100,000 + $17,000 + $2,500

Total Budget = $119,500

So, the current budget for the project is $119,500.

261. The PMIS serves as a valuable tool for the project manager, acting as a communication mechanism throughout the project's duration. The communication plan is often treated as a sub-project within the PMIS development milestone. It is designed to collect, integrate, and disseminate the _____ of the Project Management processes.
a) Outputs
b) Tools
c) Deliverables
d) Scope changes

Explanation: The communication plan is frequently a sub-project of the PMIS development milestone that is used to gather, integrate, and disseminate the outputs of the Project Management processes.

262. While executing a project to construct a road bridge, your team discovered a significant flaw in the technical drawings. To prevent delays and address technical issues, they improvised a workaround on an ad-hoc basis. What should be your next course of action?
a) The workaround was performed on a technical level only. As long as it does not influence the function of the bridge or the organizational configuration of the project, a change request will not be necessary.
b) A retroactive formal change request side is not reasonable. Create some additional documents describing the workaround and sign them yourself, you can present them during the next regular CCB meeting.

c) It is normal in projects that during project execution inconsistencies arise between planning documents and actual implementation. This is no problem as long as the functional status of the product is maintained.
d) **Meticulously document the problem and the workaround to create a requested change to the project management plan, which will then be passed to the body responsible for the change control decision.**

Explanation: A Meticulously document the problem and the workaround to create a requested change to the project management plan, which will then be passed to the body responsible for the change control decision.

263. The level of uncertainty, characterized by a lack of information in a project, impacts the precision with which cost estimators can determine the total cost. One approach to address this uncertainty is influence diagramming, employed to assess _____.
   a) **Alternative outcomes, risk factors, and given quantities.**
   b) Risk factors, price, and alternatives.
   c) Price, quantities, and time of purchase.
   d) Price, net worth, and net present value.

   Explanation: Influence diagramming is a technique used in project management to assess and analyze alternative outcomes, risk factors, and given quantities. It helps project managers understand the relationships between different variables and factors that may impact the project's outcomes. By visualizing these relationships in an influence diagram, project teams can better identify and manage uncertainties and make informed decisions to address them.

264. Over an extended period, CHINA SPACE worked on the development of a space rocket. At the project's initiation, engineers had to create blueprints based on their vision of the final product. Emphasizing the result is crucial for successful Project Management. With each revision of the blueprints before final design approval, the team faced additional challenges, yet they recognized that:
   a) **Progressive elaborations of the blueprints accurately defined the final deliverable.**
   b) New subject matter experts would be needed.
   c) The project sponsor would develop a change control document.
   d) Scope creep was common in a large-scope project.

   Explanation: Progressive Elaboration. The iterative process of increasing the level of detail in a project management plan as greater amounts of information and more accurate estimates become available.

265. Confronted with a decision regarding an identified risk, a project manager is addressing a concern related to the design of a bicycle. The risk involves the potential corrosion of the bicycle's neck in a high salt environment, where the steering bearing is located and the two supporting bars of the frame converge. In

response, the project team determined that the bicycle's design should be altered by incorporating corrosion-resistant materials into the neck's design. This modification aims to eliminate the risk from consideration. This technique is commonly known as:

a) Risk acceptance
b) Risk rejection
c) Risk deflection
d) Risk avoidance

Explanation: Risk avoidance involves taking actions to eliminate the risk from consideration or to remove the risk entirely from the project. In this scenario, the project team modifies the design of the bicycle by incorporating corrosion-resistant materials into the neck's design to prevent the potential corrosion risk in a high salt environment. By making this modification, the team aims to avoid the risk altogether rather than accepting, rejecting, or deflecting it.

266. In a project budget, there are typically two types of reserves. The _____ reserve is allocated for cost variances resulting from inaccurate estimates, pricing discrepancies, or cost overruns. Meanwhile, the _____ reserve is set aside to cover costs related to project work that was inadvertently omitted from the plan due to errors or oversights.

a) Management; contingency allowance.
b) Management; mismanagement allowance.
c) Project; budget overrun allowance.
d) Engineering; management allowance.

Explanation: Contingency reserves are associated with the known-unknowns, which may be estimated to account for this unknown amount of rework. Management reserves are intended to address the unknown-unknowns that can affect a project. Management reserve is not included in the schedule baseline, but it is part of the overall project duration requirements. Depending on contract terms, use of management reserves may require a change to the schedule baseline.

267. You find yourself in a joyous state today, having successfully completed all the deliverables in a complex project fraught with challenges. Thanks to effective management and a motivated team, everything has been accomplished. As a project manager, what should be your next course of action?

a) Verify scope.
b) Disband the team.
c) Perform Quality Control.
d) Update lessons learned.

Explanation: After successfully completing all project deliverables, the next course of action for a project manager should be to update lessons learned. This

involves documenting what went well during the project, what challenges were faced, and what lessons can be learned for future projects**.**

268. Meetings serve various purposes and involve different parties. They can be categorized into types such as information sharing, problem-solving, decision-making, planning, and evaluation. The definition of a problem-solving meeting could be _____.
    a) Organized exchange of data and information.
    b) Selection of a course of action from options and alternatives.
    c) Future options and alternatives.
    d) Options and alternatives reviewed.

    Explanation: Problem-solving meetings are oriented around solving either a specific or general problem, and are perhaps the most complex and varied type of meeting. At problem-solving meetings, the outcome is often an important decision and thus these meetings can be crucial to the development of a team or product.

269. The financing plan for a project should outline the strategies for acquiring and allocating capital throughout the entire project duration. While formulating the plan and taking into account risk and reward, management should guarantee the preservation of participants' credit standing, utilize maximum leverage, optimize tax benefits, incorporate flexibility into ownership, and _____.
    a) Additional tax relief is pursued.
    b) Lines of credit are opened.
    c) Access to sources of new capital is maintained.
    d) Maximum financing is used.

    Explanation: In the financing plan for a project, it is essential to ensure access to sources of new capital to support the project's ongoing funding requirements and potential expansion or unforeseen circumstances. This helps maintain financial stability and flexibility throughout the project duration.

270. Regardless of the contract type, the owner (buyer) consistently prioritizes the project's cost and the contractor's (seller's) expenditure rates. While a fixed-price contract shifts a significant portion of the risk to the contractor, the owner still faces substantial risks in achieving objectives. One of the most effective methods for controlling contractor expenditure rates under any contract type is to _____.
    a) Have an owner's representative" monitor and track all expenditures and have the power to disapprove any expenditure that is not justified to him/her.
    b) Track and pay the costs of the contract by phases or stages and evaluate the costs at each review/decision point.
    c) Hold back a portion of the payment until all work is completed to the satisfaction of the technical inspectors.

d) Ensure that only fixed-price contracts are awarded, and that payment is only upon satisfactory completion of the project.

Explanation: Tracking, paying, and auditing the invoices is a regular task performed by the buyer irrespective of the type of contract.

271. Your team has recently secured a software development project, eliciting great satisfaction from the executive management due to its prestigious nature. However, your apprehension grows due to the project's complex nature. In response, you take proactive measures to identify potential risks by convening all relevant stakeholders for extensive discussions. Despite collaboratively pinpointing several risks, your experience leads you to believe that not all risks have been identified. Consequently, you make the decision to allocate a budget for unidentified yet anticipated risks. The question arises: where does this budget allocation originate?
    a) Management Reserve.
    b) Cost Management Plan.
    c) Cost Baseline.
    d) Contingency Reserves.

    Explanation: Management reserves are intended to address the unknown-unknowns that can affect a project. Management reserve is not included in the schedule baseline, but it is part of the overall project duration requirements. Depending on contract terms, use of management reserves may require a change to the schedule baseline.

272. The WBS and WBS dictionary have been finalized, and the project team has initiated the identification of risks. The sponsor reaches out to the project manager, asking for the responsibility assignment matrix. The project, with a budget of $100,000, spans three countries and involves 14 human resources. The anticipated risk for the project is minimal, and the project manager has experience managing similar projects. What should be the next step?
    a) Make sure the project scope is defined.
    b) Complete risk management and issue the responsibility assignment matrix.
    c) Understand the experience of the sponsor on similar projects.
    d) Create an activity list.

    Explanation: After the creation of the WBS, the deliverables have to be decomposed into activities.

273. Engaged in a bridge construction project, you have reached the intended halfway point. The total planned cost at this juncture amounts to five hundred dollars. The tangible progress achieved thus far is valued at $400. However, the expenses incurred on the project have already reached one thousand dollars. What is the CPI (Cost Performance Index)?
    a) 0.8

b) 1.2
c) 0.5
d) 0.4

Explanation: The Cost Performance Index (CPI) is calculated as the ratio of Earned Value (EV) to Actual Cost (AC).

CPI = earned value (EV) / actual cost (AC)

So, the Cost Performance Index (CPI) is 0.4.

274. You are currently in the Source Selection process for a contract related to your project. In this context, you intend to employ a weighing process, with two crucial factors being price (weighted at 40) and quality (weighted at 60). Four vendors are under consideration, and their computed scores by your team are presented in the table below:
    - Vendor 1 (price, quality): 8, 12
    - Vendor 2 (price, quality): 10, 10
    - Vendor 3 (price, quality): 12, 8
    - Vendor 4 (price, quality): 11, 11

    Given this information, which vendor among the four should be selected?

    a) Vendor 1
    b) Vendor 2
    c) Vendor 3
    d) Vendor 4

    Explanation: To determine which vendor to select, we need to calculate the overall score for each vendor based on the weighted factors of price and quality.

    Weight for price = 40

    Weight for quality = 60

    For Vendor 1:

    Overall score = (Price score * Weight for price) + (Quality score * Weight for quality)

    = (8 * 40) + (12 * 60)

    = 320 + 720

    = 1040

    For Vendor 2:

    Overall score = (10 * 40) + (10 * 60)

= 400 + 600

= 1000

For Vendor 3:

Overall score = (12 * 40) + (8 * 60)

= 480 + 480

= 960

For Vendor 4:

Overall score = (11 * 40) + (11 * 60)

= 440 + 660

= 1100

Vendor 4 has the highest overall score of 1100, so Vendor 4 should be selected.

275. You are overseeing the project to organize a party for your company's employees. A risk has been identified that the employees may not attend the party. Assessing the low likelihood of this occurrence, you opt not to implement any specific actions in response. What risk response strategy are you employing?
    a) Avoidance
    b) Mitigation
    c) Transference
    d) Acceptance

    Explanation: Accepting risk, or risk acceptance, occurs when a business or individual acknowledges that the potential loss from a risk is not great enough to warrant spending money to avoid it.

276. In need of outsourcing work, a project manager encounters a considerable level of risk associated with the task, making it challenging to find a contractor willing to undertake the job. Among the following types of contracts, which would provide the highest incentive to the contractor?
    a) Cost plus fixed fee.
    b) Cost plus incentive fee.
    c) Firm fixed price.
    d) Cost plus award fee.

    Explanation: Cost plus award fee (CPAF). The seller is reimbursed for all legitimate costs, but the majority of the fee is earned based on the satisfaction of certain broad subjective performance criteria that are defined and incorporated into the contract. The determination of fee is based solely on the subjective

determination of seller performance by the buyer and is generally not subject to appeals.

277. Under a fixed-price contract, a contractor is committed to receiving a single lump-sum payment upon the satisfactory completion of the contract. At the midway point, the contractor's project manager communicates financial challenges, hindering their ability to compensate employees and subcontractors. Seeking a partial payment for completed work, the contractor requests intervention. Among the buyer's potential actions, which is most likely to pose problems for the project?
    a) Adhering to the contract and making no early payments.
    b) Paying for all work accomplished to date only.
    c) Demanding penalty-free options to make early payments.
    d) Starting partial payments for work accomplished.

    Explanation: Adhering to contract at this scenario is not the best option since this will create delays and bureaucracy in the project.

278. As the Project Manager tasked with constructing a next-generation vehicle, you recognize a dependency: the vehicle must obtain approval from the Government regulatory authority before it can be sold. This is an example of which type of dependency?
    a) Discretionary
    b) Soft Logic
    c) Mandatory
    d) External

    Explanation: External dependencies. External dependencies involve a relationship between project activities and non-project activities. These dependencies are usually outside of the project team's control. For example, the testing activity in a software project may be dependent on the delivery of hardware from an external source, or governmental environmental hearings may need to be held before site preparation can begin on a construction project. The project management team determines which dependencies are external during the process of sequencing the activities.

279. Recently appointed as the Project Manager for the ABC project, you encounter difficulty comprehending certain components of the Work Breakdown Structure (WBS). What should be your initial step to gain a better understanding?
    a) Refer to the scope statement.
    b) Refer to the Project charter.
    c) Contact the previous Project Manager.
    d) Refer to the WBS dictionary.

    Explanation: The WBS dictionary will contain all of the detailed information related to each of the WBS work packages.

280. To manage the schedule effectively, a project manager is reanalyzing the project to forecast project duration. This involves scrutinizing the sequence of activities with minimal scheduling flexibility. Which technique is she employing?
   a) Flowchart
   b) Precedence diagramming
   c) Work breakdown structure
   d) Critical path method

   Explanation: The critical path method is used to estimate the minimum project duration and determine the amount of schedule flexibility on the logical network paths within the schedule model.

281. As the project manager of the ABC project, you encountered a mid-project discovery of a defective key component, which was not anticipated. The team convenes after this event and successfully ensures the product functions without the defective component. This is an example of:
   a) Transfer of risk
   b) Workaround
   c) Accepting the consequences passively
   d) Risk mitigation

   Explanation: Workarounds are responses to unidentified risks or passively accepted risks. Put simply, if any unidentified risk occurs, you will manage it through a workaround.

282. Your company has recently secured a project involving the layout and routing of electric cables for a large shopping complex, and you've been assigned to provide the cost estimates. Drawing on your past experience with similar projects, you break down the costs into the cost per foot of cables and the unit cost of cable trays. What type of cost-estimating technique are you employing?
   a) Analogous cost estimate
   b) Top-down cost estimate
   c) Bottom-up cost estimate
   d) Parametric estimates

   Explanation: A parametric estimate is an estimate of cost, time, or risk that is based on a calculation or algorithm. As the name suggests, parametric estimates are based on parameters that define the complexity, risk, and costs of a program, project, service, process, or activity.

283. Investors are frequently utilized to secure essential financing for significant projects, potentially leading to the transfer of ownership to the investors. Subsequently, the system is leased back for operation and use. The primary benefit for the investors is _____, while the primary advantage for the system user is _____.
   a) Control of a business; low risk for operation and maintenance

b) High internal rate of return; favorable tax write-off
c) Favorable tax write-off; new source of funds
d) Ownership without risk of operation; favorable tax write-off

Explanation: The primary benefit for investors is a favorable tax write-off through deducting lease payments, while the system user gains a new source of funds by leasing back the system, enabling immediate access to capital while retaining operational control.

284. Imagine the project manager is called to an information meeting with the customer, where a significant problem is brought up. This issue has substantial implications for the project manager's company, but the customer is eager to explore a solution during the meeting. In this situation, the project manager should _____.
a) Collect as much information on the problem without committing his/her company to a solution.
b) Tell the customer that he/she will not address any problems because this is only an information exchange meeting.
c) Give the customer a range of solutions that might be acceptable to his/her company.
d) Give the customer an interim solution that must be approved by his/her boss.

Explanation: At this stage the Project Manager has to collect client's feedback and then analyze it internally. After the internal impact assessment on cost, schedule, scope, he/she can come back with a proposed approach.

285. Entering as a project manager in the later stage of the project, you discover numerous defects during the final testing phase. To address this, you decide to prioritize and fix defects based on the schedule. In this scenario, which of the following tools would be the most effective for you to use?
a) Flow Chart
b) Pareto Chart
c) Control Chart
d) Run Chart

Explanation: A Pareto chart is a histogram that divides discrete observations into several categories to identify the “vital few.” The phrase “vital few” is used to describe the elements that have the maximum impact on the solution.

286. You've been designated as the project manager for an international undersea cable project, and a specific segment of the project has been subcontracted to another company. What is the MOST probable result of this collaboration?
a) Number of communication channels will increase
b) Performance report will be more inclusive and detailed
c) Informal written communication will increase

d) Formal written communication will increase

Explanation: Since there is involvement of an external company this means that there will be added more stakeholders in the project and thus, the communications channels will increase.

287. A project is initially planned for 6 months with a budget of $9,000,000. By the end of the 2nd month, the project is 30% complete, and the cost incurred is $3,250,000. What is the Cost Performance Index (CPI) of the project?
   a) 0.831
   b) 0.746
   c) 1.046
   d) 0.946

   Explanation: The Cost Performance Index (CPI) is a method for calculating the cost efficiency and financial effectiveness of a specific project through the following formula: CPI = earned value (EV) / actual cost (AC).

288. As the project manager overseeing a construction project spanning 3 years, you're six months into the project when one of your contractors reveals a risk that was previously unidentified and omitted from the risk register. In response, you engage with an expert to identify a workaround for the project. Which process are you currently implementing?
   a) Perform Quantitative Risk Analysis
   b) Plan Risk Responses
   c) Control Risks
   d) Perform Qualitative Risk Analysis

   Explanation: Control Risks is the process of implementing risk response plans, tracking identified risks, monitoring residual risks, identifying new risks, and evaluating risk process effectiveness throughout the project.

289. A project team is diligently working towards project delivery, having dedicated over a year and approaching the completion of the execution phase. During this period, one team member uncovers a risk that was not previously identified. What would be the BEST course of action for the manager of the project team?
   a) Assess the Risk
   b) Update the Risk Management Plan
   c) Ignore the Risk
   d) Find ways to Mitigate the Risk

   Explanation: When a risk is first identified, the next course of action is to assess the impact of the risk.

290. Michael serves as the project manager for a multinational project, with the team distributed across four locations. In navigating this virtual team structure, Michael has decided to extensively utilize video-conferencing technologies for

management purposes. Identify two true statements regarding the usage of video-conferencing technologies for team management.

a) **Body language and tone of voice play a vital role in effective communication**.
b) Video-conferencing technology ensures faster decision-making.
c) Video-conferencing tools will eliminate the need for co-location of the project team.
d) Face-to-face communications have a better impact.

Explanation: Teams have been demonstrated to communicate better face-to-face, and there is also evidence of the criticality of being able to assess body language and tone of voice for communication to be effective.

291. In your role as a project manager for a construction project, you've crafted the initial project plan and presented it to the customer. The customer, however, notifies you that the deadline date is incorrect, expressing the desire for the first phase delivery to be completed within the initial 6 months. They explicitly state that if the delivery cannot be achieved within this timeframe, they will not proceed with the project. In which section of the project plan will you incorporate this information?

a) Project Constraints
b) Project Assumptions
c) Project Milestones
d) Project Goals

Explanation: Project constraints are limiting factors for your project that can impact quality, delivery, and overall project success.

292. As the project manager overseeing the development of a new product with a four-month project duration, the product building stage is presently underway. A new regulation, applicable to this product, is set to take effect in six months. Failure to comply with this regulation could necessitate the halt of product production. In this situation, what is the responsibility of the project manager?

a) Continue with the original project plan and complete development of the product.
b) Schedule a meeting with the key stakeholders. If it is agreed, issue a change request for scope and other changes to comply with the new law.
c) Inform the team to change the product design to consider compliance requirements as per new law.
d) Meet with the legal team and seek guidance to claim exemption from the requirements as per new law.

Explanation: Addressing regulatory requirements may require the implementation of changes. The change control procedures should be followed.

293. In the development project for a solar energy park, the budget has been determined by factoring in the government subsidy. However, during the project's build stage, it becomes apparent that, due to a specific clause in the subsidy policy, the project is ineligible for the anticipated subsidy. What should be the project manager's immediate next step?
   a) Log it as an issue in the issue log and schedule a meeting with the key stakeholders.
   b) Raise a change request to reduce the scope of the project to fit within the available funds.
   c) Raise a change request to revise the project cost baseline.
   d) Recommend termination of the project as it cannot be completed without receipt of subsidy.

   Explanation: As this is a current issue, it should be handled, and appropriate action taken by the project management team.

294. In a project with an approved budget of USD 400,000, the monthly status report provides the following information: Planned value - USD 250,000, Actual cost - USD 275,000, Earned value - USD 240,000. Which of the following statements holds true for this project?
   a) It is over budget and behind schedule.
   b) It is under budget and ahead of schedule.
   c) It is over budget and ahead of schedule.
   d) It is under budget and behind schedule.

   Explanation: When actual cost is more than earned value, the project is over budget; and when planned value is more than earned value, the project is behind schedule.

295. Shortly after the issuance of a project charter, the business analyst alerts the project manager to a new regulation by the local government that could potentially affect the financial viability of the project. What should be the project manager's immediate course of action?
   a) Update the risk register with the new risk identified.
   b) Gather more details about the regulation and present your findings to the sponsor and key stakeholders.
   c) Inform the sponsor to cancel/terminate the project.
   d) Go ahead with planning activities as the project has been authorized and a project charter is issued.

   Explanation: Any issue or concern relating to the project must be addressed by collecting more information and assessing the impact on project objectives.

296. You've been designated as the project manager for a recently approved strategic opportunity, and while the project charter is pending issuance, you have received information about the high-level scope, key milestones, and risks of the proposed

project. At this stage, what other crucial documents or information should you examine?

a) The reasons for selecting this project, the business case, and information on key stakeholders.
b) The project management software and the knowledge repositories should be used for managing the project.
c) The scope, schedule, and cost baselines.
d) The project management plan templates and lessons learned from previous projects.

Explanation: The project charter will require a list of assumptions, the preliminary scope, and some of the restrictions. Additionally, the project goals and business case will also be needed.

297. The project team is engaged in upgrading banking software, with one of the key constraints being the high level of data security needed. In addition to this, the project team must ensure compliance with requirements about banking data security. How should this requirement be addressed?

a) Organize more frequent unscheduled security audits and check for compliance failures if any.
b) Develop data security policies and standards to be followed by the project team. Create awareness of the policies and standards, and ensure compliance.
c) Outsource the project to a banking technology specialist organization.
d) Inform the stakeholders that the team will do its best and that they will not be accountable for any security breaches.

Explanation: The quality management plan should identify the compliance requirements and detail the processes and procedures to be adhered to.

298. The project management plan must incorporate details on how compliance requirements are addressed. This may involve creating a separate project compliance plan or integrating relevant sections into the subsidiary plans. Which of the following is not specified as a compliance requirement in this plan or section?

a) Team norms and ground rules
b) Quality and risk
c) Corporate policies
d) Government regulations

Explanation: Team norms and ground rules are typically addressed in the team charter or team agreement, rather than in a compliance plan. Compliance requirements usually focus on adherence to corporate policies, government regulations, quality standards, and risk management protocols within the project.

299. In the process of project planning, the project management team is required to formulate a plan for handling compliance requirements. Regarding project compliance management, what is the most crucial aspect to evaluate when processing changes requests?
   a) The category and the type of change request.
   b) The consequences of non-compliance.
   c) The organization that requires compliance.
   d) The knowledge of the project team in the relevant area.

   Explanation: It is important to assess the consequences of non-compliance as they may have a huge impact on the project.

300. You are managing a project involving the construction of a skyscraper. However, in adherence to regulations, no work can commence until the government body approves and provides the necessary clearance. This situation exemplifies what type of dependency?
   a) Discretionary Dependency
   b) External Dependency
   c) Mandatory Dependency
   d) Hard Logic

   Explanation: External dependencies involve a relationship between project activities and non-project activities. These dependencies are usually outside of the project team's control. For example, the testing activity in a software project may be dependent on the delivery of hardware from an external source, or governmental environmental hearings may need to be held before site preparation can begin on a construction project. The project management team determines which dependencies are external during the process of sequencing the activities.

301. Your organization is initiating a project that involves an investment of $1,000,000. The projected revenues from the project are estimated to be $250,000 in the first year after the project concludes and $420,000 in each of the two subsequent years. What holds true for the net present value of the project over the three-year cycle at a discount rate of 10%?
   a) The net present value is negative, which makes the project unattractive.
   b) The net present value is positive, which makes the project attractive.
   c) The net present value is positive, which makes the project unattractive.
   d) The net present value is negative, which makes the project attractive.

   Explanation: The NPV formula. It's important to understand exactly how the NPV formula works in Excel and the math behind it. NPV = F / [(1 + r)^n] where, PV = Present Value, F = Future payment (cash flow), r = Discount rate, n = the number of periods in the future is based on future cash flows.

302. Alice is in charge of a project with a projected budget at completion of $500,000. According to the schedule, her team should have accomplished 40% of the work by now. Nevertheless, in the recent status meeting, the team disclosed that only 35% of the work has been completed. The project has incurred expenses totaling $225,000. As a supervisor overseeing Alice's project, how would you MOST accurately characterize the project?
   a) The project is behind schedule and over its budget.
   b) The project is behind schedule and within its budget.
   c) The project is ahead of schedule and within its budget.
   d) The project is ahead of schedule and over its budget.

   Explanation: SPI<1 indicates that the project is behind schedule, while CPI<1 indicates that the project is over its budget. Therefore, both SPI and CPI being less than 1 accurately characterize the project as behind schedule and over its budget.

303. Project A holds a net present value of $80,000 with an estimated completion time of six months. Project B, with a net present value of $120,000, is projected to take two years to finish. Project C boasts a net present value of $30,000 and a benefit-cost ratio of 5:2. Lastly, Project D has a net present value of $45,000 and a completion duration of 3 months. Which project would be the optimal choice?
   a) Project A
   b) Project B
   c) Project C
   d) Project D

Explanation: Project B has the highest net present value ($120,000), indicating the highest potential profitability among the options presented. While Project A has a higher NPV per month ($80,000 / 6 months = $13,333) compared to Project B ($120,000 / 24 months = $5,000), Project B still offers the highest total NPV. Therefore, based on maximizing net present value, Project B would be the optimal choice.

304. A project commences in accordance with the approved scope. During the implementation of the project design, a team member raises concerns that the product will not meet the requirements in specific situations. Rectifying this issue would not only lead to an expansion of scope and increased costs but would also result in missing the project deadline. Additionally, the project sponsors lack the necessary additional funding to support the project. Given this information, what course of action should the Project Manager take?
   a) Enact the perform integrated change control process.
   b) Resolve the issue to satisfy the customer.
   c) Contact the customer and negotiate for risk acceptance.
   d) Conduct a risk analysis and inform the project sponsor.

Explanation: The risk register exists for the project team and stakeholders to understand potential problems with the project. Since the team member may not have all the answers, it is necessary to first perform a full risk analysis and inform the project sponsor before making any decisions.

305. The project manager recognizes that conflicting stakeholder interests are likely to arise in a project and opts to employ a stakeholder engagement assessment matrix. The matrix identifies a key stakeholder who holds a resistant stance towards the project. In order to align with this stakeholder's expectations, what actions should the project manager take to achieve this objective?
    a) Convey the stakeholder engagement plan.
    b) Gain consensus on project objectives by meeting with all stakeholders.
    c) Ensure that the stakeholder register is complete and updated.
    d) Produce a stakeholder engagement plan.

    Explanation: Plan Stakeholder Engagement is the process of developing approaches to involve project stakeholders based on their needs, expectations, interests, and potential impact on the project. The key benefit is that it provides an actionable plan to interact effectively with stakeholders. PMBOK Guide Sixth Edition (2018) PMI/PMI/13.2/516

306. A medium-sized company is venturing into a new market without prior experience. In this context, a Project Manager is tasked with collecting requirements. What tool or technique should the Project Manager employ for this purpose?
    a) Prototypes
    b) Expert judgment
    c) Product analysis
    d) Create requirements traceability matrix

    Explanation: Experts come with many lessons learned. Given the increased risks of a new and potentially complex market, using expert judgement is recommended.

307. A project manager oversees a project with a mix of local and virtual team members. To address scheduling priority conflicts among team members, the project manager arranges a video conference call. Which statement accurately describes this scenario?
    a) It is an example of the Collaborate/Problem Solve conflict resolution technique.
    b) It demonstrates the Force/Direct conflict resolution technique.
    c) It is an example of the Compromise/Reconcile conflict resolution technique.
    d) It demonstrates the Smooth/Accommodate conflict resolution technique.

    Explanation: Collaborate/problem solve. Incorporating multiple viewpoints and insights from differing perspectives

308. A high-level manager within an organization expresses the desire to utilize a critical project resource for a new project. What course of action should the project manager pursue?
   a) Request that the high-level manager provides a replacement.
   b) Use the Legitimate form of power to deny the request.
   c) Tell the team and reassign that resource's responsibilities to other team members.
   d) Contact the program manager to resolve the issue.

   Explanation: The project management team's ability to influence others plays an important role in negotiating resource allocation, as does the politics of the organizations involved. If getting to an agreement is not possible, the issue needs to be escalated for resolution. PMBOK Guide Sixth Edition (2018) PMI/PMI/3.3/52

309. To secure a stakeholder's approval when the stakeholder perceives that a project's deliverables were not met, what steps should the project manager take?
   a) Review the project charter thoroughly with this stakeholder on a conference call.
   b) Ensure that the stakeholder understands the quality management plan.
   c) Perform a risk analysis after obtaining the stakeholder's new requirements.
   d) Review the project scope while analyzing the stakeholder's needs and expectations.

   Explanation: Monitor Stakeholder Engagement is the process of monitoring project stakeholder relationships and tailoring strategies for engaging stakeholders through modification of engagement strategies and plans. The key benefit of this process is that it maintains or increases the efficiency and effectiveness of stakeholder engagement activities as the project evolves and its environment changes. This process is performed throughout the project. PMBOK Guide Sixth Edition (2018) PMI/PMI/13.4/530

310. During a status meeting, an employee from the client's side requested an additional new product feature from the project manager. In order to identify the individual with the authority to approve this request, which document should the project manager consult first?
   a) Change management plan
   b) Responsible, accountable, consult, and inform (RACI) matrix
   c) Change request document
   d) Change log

   Explanation: The change management plan describes the process for submitting, evaluating and implementing changes to the project. PMBOK Guide Sixth Edition (2018) PMI/PMI/4.2.3.1/87

311. Review the following steps:

1. Successful completion of a project.

2. The project manager is in the process of formal closure.

3. _____.

What is Step 3?

a) The project manager should obtain stakeholder acceptance.
b) The project manager should reassign the project team.
c) The project manager should reallocate the unused funds.
d) The project manager should update the bid documents.

Explanation: Deliverables that meet the acceptance criteria are formally signed off and approved by the customer or sponsor. Formal documentation received from the customer or sponsor acknowledging formal stakeholder acceptance of the projects deliverables is forwarded to the Close Project process. PMBOK Guide Sixth Edition (2018) PMI/PMI/5.5.3.1/166

312. Review the following project characteristics:

- The project is currently in its initiation phase.
- Due to insufficient funding, the project faces termination.
- The project manager is required to close the project.

What is the initial action the project manager should undertake?

a) Examine the organization's project closure guidelines.
b) Determine the potential impact of terminating the project by performing a risk analysis.
c) Close all project procurement processes.
d) Rely on the communications management plan to properly inform all stakeholders.

Explanation: Project closure guidelines or requirements will contain the organization's procedures to follow. PMBOK Guide Sixth Edition (2018) PMI/PMI/2.3.1/41

313. A project exhibits the following characteristics:

- A functional manager serves as a project stakeholder.
- Due to time constraints, the functional manager delegates project responsibilities to another member of the functional team.
- The newly assigned stakeholder is not actively participating in meetings.

To involve this new stakeholder, what actions should the project manager take?

a) Present the project schedule and the specific, assigned tasks to the new stakeholder.
b) Request that the functional manager briefs this new stakeholder and ask that they send them to all relevant meetings.

c) Ask the new stakeholder to pair with a more experienced stakeholder to learn responsibilities.
d) Examine the project charter with the new stakeholder and assess their level of engagement.

Explanation: The first version of the stakeholder engagement plan is developed after the initial stakeholder community has been identified by the Identify Stakeholder process. The stakeholder engagement plan is updated regularly to reject changes to the stakeholder community. PMBOK Guide Sixth Edition (2018) PMI/PMI/13.2.3.1/522

314. During a periodic review with stakeholders, the project manager receives a query from the project sponsor regarding the verification and acceptance process for deliverables. Where can the project manager locate this information within the project documentation?
a) Requirements management plan
b) Quality management plan
c) Work performance reports
d) Scope management plan

Explanation: The quality management plan identifies the quality requirements and/or standards for the project and its deliverables, and documents how the project will demonstrate compliance with quality requirements and/ or standards. PMBOK Guide Sixth Edition (2018) PMI/PMI/8.1.3.1/324

315. In the planning phase of a project set in a remote area with restricted vehicle and equipment access, the project manager suggests that the company handle the delivery of all heavy equipment, even though it entails substantial costs. The project manager is willing to assume full responsibility for this undertaking. What category of risk response is the project manager showcasing?
a) Accept
b) Avoid
c) Transfer
d) Mitigate

Explanation: Risk acceptance is a risk response strategy whereby the project team decides to acknowledge the risk and not take any action unless the risk occurs. PMBOK Guide Sixth Edition (2018) PMI/PMI/11.5.2.4/443

316. Upon receiving the project charter, a project manager initiates the preparation of the project management plan. In this context, the project management office (PMO) lacks historical information. What should be the subsequent step for the project manager in this scenario?
a) Utilize analogous estimating.
b) Seek advice from experts in the field.
c) Organize an initial kick-off call.

d) Request that the project sponsor provides input.

Explanation: If there is no historical information available, then consulting with subject matter experts is the next best thing a project manager can do to prepare the project plan. PMBOK Guide Sixth Edition (2018) PMI/PMI/8.1.2.2/281

317. The marketing team is in the process of creating a new website and has established a target audience. The next step involves identifying the style preferences of the target audience. In this situation, what would be a suitable tool or technique to utilize?
    a) Brainstorming
    b) Product analysis
    c) Benchmarking
    d) Focus groups

    Explanation: Focus groups bring together prequalified stakeholders and subject matter experts to learn about their expectations and attitudes about a proposed product, service, or result. PMBOK Guide Sixth Edition (2018) PMI/PMI/5.2.2.2/180

318. A client introduces a Project Manager to a project during its execution stage. The Project Manager examines the project scope for cost estimating and observes that certain completed deliverables were omitted in the work breakdown structure (WBS). What should be the subsequent step for the Project Manager in this situation?
    a) Include the work in the WBS, per the 100% rule, and recalculate the total project cost.
    b) Determine whether these deliverables are necessary by verifying with experts in the field.
    c) Verify with the stakeholders to see if these missed deliverables should be added.
    d) Ask that the project management office (PMO) approve additional funds for this work.

    Explanation: The 100% rule states that any work package in the WBS should consist of 100% of the work for that work package. Since the WBS is a work package itself (the project) and since the WBS should contain only work packages that are in-scope, any missing work packages should be added to the WBS and recalculated.

319. A project manager is collaborating with three crucial stakeholders, each holding a distinct perspective on the project and its objective. These stakeholders are firm in their opinions. The project manager must secure approval for the project charter. What actions should the project manager take in this situation?
    a) Request that the project sponsor resolve the conflict.
    b) Use the project kick-off meeting to settle the project goals.

c) Conduct a stakeholder analysis.
d) Use negotiation to ensure the three stakeholders are in agreement.

Explanation: Projects experiencing a high degree of change require active engagement and participation with project stakeholders. To facilitate timely, productive discussion and decision making, adaptive teams engage with stakeholders directly rather than going through layers of management. PMBOK Guide Sixth Edition (2018) PMI/PMI/12.2.2.5/488

320. Examine the following project attributes:
- A sponsor gives the green light to a project.
- The project manager assembles chosen team members, key stakeholders, and the sponsor for a meeting.
- The principal objective of the meeting is to produce a document that will assist the team in outlining the primary project deliverables.

What will be the ultimate outcome of this meeting?

a) Scope management plan
b) Requirements management plan
c) Milestone list
d) Work breakdown structure (WBS) dictionary

Explanation: The scope management plan is a component of the project management plan that describes how the scope will be defined, developed, monitored, controlled, and validated. PMBOK Guide Sixth Edition (2018) PMI/PMI/5.1.3.1/137

321. A project is currently in the development stage, and during this phase, sensitive client data will be handled. The compliance manager raises a concern about project non-compliance. What is the initial document the project manager should examine?
a) Issue log
b) Business case
c) Risk register
d) Statement of work (SOW)

Explanation: After a new issue is raised, the issue log should be reviewed prior to taking action. PMBOK Guide Sixth Edition (2018) PMI/PMI/4.3.3.3/487

322. A project is in progress and has reached 50% completion. A crucial project team member requests a 1.5-month leave for personal reasons. What should be the project manager's initial course of action?
a) Discuss the issue with the functional manager and request a replacement.
b) Assess how the leave will impact the project.
c) Refer to the resource management plan.
d) Check with this team member to see if the leave can be postponed.

Explanation: The FIRST thing the project manager needs to do is to understand what impact (if any) this absence will have on the project. PMBOK Guide Sixth Edition (2018) PMI/PMI/11.3.2.3/461

323. A project manager is employed by a company gearing up to create a new product. The development of this new product necessitates skills that are currently unavailable. What steps should the project manager take to kickstart this project?

    a) Focus on quality control
    b) Focus on the assessment of high-level risks
    c) Develop a competitor analysis
    d) Develop detailed project requirements

    Explanation: If you plan to deliver a product and do not possess the skills required, this is a risk. PMBOK Guide Sixth Edition (2018) PMI/PMI/11.1.1.4/441

324. A project manager, facing uncertainty about the duration of a new product, seeks input from various groups of subject matter experts (SMEs). The first group suggests a completion time of 25 days, the second group identifies risks that may extend it to 40 days, and the third group proposes a new development method that could shorten the time to 10 days. What is the projected development duration for the new product?
    a) 30 days
    b) 23 days
    c) 25 days
    d) 27 days

    Explanation: Using the PERT formula: [10 + 4(25) + 40]/6 = 25. PMBOK Guide Sixth Edition (2018) PMI/PMI/7.2.2.5/245

325. In a new product development project with three main tasks, Task A must be completed four weeks before Task C begins. Once Task B is finished, Task C will commence. What type of relationship exists between Task A and Task B?
    a) Finish-to-start (FS)
    b) Finish-to-finish (FF)
    c) Start-to-start (SS)
    d) Start-to-finish (SF)

    Explanation: While there is no DIRECT relationship between A and B, both must be completed before C can start, so, of the choices, the only response that makes sense is FF. PMBOK Guide Sixth Edition (2018) PMI/PMI/6.3.2.1/190

326. While a product deliverable undergoes acceptance testing, a senior-level stakeholder expresses concern about the absence of a crucial product feature for their department. To prevent such issues in the future, what is one approach the project manager can take?

a) Prepare a change management plan.
b) Ensure a scope management plan gets approved.
c) Hold a product demo with stakeholders.
d) Properly manage stakeholder expectations.

Explanation: A scope management plan documents how the project and product scope will be defined, validated, and controlled. PMBOK Guide Sixth Edition (2018) PMI/PMI/5.1.3.1/137

327. A skincare company observes a growing number of customer complaints about a product being underweight. To assess if changes are necessary in the production line, what tool or technique can the project manager employ for this task?
a) Scatter diagram
b) Affinity diagram
c) Control chart
d) Histogram

Explanation: Control charts are used to determine whether or not a process is stable or has predictable performance. PMBOK Guide Sixth Edition (2018) PMI/PMI/8.3.2.5/304

328. A product meets 96% of the functional requirements outlined in the scope. However, the customer refuses to accept the product, insisting on 100%. The project manager is reviewing what could have been done differently to prevent this situation. What action should the project manager have taken?
a) Reviewed the requirements documentation.
b) Assessed and evaluated the test management plan.
c) Created the work breakdown structure (WBS).
d) Ensured that the quality metrics were defined.

Explanation: Quality metrics are a description of a project or product attribute and how to measure it. PMBOK Guide Sixth Edition (2018) PMI/PMI/8.1.3.2/287

329. A Project Manager is involved in a data center migration project and is formulating the schedule management plan. As per the project sponsor's directive, the schedule must encompass the plan for all tasks at the earliest opportunity. Additionally, the Project Manager is presented with modified resource availability that should not violate schedule constraints. The Project Manager needs to select the appropriate leads and lag relationships for this situation. Which type should the Project Manager choose?
a) Early start equals the late start in some activities.
b) Early start equals the late start in all activities.
c) Early start is smaller than the late start in some activities.
d) Early start is smaller than the late start in all activities.

Explanation: The time between the early start and the late start is the amount of float, or extra time before the activity starts where the critical path will not be impacted. Since the sponsor has indicated they want all tasks to start as soon as possible, there should be no float, which will make every task a part of the critical path.

330. In the process of drafting a project charter for a project focused on automating 30 percent of product testing, the project manager receives guidance from the project sponsor to engage senior test engineers in identifying automatable tests. However, the test department manager is uncooperative due to concerns about potential downsizing. In response, what should be the initial action of the project manager?
    a) Request for the project sponsor to provide a rationale for test automation.
    b) Present data-driven, cost-benefit evidence of test automation to the test department manager.
    c) Raise the risk that the senior testing engineers will be unavailable for the testing.
    d) Have a meeting with the test department manager to review the statement of work (SOW).

    Explanation: The FIRST thing the PM need to do is to make sure the project goals are clearly understood by the stakeholders. PMBOK Guide Sixth Edition (2018) PMI/PMI/13.3/523

331. A company is gearing up to implement a new product incorporating cutting-edge technologies and processes. The project manager has ensured that the individual work packages for each component are clearly defined, and constraints for each component have been pinpointed. Which estimating technique should the project manager employ to achieve an accurate cost estimate for the project?
    a) Three-point
    b) Bottom-up
    c) Parametric
    d) Analogous

    Explanation: Bottom-up estimating is a method of estimating project duration or cost by aggregating the estimates of the lower-level components of the WBS. PMBOK Guide Sixth Edition (2018) PMI/PMI/6.4.2.5/202

332. A project exhibits the following features:
    - It is currently in the planning phase.
    - A crucial stakeholder is set to be unavailable for 2.5 weeks during the project.
    - Upon discovering the key stakeholder's upcoming unavailability, the project manager must determine the initial course of action.

    What is the appropriate step to take?

a) Request that the finance department provide a replacement during the stakeholder's absence.
b) Organize a meeting with the sponsor to evaluate the situation and identify alternatives.
c) Continue with planning and acknowledge the impact the change may have on the project schedule.
d) Ensure that the issue log is updated and inform the other stakeholders.

Explanation: The FIRST thing the PM needs to do is to capture this as an issue and communicate this to the other stakeholders. Until then, no action can or should be taken. PMBOK Guide Sixth Edition (2018) PMI/PMI/4.3.3.3/487

333. A project possesses the following attributes:
- It is currently in the execution phase.
- A regulatory law has recently been approved, necessitating immediate compliance.
- This law will have implications for the project's scope, schedule, and cost.
- The project manager must decide the initial course of action.

What is the appropriate step to take?

a) Ensure the project management plan is updated.
b) Escalate the issue to senior management.
c) Begin the change request process.
d) Initiate mitigation of the risk.

Explanation: Since compliance is mandatory, a change request needs to be submitted. PMBOK Guide Sixth Edition (2018) PMI/PMI/4.3.3.4/96

334. The deliverable on the critical path of a project is falling behind schedule. The cost performance index (CPI) is 1.3. The Project Manager can restore the project to its original schedule. What is the most probable result for the project?
a) There will be a revised scope.
b) There will be an increase in costs and risks.
c) There will be rework.
d) There will be activities performed in parallel.

Explanation: The cost performance index (CPI) is above 1, therefore the project is spending money at a rate lower than planned. Since the project is behind schedule, its schedule performance index (SPI) is less than 1. In order to increase the SPI to 1, there is most likely an increase in cost (perhaps from crashing or fast-tracking) that will lower the CPI to 1. This will most likely increase risks associated with the project when trying to do more in less time.

335. In a complex project with key stakeholders situated in the United States, France, and China, the project manager learns that a senior manager has requested

project information from a team member. What is the suitable next step to take?

a) Read the communications management plan.
b) Rely on the information in the stakeholder register.
c) Refer to the project management information system (PMIS).
d) Ensure that the senior manager attends the next team meeting.

Explanation: A communications management plan is developed to ensure that the appropriate messages are communicated to stakeholders in various formats and various means as defined by the communication strategy. PMBOK Guide Sixth Edition (2018) PMI/PMI/10.1.3.1/377

336. Seven months into the project execution, the project manager calculates a cost performance index (CPI) of 0.80. Upon conducting a trend analysis, it is observed that the CPI has a downward trend. What should be the project manager's next course of action?
a) Ask for additional resources.
b) Crash the project schedule.
c) Utilize the management reserve.
d) Issue a change request for cost re-baselining.

Explanation: Since the project is over budget and the trend analysis show that will continue, the best course of action is to re-baseline the cost of the project and manage to that. PMBOK Guide Sixth Edition (2018) PMI/PMI/7.3.3.1/254

337. The project is in the implementation phase when the research and development department inform the project manager about unexpected changes in the product's requirements. These changes, unforeseen initially, could have a substantial impact on potential sales numbers and may affect the project scope. What would be the suitable action for the project manager in this situation?
a) Use forecasting methods.
b) Plan to use the contingency reserve and update the risk mitigation plan.
c) Utilize methods such as earned value management (EVM).
d) Anticipate risk responses and modify the risk register accordingly.

Explanation: These new requirements may pose additional risks. These risks need to be documented and appropriate responses need to be identified. PMBOK Guide Sixth Edition (2018) PMI/PMI/11.2.3.1/417

338. The project is currently in the execution phase, and due to the urgency for a swift turnaround, the project manager must promptly allocate resources to form the team and fulfill project requirements. What should be the next action for the project manager?
a) Host a webinar.
b) Build library services.
c) Deliver workshops.

d) Train the new team.

Explanation: Develop Team is the process of improving competences, team member interaction, and overall team environment to enhance project performance. Training is a Tool & Technique of the Develop Project Team process. PMBOK Guide Sixth Edition (2018) PMI/PMI/4.5/602

339. A new Project Manager takes over after the departure of the previous one and aims to steer the team towards project success. What should be the initial step for the new Project Manager?
    a) Review the project diary and look for notes made by the former Project Manager.
    b) Modify the project schedule, adding a new activity related to the knowledge transfer process.
    c) Revisit the project budget to determine if extra time is required.
    d) Look over the project charter to identify goals and deliverables.

    Explanation: The project charter is one of the most important project documents. It defines many aspects of the project which are critical to the new Project Manager's understanding of the project strategy as well as the tactical execution. Any other actions before reading the official project charter will be made with a bias.

340. Project team members are unusually occupied and express their inability to update lessons learned due to other project commitments. How should the project manager address this concern?
    a) Ensure there is an agenda item for discussion of lessons learned from project activities as part of all project meetings.
    b) File a change request to the project management office (PMO) to change how often lessons learned is collected.
    c) Bring in a consultant to assist with collecting lessons learned to help reduce the burden on project team.
    d) Require submission of lessons learned documentation after each project deliverable is accepted by the customer.

    Explanation: The lessons learned collection must be present throughout the project, using different facilitating techniques. PMBOK Guide Sixth Edition (2018) PMI/PMI/4.4.3.1/104

341. A Construction Project Manager entered into a firm-fixed contract with a supplier to acquire diverse construction materials. The supplier fulfilled the agreement by delivering all the materials to the construction site on the agreed-upon delivery date. What steps should the Project Manager take next?
    a) Pay the supplier and close the contract
    b) Start work using part of the materials while inspecting the rest

c) **Inspect the materials to ensure their compliance with the requirements stipulated in the contract**
d) Start work using the provided materials to avoid having any delays

Explanation: The Project Manager should inspect the materials to ensure that they meet the requirements stipulated in the contract. This is important to ensure that the materials are of the right quality and quantity and that they meet the project's specifications

342. A Project Manager is leading a team of engineers in constructing a packaging machine. The machine underwent thorough internal verification. However, upon the client receiving it, they contacted the Project Manager to inquire whether the machine met the specified requirements or not. What steps should the Project Manager take next?
   a) **Collaborate with the client to get their formal approval that the product complies with the requirements**
   b) Send an engineer to the client's site to double-check the product requirements
   c) Confirm that the product was internally tested and formally close the project
   d) Perform the control quality process and use the contingency reserve to fix any detected issues

   Explanation: The Project Manager should collaborate with the client to get their formal approval that the product complies with the requirements. This is because the client is the ultimate authority on whether the product meets their requirements or not. The Project Manager should work with the client to understand their concerns and address them as necessary.

343. TechEdge, a San Francisco-based IT company, is currently engaged in a project to develop a mobile app for online training. The Project Manager chooses to construct a prototype for the app. What advantages do prototypes provide? (Choose two)
   a) Allow the project to be completed faster.
   b) **Allow for early feedback on the requirements.**
   c) **Help to address ambiguities.**
   d) Reduce project costs.

   Explanation: Prototypes provide added value by allowing for early feedback on the requirements and helping to address ambiguities.

344. A Project Manager is overseeing a project utilizing a hybrid approach. In a cost management planning meeting attended by the Project Sponsor and the CFO, the Project Manager declared that they would document rounding to the nearest thousand and utilize weeks for resource estimation. The Project Manager's statement encompasses two elements of the cost management plan, which are _____.

a) **Level of accuracy and Units of measure**
b) Units of measure and Control thresholds
c) Units of measure and Control thresholds
d) Level of accuracy and Rules of performance

Explanation: The Project Manager's statement describes two elements of the cost management plan, which are level of accuracy and units of measure.

345. A Project Manager is overseeing a construction project for a shopping center. After an inspection, the customer declined acceptance of certain deliverables citing non-conformance to the acceptance criteria. What steps should the Project Manager take next?
a) Issue a Change Request
b) **Use their soft skills to convince inspectors to accept the deliverables**
c) Document the reasons for nonacceptance
d) Assess the impact on project constraints

Explanation: Documenting the reasons for nonacceptance is the right answer because it is important to understand why the deliverables were rejected and what needs to be done to correct the non-conformance. This information will be used to update the project management plan and to issue a change request if necessary.

346. Working within a predictive environment, a Project Manager has chosen to employ a three-point estimation with a beta distribution to assess the cost of project activities. Which of the subsequent formulas will they utilize?
a) (Optimistic + Pessimistic + Most likely) ÷ 7
b) **(Optimistic + Pessimistic + (4 × Most likely)) ÷ 6**
c) (Optimistic + Pessimistic + (3 × Most likely)) ÷ 5
d) (Optimistic + Pessimistic + (2 × Most likely)) ÷ 4

Explanation: The formula for three-point estimation with a beta distribution in a predictive environment is (Optimistic + Pessimistic + (4 × Most likely)) ÷ 6. This formula gives more weight to the most likely estimate, which is more accurate than the optimistic and pessimistic estimates.

347. An organization designated a new Project Manager to take over the role of the one who recently left. What should be the first action of the new Project Manager?
a) **Consult the project charter to understand the project goals and its business case**
b) Consult the issues log and the lessons learned register to check if there is any serious problem
c) Consult the project management plan to learn about the project baselines
d) Consult the stakeholders register to start interacting with the different parties involved in the project

Explanation: The new Project Manager should first consult the project charter to understand the project goals and its business case. This will provide a clear understanding of the project's objectives and help the Project Manager align their efforts with the organization's strategic goals.

348. A Product Owner creates a release plan based on an estimated team velocity of 50 story points. Nonetheless, in the initial two sprints, the team accomplishes 37 and 35 story points. What actions can the Agile Project Manager take in this scenario? (Choose two)
    a) Extend the duration of the sprint until completing 50 story points
    b) Inform the product owner that, based on available empirical data, the release plan could not be achieved
    c) Study with the product owner the possibility of adding more resources to the team
    d) Use their leadership skills to motivate their team to reach the estimated velocity

    Explanation: The Agile Project Manager can inform the Product Owner that, based on available empirical data, the release plan could not be achieved and study with the product owner the possibility of adding more resources to the team.

349. Upon concluding a release, the Scrum team is reviewing its burndown chart. Despite no alterations in estimates, the chart indicates a rise in story point value during the third sprint. How can this be elucidated?
    a) Work was removed from the Product Backlog
    b) The team's velocity decreased
    c) The team's velocity increased
    d) Work was added to the Product Backlog

    Explanation: The increase in story point value during the 3rd sprint can be explained by the addition of work to the Product Backlog, which increased the total amount of work to be completed.

350. A Project Manager is leading a team of engineers through a two-year project. After a few months, the Project Manager observed a decline in the team's interest in the project. What actions can the Project Manager take to sustain motivation?
    a) Align the project with the organization’s goals
    b) Align work assignments with the team members’ personal ambitions
    c) Offer an immediate salary raise
    d) Offer a bonus at the end of the project

    Explanation: Aligning work assignments with the team members’ personal ambitions is a way to maintain motivation. This will help the team members to feel more connected to the project and will increase their interest in the project.

351. Despite establishing an agenda and timeboxing for the meetings they conduct, the Project Manager struggles to engage all attendees. What steps can they take to rectify this issue?
   a) Change to remote meetings to make everyone feel more comfortable
   b) Make meetings shorter to prevent boredom
   c) Encourage all attendees to participate and share their thoughts by asking for their opinions on the discussed matters
   d) Limit the interventions of the most talkative attendees to allow others to participate

   Explanation: Encouraging all attendees to participate and share their thoughts by asking for their opinions on the discussed matters is the right answer because it helps to engage all attendees and make them feel valued. This can lead to better collaboration and decision-making.

352. A Project Manager is overseeing a team with members spread across various countries. Beyond facing cultural barriers, the team appears disconnected, resulting in numerous instances of rework. What actions could the Project Manager take to improve communication within the team?
   a) Request from the team to only use emails for communication to avoid any misunderstanding or misinterpretation
   b) Hold short daily meetings with all team members to sync on what everyone is doing
   c) Offer team members training about overcoming cultural barriers
   d) Collaborate with senior management to co-locate the team

   Explanation: Holding short daily meetings with all team members to sync on what's everyone is doing is the right answer because it helps to improve communication within the team and ensure everyone is on the same page.

353. In a workshop about the Agile approach, the Project Manager asserted that Agile teams ought to be cross-functional and self-organizing, clarifying that _____.
   a) The Agile team should have complementary skill sets and be able to organize the backlog by themselves
   b) The Agile team should self-organize to acquire the necessary skills to be cross-functional
   c) Each member of the team should be cross-functional and self-organized
   d) The Agile team should have all the required skills to deliver the product on their own

   Explanation: The Agile team should have all the required skills to deliver the product on their own. This statement is correct because Agile teams are expected to be self-sufficient and have all the necessary skills to deliver the product without relying on external resources.

354. Amid a financial crisis, an organization opted to pause all internal projects to an undefined date. Considering that a Project Manager within this organization holds a weekly meeting with a subcontractor who supplies the majority of the project's physical resources, what is the most effective form of communication to relay the decision to discontinue the contract to this subcontractor?
   a) Informal verbal
   b) Formal verbal
   c) Formal written
   d) Informal written

   Explanation: Formal written communication is the best type of communication to inform the subcontractor of the contract discontinuation decision because it provides a clear and documented record of the communication. This is important in case of any legal disputes or misunderstandings in the future.

355. A Project Manager is overseeing a biodegradable packaging project. A team member notified the Project Manager that he identified a risk wherein the printing machine might require maintenance before the project concludes. The Project Manager assesses the risk's probability of occurrence as 10% and its cost as $5,000. The expected monetary value of this risk is _____.
   a) 4500
   b) -4500
   c) 500
   d) -500

   Explanation: The Expected Monetary Value (EMV) of a risk is calculated by multiplying the probability of the risk occurring by the cost of the risk. In this case, the EMV is 10% x $5,000 = -$500 since it represents a potential cost to the project.

356. A Project Manager is tasked with overseeing the project quality assurance report. While examining a control chart, they noticed nine consecutive data points on one side of the mean. What actions should they take in this scenario?
   a) Identify the assignable cause
   b) Make the necessary changes to the chart to reflect the new mean
   c) Carry out a design of experiments
   d) This is the rule of seven, meaning it can be disregarded

   Explanation: The right answer is to identify the assignable cause. This is because nine data points in a row on one side of the mean is an indication of an out-of-control process, which means that there is an assignable cause that needs to be identified and addressed.

357. A senior Project Manager joins an ongoing project after the resignation of its former Project Manager. What should be the initial action for the new Project Manager?

a) Add a new activity to the project schedule concerning knowledge transfer
b) Review the project budget to determine if more funding is required
c) Examine the project charter to understand the project goals and deliverables
d) Go through the issue log to check the previous and current difficulties the project is facing

Explanation: The new Project Manager should first examine the project charter to understand the project goals and deliverables. This will help the new Project Manager get a clear understanding of the project's objectives and ensure that the project is aligned with the organization's strategic goals.

358. A Project Manager is leading a mobile app development project. Faced with quality issues, the project team decided to employ a(n) _____ to prioritize quality problems.
a) Control chart
b) Pareto chart
c) Ishikawa diagram
d) Scatter diagram

Explanation: A Pareto chart is a quality management tool used to prioritize problems by identifying the most frequent or significant issues. It helps the project team to focus on the vital few problems that have the most significant impact on the project's quality.

359. A Project Manager is overseeing a project outside her expertise field. In a planning meeting, two team members engage in a heated debate regarding the appropriate way to implement a specific project requirement. What should the Project Manager do in this scenario?
a) Help team members decide the best way to implement the concerned requirement
b) Put an end to the discussion to avoid getting the meeting out of control
c) Let the most experienced team member decide
d) Inform team members that the final decision should be taken by her after listening to their opinions

Explanation: The Project Manager should facilitate the discussion and help team members decide the best way to implement the concerned requirement. This approach promotes collaboration and encourages team members to share their ideas and opinions, which can lead to a better outcome for the project.

360. A Project Manager is in the process of identifying and evaluating risks for a construction project. Encountering a risk that is complex and ambiguous, the Project Manager opts to transfer it to the Program Manager for the latter to make the appropriate decision. How did the Project Manager address this risk?
a) Mitigate

b) Avoid
c) Transfer
d) Escalate

Explanation: The Project Manager responds to the risk by escalating it to the Program Manager for decision-making, which is a valid risk response strategy when the risk is beyond the Project Manager's authority or expertise.

361. A Project Manager is overseeing a project that adopts a hybrid approach. Despite ensuring that all iterations maintained a balanced workload, the Project Manager observed that a specific team member was becoming overwhelmed, while others appeared to be managing well. How should the Project Manager handle this issue?

a) Track the stressed team member's tasks in a separate backlog for additional analysis and reporting
b) Inform upper management and request additional resources for the project
c) Personally meet with the stressed team member to discuss ways to better manage their time
d) Raise the issue with the team in the daily stand-up meeting

Explanation: The Project Manager should personally meet with the stressed team members to discuss ways to better manage their time. This approach allows the Project Manager to understand the root cause of the team member's stress and work with them to find a solution that works for both parties. It also shows that the Project Manager is invested in the well-being of their team members, which can help build trust and improve team morale.

362. While working on a new feature, a team member discovers that the original code does not adhere to the organization's coding standards. What actions should the team member take in this situation?

a) Ask the Product Owner to get a new item added to the product backlog dedicated to technical debt
b) Ignore the code compliance work as it will be considered as Gold Plating
c) Discuss with the rest of the team members and add a new technical task to the sprint backlog
d) Start refactoring the code and update the time estimate of the feature

Explanation: The team member should ask the Product Owner to get a new item added to the product backlog dedicated to technical debt. This will ensure that the technical debt is tracked and addressed on time.

363. A Project Manager is executing a project using the agile approach with two-week sprints. At the one-week mark in the current sprint, the team has not finished any of the sprint backlog items, rendering the attainment of the sprint goal impossible. What steps should the Project Manager take?

a) Discuss with the self-organizing team the root cause of their poor performance during the standup meeting
b) Cancel the sprint, hold an early retrospective meeting, and start over with a new energy
c) Complete the sprint even though it will not achieve its goal, and help the team overcome the encountered problems
d) Discuss with the product owner the possibility of updating the sprint goal to make it more achievable

Explanation: The Project Manager should work with the team to identify the root cause of the issues and help them find solutions to overcome them. This will help the team learn from their mistakes and improve their performance in future sprints.

364. A Project Manager is overseeing a one-year Agile project. A few months into the project, the Sponsor expressed a desire to understand the overall progress of the project, particularly the status of implemented vs. pending features. Which of the following tools would provide this information? (Choose two)
a) Feature chart
b) Cumulative flow diagram
c) Iteration burn-down chart
d) Kanban board

Explanation: Cumulative flow diagrams and Feature charts are Agile tools that provide information about the big picture of the project progress, more specifically, the implemented vs. pending features.

365. Despite being estimated at 2 and 3 story points consecutively, two user stories each took two days to implement. How many story points do these two user stories contribute to the team velocity when calculated at the end of the iteration?
a) 3 story points
b) 2 story points
c) 5 story points
d) 4 story points

Explanation: The two user stories were estimated at 2 and 3 story points, which totals to 5 story points. However, their implementation took two days each, which means that each story point took 2 days to complete. Therefore, the two user stories contribute 5 story points to the team velocity.

366. A Project Manager presents the project plan to the stakeholders during the kick-off meeting of their first project. While delving into the details, the Project Manager observes that a majority of stakeholders appear dissatisfied with the plan. One stakeholder, in particular, conveys frustration, citing the plan's inaccuracy in capturing their expectations. What is the probable cause for the Project Manager providing an inaccurate depiction of stakeholders' requirements?

a) The Project Manager did not explain the project well enough. A better presentation of the plan would have prevented this misunderstanding.
b) The Project Manager did not engage the project stakeholders enough to be able to accurately capture their requirements.
c) The Project Manager did not use the project scope as a guide when creating the project plan.
d) The Project Manager only presented the first draft of the project plan, so it can always be improved.

Explanation: The Project Manager did not engage the project stakeholders enough to be able to accurately capture their requirements. This is the most likely reason behind the inaccurate description of stakeholders' requirements. The Project Manager should have engaged the stakeholders more to understand their requirements and expectations.

367. The Project Manager finds out they were not assigned the promised human resources during the planning phase. Instead, they received a team lacking the necessary skills for project execution. Given that they operate in a weak matrix organization, what course of action should the Project Manager take?
a) Issue a change request to get a more qualified team
b) Negotiate with the Functional Manager to obtain the required team members
c) Use their legitimate power by informing the organization that they will not perform the project without the needed resources
d) Take the matter to the Project Sponsor

Explanation: The Project Manager should negotiate with the functional manager in order to obtain the required team members. In a weak matrix organization, the Project Manager has limited authority over the project team, and the Functional Manager has more control over the team members. Therefore, the Project Manager needs to work with the Functional Manager to obtain the necessary resources.

368. While planning a call center project, a stakeholder said that since the used internet cable (DSL) can be slow, it should be considered a high risk to the project. However, the Project Manager, drawing from past projects, believes the cable internet would suffice. Another stakeholder steps in to clarify that internet cable speed and reliability fluctuate based on geographical location. What should the Project Manager's next course of action be?
a) Perform a quantitative risk analysis to determine the risk exposure
b) Perform a qualitative risk analysis by developing a risk mitigation plan
c) Set a contingency reserve to install fiber Internet cables if the threat occurs
d) Analyze and prioritize the risk by using a probability and impact matrix

Explanation: The Project Manager should analyze and prioritize the risk by using a probability and impact matrix. This will help the project manager to determine the likelihood and impact of the risk and prioritize the response accordingly.

369. Upon receiving a document from the customer outlining the requirements for the new project, the Project Manager encounters difficulty in understanding certain requirements, outputs, and, crucially, the project's ultimate goal. What should be the Project Manager's next step?
    a) Acquire a team to help analyze the project requirements
    b) Ask the customer for clarifications
    c) Start planning the first iteration
    d) Reject the new project as it involves a lot of ambiguity

    Explanation: The Project Manager should ask the customer for clarifications to gain a better understanding of the project requirements and goals.

370. While executing the project, the project team faced an issue with an unreliable vendor. Given that the incident significantly impacted the project scope, the Project Manager opted to terminate the vendor's contract and abstain from future dealings with them. Where should the Project Manager record this issue?
    a) Change log
    b) Lessons learned register
    c) Issue log
    d) Risk register

    Explanation: The Project Manager should record this issue in the issue log, which is used to document and monitor issues that arise during project execution. The issue log is a tool used to track and manage issues, including their status, priority, and resolution. By recording the issue in the issue log, the Project Manager can ensure that the issue is properly tracked and addressed.

371. As part of his role and through multiple visits to various countries, a Project Manager realized the importance of avoiding ethnocentrism. What does ethnocentrism mean?
    a) The initial shock of arriving in a foreign country
    b) The belief that your own culture is superior to others
    c) Inability to get adjusted to the culture of a new place
    d) Being unfriendly to strangers

    Explanation: Ethnocentrism is the belief that your own culture is superior to others. This can lead to a lack of understanding and respect for other cultures, which can be detrimental to project success in a global environment.

372. A team member has demonstrated outstanding technical skills and consistently completed tasks promptly. Consequently, the company's CEO chose to promote

the team member to a Project Manager role. The CEO's behavior is an example of _____.

a) Halo Effect
b) Herzberg Theory
c) McGregor's Theory of X and Y
d) Expectancy Theory

Explanation: The CEO's behavior is an example of the Halo Effect, which is a cognitive bias where a person's overall impression of a person, company, brand, or product influences their feelings and thoughts about that entity's character or properties. In this case, the CEO is promoting the team member based on their exceptional technical skills and timely task completion, which is an example of the Halo Effect.

373. A project manager ensures a healthy work environment for their team by offering secure working conditions, job stability, and competitive salaries, with a focus on recognizing and appreciating their work. According to Herzberg, which of the following options represent examples of hygiene factors? (Choose three)
a) Appreciation
b) Job security
c) Safe working conditions
d) Salary

Explanation: According to Herzberg's Two-Factor Theory, hygiene factors are the basic necessities that must be met in the workplace to prevent dissatisfaction. These factors include job security, salary, and safe working conditions. They are not motivators, but their absence can lead to dissatisfaction among employees.

374. What is the primary purpose of the Project Charter in project management?
a) To outline the detailed project plan
b) To formally authorize the project and give the project manager the authority to apply organizational resources to project activities
c) To define the project schedule and budget
d) To describe the project deliverables in detail

Explanation: The Project Charter formally authorizes a project and grants the project manager authority to use organizational resources. It outlines the project's objectives and identifies key stakeholders. Detailed project plans, schedules, and budgets are developed later in the planning process.

375. A Project Manager is overseeing a pharmaceutical manufacturing project utilizing a predictive approach. The project management team consistently communicates with stakeholders to gather feedback and evaluate their involvement. In which process group do stakeholders exert the most influence over the project?
a) Closing

b) Initiating
c) Planning
d) Executing

Explanation: In the Initiating process group, the project manager identifies stakeholders and assesses their engagement level. This process group is where stakeholders have the most influence over the project because their feedback and input can shape the project's direction and objectives.

376. As per the client's request, the project manager for an e-commerce website development project agreed to extend the scope while maintaining the original delivery date. To meet this commitment, the project team must work 24/7. What is the optimal course of action for the Project Manager?
   a) Crashing the project
   b) Using the resource leveling technique
   c) Hiring virtual teams that work in different time zones and schedules
   d) Starting a night shift for the collocated team members

   Explanation: Hiring virtual teams that work in different time zones and schedules is the best course of action for the Project Manager because it allows the project team to work around the clock without overburdening the collocated team members. This approach also provides the project manager with the flexibility to scale up or down the team size as needed.

377. Invited to assist an organization's transition to the Scrum framework, an agile coach discovers that, upon assessing the current situation, the cross-functional team lacks a Single Point Of Contact (SPOC) for product-related inquiries. Which of the following roles is absent in this scenario?
   a) Project Manager
   b) Product Owner
   c) Sponsor
   d) Scrum Master

   Explanation: The Product Owner is responsible for defining and prioritizing the product backlog items. The Product Owner is the single point of contact for the team to refer to in case they have any questions about the product. Therefore, the missing role in this situation is the Product Owner.

378. A Project Manager is overseeing a project utilizing a hybrid approach. After the first week of the initial iteration, the customer conveyed dissatisfaction with the deliverables. What should be the Project Manager's next course of action?
   a) Investigate the cause of dissatisfaction and verify the deliverables
   b) Ask the customer to submit a change request so that their dissatisfaction can be addressed
   c) Use soft skills to convince the customer that the deliverables conform to project specifications

d) Implement adjustments and improvements in the next iteration

Explanation: The Project Manager should investigate the cause of dissatisfaction and verify the deliverables to ensure that they meet the project specifications. This will help the Project Manager identify the issues and make necessary adjustments to improve the deliverables in the next iteration.

379. The project team comprises the following members:

Member A: enthusiastic about the project and willing to exert additional effort if necessary.

Member B: consistently exceeds expectations.

Member C: well-liked by the customer and ready to invest extra effort as required.

Member D: consistently meets the project's requirements.

Who among these members is the most valuable resource to the project?

a) Member A
b) Member B
c) Member C
d) Member D

Explanation: Member D consistently achieves the project's requirements, making them the most valuable resource to the project.

380. At the end of an iteration, a graphic designer working on a dairy brand project informs the Project Manager that he couldn't complete one of his assigned tasks due to a laptop issue. To avoid a recurrence of such situations in the future, the Project Manager should _____.

a) Handle the issue during the next daily standup meeting
b) **Discuss the issue during the retrospective meeting**
c) Address the issue during the following iteration planning meeting
d) Discuss the issue during the demonstration session

Explanation: The retrospective meeting is the best place to discuss issues that occurred during the iteration and identify ways to prevent them from happening again in the future.

381. In the planning phase, to acquire physical resources for a high-tech project, a Project Manager received numerous proposals from qualified suppliers they had collaborated with in previous projects, as well as proposals from other suppliers. To evaluate and make the optimal choice among the various suppliers, which of the following tools or techniques should the Project Manager utilize?

a) Selection criteria
b) Procurement performance review

c) Make or buy analysis
d) Procurement audit

Explanation: Selection criteria is a tool and technique used in the Plan Procurement Management process to evaluate and rank proposals from potential sellers. It is used to compare different proposals and make the best choice based on predetermined criteria.

382. Upon receiving proposals from vendors for the new project they're overseeing, and to reduce the impact of personal prejudice on source selection, a Project Manager employs a method of quantifying qualitative data. This method is known as _____.
a) Selecting system
b) Screening system
c) Weighting system
d) All of the above

Explanation: The right answer is C because a weighting system is a quantifying method for qualitative data that assigns numerical values to each criterion and then calculates a total score for each vendor's proposal. This method helps to minimize the effect of personal prejudice on source selection.

383. While overseeing the performance of a construction project, the Project Manager observes that the project is at risk of deviating from the planned schedule. Consequently, they implement _____ to guarantee that the project aligns with the established project baselines.
a) Defect repairs
b) Preventive actions
c) Corrective actions
d) Change request

Explanation: Preventive actions are taken to prevent a problem from occurring. In this case, the Project Manager is taking preventive actions to ensure that the project does not drift away from the planned schedule.

384. Overseeing an electric car manufacturing project, a Project Manager encounters a complex quality issue for which they lack an understanding of its source and causes. They schedule a meeting with their team to address the matter and initially trace the problem back to its root cause. For this, they use _____.
a) Why Why diagram
b) Pareto diagram
c) Scatter diagram
d) Chain diagram

Explanation: The Why Why diagram is a tool used to trace a problem back to its root cause. It is a simple and effective tool that helps identify the underlying

causes of a problem. By asking 'why' repeatedly, the team can identify the root cause of the problem and develop a solution to address it.

385. A Project Manager was tasked with coordinating a national chess tournament. The tournament project has been planned and approved for execution. Three months stand between the Project Manager and the major event, during which they will adhere to the Scrum approach with 2-week long sprints. Prior to commencing any work, they convene a meeting with the sponsor, project team, stakeholders, and key contractors. What type of meeting did the Project Manager conduct?
    a) Status meeting
    b) Sprint planning meeting
    c) Scoping meeting
    d) Kick-off meeting

    Explanation: The Project Manager held a kick-off meeting to initiate the project and align all stakeholders on the project objectives, scope, and expectations.

386. A Project Manager is overseeing a project involving the implementation of an accounting application for a pet store. In a project performance review meeting, they presented the following figures: AC = $4,000, PV = $5,000, and EV = $5,500. What does this signify about the project?
    a) Since the SV is negative, the project is behind schedule
    b) Since the CV is negative, the Project Manager has probably spent more than they initially planned to
    c) Since both the CV and SV are positive, the project is under budget and ahead of schedule
    d) These numbers are insufficient to calculate the project SPI and CPI

    Explanation: The correct answer is C because the CV and SV formulas are: CV = EV - AC and SV = EV - PV. In this case, CV = $5,500 - $4,000 = $1,500 and SV = $5,500 - $5,000 = $500. Since both the CV and SV are positive, the project is under budget and ahead of schedule.

387. An organization is considering two prospective projects with identical payback periods. Following a benefit measurement analysis, the Project Manager discovers that Project A has a lower Internal Rate of Return (IRR) compared to Project B. Which project should the organization opt for?
    a) Project A
    b) Project B
    c) The available information is not enough to make a decision
    d) There is no difference

    Explanation: Project B has a higher IRR, which means it will generate more cash flows over the life of the project. Therefore, it is the better choice.

388. A Project Manager is leading a complex project with a highly demanding Sponsor. The Sponsor seeks to be informed about every update on the project's progress and all planned subsequent work. What is the most effective method for the Project Manager to communicate with the Sponsor?
   a) Schedule a fortnightly meeting and give the Sponsor access to the project management software so they can check real-time updates concerning the project status
   b) Schedule a monthly face-to-face meeting to discuss the project status
   c) Schedule a meeting once or twice a week and let the Sponsor know that they can also attend daily standups
   d) Send a comprehensive monthly report detailing all project progress aspects regarding scope, schedule, cost, risks, etc.

   Explanation: Option C is the right answer because it allows the Sponsor to stay informed about the project progress without being too intrusive. The Project Manager can provide regular updates, and the Sponsor can attend daily standups if they wish to be more involved.

389. A Project Manager has recently assumed responsibility for an in-progress project focused on establishing a brand for newborn essentials. Fabric-based items are produced by external vendors. To familiarize themselves with all vendors participating in the project, the Project Manager should explore _____.
   a) Stakeholder register
   b) Vendors' proposals
   c) Stakeholder engagement plan
   d) RFQs

   Explanation: The stakeholder register contains information about all stakeholders involved in the project, including vendors. Therefore, it is the best option for the Project Manager to check out to get to know all vendors involved in the project.

390. An Agile Project Manager observed a loss in interest from multiple stakeholders who rarely contribute input, provide feedback, or attend meetings for the project. What actions can the Project Manager take to resolve this issue? (Choose two)
   a) Inform stakeholders that they can provide their feedback anonymously
   b) Send a reminder before each meeting
   c) Demonstrate the working increments
   d) Value and show appreciation of everyone's ideas

   Explanation: The project manager can resolve this problem by valuing and showing appreciation for everyone's ideas and demonstrating the working increments. This will help to re-engage the stakeholders and encourage them to participate more actively in the project.

391. As a part of the project, an outdoor ceremony event is scheduled. However, a day before the event, weather forecasts indicate continuous rain for the entire week. Understanding that changing the event date is not feasible, what actions can the project manager take to mitigate the situation? (Choose two)
   a) Ask all invitees to take the needed precautions by bringing their raincoats, umbrellas, etc.
   b) Get waterproof tents
   c) Change the event location to an indoor space and inform all invitees of the new location
   d) Apply the risk response previously elaborated in the risk register

   Explanation: Getting waterproof tents and changing the event location to an indoor space are both valid risk responses to the situation. The project manager can use the contingency plan to mitigate the risk of rain and ensure the event is successful.

392. In the process of identifying risks, the Project Manager recognized resource attrition as a major risk. In response, the Project Manager chose to implement financial incentives to motivate team members. What type of risk response is being employed?
   a) Avoidance
   b) Acceptance
   c) Mitigation
   d) Acceptance

   Explanation: The Project Manager is opting for a mitigation response by providing financial incentives to motivate their team members. This response aims to reduce the probability and/or impact of the risk.

393. A Project Manager was assigned to a new project. Before the project kick-off, the sponsor introduces the Project Manager to the organization's C-level executives and assures them that the project is destined for success. Who is accountable for the project's success?
   a) Project Manager
   b) Project Sponsor
   c) C-level Executives
   d) All of the above

   Explanation: The Project Manager is ultimately responsible for the success of the project, as they are the ones leading and managing the project team and making decisions that impact the project's outcome.

394. A Project Manager is overseeing an agricultural project with three high authority stakeholders. After analyzing each stakeholder, the Project Manager determines that Stakeholder A has the greatest interest in the project. What approach should the Project Manager employ to manage Stakeholder A?

a) Keep informed
b) Monitor
c) Keep satisfied
d) Manage closely

Explanation: Stakeholder A has high interest in the project, which means that the Project Manager should manage this stakeholder closely to ensure that their needs and expectations are met.

395. A Project Manager is overseeing a construction project. During the initial resource acquisition phase, the Project Manager requested the talent acquisition manager to conduct an MBTI (Myers–Briggs Type Indicator) Personality Test for the procurement manager position. The Project Manager specifically instructed the talent acquisition manager to focus on candidates comfortable with approaching providers and effective communication. To which personality type is the Project Manager referring?
a) ISTJ
b) ESTP
c) INFJ
d) INFP

Explanation: The Project Manager is looking for a candidate who is comfortable approaching providers and communicating with them. ESTP personality types are known for being outgoing, confident, and persuasive, making them ideal for this role.

396. Despite having experience, a Project Manager is overseeing a remote team for the first time. In meetings, the Project Manager occasionally needs to reiterate their statements due to a poor internet connection or headphone issues. In a standard communication context, factors that disrupt the transmission and comprehension of the Project Manager's message are referred to as _____.
a) Constraint
b) Medium
c) Noise
d) Decode

Explanation: Noise refers to any element that interferes with the transmission and understanding of a message. In this case, the bad Internet connection and headphone issues are causing noise in the communication process.

397. A Project Manager is overseeing the construction of a new bridge in Chicago. One of the Subject Matter Experts informed them that, in next September, all bridge construction work will need to stop, considering the historical occurrence of river flooding during the hurricane season. Consequently, they concurred with the expert and arranged to suspend work during that month. This decision is regarded as _____.

a) Mitigation
b) Acceptance
c) Transfer
d) Avoidance

Explanation: This decision is considered as avoidance because the Project Manager has decided to stop the work during the month of July to avoid the risk of river flooding due to hurricane season.

398. A Project Manager works together with the project team to develop the Work Breakdown Structure (WBS) for their project. What functions does this document fulfill? (Choose two)
   a) Help the project team identify the project requirements
   b) Help the project team define the project scope
   c) Help the project team organize the project scope
   d) Help the project team visualize the project work

   Explanation: The WBS helps the project team visualize and organize the project work into manageable sections or components.

399. Upon reviewing the project's progress, a Project Manager discovers that the schedule variance is $0.00. Which of the following statements is true about the status of this project?
   a) The project's cost variance is $0.00 too
   b) The project's EV equals its PV
   c) The project's EAC equals its BAC
   d) The project's CPI equals 1

   Explanation: Option B is correct because the Schedule Variance (SV) is calculated by subtracting the Earned Value (EV) from the Planned Value (PV). If the SV is zero, then the EV must equal the PV.

400. Three months into project execution, the Sponsor faces a financial crisis and instructs the Project Manager to immediately terminate the project. What should be the Project Manager's next course of action?
   a) Start the close project process
   b) Start the close procurement process
   c) Talk to their management about the situation
   d) Try to persuade the client to complete the project

   Explanation: The right answer is A because the Project Manager should follow the project closure process to ensure that all project activities are completed and the project is closed in an orderly manner. This process includes obtaining formal acceptance of the project deliverables, archiving project documents, and releasing project resources.

401. A Project Manager is overseeing a construction project for a children's hospital. The Project Manager receives an email from the quality control team, indicating numerous issues with one of the project vendors. What steps should the Project Manager take in response?
   a) Raise the issue to the sponsor
   b) **Set up a meeting with the vendor and the control quality team**
   c) Penalize the vendor
   d) Terminate the contract and discuss with the control quality team the selection criteria of a new vendor

   Explanation: Setting up a meeting with the vendor and the control quality team is the best course of action as it allows the Project Manager to understand the issues and work with the vendor to resolve them. Penalizing the vendor or terminating the contract without understanding the issues may not be the best course of action. Raising the issue to the sponsor may be necessary, but it should be done after the project manager has a better understanding of the situation.

402. The Management informed the Project Manager that they are dispatching a team of experts to perform a risk audit. Despite the Project Manager having already completed a risk review process without identifying any shortcomings, the Management insisted on proceeding with the risk audit. Which of the following statements are true risk audits and risk reviews? (Choose two)
   a) A risk review is conducted to investigate whether the risk response plan is being followed
   b) A risk audit is conducted to examine the risk response plan
   c) A risk review is conducted to examine the risk response plan
   d) A risk audit is conducted to investigate whether the risk response plan is being followed

   Explanation: A risk audit examines the effectiveness of risk responses, while a risk review assesses the risk management process and its effectiveness.

403. One of the crucial members of the project team left for personal reasons. The Project Manager hadn't planned for such a scenario, but they managed to find a replacement with the help of a staffing agency. However, they must pay them a higher salary. This will lead to additional expenses categorized under the project's _____.
   a) Cost
   b) Management reserve
   c) Budget
   d) Contingency reserve

   Explanation: The replacement cost falls under the management reserve, which is used to cover unforeseen events that are not part of the project budget.

404. Recognizing the importance of leadership in project success, a Project Manager enrolls in a training program to acquire the skills necessary to become an effective leader. Which of the following activities should the Project Manager learn to employ to achieve their desired leadership role?
   a) Learning more about project management by obtaining a PMP certification
   b) Learning more about the specific domain of their project to be able to technically support the project team
   c) Improving their project's performance
   d) Guiding, motivating, and directing the project team

   Explanation: Leadership involves guiding, motivating, and directing the project team to achieve project success. Therefore, the Project Manager should learn how to apply these activities to become a good leader.

405. A Project Manager oversees a project with team members located globally. The project incorporates innovative technology that has never been utilized before. In this context, what is the primary advantage of having a virtual team?
   a) Allows work to continue around the clock
   b) Reduces the cost of setting up a workplace
   c) Widens the potential resource pool
   d) Promotes cultural inclusiveness

   Explanation: Having a virtual team widens the potential resource pool, as the Project Manager can select the best talent from around the world without being limited by geographical location.

406. A Project Manager employs various forms of power to gain support from project stakeholders. Which of the following represent types of power? (Choose three)
   a) Seeking consensus
   b) Coercive
   c) Relational
   d) Situational

   Explanation: The three forms of power used by a Project Manager to gain support from project stakeholders are relational, situational, and coercive. Relational power is based on the Project Manager's personal relationships with stakeholders. Situational power is based on the Project Manager's position or role in the project. Coercive power is based on the Project Manager's ability to punish or reward stakeholders.

407. A Project Manager gained insights from professional experience that, when interacting with the project team, it's crucial to consider the distinct characteristics of each member. These individual differences in characteristic patterns of thinking, feeling, and behaving are referred to as _____.
   a) Personality
   b) Interpersonal skills

c) Integration
d) Seeking consensus

Explanation: Personality refers to the individual differences in characteristic patterns of thinking, feeling, and behaving. A Project Manager should take into consideration the personality of each team member when interacting with them.

408. A project manager is encountering challenges after joining a new company. His superiors observed that during a kick-off meeting, he had difficulty addressing questions about how the project aligns with the organization's objectives. What essential skill is the project manager lacking?
    a) Communication skills
    b) Business acumen
    c) Power skills
    d) Ways of working

    Explanation: The project manager lacks business acumen, which is the ability to understand and align the project with the organization's objectives.

409. A project manager is tasked with an eco-friendly packaging project. To effectively handle the involved stakeholders, the project manager begins by assessing their power, urgency, and legitimacy. Which analytical method is the project manager employing?
    a) Power/influence grid
    b) Power/interest grid
    c) Salience model
    d) Stakeholder cube

    Explanation: The project manager is applying the salience model to analyze the power, urgency, and legitimacy of stakeholders. The salience model is a technique used to identify and prioritize stakeholders based on their power, urgency, and legitimacy.

410. A project manager observes that his colleague, also a project manager, arrives at the office with new high-tech gadgets regularly. This triggers his suspicions that his colleague might be receiving gifts from hardware vendors who will be bidding on one of their company's upcoming multimillion-dollar contracts. What action should the project manager take?
    a) Convince his colleague to return the items and stop accepting any gifts from vendors
    b) Report his colleague to the organization so that a conflict-of-interest investigation can take place
    c) Warn the concerned colleague that such gifts aren't appropriate and leave it at that
    d) Directly ask his colleague whether these items were gifts from vendors or he purchased them himself

Explanation: Directly asking the colleague whether the items were gifts from vendors or he purchased them himself is the right answer because it is the most appropriate way to address the situation. It is important to clarify the situation and ensure that there is no conflict of interest or ethical violation.

411. Deciding to enhance the organization's project management capabilities, the CEO of ChicLife, a luxury lifestyle gadgets company, established a PMO to enforce compliance with a set of project management standards. What type of PMO is the CEO implementing?
    a) Controlling
    b) Agile
    c) Directive
    d) Supportive

    Explanation: The CEO is implementing a controlling PMO because the PMO is established to ensure compliance with a set of project management standards.

412. A project manager is hired to lead the marketing department of a financial institution. Initially, the project manager meets with the project management team to assess the project portfolio. Following this, they concentrate on understanding the governance framework in place, which is regarded as:
    a) Agile Manifesto principles
    b) Organizational process assets
    c) Project management framework
    d) Enterprise environmental factors

    Explanation: The governance framework in place is considered to be an enterprise environmental factor, which refers to external factors that can impact the project. In this case, the governance framework is an external factor that can impact the project.

413. A project manager is employed by a construction company that has recently undergone significant organizational changes. The new manager of the organization appears to be closely monitoring employees' arrival and departure times, expressing concern about the team's motivation. What type of management style is demonstrated by the new manager?
    a) Theory X
    b) Theory Y
    c) Theory Z
    d) Theory XY

    Explanation: The new manager is exhibiting Theory X management style, which assumes that employees are inherently lazy and need to be closely monitored and controlled to ensure they are working. This is evident from the manager's concern about the team's lack of motivation and closely monitoring when employees arrive and leave work.

414. The risk manager is working alongside the project manager to conduct a risk analysis. Which tool can be employed to simulate thousands of potential scenarios and determine the most likely project completion date, taking into account known risks?
   a) Influence diagrams
   b) Sensitivity analysis
   c) Monte Carlo analysis
   d) Decision-tree analysis

   Explanation: Monte Carlo analysis is a simulation technique used to estimate the probability of completing a project within a certain time frame. It is used to determine the most likely project completion date based on known risks.

415. A project manager following the servant leadership style is onboarding a new team member to manage skill-level gaps in the current project team. The project manager provides the new team member with a copy of the project charter and commits to share with them a document that covers team values, ground rules, and working agreements, known as _____.
   a) The team charter
   b) The communication management plan
   c) The resource management plan
   d) The employment contract

   Explanation: The team charter is a document that outlines team values, ground rules, and working agreements. It is a critical document that helps to ensure that all team members are aligned and working towards the same goals. Providing the new team member with a copy of the team charter is an excellent way to onboard them and ensure that they are aware of the team's values and expectations.

416. With the project expanding significantly in size, involving multiple teams, a lengthy duration, and a substantial budget, the project manager finds the sponsor replaced by a steering committee. What steps should the project manager take to accommodate this change?
   a) Share an updated status report with the new steering committee
   b) Acknowledge that the stakeholders' structure has changed and tailor communication accordingly
   c) Hold a meeting with the new steering committee and continue working on the project
   d) Work according to the original plan while keeping in mind that the project may now include additional stakeholders

   Explanation: The project manager should acknowledge that the stakeholders' structure has changed and tailor communication accordingly. The steering committee is a new stakeholder group that has replaced the sponsor. The

project manager should communicate with the steering committee in a way that is appropriate for their level of authority and responsibility. This may include providing more detailed reports, attending more meetings, and providing more frequent updates.

417. In choosing between two projects with same budgets but varying returns and risk levels, which tool or technique should the company employ to ensure the correct decision is made?
    a) Tornado diagram
    b) Sensitivity analysis
    c) Decision tree
    d) SWOT Analysis

    Explanation: A decision tree is a tool used to evaluate multiple options with different outcomes and risks. It is a useful tool for selecting the best option when there are multiple alternatives with different risks and returns.

418. A government contractor aims to manage costs effectively in their new project, shifting the risk to the vendor. Additionally, they seek to enforce legal obligations on the seller for contract completion to mitigate potential financial liabilities. What type of contract aligns with the contractor's requirements and minimizes risks as a buyer?
    a) Time and Materials (T and M)
    b) Cost Plus Incentive Fee (CPIF)
    c) Cost-reimbursable
    d) Fixed-price

    Explanation: A fixed-price contract is a type of contract where the buyer agrees to pay the seller a set amount of money for a specific product or service. In this type of contract, the seller bears the risk of cost overruns and is legally obligated to complete the contract. Therefore, a fixed-price contract best conforms to the contractor’s needs and represents the least risk for them as a buyer.

419. A project manager is overseeing a small team of developers using project management software that includes a Kanban board. In response to numerous inquiries from team members regarding the creation, review, and validation of tasks or issues, the project manager decides to generate the following chart. This chart is referred to as:
    a) Scatter diagram
    b) Flowchart
    c) Pareto chart
    d) Context diagram

    Explanation: The chart described in the question is a flowchart, which is a graphical representation of a process or system that shows the steps involved and their sequence. Kanban boards are often used in Agile project management

to visualize the flow of work and identify bottlenecks, and flowcharts can be used to further clarify the process.

420. Tasked with a 3D home design project, a project manager encounters challenges in scheduling daily standup meetings due to inconvenience for some remote team members. In addressing time zone differences, which conflict resolution technique should the project manager employ?
    a) Withdrawal
    b) Avoid
    c) Accommodate
    d) Compromise

    Explanation: Compromise is the right answer because it involves finding a solution that partially satisfies both parties. In this case, the project manager can adjust the meeting time to a more convenient time for the remote team members while still ensuring that the meeting is held daily.

421. A project manager at CreatiVista, an innovative design, web, and marketing agency, suggests creating a drop shipping e-commerce project, citing its current trend and profitability. However, his manager declines, pointing out CreatiVista's lack of experience in this field. Excited about the prospect of exploring this new venture and confident in his project management skills, what steps should the project manager take to persuade his manager to select the project?
    a) Calculate the Net Present Value (NPV)
    b) Conduct a what-if analysis
    c) Conduct a cost-benefit analysis of the business case
    d) Calculate the Return On Investment (ROI)

    Explanation: Conducting a cost-benefit analysis of the business case is the right answer since it will help the project manager to present the potential benefits of the project and compare them with the costs. This will help the manager to make an informed decision based on the analysis.

422. A project manager oversees a project involving both remote and local resources. The remote team expresses dissatisfaction, claiming insufficient information on the project's status, leading to rework. In response, the project manager arranges a conference call between the two teams and decides to switch to a more advanced enterprise chat platform to enhance communication. Which conflict resolution technique did the project manager employ?
    a) Compromise
    b) Force
    c) Smooth
    d) Collaborate

    Explanation: The project manager used the Collaborate conflict resolution technique by involving both teams in a conference call and switching to a

sophisticated enterprise chat platform to optimize communication. This technique aims to find a mutually beneficial solution that satisfies both parties.

423. In gaining a deeper comprehension of the project's system flows and representing the scope in abstraction, facilitating communication among stakeholders regarding both the system and its external boundaries, a project manager employs a visual depiction illustrating how people and other systems interact with the new system. Which tool is utilized by the project manager?
    a) Influence diagram
    b) Context diagram
    c) Affinity diagram
    d) Flowchart

    Explanation: The project manager uses a context diagram to better understand how the project's system flows, depicting the scope in abstraction, and enabling stakeholders to communicate about it, and what lies beyond its boundaries.

424. Noticing that her project is deviating from initial estimates, a project manager suspects that Student Syndrome may be a contributing factor. What does this syndrome indicate?
    a) People who begin to apply themselves only at the last possible moment before the deadline
    b) People who experience a learning curve whenever they are working on a project
    c) Professional project managers who consider themselves continuous learners
    d) People who proceed with their studies during their professional careers but are unable to balance both

    Explanation: Student Syndrome implies that people begin to apply themselves only at the last possible moment before the deadline. This is the right answer because Student Syndrome is a term used to describe the tendency of people to procrastinate and delay their work until the last minute, just like students who wait until the night before an exam to study.

425. A project manager, implementing the Scrum framework for a project, gets a call from the product owner requesting a meeting with the team to discuss potential approaches for implementing user stories and provide initial size estimations. What kind of meeting is the product owner alluding to?
    a) Story mapping
    b) Backlog refinement
    c) Sprint planning
    d) Brainstorming

    Explanation: The product owner is referring to a Backlog Refinement meeting, which is a Scrum event that occurs regularly to discuss and estimate user stories.

426. Three manufacturing lines are manufacturing cylindrical steel bars with a diameter of 5 mm. To conduct a quality inspection, the project manager randomly selects 10 bars from each line. The measurements in mm for Line 1 inspection results can be characterized as:
   a) Precise
   b) Accurate
   c) Both precise and accurate
   d) Neither precise nor accurate

   Explanation: Precision refers to consistency in measurements, and accuracy refers to being close to the target value .

427. A project manager is employed in a large organization with a matrix structure, characterized by intricate and cross-functional communication channels. Each day, the project manager engages in communication with the executive team and reports back to the manager. What type of communication is being described here?
   a) Triangular communication
   b) Vertical communication
   c) Parallel communication
   d) Horizontal communication

   Explanation: Option B is the right answer because the project manager is communicating with the executive team and reporting back to the manager, which is an example of vertical communication.

428. A project manager encounters difficulty in getting a team member to complete assigned tasks. To address this, the project manager invites the concerned member to watch a football game and subsequently discusses his performance issues after the game. What type of communication is the project manager employing?
   a) Visual
   b) Para-lingual
   c) Informal verbal
   d) Formal verbal

   Explanation: The project manager is using informal verbal communication by inviting the team member to watch a football game and then discussing his performance issues. This type of communication is not planned or structured and is used to build relationships and trust with team members.

429. Upon the completion of a rebranding project's execution, several key stakeholders, including the sponsor, hesitated to accept deliverables and officially close the project. What actions should the project manager take in this situation? (Choose three)

a) Formally close the project. Stakeholders will get well-acquainted with the new brand by that time.
b) Identify and resolve any open issues. Then, be firm on formal closure.
c) Identify and openly discuss the reasons for reluctance.
d) Invite earnest feedback from all sides and try to identify any misunderstanding.

Explanation: The project manager should identify and resolve any open issues, openly discuss the reasons for reluctance, and invite earnest feedback from all sides to identify any misunderstanding. This will help to ensure that all stakeholders are satisfied with the deliverables and that the project can be formally closed.

430. A project manager holds the view that being an active listener is challenging. She perceives active listening as more of an art than a skill, prompting her to frequently advise her team members to utilize a checklist outlining the key elements of active listening before crucial meetings with stakeholders. Which of the following are considered elements of active listening? (Choose three)
a) Paraphrasing
b) Making eye contact
c) Interrupting when appropriate
d) Interpreting information

Explanation: These elements help the listener to understand the speaker's message and respond appropriately.

431. Typically, the company executives make the ultimate decision on project selection for execution. Nevertheless, a project manager consistently leverages their experience to aid the organization's decision-makers in selecting the most suitable project using an appropriate benefit measurement project selection method. Which of the following is not considered a benefit measurement method?
a) Scoring model
b) 5 Whys
c) Internal Rate of Return
d) Scoring model

Explanation: The 5 Whys is a root cause analysis technique and not a benefit measurement method.

432. A project manager at Steel Foundations has joined the project in its initial phases and has been tasked to write its charter. In the last week, the project manager met with several stakeholders, gathered their input, and is now prepared to have the charter signed. Who holds the responsibility for signing the project charter?
a) Project stakeholders
b) Senior Management
c) Project Sponsor

d) Project Manager

Explanation: The project sponsor is responsible for signing the project charter. The project manager is responsible for developing the charter, but the sponsor is the one who authorizes it.

433. The authority and influence of a project manager vary based on whether they operate within a functional, matrix, or projectized organization. Upon assuming the role of a project manager, certain inherent forms of legitimate power are conferred upon them. What are the three manifestations of legitimate power?
   a) Formal, reward, and penalty
   b) Absolute, direct, and indirect
   c) Formal, informal, and persuasive
   d) Formal, reward, and direct

   Explanation: The right answer is option A because the three forms of legitimate power are formal, reward, and penalty.

434. While discussing the decision to utilize remote resources with the project sponsor, a project manager listed numerous benefits of collaborating with remote teams. They highlighted all of the following advantages except:
   a) Reduction of time spent commuting
   b) Utilization of a war room
   c) Less travel and relocation expenses
   d) Access to more skilled resources

   Explanation: The project manager did not mention utilization of a war room as an advantage of working with remote teams.

435. A project manager is overseeing a predictive project. After identifying a risk, they choose to transfer it to another party by:
   a) Asking the sponsor to deal with the risk
   b) Accepting a lower profit in case of cost overrun
   c) Eliminating the risk through beta testing
   d) Buying insurance

   Explanation: Transferring risk to another party can be done through buying insurance, which is a common method of transferring risk.

436. A project manager gives team members the autonomy to make decisions, anticipating that they will solve issues on their own. Which of the following leadership approaches is adopted by the project manager?
   a) Interactional leadership
   b) Laissez-faire leadership
   c) Transactional leadership
   d) Servant leadership

Explanation: The project manager adopts the Laissez-faire leadership approach, which is characterized by giving team members the freedom to make decisions and solve issues on their own.

437. After being assigned to oversee a distressed project, a project manager examines the schedule and identifies that the previous project manager failed to account for procurement delays. Which of the following options represents this situation?
    a) Assumption
    b) Constraint
    c) Schedule variances
    d) Critical path activity

    Explanation: The scenario represents an assumption because the previous project manager did not consider procurement delays while creating the schedule, which is an assumption that turned out to be incorrect.

438. A project manager is tasked with overseeing a software development project. Upon encountering a key stakeholder from the company in the hallway, the project manager took advantage of the opportunity to seek their feedback. The key stakeholder, impressed with the project's progress, urgently asked the project manager to add a new user story for marketing the product. How should the project manager respond to this request?
    a) Refuse to create the user story since it’s the product owner’s responsibility to maintain and refine the product backlog
    b) Add the user story to the product backlog and let the product owner decide its priority
    c) Welcome the stakeholder’s request and add the user story to the sprint backlog
    d) Add the user story to the product backlog and schedule it for the next sprint to protect the team from disruptions

    Explanation: The project manager should add the user story to the product backlog and let the product owner decide its priority. The product owner is responsible for maintaining and refining the product backlog, and they are the ones who should decide the priority of the user story. The project manager should not make decisions that are outside their scope of responsibility.

439. You, as a Project Manager working under a strict deadline, come across a complex work package on the critical path with the potential to delay the final product delivery. After consulting your customer, you decide to remove this work package from the project. What risk response strategies do you employ in this case?
    a) Avoid
    b) Sharing
    c) Retention

d) Reduction

Explanation: Dropping the work package is an example of avoiding the risk. This strategy is used when the risk is too high, and the project team decides to eliminate it completely.

440. When developing the probability and impact matrix, what factors should the team and Project Manager consider to support the project risk analysis?

a) Project scope and project deliverables
b) Possible effects on the objectives of the project
c) Contingency reserve for budget and schedule
d) Project quality and resources

Explanation: The team and project manager should consider the possible effects on the objectives of the project while creating the probability and impact matrix to support the project risk analysis. This helps in identifying the risks that could have a significant impact on the project objectives and prioritize them accordingly.

441. As a result of analyzing risks with the project team, you, as the Project Manager, opt to establish a contingency reserve to address the potential threat. What category of strategy does this represent?
a) Active mitigation
b) Passive acceptance
c) Active acceptance
d) Passive transference

Explanation: Establishing a contingency reserve is an active acceptance strategy. This strategy involves taking proactive steps to prepare for a risk event and having a plan in place to deal with it if it occurs.

442. What Processes should be employed to implement agreed-upon risk response plans? The advantage of this procedure is that it guarantees the execution of agreed-upon risk responses as intended.
a) Implement risk responses
b) Plan risk management
c) Monitor risk
d) Identify risk

Explanation: The Implement Risk Response process is used to execute agreed-upon risk response plans, ensuring that they are carried out as planned.

443. A company faces recurring quality problems with a sole-source supplier. Late deliverables on a new project will result in substantial penalties from the customer. What actions should the Project Manager take?
   a) Identify high-level risks prior to developing contingency plans
   b) Escalate the issue to top management
   c) Negotiate with other vendors with better quality-control methods.
   d) Review incoming inspection reports and quality logs

   Explanation: Escalating to top management ensures that the problem receives immediate attention and resources, allowing for strategic decisions to mitigate the impact on the project and avoid substantial penalties.

444. A buyer, upon sending the proposal (RFP) to potential vendors, received inquiries from some vendors regarding the terms and conditions after a few days. As a Project Manager, what steps should you take to ensure that all vendors have a clear and common understanding?
   a) Conduct one-to-one meetings with vendors
   b) Resubmit the RFP
   c) Send a clarification to the vendors via email
   d) Conduct a bidder conference

   Explanation: Conducting a bidder conference is the best way to ensure that all vendors have a clear and common understanding of the terms and conditions of the RFP. This allows all vendors to ask questions and receive answers at the same time, ensuring that everyone has the same information.

445. The Project Manager finds that the concrete for a project arrives on time but realizes that its composition is incorrect. Which document should the Project Manager review to confirm the concrete's adherence to performance and quality requirements?
   a) Procurement management plan
   b) Quality management plan
   c) Request for Proposal (RFP)
   d) Procurement statement of work
   Explanation: The procurement statement of work outlines the specific requirements for the product or service being procured, including performance and quality requirements. Therefore, the project manager should review the procurement statement of work to verify the concrete's performance and quality requirements.

446. Upon receiving an email from a vendor indicating a one-month delay in key deliverables, what should be the Project Manager's initial course of action?
   a) Meet with the team to identify ways to absorb this delay
   b) Update the management and obtain their advice
   c) Assess the impact of the delay on the project plan
   d) Call the vendor to obtain a justification for the delay

   Explanation: Assessing the impact of the delay on the project plan is the first thing the project manager should do. This will help the project manager understand the extent of the delay and determine what actions need to be taken to mitigate the impact of the delay on the project.

447. A project is on a Cost-Plus-Fixed-Fee (CPFF) contract with a fixed fee of US $5,000. The estimated cost is US $50,000. If the project ends up costing US $60,000 without any changes to the project scope, what will be the total cost of the contract?
   a) US$65,000
   b) US$80,000
   c) US$55,000
   d) US$75,000

   Explanation: The total cost of the contract is the sum of the fixed fee and the actual cost. Therefore, the total cost of the contract is US$65,000 (US$60,000 + US$5,000).

448. A seller has completed all the work according to the contract, but the buyer contends that the seller's performance falls outside the acceptable range for one of the buyer's internal metrics. Consequently, the buyer opts to deduct a portion of the payment to the seller. In accordance with the contract terms, what is the buyer allowed to do?
   a) The buyer must pay the seller for the work performed
   b) The buyer is allowed to deduct a small sum from the payment to the seller due to non-compliance with the buyer's internal metric value range
   c) The buyer should go to court, or else the seller may sue them for non-payment
   d) The seller should complete the work again to meet the buyer's internal metric value range

   Explanation: As per the contract, the buyer must pay the seller for the work performed.

449. As a Project Manager representing the buyer, the CEO of a company inquires whether the vendor is meeting the deliverables to assess if the contract should be terminated early. What information should be provided to the CEO?
   a) Performance requirements
   b) Procurement management plan
   c) Proposal evaluation techniques
   d) Seller performance evaluation

   Explanation: The right answer is d because the seller performance evaluation provides information on whether the vendor is meeting the deliverables or not. This information can be used to determine if the contract should be terminated early.

450. As a Project Manager representing the buyer, you've given a contract to a small startup company. During the project, the supplier informs you that they won't be able to deliver all the required deliverables. The regular monthly payment for this supplier is scheduled for this week. What should you do next as a Project Manager?
   a) Update the schedule baseline to incorporate the delay
   b) Delay the payment to the supplier
   c) Hold a meeting with the supplier to understand delivery problems and possibly solve them.
   d) Remove the supplier from your project

   Explanation: Holding a meeting with the supplier to understand delivery problems and possibly solve them is the right answer. As a project manager, it is important to understand the root cause of the problem and work with the supplier to find a solution. Delaying the payment or removing the supplier from the project may not solve the problem and could lead to further delays and complications. Updating the schedule baseline is not the right answer, as it does not address the root cause of the problem.

451. As the Project Manager, you've faced quality issues in your project, primarily due to a supplier. Despite a final invoice of $20,000, including rework prompted by their quality problems, you revised the payment to $18,000. After sending this payment with justification, the supplier issued a new invoice, indicating a balance due of $2,000. What should be your next course of action?
   a) Alternative dispute resolution (ADR)

b) Ignore the supplier's second invoice and consider your payment of $18,000 payment in full
c) Negotiate with the supplier
d) Contact the legal team about pursuing legal action against the supplier

Explanation: Negotiating with the supplier is the best course of action. The supplier has billed for rework that was created by their quality issues, and the project manager has arrived at a revised amount of $18,000 to be paid. However, the supplier has reinvoiced with a balance due of $2,000. Negotiating with the supplier is the best course of action to resolve the issue and come to a mutually agreed-upon solution.

452. After concluding procurements and issuing a formal written notice stating that the contract has been fulfilled, what should be the subsequent actions taken by the Project Manager?
a) Inform the stakeholders
b) Write the final project report
c) Update the lessons learned register
d) Initiate procurement audits

Explanation: Updating the lessons learned register is the next step after closing procurements and providing a formal written notice that the contract has been completed. This is because the lessons learned register is a key output of the project closure process, and it captures the knowledge and experience gained during the project, including procurement-related lessons learned.

453. As a Project Manager, to verify if your project still aligns with the organizational goals, which document would be beneficial?
a) Business Case
b) Business need
c) Product scope description
d) Issue log

Explanation: The Business Case document outlines the justification for the project and how it aligns with the organizational goals. Reviewing the Business Case can help the project manager ensure that the project is still aligned with the organization's goals.

454. The Project Manager notices a delay in the project due to differences in views between the organization's senior management and external

stakeholders regarding contracts. What actions should the Project Manager take in response?

a) Rebaseline the scope and project's schedule to take into account delays in decision making
b) Review the sufficiency of the project's governance and ensure that an appropriate structure is in place
c) Isolate the project team members from senior management and stakeholders so that decisions can be made more quickly
d) Review the stakeholder register and ensure that communications plans are being followed

Explanation: The project manager should review the sufficiency of the project's governance and ensure that an appropriate structure is in place. This will help to identify the root cause of the delay and take corrective actions to address the issue.

455. In a project status meeting, a team member notifies the Project Manager about an unforeseen rise in the price of a crucial component needed to complete the deliverable. What steps should the Project Manager take in response?

a) Analyze the impact of the price increase on the project
b) Include the increased price in the project budget
c) Update the risk management plan to adapt to the new scenario
d) Refer to the procurement management plan

Explanation: The project manager should analyze the impact of the price increase on the project to determine the best course of action. This includes assessing the impact on the project budget, schedule, and scope and identifying any potential risks or issues that may arise as a result of the price increase.

456. In a project with multiple iterations, a Project Manager is facing challenges in obtaining consensus on the issues encountered by the delivery team. These issues are extensive and involve the majority of the project and stakeholder teams. What actions should the Project Manager take to gain consensus?

a) Conduct a retrospective
b) Conduct a stakeholder barnstorming session
c) Hold a stakeholder analysis meeting
d) Hold a lesson's learned session

Explanation: A retrospective is a meeting held at the end of an iteration to discuss what went well, what didn't go well, and what can be improved. It is a great way to gain consensus on issues and to identify areas for improvement.

457. You are a Project Manager responsible for a crucial project in your company. Within your project team, there are frequent technical disagreements among members and the lead. What conflict management technique should be used by the Project Manager to resolve this?
    a) Withdraw/avoid
    b) Compromise/reconcile
    c) Force/direct
    d) Collaborate/problem solve

    Explanation: Collaboration is the best conflict management technique to use in this situation because it involves working together to find a mutually beneficial solution. This technique is especially useful when there are technical disagreements among team members and the lead.

458. You're in the norming stage of the Tuckman model, and there's a tough choice needs to be made about a scope change. The senior team lead prefers option X because of their experience, while the rest of the team prefers option Y for its new technology that will reduce the complexity of the implementation. You decide to go with the senior member's suggestion. What is this an example of?
    a) Forcing
    b) Compromising
    c) Withdrawal
    d) Win-win

    Explanation: This is an example of forcing because the project manager is using his or her authority to make a decision without considering the input of the team.

459. In a team meeting, team members frequently deviate from the topic and interrupt each other, leading to the meeting exceeding the planned duration. What actions should the Project Manager take in response to this situation?
    a) Direct the meetings
    b) Better manage the communication channels
    c) Conduct the ground rules
    d) Co-locate the team members in a place that is suitable for the team

Explanation: Conducting ground rules is the right answer because it helps to establish guidelines for the meeting and ensure that everyone stays on topic and respects each other's opinions. This can help to make the meeting more efficient and effective.

460. The project is behind schedule and over budget, leading to increased stakeholder pressure and placing the project team under serious stress. This results in a conflict between the quality and production managers, each accusing the other of the delay and cost overruns. What actions should the Project Manager take in response?
    a) **Meet with both managers and ask them to immediately end the conflict to avoid negatively impacting the project**
    b) Discuss the situation with the stakeholders and follow their suggestions for dealing with the conflict
    c) Meet with the managers' supervisors to discuss possible actions
    d) Refer to the team charter for guidelines on how to deal with this conflict

    Explanation: The project manager should meet with both managers and ask them to immediately end the conflict to avoid negatively impacting the project. This is the best course of action as it addresses the conflict directly and encourages the managers to work together to find a solution. It also helps to reduce the stress on the project team and can help get the project back on track.

461. As the Project Manager for an industrial design project, concerns about inadequate project communications management are raised by some stakeholders during project execution. Despite thorough preparation of project management plans, particularly the Communications Management Plan, you are surprised by these claims, considering your efforts. In this scenario, who is currently accountable for managing the expectations of these stakeholders?
    a) Project Team
    b) Project Sponsor
    c) **Project Manager**
    d) Project Stakeholders

    Explanation: The Project Manager is responsible for managing stakeholder expectations, including those related to project communications management. The Project Manager should investigate

the concerns raised by stakeholders and take appropriate action to address them.

462. A new influential stakeholder enters the organization and will be accountable for certain aspects of the project. What should the Project Manager prioritize as the initial step to effectively engage with this stakeholder?
   a) Schedule a meeting with the new project stakeholder
   b) Submit a change request to update the stakeholder engagement plan
   c) Revise the stakeholder engagement assessment matrix
   d) Update the RACI chart with the responsibilities of the new stakeholder

   Explanation: The Project Manager should schedule a meeting with the new stakeholder to introduce themselves, explain the project, and discuss the stakeholder's role and expectations. This will help to establish a positive relationship and ensure that the stakeholder is engaged and committed to the project.

463. As a Scrum Master leading a retrospective for a sprint that didn't meet its goals, the development team and Product Owner are arguing about who is responsible for the failure. You uncover the conflict's source after performing a root cause analysis. What is the most probable discovery among the following options?
   a) The development team decided how many user stories were to be completed during the sprint
   b) The Gantt chart did not include enough story points
   c) You, as the Scrum Master, failed to assign user stories to the development team
   d) The Product Owner determined how many user stories were to be completed during the sprint

   Explanation: The Product Owner is responsible for determining the user stories to be completed during the sprint. Therefore, if the sprint failed to meet its objectives, it is likely that the Product Owner did not provide clear or realistic goals for the development team to work towards.

464. You are collaborating with research associates to assess the potential benefits of implementing a WiFi system in the residents. As part of this initiative, you will work with authorities to obtain relevant government documents. Additionally, you will conduct an extensive survey on how people use the system and generate reports based on the survey. What term is commonly used to describe this effort?

a) A survey
b) An operation
c) A research study
d) A project

Explanation: This effort involves working with a team to achieve a specific goal, which is to determine if implementing a WiFi system in the neighborhood is going to benefit residents. This effort is temporary, unique, and has a defined start and end date, which makes it a project.

465. The business case outlines the business needs and the cost-benefit analysis that supports the project. Which of the following most accurately describes the Project Manager's responsibility in the project business case?

a) Development of the project business case

b) The Project Manager may make recommendations to keep the project business case in alignment with the goals and objectives of the organization

c) Making changes on the business case if necessary

d)Update the business case at the end of each phase

Explanation: The Project Manager may make recommendations to keep the project business case in alignment with the goals and objectives of the organization.

466. As a part-time manager in a weak matrix organization, your Functional Manager increases your workload by appointing you as the Project Manager for a new business idea. Upon reviewing the business case, you realize that recruiting new personnel is necessary to execute the project. What constraints will you be operating under in terms of hiring additional staff?
a) You, as project manager, would be responsible for hiring staff
b) Senior management would be responsible for hiring staff
c) The functional manager would be responsible for hiring staff
d) All stakeholders would be responsible for hiring staff

Explanation: As per the weak matrix organization, the Functional Manager has more control over the project than the Project Manager. Therefore, the Functional Manager would be responsible for hiring staff.

467. You are the Project Manager for a project where team members maintain a dual reporting structure. In addition to reporting to you, they also report to

their functional manager. Moreover, team members are required to balance project tasks with departmental responsibilities. In what form of organization are you working?

a) Matrix
b) Tight Matrix
c) Project-oriented
d) Functional

Explanation: In a matrix organization, team members have a dual reporting structure where they report to both the project manager and their functional manager. This structure allows team members to work on both project work and department work.

468. During your project meetings, you consistently observe that project staff nod their heads but do not speak much or ask questions. Which of the following training options will be beneficial in addressing this situation?

a) Interpersonal skills
b) Presentation skills
c) Active listening skills
d) Body language training

Explanation: Active listening skills will help improve this situation because it will help project staff to better understand the information being presented and ask relevant questions.

469. A Portfolio Manager notifies the Project Manager of the termination of their project. The Project Manager intends to communicate this information to stakeholders and team members. Which document should the Project Manager refer to for guidance?

a) Change Management Plan
b) Communication management plan
c) Responsibility assignments matrix (RAM)
d) Resources Management Plan

Explanation: The Project Manager should refer to the Communication Management Plan to inform stakeholders and team members about the termination of the project. This plan outlines the communication requirements and expectations for the project, including who needs to be informed, what information needs to be communicated, and how it should be communicated.

470. Micheal, serving as the project manager for a research and development project, has faced challenges in recent meetings where more time was dedicated to arguments and less to decision-making, leading to prolonged sessions. What actions should be taken to prevent this situation?
   a) Use listening techniques
   b) Invite only team members boss
   c) Set a virtual conversation
   d) Set a meeting agenda before the meeting

   Explanation: Setting a meeting agenda before the meeting helps to keep the meeting on track and focused on the topics that need to be discussed. This can help prevent arguments and ensure that decisions are made efficiently.

471. You have just finished the initial phase of a multi-phase project, and after calculating earned value measurements, you discovered that the current CPI is 0.79 and the current SPI is 1.08. In planning the next phase, the primary focus should be on which element of the project?
   a) Cost
   b) Schedule
   c) Scope
   d) Quality

   Explanation: The current CPI of 0.79 indicates that the project is over budget, while the SPI of 1.08 indicates that the project is ahead of schedule. Therefore, the next phase plan should focus on cost management to bring the project back on track and within budget.

472. While gathering requirements and defining the project's products, you've hired an external consultant to offer advice on enhancing the process and boosting the standards for associated methods. The consultant possesses extensive knowledge of industry best practices in this regard. What will be the consultant's primary focus?
   a) Administer a questionnaire with key stakeholders in your project about their best practices
   b) Teach requirements analysis techniques to the project management team
   c) Perform benchmarking by recommending best-in-class practices in the industry related to collecting requirements and defining products

d) Brainstorm among your team members to define the best practices for requirements gathering and product definition

Explanation: The consultant is mainly going to perform benchmarking by recommending best-in-class practices in the industry related to collecting requirements and defining products. This will help the project management team to raise the standard for their associated methods.

473. While running a software development project, the project stakeholders are currently in the process of accepting deliverables. As they perform this task, they wish to assess whether the deliverables meet non-functional requirements such as the level of service, performance, and supportability. Where are these non-functional requirements usually documented?
    a) Issue Log
    b) Requirements Management Plan
    c) Work Breakdown Structure
    d) Requirements Documentation

    Explanation: Non-functional requirements such as level of service, performance, and supportability are usually documented in the Requirements Documentation.

474. As a Project Manager organizing a project for a new payroll system, you are presently engaged in identifying, documenting, and managing stakeholder needs to achieve project objectives. One of the stakeholders has proposed incorporating a visual representation illustrating how individuals and other systems will interact with the new payroll system. You find the suggestion valuable and wish to incorporate it. How can you best implement this stakeholder's suggestion?
    a) Build a matrix diagram
    b) Design an affinity diagram
    c) Develop a context diagram
    d) Create a fishbone diagram

    Explanation: A context diagram is a visual representation of the system and its environment, showing the system as a single process and its interactions with external entities. It is used to show the scope of the system and its boundaries, and how it interacts with other systems and people. Therefore, it is the best way to implement the stakeholder's suggestion.

475. A Project Manager has successfully delivered a significant project deliverable. However, a week after the release, the customer expressed dissatisfaction with the deliverable. What should be the Project Manager's next course of action?

   a) Do a scope verification of this deliverable to check if it satisfies project objectives
   b) Continue with the next deliverable, as the project has to be finished on time
   c) Talk to the Senior Manager to discuss the customer's concern and resolution
   d) Ensure that the next deliverables have the features that the customer specified and exceed his expectations

   Explanation: The Project Manager should do a scope verification of the deliverable to check if it satisfies project objectives. This is because scope verification is the process of formalizing acceptance of the completed project deliverables. It is important to ensure that the deliverable meets the project objectives and requirements before it is accepted. If the deliverable does not meet the requirements, the Project Manager can take corrective action to address the customer's concerns.

476. The Project Manager is presently involved in identifying, documenting, and managing stakeholder needs and requirements. Given the sensitive nature of the project, there is a worry that certain stakeholders may withhold honest feedback unless confidentiality is maintained. What is the optimal course of action for the Project Manager?

   a) Conduct interviews to elicit information from stakeholders by talking to them directly
   b) Hold a focus group to learn about stakeholder expectations and attitudes about the project
   c) Use multicriteria decision analysis to provide a systematic analytical approach for establishing criteria
   d) Use brainstorming to generate and collect multiple ideas related to project requirements

   Explanation: Conducting interviews to elicit information from stakeholders by talking to them directly is the best course of action for the Project Manager. This approach allows the Project Manager to obtain candid feedback from stakeholders while maintaining confidentiality.

477. A project scope statement is created outlining project deliverables, constraints, and assumptions related to the project objectives. What is the subsequent step for the Project Manager?
   a) Take a formal acceptance of the completed project deliverables
   b) Establish a hierarchical breakdown of the total scope to be carried out by the project team
   c) Determine the products, services, and results to be delivered
   d) Monitoring the status of the project and product scope and managing changes to the scope baseline

   Explanation: Establishing a hierarchical breakdown of the total scope to be carried out by the project team is the next step after defining the project scope statement. This is done through creating a Work Breakdown Structure (WBS), which is a hierarchical decomposition of the total scope of work to be carried out by the project team to accomplish the project objectives and create the required deliverables.

478. A Project Manager possesses a good understanding of the Project Stakeholders' expectations. What should be the Project Manager's next step to plan the project effectively?
   a) Assess required competencies against resource availability
   b) Discover needs and decompose them into requirements
   c) Determine the project budget and funding sources
   d) Conduct a kick-off meeting with the team

   Explanation: Discovering needs and decomposing them into requirements is the next step in the project planning process after understanding the expectations of the project stakeholders. This helps to ensure that the project requirements are clearly defined and understood by all stakeholders.

479. You are responsible for the construction of a high-rise residential building. After creating a network diagram, you've identified the critical path. While work has started, there is an unexpected delay in the plinth work due to inclement weather. The plinth work activity was on the critical path. As the manager driving this project, would this situation concern you, and if so, why?
   a) Yes - you will now have to relay this bad news to the client
   b) No - It will not matter if the Plinth work is delayed.
   c) Yes - Any delay in activities on the critical path would lead to a delay in the entire project

d) No - Activities on the critical path do not impact the schedule

Explanation: The right answer is C because any delay in activities on the critical path would lead to a delay in the entire project. The critical path is the longest path in the network diagram, and any delay in the activities on the critical path will delay the project's completion date.

480. As the Project Manager leading a crucial project for your company, a contractor submits a high-level project schedule for a three-year project, which includes a detailed design-phase schedule. What does this describe?
   a) Schedule management plan
   b) Expert judgment
   c) Rolling wave planning
   d) Decomposition

   Explanation: Rolling wave planning is a technique used in project management to plan and execute projects in waves as work is being done. It is a form of progressive elaboration, where the project is planned in phases, and each phase is planned in more detail as the project progresses. In this case, the contractor has submitted a high-level project schedule for a three-year project, including a detailed design-phase schedule, which is an example of rolling wave planning.

481. The specified upper and lower control limits are 12 and -12, respectively, and you have recorded the following data point readings: 1.6, 3.7, 7.8, 9.1, -6.8, 14, 8.9, 4.2, 0.0, -5.2. How would you assess this process?
   a) Out of control
   b) In control
   c) It has a predictable performance
   d) Can't be determined

   Explanation: The process is out of control because some of the data points are beyond the established control limits.

482. During the construction of a new water treatment plant, the government introduced more stringent water quality requirements for treatment plants. What should be the subsequent steps for the Project Manager?
   a) Submit a change request to update the quality management plan
   b) Ask the project sponsor to update the project charter
   c) Update the scope baseline to reflect the changes
   d) Submit a change request to update the scope management plan

Explanation: The Project Manager should submit a change request to update the quality management plan because the new water quality requirements will impact the quality standards of the project.

483. The vendor you chose for supplying materials to your project is facing numerous issues with the quality control department, and there has been significant conflict with them in the past. As a Project Manager, what should be your next course of action?
    a) Perform quality audits
    b) Raise the issue to sponsor
    c) Directing the Quality Department to reject all materials related to the vendor
    d) Bring the vendor to meet the control quality department and follow the quality procedures

    Explanation: As a Project Manager, it is important to ensure that all stakeholders are aligned and working together towards the project objectives. In this case, the project vendor has had issues with the quality control department in the past, so it is important to bring them together to ensure that they follow the quality procedures and work together to resolve any issues that may arise. This will help to ensure that the project materials meet the required quality standards and are delivered on time and within budget.

484. To enhance the quality of project deliverables, the Project Manager decides to engage the project team in establishing project standards, policies, and guidelines. However, the team perceives this as unnecessary effort and prefers to proceed with the project as originally planned. What should the Project Manager do next?
    a) Organize a project team meeting to create a shared vision.
    b) Meet with individual team members and force them to comply.
    c) Produce the quality management plan and distribute it to stakeholders.
    d) Log an issue with the team and discuss it with the human resources department.

    Explanation: The project manager should organize a project team meeting to create a shared vision. This will help the team understand the importance of defining project standards, policies, and guidelines and how it can improve the quality of project deliverables. By involving the

team in the process, they will feel more invested in the outcome and be more likely to comply with the new standards.

485. The end-user rejects certain deliverables, even though many requirements are fulfilled. After conducting a root cause analysis, the Project Manager discovers that a particular quality control process was omitted due to resource constraints. What actions could the Project Manager have taken to prevent this situation?
   a) Held quality control process meetings
   b) Created quality checklists
   c) Requested more resources for quality control activity
   d) Ensured that quality was managed

   Explanation: The Project Manager should have ensured that quality was managed to avoid this issue. This includes performing quality control processes and ensuring that all requirements are met.

486. During a software project, the Project Manager observes that one developer's coding quality is statistically lower than another developer assigned the same task. Which tool should the Project Manager employ to identify the actionable root cause of this problem?
   a) Scatter diagram
   b) Ishikawa diagram
   c) Histogram
   d) Control chart

   Explanation: The Ishikawa diagram is a tool used to identify the root cause of a problem. It is also known as a fishbone diagram or cause-and-effect diagram. By using this tool, the Project Manager can trace the problem's source back to its actionable root cause.

487. A Project Manager is managing a project with an expectation of extreme weather. The project is exceeding its budget, and the project sponsor is unwilling to allocate additional funds. To reach the milestone before the extreme weather begins, which of the following techniques should the Project Manager employ?
   a) What-if scenario analysis
   b) Crashing
   c) Resource leveling
   d) Fast-tracking

Explanation: Fast-tracking is a technique used to shorten the project schedule by overlapping activities that would normally be done in sequence. In this case, the Project Manager can use fast tracking to complete the remaining work before the extreme weather begins, without additional funding.

488. Sam, the Project Manager of a software project initially projected to be finished in 12 months, receives reports of a higher-than-expected failure rate during the testing phase. What should Sam do in response?
    a) Create a fishbone diagram
    b) Create a product flowchart
    c) Perform integrated change control
    d) Create Scatter diagram

    Explanation: Creating a fishbone diagram is a tool used in quality management to identify the root cause of a problem. In this case, the higher-than-expected failure rate during testing is a problem that needs to be addressed. By creating a fishbone diagram, Sam can identify the root cause of the problem and take corrective action to address it.

489. A project faced a significant team conflict during the final sprint. However, the Agile team succeeded in delivering a product that was accepted by the customer. What should the Project Manager do at the end of the sprint?
    a) Meet individually with the involved team members to express concerns regarding their behavior
    b) Conduct a retrospective meeting with the team to discuss what happened
    c) Disregard the conflict since the team succeeded in achieving the project goals
    d) Disregard the conflict since the project is coming to an end

    Explanation: The Project Manager should conduct a retrospective meeting with the team to discuss what happened. This will help the team to identify the root cause of the conflict and come up with ways to prevent it from happening in future projects.

490. In a retrospective meeting, certain Agile team members mentioned that it can be confusing to track who is handling each work item and at which stage they are. What steps can the Project Manager take to address this issue?
    a) Guide the team on how to use the Kanban board and update it regularly

b) Coach the team on how to be more attentive during the daily standup meeting
c) The Project Manager shouldn't do anything since agile teams are self-organizing
d) Send a daily status report to all team members to remove the impediment

Explanation: The Kanban board is a visual tool that helps the team to track the progress of work items. By updating the board regularly, the team can easily see who is working on each item and at what stage they are. This will help to eliminate confusion and improve transparency.

491. The project team's morale has been low ever since a key project stakeholder, who provided significant support and motivation, left. What should the project manager do as the first step?
a) Perform team-building activities to strengthen team bonds
b) Set objectives and encourage the team to attain them
c) Understand the team's needs and what motivates them
d) Try to prevent the departure of the key stakeholder

Explanation: The Project Manager should first understand the team's needs and what motivates them to address the low morale issue. This will help the Project Manager identify the root cause of the problem and develop a plan to address it.

492. To guarantee the cross-functionality of their Agile team, a Project Manager hires in a new team member with a focus on testing tasks. After a few days, the new member informs the Project Manager that they are struggling to perform their tasks effectively due to lacking the necessary permissions to access a section of the code. What should the Project Manager do in response?
a) Tell the team member to adjust the acceptance criteria of their pending tasks and then inform the Product Owner
b) Explain to the team member that they cannot access the whole code due to security restrictions
c) Ask the team member to raise the issue during the retrospective meeting
d) Reach out to the appropriate stakeholder to either confirm or update the team member's privilege

Explanation: The Project Manager should reach out to the appropriate stakeholder to either confirm or update the team member's privilege. This

is the right answer because the project manager needs to ensure that the team member has the necessary permission to access the code to perform their tasks effectively.

493. In the middle of the sprint, an Agile team encountered so many obstacles that it became impossible to produce any working increments. What steps should the Project Manager take in response?
   a) Cancel the sprint
   b) Continue supporting the development team even though the sprint goal will not be achieved
   c) Ask the development team to work on the less difficult work items first
   d) Proceed with resource-leveling or smoothing

   Explanation: The Project Manager should continue supporting the development team even though the sprint goal will not be achieved. This is because the Agile methodology encourages continuous improvement and learning, and the team can use the experience gained from the failed sprint to improve future sprints.

494. During sprint zero, a Scrum team undertakes some preliminary work before officially starting a new project. Which of the following items cannot be defined by the Scrum team during this stage?
   a) Product vision statement
   b) Release plan
   c) Baselined product backlog
   d) Product roadmap

   Explanation: During the sprint zero, the Scrum team can define the product roadmap, product vision statement, and release plan. However, the product backlog is not baselined during this stage.

495. While discussing the technical choices for their new project during the zero sprint, an Agile team had a senior member recall a previous project with significant challenges, resulting in exceeding the budget and falling behind schedule. What should the Project Manager's next step be?
   a) Since the mentioned past project followed a predictive approach, its performance details are irrelevant
   b) Look for the performance details of the mentioned past project in the Organizational Process Assets (OPAs)

c) Look for the performance details of the mentioned past project in the issue log
d) Look for the performance details of the mentioned past project in the Enterprise Environmental Factors (EEF)

Explanation: The Project Manager should look for the performance details of the mentioned past project in the Organizational Process Assets (OPAs) to learn from the past project's challenges and avoid repeating the same mistakes.

496. During one of the initial project meetings, a stakeholder conveyed to the Project Manager that the video editing software they intend to create should incorporate a free subscription plan. They specified that this plan should only permit users to edit videos of less than 5 minutes and that all exported videos must be watermarked. Where should this information be documented? (Choose two)
a) Scope management plan
b) Project charter
c) Product backlog
d) Project scope statement

Explanation: The information provided by the stakeholder should be captured in the project scope statement and the product backlog. The project scope statement defines the project scope and the product backlog contains the requirements for the product. The stakeholder's request for a free subscription plan with certain limitations is a requirement that should be captured in the product backlog. The project scope statement should also capture this requirement as it defines the project scope.

497. While reviewing an important Epic, members within the self-organizing team had varying interpretations of how it should be decomposed, highlighting a lack of shared understanding regarding the project's purpose. What steps should the Project Manager take?
a) Ask the team to check the release plan
b) Remind the team of the project vision
c) Ask the team to check the acceptance criteria of the Epic
d) Remind the team of the sprint goal

Explanation: The Project Manager should remind the team of the project vision to ensure that everyone has a common understanding of the

project's purpose. This will help the team to align their interpretations and work towards the same goal.

498. With 8 years of experience in the same role, a Project Manager is promoted to the position of Portfolio Manager. What is the definition of a portfolio among the following options?
    a) Creates a common vision and helps senior management to see the potential strategic implications of corporate decisions
    b) Following methods, processes, and tools to ensure that projects are managed as designed
    c) A number of projects, programs, or operations that are all managed as a collection with the aim of achieving the organization's strategic goals
    d) Analyzing IT requirements and ensuring regulatory requirements are followed

    Explanation: A portfolio is a collection of projects, programs, or operations that are managed as a group to achieve the organization's strategic goals. This is the correct definition of a portfolio.

499. Before initiating a new project, a Project Manager conducts a cost-benefit analysis to evaluate potential costs and revenues. The outcome of the analysis will decide if the project is likely to be profitable or if the organization should consider an alternative project. The Project Manager discovers that the project's Benefit-Cost Ratio is 1.3. What does this indicate?
    a) The project's revenue is 1.3 times its cost
    b) The project's revenue is 1.3 times its profit
    c) The project's profit is 1.3 times its cost
    d) The project's cost is 1.3 times its profit

    Explanation: The Benefit-Cost Ratio (BCR) is a financial metric that compares the benefits of a project to its costs. A BCR of 1.3 means that for every dollar invested in the project, the organization will receive $1.30 in benefits. Therefore, the project's revenue is 1.3 times its cost.

500. In a weak matrix organization, employees collaborate on multiple projects and across various company departments to enhance interaction and foster a team spirit. If additional employees are required, who is responsible for the hiring process in this matrix structure?
    a) The Project Manager
    b) The Functional Manager

c) The Sponsor
d) The Project Team

Explanation: In a weak matrix organization, the Functional Manager is responsible for the hiring process. This is because the Functional Manager has control over the resources and personnel within the department, and can allocate them as needed to various projects.

501. The Project Manager in charge of a digital marketing project has some stakeholders who don't support the project and others who want the entire project planned out before it starts. Meanwhile, the Sponsor aims for a quick win to support majority stakeholder backing and ensure the project's continuation. Considering these circumstances, what approach should the Project Manager adopt?
    a) An Agile approach, as it would please the majority of stakeholders
    b) A predictive approach, since Agile is only used for software development projects
    c) A predictive approach, as it would give more clarity to the project
    d) An Agile approach, as it would produce functioning deliverables quickly

    Explanation: An Agile approach would produce functioning deliverables quickly, which would please the Sponsor and ensure the project's continuation.

502. A Project Manager was given the task of creating a mobile app with a team of five members. Which of the following project lifecycles can a project manager follow?
    a) Wireframing, prototyping, designing, developing, testing, and deploying
    b) Development, introduction, growth, maturity, and decline
    c) Initiation, planning, executing, monitoring and controlling, and closing
    d) Sprint planning, development, testing, review, and iteration

    Explanation: The best project lifecycle for creating a mobile app is "Initiation, planning, executing, monitoring and controlling, and closing." This structured approach covers all essential phases of project management, ensuring effective oversight of scope, timeline, and resources.

503. An organization underwent a transformation to adopt the Kanban approach for managing its new projects. The Kanban board consists of four columns: To do, Doing, Testing, and Done. How should the Project Manager handle the items in these various lists?

a) Increase the number of items in the Doing list
b) Limit the items in the Testing list
c) Limit the items in the To-do list
d) Increase the number of items in the Testing list

Explanation: The Project Manager should limit the items in the Testing list to ensure that the team focuses on completing the work in progress before starting new work. This helps to prevent multitasking and improves the flow of work through the system, which is a key principle of the Kanban approach.

504. Shortly after introducing a new release to the market, the Product Owner received a user complaint about an irritating defect in the product. What should the Product Owner's next steps be?

a) Estimate the effort required to fix the defect, then decide what to do accordingly
b) Ask the development team to fix the defect immediately since the product has already been released
c) Ask the development team to roll back the last release in order to work more on exploring undetected defects
d) Create a task to fix the defect, add it to the product backlog, then prioritize it

Explanation: The Product Owner should create a task to fix the defect, add it to the product backlog, and then prioritize it. This is the right answer because the Product Owner should not interrupt the development team's work on the current release. Instead, the Product Owner should prioritize the defect and add it to the product backlog to be addressed in the next release.

505. Midway through a sprint, the development team recognized that they had scheduled more work than they could complete. What guidance should the Project Manager advise to the development team for their next steps?

a) Inform the Product Owner during the sprint review that some features could not be completed
b) Ask the Product Owner to remove some work items from the sprint backlog

c) Collaborate with the Product Owner to reprioritize the product backlog items
d) Continue work and put off discussing this issue to the sprint retros

Explanation: The Project Manager should advise the development team to ask the Product Owner to remove some work items from the sprint backlog. This will help the team to focus on the most important work items and complete them within the sprint timeline.

506. The Project Manager is leading a software project, and the project team completed all tasks in the iteration backlog ahead of schedule. They decided to focus on code clean-up for the rest of the iteration with the aim of reducing technical debt, ultimately saving the customer maintenance fees in the future. However, both the customer and the Project Manager appear dissatisfied. What could be the cause of the customer's dissatisfaction?
   a) They wanted to avoid gold plating
   b) They wanted to minimize costs by ending the iteration earlier
   c) They wanted to avoid scope creep
   d) They wanted to prioritize the scope

   Explanation: The customer wanted to prioritize the scope, which means that the team should have focused on delivering the agreed-upon scope instead of working on code clean-up.

507. A contracted Project Manager is leading a project using a predictive approach. The Project Sponsor expresses disappointment with the current progress, claiming that a crucial deliverable has been overlooked. Despite approving all project stages thus far, they request the Project Manager to pause the project for a reevaluation. To facilitate a resolution, the Project Manager can employ arbitration and mediation as which of the following techniques? (Choose two)
   a) Direct negotiation techniques
   b) Dispute resolution techniques
   c) Coaching techniques
   d) Assisted negotiation techniques

   Explanation: Arbitration and mediation are two of the most common dispute resolution techniques used in project management. They are used to resolve conflicts between parties in a neutral and impartial way without the need for litigation. By using these techniques, the Project Manager can

help the Sponsor and other stakeholders reach an agreement and move forward with the project.

508. Upon recently joining a new organization, a Project Manager observed a lack in collaboration within the team. What is the optimal approach to foster a collaborative work environment?
   a) Conducting soft skills development training
   b) Organizing team-building activities
   c) Opting for a pull-based system for work assignments
   d) Adopting an Agile work approach

   Explanation: Organizing team-building activities can help create a cooperative work environment by fostering trust, communication, and collaboration among team members.

509. A Project Manager aims to establish a work environment that encourages creativity, innovation, and the sharing of ideas. They schedule regular meetings with the team to address project risks and issues. What type of leadership is the Project Manager employing?
   a) Interactional
   b) Laissez-faire
   c) Transactional
   d) Transformational

   Explanation: Here, the Project Manager is applying transformational leadership by promoting creativity, innovation, and idea-sharing. They are also setting up regular meetings with the team to discuss project risks and issues, which is a characteristic of transformational leadership.

510. A Project Manager working for a kitchen equipment manufacturer has been instructed by their superiors to prepare for the adoption of Lean practices. What type of power is being utilized by the Project Manager's superiors?
   a) Situational
   b) Referent
   c) Legitimate
   d) Expert

   Explanation: The right answer is C because the superiors are using their formal authority to make a decision and direct the Project Manager to prepare for the transition to Lean practices.

511. During a retrospective meeting, the development team expressed their interest in receiving training in Python programming. While the Project Manager was exploring established training providers, they came across a new one offering the same training at appealing prices. However, the Project Manager hesitated due to uncertainty about the new provider's quality. What actions should the Project Manager take to mitigate the risk?
   a) Share the risk with senior management and let them decide
   b) Get the training for all of the team since it’s not expensive
   c) Look for another accredited training agency that has a good reputation
   d) Enroll one developer to assess the training quality before enrolling the rest of the team

   Explanation: The Project Manager should enroll one developer to assess the training quality before enrolling the rest of the team. This approach will help mitigate the risk of getting low-quality training for the entire team. By enrolling one developer, the Project Manager can evaluate the quality of the training and decide whether to continue with the new provider or look for another accredited training agency.

512. Many quality issues have been affecting a project, posing a risk. What type of meeting should the Project Manager organize with the team to discuss how to prevent or handle such problems in the future?
   a) Team building
   b) Kick-off
   c) Lessons learned
   d) Risk review

   Explanation: Lessons learned meetings are conducted at the end of a project to identify what went well and what didn't go well. The team can then use this information to improve future projects. In this case, the Project Manager should conduct a lessons learned meeting to identify the quality issues that have hampered the project and figure out how to avoid or deal with such problems in the future.

513. At the end of a sprint retrospective, a team member stated that the meeting's outcomes were nearly identical to previous ones, proposing either cancelling it or reducing its frequency. What actions should the Scrum Master take in this situation?
   a) Keep the same frequency of retrospectives and reflect on how to make them more relevant and actionable

b) Extend the sprint timebox in order to allow for more improvements to be discussed in the retrospective
c) Reduce the frequency of retrospectives to once every two sprints
d) Go with voting, and if all team members agree, then skip retrospectives

Explanation: The right answer is A because the Scrum Master should not cancel or reduce the frequency of retrospectives based on one team member's feedback. Instead, the Scrum Master should reflect on how to make the retrospectives more relevant and actionable.

514. A Project Manager chooses a seller to deliver specific services, and both parties decide to utilize a Fixed Price Incentive Fee contract (FPIF). What aspects should be defined next? (Choose two)
a) What hourly, daily, or weekly rate the seller will charge
b) What legitimate costs will be reimbursed by the buyer
c) The extra financial incentive the seller will get and when they will get it
d) How much the seller will get for their predetermined services

Explanation: The financial incentive and the amount to be paid for the predetermined services are the two key elements that need to be defined in a Fixed Price Incentive Fee (FPIF) contract.

515. While leading a retrospective meeting, a Scrum Master receives a suggestion from a team member to change the sprint timebox from three weeks to two weeks. What steps should the Scrum Master take in response to this suggestion?
a) Accept the implementation of any suggestion provided by self-organizing team members
b) Investigate the cause behind the team member's suggestion to tailor the process
c) Decline the team member's suggestion since it can reduce the team's velocity
d) Decline the team member's suggestion since the sprint timebox is set at the beginning of the project and should never be changed

Explanation: The Scrum Master should investigate the cause behind the team member's suggestion to tailor the process. This is because the Scrum Master should facilitate the team's decision-making process and encourage self-organization. The Scrum Master should not make

decisions for the team, but rather help the team make informed decisions.

516. In one of the meetings with their remote team, a Project Manager faced difficulty hearing a specific team member. Initially, the Project Manager thought it to be internet or equipment issues. However, upon inquiry, the team member turned off their cooling fan, resulting in clearer communication. What kind of knowledge did the team member employ to address the issue? (Choose two)
    a) Tacit knowledge
    b) Implicit knowledge
    c) Explicit knowledge
    d) Codified knowledge

    Explanation: The team member is using tacit knowledge to fix the problem by turning off the cooling fan, which is an unspoken understanding or skill that is not easily transferred to others. Additionally, the team member is using implicit knowledge, which is knowledge that is not easily articulated or codified, to identify the issue and resolve it.

517. A Project Manager is tasked to lead a large construction project. In the procurement process, they aim to guarantee that all potential providers have a thorough and clear understanding of the project's procurement requirements and specifications. How can the Project Manager accomplish this?
    a) Conduct face-to-face meetings with each seller
    b) Conduct a negotiation meeting with each seller
    c) Conduct a bidder conference
    d) Conduct a virtual meeting with each seller

    Explanation: A bidder conference is a meeting between the project team and all potential sellers to ensure that they fully understand the project procurement requirements and specifications. This is an effective way to ensure that all potential providers fully and clearly understand the project procurement requirements and specifications.

518. A stakeholder included a user story in the product backlog, and the Scrum team found it clear. However, during execution, the team member assigned to it becomes confused about one of its implementation scenarios. What should the development team member do in this situation?
    a) Ask the Product Owner for clarification
    b) Ask the Stakeholder for clarification

c) Wait for the next standup meeting to report the issue to the Scrum Master or Product Owner if they do attend the meeting
d) Convey their confusion to the Scrum Master, who will help get clarification from the Product Owner

Explanation: The Product Owner is responsible for the product backlog and is the best person to provide clarification on the user story. The development team member should ask the Product Owner for clarification to ensure that the user story is implemented correctly.

519. In the planning phase, the Project Manager speculated no alterations in the composition of the project steering committee. However, recognizing the potential significance of any changes, the Project Manager designated a team member to monitor for triggers. Where should the name of the person responsible for monitoring the steering committee be recorded?
a) Stakeholder register
b) Change log
c) Issue log
d) Risk register

Explanation: The name of the person responsible for monitoring the steering committee should be recorded in the risk register because any change in the composition of the steering committee could be a risk to the project.

520. A senior software engineer has been newly assigned to manage a project using the Agile approach. To ensure successful project delivery, the Project Manager should initially _____.
a) Define quality metrics
b) Define the project success criteria
c) Identify all risks
d) Determine all of the sprints' activities

Explanation: Defining the project success criteria is the first step in Agile project management. This helps the team to understand what they are working towards and what the end goal is. It also helps to ensure that the project is aligned with the organization's goals and objectives.

521. To guarantee that the project aligns with the organization's requirements, the Project Manager conducts a tailoring process. In this context, what is Tailoring?

a) **The selection of the appropriate processes, inputs, tools, techniques, and life cycle phases for managing a project**
b) The application of knowledge, skills, tools, and techniques needed to meet a program's requirements
c) The knowledge and practices which are applicable to most projects in most cases
d) The application of a set of knowledge, skills, tools, and techniques to the project activities in order to meet its requirements

Explanation: Tailoring is the selection of the appropriate processes, inputs, tools, techniques, and life cycle phases for managing a project. This ensures that the project is performed in accordance with the organization's requirements.

522. A team member observes that some team members confuse projects and operations, and oftentimes treat operations as continuous projects. In response, the Project Manager clarifies that there is no concept of an ongoing project, highlighting the distinction between a project and an operation by defining operations as _____.
a) The application of a set of skills, knowledge, tools, and techniques in a project to meet its requirements
b) An undertaken temporary endeavor with the aim of making a distinctive product, service, or result
c) **Ongoing endeavors that generate repetitive outputs**
d) A collection of programs that have been grouped to achieve strategic business goals

Explanation: The right answer is C because operations are ongoing endeavors that generate repetitive outputs, while projects are temporary endeavors with a defined beginning and end that aim to create a unique product, service, or result.

523. A Project Manager is in a meeting, presenting his project to the company's committee for approval. The committee, consisting of Senior Managers and Subject Matter Experts, poses challenging and critical questions. What project selection technique is exemplified in this scenario?
a) Scoring model
b) Benefit analysis
c) Brainstorming
d) **Murder board**

Explanation: This scenario describes the murder board technique, which is a process where a project manager is subjected to a rigorous and challenging questioning session by a group of experts to identify potential weaknesses in the project plan and to ensure that the project is viable and feasible.

524. An Agile team desires to examine a visual summary of all the required features and functionality of the product. Therefore, the Project Manager recommends to look at _____.
    a) Story map
    b) Themes
    c) Product backlog
    d) Epics

    Explanation: A story map is a visual representation of the product backlog that helps the team understand the product's features and functionality. It is a great tool for Agile teams to examine the visual overview of all of the product's required features and functionality.

525. An Agile Project Manager is worried about the duration of time their team spends planning. Which of the following planning activities should the team refrain from participating in?
    a) Daily planning
    b) Portfolio planning
    c) Release planning
    d) Iteration planning

    Explanation: Portfolio planning is not a planning activity that the team should be involved in. It is a high-level planning activity that is typically performed by senior management or executives.

526. In the process of hiring an electrical engineer, a Project Manager conducted only one behavioral interview. Subsequently, it was found that the candidate's technical skills did not align with the job requirements. Therefore, the decision was made to enhance the recruitment process by incorporating a technical test and an additional technical interview. Which risk response strategy did the Project Manager employ?
    a) Avoid
    b) Accept
    c) Mitigate
    d) Transfer

Explanation: The Project Manager applied the mitigate risk response strategy by adding a technical test and an additional technical interview to the recruitment process to reduce the probability of hiring a candidate with inadequate technical skills.

527. A Project Manager was tasked with leading a project for the development of a flight booking website. The airline specified that the website should be capable of accommodating one thousand visitors per hour. This requirement is referred to as _____. (Choose two)
   a) Acceptance criterion
   b) Requirement
   c) Deliverable
   d) Scope

   Explanation: The condition of supporting one thousand visitors per hour is a requirement that the project manager must meet. It is also an acceptance criterion that the project must satisfy to be considered successful.

528. After product release, the organization encountered customer claims regarding poor quality. What types of non-conformance costs might the organization incur?
   a) External costs
   b) Internal costs
   c) Risks and issue response costs
   d) Appraisal costs

   Explanation: The organization might incur external costs as a result of poor quality claims from the customer. External costs are costs that are incurred after the product has been delivered to the customer.

529. A Project Manager is leading a high-risk project that has recently experienced varying performance and a high defect rate. In a meeting, the Sponsor requests an overview of work performance and the current status of the project. Which of the following documents should the Project Manager provide to the Sponsor?
   a) Project management plan
   b) Quality report
   c) Risk report
   d) Status report

Explanation: The Project Manager should send the status report to the sponsor as it provides an overview of the project's current standing and performance. This report includes information on the project's progress, budget, schedule, and risks.

530. Zenith Bank is renowned for providing outstanding online services to its customers. As a part of their new project, customers will have the ability to electronically submit a loan request. During the third week of project execution, the Project Manager discovers that a crucial resource has been overallocated. This project is in need of _____.
    a) Resource loading
    b) Resource calendar
    c) Resource leveling
    d) Resource smoothing

    Explanation: Resource leveling is the process of resolving resource conflicts by delaying tasks or splitting them into smaller parts. This technique is used to balance the workload of resources and ensure that no resource is over-allocated. In this scenario, the Project Manager has discovered that an important resource has been over-allocated, so resource leveling is the appropriate technique to use.

531. Experienced Project Managers always conduct a kick-off meeting before the start of a project. What's the purpose of conducting such a meeting?
    a) Ensuring the allocation of the required project resources
    b) Obtaining approval to proceed with project execution
    c) Reassuring the stakeholders that execution is on track
    d) Informing and engaging stakeholders to ensure their commitment

    Explanation: The purpose of conducting a kick-off meeting is to inform and engage stakeholders to ensure their commitment to the project.

532. A Product Owner picked several user stories totaling 110 points for the upcoming release. Considering the team's velocity is 20 story points, what is the estimated duration to finish the release?
    a) 6 sprints
    b) 5 sprints
    c) 5.5 sprints
    d) It's impossible to calculate the estimated number of sprints to complete a release in Agile

Explanation: The estimated duration to complete the release is calculated by dividing the total number of story points by the team's velocity. In this case, 110/20 = 5.5 sprints. Since the team cannot complete half a sprint, the answer is rounded up to 6 sprints.

533. A Product Owner, who is new to the agile approach, prepared the following user story: "As a customer, I want a new functionality so that I can achieve a 50% increase in sales." The Project Manager identified shortcomings in the user story and contacted the Product Owner to clarify that _____.
    a) The user story should not include financial values
    b) The user story should be time-bounded
    c) The user story should be specific and testable
    d) The user story should follow a common structure

    Explanation: The user story should be specific and testable because it should be clear what the customer wants and how it can be tested to ensure that the functionality meets the customer's needs.

534. A Project Manager is examining the investment efficiency of two potential projects. Recognizing that the discount rate is unknown, what action should the Project Manager take?
    a) Select the project with the highest NPV
    b) Select the project with the lowest NPV
    c) Select the project with the highest ROI
    d) Select the project with the lowest ROI

    Explanation: Selecting the project with the highest ROI is the best option when the discount rate is unknown because ROI is a relative measure that does not depend on the discount rate.

535. A Project Manager is leading an Agile project with a virtual team, and their daily 15-minute standup is scheduled for 9 a.m. In the last meeting, the Project Manager observed a 5-minute delay. Following that, the team engaged in 5 minutes of casual conversation before officially commencing the meeting. Towards the end of the standup, an extra 5 minutes were spent brainstorming an issue that had arisen the previous day. What is the total cycle time for this occurrence?
    a) 15 minutes
    b) 30 minutes
    c) 10 minutes
    d) 25 minutes

Explanation: The total cycle time of the event is 30 minutes. The standup was delayed by 5 minutes, followed by 5 minutes of chit-chatting, and then an additional 5 minutes were spent brainstorming. Therefore, the total cycle time is 5 + 5 + 5 + 15 = 30 minutes.

536. Encountering several quality issues, the Project Manager decided to identify high-priority defects using the 80/20 rule (also known as the Pareto Chart). What should be the Project Manager's next course of action?
    a) **Find out the frequency of each quality issue**
    b) Find out the impact of each quality issue
    c) Find out the cause of each quality issue
    d) Find out the urgency of each quality issue

    Explanation: The Project Manager should find out the frequency of each quality issue to identify the top priority defects by applying the 80/20 rule (aka the Pareto Chart). This is because the Pareto Chart helps to identify the vital few from the trivial many, and the frequency of each quality issue helps to determine which issues are occurring most frequently and, therefore, require the most attention.

537. A Project Manager is leading a project using an adaptive approach. In the sprint review, the Product Owner requests an acceptance document for signing off to demonstrate approval of the deliverables. What should be the Project Manager's next course of action?
    a) **Ask the Product Owner to create a task in the product backlog and prioritize it so that the development team can create the acceptance document**
    b) Create an acceptance document and share it with the Product Owner to sign off
    c) Inform the Product Owner that an acceptance document is unnecessary
    d) Ask the Product Owner to create the acceptance document and sign it off

    Explanation: In an adaptive approach, it is unusual to require formal acceptance documents; however, if the Product Owner requests it, it should be handled as a backlog item to ensure it is addressed appropriately.

538. A Project Manager is leading a team unfamiliar with the adaptive approach. After several weeks, the project has reached a stage where there is an accumulation of Work In Progress (WIP). What steps should the Project

Manager take to determine whether the team is adhering to procedures appropriately?

a) Check relevant Key Performance Indicators (KPIs)
b) Review the notes of the previous retrospective meetings
c) Check the Kanban board
d) Perform a process evaluation

Explanation: Performing a process evaluation is the right answer because it will help the Project Manager to identify whether the team is following the procedures or not. The process evaluation will help the Project Manager to identify the root cause of the accumulation of WIP and take corrective actions.

539. In the middle of the project, the information security department informs the Project Manager that they have recently become aware of the design changes. They also express concerns about the compliance of these changes with the organization's standards, as they were not involved when the changes were made. How should the Project Manager overcome this issue?

a) Conduct a meeting with the information security department to go over all design changes and ensure their compliance with the organization's standards
b) Undo all design changes since the information security department wasn't involved in the process
c) Ask the security department to raise the issue with the Product Owner
d) Allow the information security department team to attend all of the following daily stand-ups to ensure all made changes comply with security requirements

Explanation: The Project Manager should conduct a meeting with the information security department to go over all design changes and ensure compliance with the organization's standards. This will help to identify any potential issues and ensure that the project is in compliance with the organization's standards.

540. In the sprint review of the project's latest iteration, the Product Owner conveyed satisfaction with the product, deeming it "fit for purpose." Nevertheless, a senior team member stated the necessity for another iteration to address bugs and enhance the user experience. The Sponsor, present at the meeting, mentioned that additional work on the product

would recommence if more funding is secured in the next quarter. What steps should the Project Manager take next?

a) Keep the project open since work could be resumed in the next quarter
b) Release all resources
c) Continue work on fixing bugs to satisfy the client, then close the project
d) Close the project but keep one developer to fix any major bugs

Explanation: Keeping the project open ensures continuity and allows for a smooth transition if additional funding is secured, avoiding the complications of reopening a closed project. This approach maintains flexibility and readiness to address the identified issues and enhancements when funding becomes available.

541. The Project Manager was notified by the Project Sponsor that there are concerns about the satisfaction of business demands by the resulting product. Consequently, there is a desire to minimize costs. What course of action should the Project Manager take?

a) Gather all requirements and execute the project using an incremental approach
b) Identify and implement the requirements for making a Minimum Viable Product (MVP)
c) Sign with the sponsor a fixed-cost contract in order to limit project costs
d) Exclude certain stakeholders from the scope definition process in order to limit requirements

Explanation: The Project Manager should identify and implement the requirements for making a Minimum Viable Product (MVP) in order to satisfy the Sponsor's concerns about costs and business demands. An MVP is a product with just enough features to satisfy early customers and provide feedback for future product development. This approach can help to keep costs low while still delivering a product that meets business needs.

542. In the midst of the project, the Project Manager observed a decline in stakeholder engagement and recognized the necessity to assess their satisfaction. Which two tools or techniques could the Project Manager employ for this purpose? (Choose two)

a) Power/interest grid

b) Engagement assessment matrix
c) Net promoter score
d) Mood chart

Explanation: Net promoter score and mood chart are tools that can be used to measure stakeholder satisfaction. Net promoter score is a metric used to measure customer loyalty and satisfaction, while mood chart is a tool used to track stakeholder emotions and feelings over time.

543. The Project Manager is leading the second phase of a project with a hybrid approach. To establish a shared understanding among stakeholders and resolve any disparities, the Project Manager moves forward by defining the project requirements through _____.
a) Inviting all concerned stakeholders to a requirements workshop to develop user stories
b) Sending a questionnaire to all of the involved stakeholders in order to determine the project requirements
c) Setting up a meeting with each stakeholder to identify their needs
d) Conducting a survey of the project's current users

Explanation: A requirements workshop is a collaborative approach to defining project requirements. It allows stakeholders to work together to develop user stories and build a common understanding of the project requirements. This approach is particularly useful in hybrid projects where there may be differences in stakeholder needs and expectations.

544. Midway through the project, the Sponsor appoints an Inspector to assess the quality of the delivered work up to that point. While the Project Manager is not personally acquainted with the inspector, they have heard that the inspector is typically inflexible and challenging to work with. How should the Project Manager best handle this situation?
a) Ignore the Inspector's designation since they are an external stakeholder
b) Ask the Sponsor to designate another Inspector
c) Use their soft skills to talk the Inspector into looking past any quality issues and not reporting them to the Sponsor
d) Collaborate with the Inspector and provide them assistance when needed

Explanation: Collaborating with the Inspector and providing them assistance when needed is the best way to handle the situation. This will help build a relationship with the Inspector and ensure that they

understand the project's goals and objectives. It will also help the Project Manager to identify any potential quality issues early on and address them before they become major problems.

545. The Project Manager received notification from a customer stating their intention to initiate a new project, assigning the same Product Owner as the current ongoing project. How should the Project Manager respond to the customer's decision?
    a) Ask the customer to reconsider their decision to avoid a conflict-of-interest situation
    b) It's ok to have the same Product Owner for both products as long as they can fulfil their duties
    c) Ask the customer to reconsider their decision since a Product Owner should handle one product at a time
    d) Ask the customer to add another Product Owner to the current project as a backup

    Explanation: It is okay to have the same Product Owner for both products as long as they can fulfil their duties. There is no rule that a Product Owner should handle only one product at a time. The Product Owner should be able to manage their time and responsibilities effectively.

546. In the product validation phase, a key stakeholder declines to approve the acceptance document, asserting that certain functionalities do not meet their expectations. What actions could the Project Manager have taken to prevent this situation?
    a) Involve the key stakeholder during the planning phase
    b) Involve the key stakeholder during the quality control phase
    c) Involve the key stakeholder during all project phases
    d) Involve the key stakeholder during the project initiation phase

    Explanation: The right answer is C because involving the key stakeholder during all project phases ensures that their expectations are met and that they are satisfied with the final product. This approach helps to avoid any surprises during the product validation phase.

547. While working on the identification of risks, a Project Manager observed a disagreement among team members regarding the optimal approach to execute a significant work item. What steps can the Project Manager take to mitigate this risk?

a) **Enroll the team in training about the technologies related to executing that work item**
b) Let the team self-organize and figure out how to execute the work item
c) Escalate the issue to the Functional Manager
d) Outsource the concerned work item

Explanation: The Project Manager can enroll the team in a training about the technologies related to executing that work item to mitigate the risk of disagreement among team members concerning the appropriate way to execute an important work item. This will help the team members to have a common understanding of the work item and how to execute it.

548. The Project Manager overseeing a software development project following an Agile approach receives a customer request to modify a requirement during project execution. What course of action should the Project Manager take?
    a) Take the change request to the Change Control Board (CCB)
    b) Block the change request
    c) **Welcome the change request**
    d) Avoid the change request

    Explanation: In Agile, change is expected and welcomed. The Project Manager should welcome the change request and work with the customer to understand the impact on the project scope, schedule, and budget.

549. A Project Manager joins a project to replace the previous Project Manager, who has suddenly departed from the organization. In the first week, they observed a lack of trust and support among team members for each other's ideas, and the work environment was counterproductive. According to Tuckman's model, in which development stage is this team?
    a) Norming
    b) Adjourning
    c) **Storming**
    d) Forming

    Explanation: The team is in the storming stage, where team members are competing for ideas and leadership. The lack of trust and counterproductive work climate are typical of this stage.

550. A Project Manager is leading a groundbreaking medical research project. In a meeting with the Sponsor, the Project Manager outlines his approach to

handling the project, expressing a desire to promote self-awareness, active listening, and coaching. The Project Manager emphasizes that facilitating the growth of the project team is one of his top priorities. What style of leadership does the Project Manager aim to adopt?

a) Transactional leadership
b) Transformational leadership
c) Authentic leadership
d) Servant leadership

Explanation: This style of leadership encourages self-awareness, listening, and coaching, which are all aimed at helping the team members grow and develop.

551. A Project Manager employed at a national bank opts to utilize the power/interest grid for planning stakeholder engagement. How should he handle a Stakeholder who shows interest but possesses limited authority?

a) Keep informed
b) Monitor
c) Manage closely
d) Keep satisfied

Explanation: The interested stakeholder with little authority should be kept informed as they have a low level of power but a high level of interest in the project. Keeping them informed will ensure they are aware of the project progress and any changes that may affect them.

552. A Project Manager overseeing a branding project for a fast-food restaurant chain encounters a disagreement between two team members regarding a design task. The Project Manager decides to hold a meeting with the two members, and after discussing the issue, they collectively agree that outsourcing the activity is the most suitable course of action. Which conflict resolution technique did the Project Manager employ?

a) Smoothing
b) Forcing
c) Collaborating
d) Compromise

Explanation: The Project Manager used the collaborating conflict resolution technique by bringing the two team members together to discuss the issue and come up with a mutually beneficial solution.

553. Organizations frequently attempt to deliver projects with constrained budgets and incomprehensive requirements. To navigate such complexities, an Agile approach can be adopted. However, effective communication is essential for this approach to realize its objectives. In this context, how should a Project Manager communicate?

a) Daily
b) Informally
c) Formally
d) Frequently

Explanation: Frequent communication is necessary to ensure that the Agile approach is successful in delivering projects with limited budgets and incomprehensive requirements. This allows the Project Manager to keep the team informed of any changes and to ensure that everyone is on the same page.

554. A senior Project Manager is employed by a medical devices firm that has recently experienced substantial organizational changes. One of the key team members conveys her apprehensions to the Project Manager regarding a new team member, expressing doubt about their technical competence to effectively fulfil assigned tasks. Given the Project Manager's trust in the team member's judgment, what action should be taken concerning the new team member?

a) Provide the new member with training and mentoring
b) Keep the new team member, but reassign their tasks to someone else
c) Release the new team member
d) Nothing should be done since the concerned team member has already been assigned

Explanation: The Project Manager should provide the new team member with training and mentoring to improve their technical proficiency. This will help the new team member to better understand their assigned work and perform better in the project.

555. Shortly after commencing a new project, a Project Manager observed that a team member, previously highly productive on their last project, exhibited decreased productivity. The Project Manager invited her to a one-on-one meeting, during which she disclosed that her frustration stemmed from the unfulfilled expectation of receiving a performance bonus in the last project. Which theory is manifested in this situation?

a) Anticipation Theory
b) Expectancy Theory
c) Theory X
d) Theory Y

Explanation: The right answer is Expectancy Theory because the team member's frustration is due to her expectation of receiving a performance bonus, which did not happen. Expectancy Theory suggests that people are motivated by their belief that their effort will lead to a certain level of performance, which in turn will lead to a specific outcome or reward.

556. One month into project execution, the Project Manager observes that the sponsor consistently arrives late for their weekly meetings. However, the Project Manager decides not to confront the sponsor's behavior directly. Instead, they document the recurring tardiness in the meeting notes. Which conflict resolution technique is the Project Manager utilizing?
a) Withdrawal
b) Information recording
c) Smoothing
d) Problem-solving

Explanation: Withdrawal (also known as avoiding) is a conflict resolution technique where the person involved chooses to avoid or withdraw from the conflict instead of addressing it directly. In this scenario, the Project Manager is not confronting the sponsor's tardiness directly but is instead documenting it without taking any immediate action to resolve the issue.

557. A Project Manager is overseeing an educational project. After assessing the power levels of the project's four stakeholders, they discovered that Emily and Alex possess high power, whereas Rahul and Sophia have low power. Seeking further insight, the Project Manager developed the stakeholder engagement assessment matrix below. Given this analysis, which stakeholder should be given priority in terms of engagement?
a) Emily
b) Alex
c) Rahul
d) Sophia

Explanation: In stakeholder management, stakeholders with high power should be given priority for engagement, as their influence can significantly impact the project's success. Both Emily and Alex have high

power, but since only one stakeholder needs to be prioritized based on the given options, either Emily or Alex would be correct.

558. An organization requiring accounting software performs a make-or-buy analysis. Opting for in-house development, the organization decides based on the analysis indicating a payback period of less than 3 years. However, during resource acquisition, the Project Manager discovers that the developer's rate considered in the make-or-buy analysis is only half the current market rate. What should the Project Manager do first?
    a) Update the make-or-buy analysis report
    b) Submit a change request
    c) Inform the sponsor about their findings
    d) Nothing; as long as the project's SPI and CPI are on track, this shouldn't be a problem

    Explanation: The Project Manager should inform the Sponsor about their findings since the make-or-buy analysis was based on an incorrect assumption. The Sponsor should be informed of the new information, and a new analysis should be conducted to determine the best course of action.

559. Two agile teams are engaged in developing the same product. Team A comprises 5 members, using 2-week sprints, and successfully finished all sprint backlog items with a velocity of 50 story points. Meanwhile, Team B, consisting of 4 members and using 3-week sprints, achieved a total of 70 story points without completing the entire sprint backlog. In terms of performance, which team performs better?
    a) Team A, taking into consideration their velocity, can achieve 75 story points in a 3-week sprint, which is better than what Team B achieved
    b) Team B, since they achieved more story points with fewer team members
    c) Team A, since they completed all sprint backlog items
    d) Team B, since they have a better velocity

    Explanation: Team A completed all sprint backlog items, which is the goal of each sprint. Therefore, they have better performance.

560. In a project status meeting, the Project Manager provided the Sponsor with an update on the project performance, stating that the SPI is 0.75, and the CPI is 1.05. What does this information signify?
    a) The project is within budget but behind schedule
    b) The project is over budget but on schedule

c) The project is behind schedule and over budget
d) The project is on schedule and within budget

Explanation: The CPI is greater than 1, indicating that the project is under budget. The SPI is less than 1, indicating that the project is behind schedule. Therefore, option 1 is correct.

561. A Product Owner and team members hold different interpretations of a specific user story. To resolve the disagreement, the Project Manager proposes the development of personas. What is the most likely reason for creating personas in this scenario?
   a) To identify and describe real users of the solution
   b) To use it as a wireframe to help clarify project outcomes
   c) To engage stakeholders who don't know much about the project
   d) To help the development team empathize with the users of the solution

   Explanation: Personas are fictional characters created to represent the different user types that might use a solution. They help the development team empathize with the users of the solution, which can help resolve disagreements and ensure that the final product meets the needs of the users.

562. Upon introducing a new foreign developer to the team, the Project Manager observed that a specific senior team member appeared to be deliberately avoiding any engagement with them. What actions should the Project Manager take to address this matter?
   a) Look past the senior team member's behavior since no conflict between the two has occurred, and work seems to be going just fine
   b) Talk to the senior team member privately to try and find out the reason behind their behavior while making it clear that discrimination is neither tolerated nor accepted
   c) Address the issue in the next team meeting by scolding the senior team member and lecturing everyone about the importance of an inclusive work environment
   d) Notify the Functional Manager of the senior team member's behavior and ask them to take the appropriate disciplinary action

   Explanation: The Project Manager should address the issue by talking to the senior team member privately to try and find out the reason behind their behavior while making it clear that discrimination is neither tolerated

nor accepted. This approach allows the project manager to understand the root cause of the issue and take appropriate action to resolve it.

563. Despite the recent transition to the agile approach, project team members consistently rely on the Project Manager to assign them work. What steps can the Project Manager take to facilitate their team in becoming self-organized?
   a) Support the team by removing encountered impediments
   b) Ensure that the team includes different functional expertise
   c) Mentor the team on how to make their own decisions
   d) Take a few weeks off to force the team to act on their own

   Explanation: The Project Manager can mentor the team on how to make their own decisions. This will help the team to be self-organized and take ownership of their work.

564. Leading a multi-year software project, a Project Manager faces a situation where, after a few months, a senior team member started losing motivation. The team member begins expressing dissatisfaction, claiming that their assigned tasks could be effectively handled by a junior team member. In this scenario, what actions should the Project Manager take?
   a) Be proactive and start looking for a replacement for the senior team member
   b) Find more challenging tasks for the senior team member to tackle
   c) Incentivize the senior team member by increasing their salary or rate
   d) Increase the engagement of the senior team member by holding team-building activities

   Explanation: Finding more challenging tasks for the senior team member to tackle is the right answer because it addresses the root cause of the senior team member's loss of motivation. By providing more challenging tasks, the senior team member will be more engaged and motivated to work on the project.

565. A Project Manager is under contractual obligation to present a monthly report detailing the accomplishments of the previous month and specifying the preventive measures that will be implemented in the upcoming month. Initially responsible for delivering the status update presentation, the Project Manager, now on vacation, had a team member step in as a volunteer. Subsequently, key stakeholders provided feedback indicating dissatisfaction with the latest presentation. How should the Project Manager handle this situation with their team member?

a) **Ask the team member what happened with the presentation that resulted in the stakeholders' dissatisfaction**
b) Send an email to the team member to use their reply as a justification for any ulterior decisions
c) Ignore the issue since it only occurred once, and it won't happen again now that they'll be back to delivering the upcoming status presentations
d) Reproach the team member to let them know how disappointed they are

Explanation: The Project Manager should ask the team member what happened with the presentation that resulted in the stakeholders' dissatisfaction. This will help the Project Manager understand what went wrong and how to prevent it from happening again in the future. It also shows that the Project Manager is willing to listen to their team member and work together to improve the situation.

566. Over the past three daily standup meetings with the team, the Project Manager observed that a new team member was facing challenges with one of their designated tasks, displaying little progress toward its completion. How should the Project Manager address this situation?
a) Ask the new team member to check the team ground rules
b) Ask an experienced team member to take over the task
c) **Check whether other team members can help their new colleague with the task after the standup meeting**
d) Express their disappointment to the new team member and encourage them to make more effort

Explanation: Option C is the right answer because the Project Manager should first check whether other team members can help the new team member with the task after the standup meeting. This approach is in line with the principle of teamwork and collaboration, which is essential for project success.

567. A Project Manager is in charge of a project within a functional organization. In the midst of project implementation, a team member left from the organization. What should be the Project Manager's initial course of action?
a) Update the project's resource breakdown structure
b) Collaborate with the functional manager to get a replacement
c) **Evaluate the impact of the team member's departure on the project**

d) Reassign their workload to the project team member

Explanation: The Project Manager should evaluate the impact of the team member's departure on the project first. This will help the Project Manager to understand the impact of the team member's departure on the project and take necessary actions to mitigate the impact.

568. Before initiating the project and identifying all individuals and groups involved or impacted by the project, the Project Manager went a step further by categorizing stakeholders based on their levels of power, interest, and influence. This step is known as _____.
   a) Stakeholder engagement
   b) Stakeholder analysis
   c) Stakeholder identification
   d) Stakeholder monitoring

   Explanation: Stakeholder analysis is the process of identifying and analyzing stakeholders according to their level of power, interest, and influence. This step is taken after identifying all of the individuals and groups involved or affected by the project and before engaging with them.

569. A Project Manager is formulating the business case for a green transportation project and opts to utilize the SWOT technique. How should the Project Manager apply this technique?
   a) Identify the project's strengths and weaknesses
   b) Assess the business model using expert judgment
   c) Follow a Plan-Do-Check-Act cycle (PDCA Cycle)
   d) Conduct a benefit/cost analysis

   Explanation: SWOT analysis is a technique used to identify the project's strengths, weaknesses, opportunities, and threats. It is a useful tool for developing the business case of a project.

570. A Project Manager overseeing an online e-learning platform is utilizing the Scrum framework with the project team. At regular intervals, they release new versions containing additional features, user interface modifications, bug fixes, security patches, etc. To prevent user confusion amid the ongoing changes, the Project Manager chose to provide all release information to platform members through the "News" section on the website. What form of communication does the Project Manager aim to employ?
   a) Interactive communication

b) Push communication
c) Proactive communication
d) Pull communication

Explanation: The Project Manager intends to use pull communication because the release information is made available to the platform members under the 'News' section on the website, and they can access it whenever they need it.

571. A Project Manager called off a lessons-learned review meeting due to time constraints. The potential consequences of this decision may involve _____.

a) The project may not be accepted by the Project Management Office (PMO)
b) Project management will get more difficult
c) Issues faced throughout the project could reoccur due to the missed opportunity to identify preventive actions
d) Project cost will be greater

Explanation: Cancelling the lessons learned review meeting will prevent the identification of preventive actions that could have been taken to avoid issues that may arise in the future. Therefore, the potential consequences of this decision might implicate that issues faced throughout the project could reoccur due to the missed opportunity to identify preventive actions.

572. Despite having a well-elaborated project management plan, the Project Manager believes that unexpected events could significantly impact the entire project. What should be the next course of action for the Project Manager?

a) Ensure that a management reserve is allotted
b) Identify these events and update the project management plan
c) Perform a qualitative risk assessment to define the priority of each of these events
d) Discuss these events with the Sponsor in order to develop a fallback plan

Explanation: The right answer is A because a management reserve is allotted to cover unforeseen events that may impact the project. This reserve is used to mitigate the impact of these events and ensure that the project stays on track.

573. In the course of a planning session, a project management team concluded that acquiring new equipment in November was necessary. The Project Manager suggested leveraging Black Friday discounts, which occur in the final week of November. How should the project team deal with this opportunity?
   a) Accept
   b) Exploit
   c) Share
   d) Enhance

   Explanation: The project team should exploit the opportunity of Black Friday discounts to purchase the new equipment in November. Exploit is a risk response strategy that aims to take advantage of opportunities that may arise during the project. In this case, the project team can take advantage of the Black Friday discounts to save money on the purchase of the new equipment.

574. An Agile team is engaged in the development of an innovative digital product. The Product Owner introduced a user story into the product backlog, specifying that it should be incorporated in the upcoming release. In a refinement session with the Product Owner, the development team expressed uncertainty about implementing the user story, citing the need for specific inputs from the current release. What should be the next step for the development team?
   a) Give the user story a final estimation based on the current understanding and available information
   b) Further study the different implementation scenarios of the user story
   c) Stop spending more time trying to figure out how to implement the user story until they collect more information
   d) Prioritize the user story since it involves a high level of ambiguity and risk

   Explanation: In Agile methodology, if a user story cannot be fully understood or implemented due to missing information, it is often best to pause further efforts on it until the necessary information is obtained. This avoids wasting time and resources on incomplete or uncertain tasks. Once the required inputs from the current release are available, the team can then reassess and proceed with a clearer understanding.

575. A Project Manager is overseeing a project with remote team members situated in various geographical areas. In a virtual bi-weekly retrospective

meeting, the Project Manager discovers that some members either forgot, missed, or did not fully comprehend the previous meeting's discussions. What actions can the Project Manager take to tackle this issue? (Choose two)

a) Change the meeting frequency to weekly rather than fortnightly so that team members do not forget what has been discussed
b) Record meeting sessions and send the recordings to everyone involved
c) Create a new group norm requiring meeting attendees to raise their hands when they don't understand something
d) Ask all team members to try to speak English with an American accent

Explanation: Recording meeting sessions and sending the recordings to everyone involved can help team members who missed the previous meeting to catch up. Creating a new group norm requiring meeting attendees to raise their hands when they don't understand something can help team members who did not fully understand what was discussed in the previous meeting.

576. A Project Manager certified with PMP possesses extensive project management knowledge and boasts over two decades of experience as a designer in the automotive industry. In her most recent role, she led a team of designers in creating a new design for a futuristic car model. Given the team's confidence in her decision-making, which of the following powers does the project manager possess?

a. Persuasive
b. Formal
c. Expert
d. Referent

Explanation: The Project Manager possesses expert power because of her extensive knowledge and experience in project management and the automotive industry. The team trusts her decisions because of her expertise.

577. After transitioning to the Agile approach, a Project Manager has been promoting self-organization within their team by granting them to decide how to carry out their assigned tasks. What constitutes the primary characteristic of a self-organizing team?

a. It gives more responsibility to the Agile team
b. It gives more responsibility to the Project Manager

c. It gives team members higher visibility of the product
d. It allows the team to deliver a working product without external dependencies

Explanation: Self-organizing teams are responsible for deciding how to execute their work, which gives more responsibility to the Agile team.

578. As the project approaches completion, a crucial stakeholder inquired with the Project Manager about any apprehension regarding the approval process for project deliverables. In response, the Project Manager expressed confidence that the customer would be satisfied with the result. What factors can guarantee the satisfaction of the project's customer?
   a) The efficiency of the warranty service
   b) How good the Project Manager's relationship is with the project's end-users and stakeholders
   c) Delivering value and conforming to project requirements
   d) The project's low running costs

   Explanation: Delivering value and conforming to project requirements is the most important factor in ensuring customer satisfaction. If the project delivers what was promised and meets the customer's requirements, they are more likely to be satisfied with the result.

579. Four months into project execution, the Project Manager observed a reduction in the performance of certain team members, while others maintained a satisfactory level of performance. What actions should the Project Manager take to realign the entire team?
   a) Identify the causes of bad performance, solicit systematic feedback, and implement adequate solutions based on findings
   b) Openly discuss the poor performance of certain team members with the whole team in order to come up with a joint solution
   c) Motivate underperforming team members by incorporating a competitive reward system that offers a bonus for top performers
   d) Avoid interfering in order to give team members a chance to improve their performance

   Explanation: Identifying the causes of bad performance, soliciting systematic feedback, and implementing adequate solutions based on findings is the right approach. This will help the Project Manager understand the root cause of the problem and take corrective actions to improve the performance of the team.

580. A Project Manager introduced a stadium construction project, anticipating significant benefits for the community. Although the mayor approved the project, the Project Manager faced substantial opposition from various stakeholders right from the start. What steps should the Project Manager take to resolve this issue?

a) Create an organizational diagram to assign each stakeholder to the appropriate project role, allowing or disallowing certain lines of communication
b) Create a Responsibility Assignment Matrix (RAM) to indicate which stakeholder is responsible for what aspects of the project and who needs to be consulted or informed
c) Schedule a meeting with the concerned stakeholders to thoroughly explain the project, discuss and establish ground rules, ensure their involvement, and identify any personal or organizational issues that might surface later on
d) Avoid contacting these stakeholders at the start of the project and, instead, create a "faits accomplish" to pressure them to support the project due to a lack of alternatives

Explanation: The Project Manager should schedule a meeting with the concerned stakeholders to thoroughly explain the project, discuss and establish ground rules, ensure their involvement, and identify any personal or organizational issues that might surface later on. This will help to address the concerns of the stakeholders and ensure their buy-in for the project.

581. A project encompasses numerous stakeholders, with the majority requiring high-level information and updates on the project status, while others necessitate more detailed information and frequent updates. In response to these varied requirements, what should the Project Manager create?

a) Salience Model
b) Power/interest grid
c) Stakeholder engagement assessment matrix
d) Stakeholder engagement plan

Explanation: The Project Manager should create a stakeholder engagement plan to respond to the different requirements of the stakeholders. This plan should outline the level of engagement required for each stakeholder, the frequency of communication, and the type of information to be shared.

582. In the sprint review, the Product Owner notifies the Scrum team that the meeting is being recorded to share it with a crucial stakeholder. A day later, the stakeholder contacts the Product Owner, expressing dissatisfaction with the product and stating that it doesn't align with their expectations. What should the Scrum Master do in response?
    a) Meet the stakeholder to review the product backlog items
    b) Ask the development team to have more frequent backlog refinement sessions with the Product Owner
    c) Facilitate a meeting between the Product Owner and the concerned stakeholder
    d) Ask the development team to collaborate with the key stakeholder

    Explanation: The Scrum Master should facilitate a meeting between the Product Owner and the concerned stakeholder to discuss the stakeholder's concerns and expectations. This will help the Product Owner to understand the stakeholder's requirements and expectations better and make necessary changes to the product backlog items.

583. A Project Manager is overseeing an adaptive project utilizing an online project management system that visually presents real-time progress for iterations and the overall project through various charts. However, a key stakeholder has a preference for receiving a weekly status report via email. In this scenario, what actions should the Project Manager take? (Choose two)
    a) Ignore the stakeholder request since they have access to the online system and, therefore, can easily check the project status
    b) Take this in charge in order to let the team focus on achieving deliverable
    c) Add a small recurring task of “1” story point in the product backlog to send the weekly report
    d) Check if there is a way to automatically send the weekly report through the project management system

    Explanation: The Project Manager should take charge of the situation and ensure that the stakeholder receives the weekly status report in their preferred format. Checking if there is a way to automatically send the report through the project management system is also a good idea to save time and effort.

584. A Project Manager consistently receives adverse feedback regarding one of the project suppliers. Recently, conflicts between the project team and the

supplier's team have escalated, negatively impacting the project. What should be the next course of action for the Project Manager?

a) Proceed with Alternative Dispute Resolution (ADR)
b) Send an email to the supplier's team manager explaining the alarming situation
c) Meet with the supplier's team manager to discuss the issue
d) Call the supplier's team manager and urge them to commit to the procurement agreement

Explanation: Meeting with the supplier's team manager is the best course of action as it allows the project manager to discuss the issue and work towards a resolution. This approach is more effective than sending an email or making a phone call as it allows for a face-to-face conversation and the opportunity to build a relationship with the supplier's team manager.

585. A Project Manager is assigned with overseeing a megaproject that demands expertise and resources from various countries. Given the unpredictability of global epidemics, the project team is concerned about encountering another event similar to the Coronavirus, potentially resulting in delays and cost overruns. What actions should the Project Manager take during the risk planning phase? (Choose two)

a) Add the risk to the issue log
b) Monitor the development of the risk
c) Add the risk to the risk register
d) Perform a qualitative risk analysis

Explanation: The Project Manager should add the risk to the risk register and perform a qualitative risk analysis during the risk planning phase. Adding the risk to the risk register ensures that it is documented and tracked throughout the project, while performing a qualitative risk analysis helps to identify the likelihood and impact of the risk on the project objectives.

586. A Project Manager is overseeing a maintenance project for an energy supply company. During project execution, an engineer discovers that a spare part needs immediate replacement to prevent an impending power outage. However, obtaining a new spare part through the organization's procurement process is expected to be time-consuming. What should the Project Manager do in this situation?

a) **Escalate the problem to the Project Sponsor**
b) Buy the spare part without following the procurement process
c) Update the organization's procurement process in order to be able to get the spare part as soon as possible
d) Follow the procurement process, regardless of the risk results

Explanation: The Project Manager should escalate the problem to the Project Sponsor, as this is a critical issue that requires immediate attention. The Project Sponsor has the authority to make decisions that can help the project team overcome the issue and avoid the power outage.

587. In the middle of the sprint, the development team faced a disruption in their work due to a sudden issue with a malfunctioning software feature. In the subsequent retrospective meeting, the Project Manager opted to employ the Five Whys method to investigate the problem. Why did the Project Manager specifically choose to utilize this technique?
a) To identify the five steps to take to resolve the issue
b) To identify the five main factors that caused the issue
c) To identify the five members responsible for the issue
d) **To identify the root cause of the issue**

Explanation: The Project Manager chose to use the Five Whys method in order to identify the root cause of the issue.

588. A Project Manager ensured the completion of all necessary tasks. Nevertheless, during an external post-completion audit, the performing organization incurred a substantial penalty. What alternative actions could the Project Manager have taken to prevent any liabilities arising from the audit?
a) **Ensure that all closing formalities were followed as per the defined contractual procedures**
b) Verify that all of the tasks indicated in the scope statement were completed
c) Verify that all identified defects were fixed
d) Ensure that all documents were updated before closing the project

Explanation: The Project Manager should ensure that all closing formalities are followed as per the defined contractual procedures to avoid any liabilities following an audit.

589. Forecasting the financial value of a theme comprising a set of features is the responsibility of:

a) Analysts
b) Product owner
c) Project manager
d) Sponsor

Explanation: Forecasting the financial value of a theme is the responsibility of the product owner.

590. A team decided to work on a set of features that are already present in a competing product and are highly desired by customers. Which of the following reasons to justify project work is exemplified in this scenario?
a) Revenue due to a new opportunity
b) Additional incremental revenue from existing customers
c) Revenue that an organization will lose if the project is not performed
d) Revenue due to streamlining a process

Explanation: In this scenario, the team is working on features that are already present in a competing product and are highly desired by customers. If the organization does not implement these features, it risks losing revenue to competitors who already offer these features. This justification aligns with the idea of preventing revenue loss by ensuring the product remains competitive in the market.

591. A product owner realizes that a software used to serve hospital customers, with some enhancements, can also be used to serve health insurers. The product owner asks the development team to make the changes and is able to create a new source of revenue to the company from the new software. Under what revenue stream is this feature classified?
a) New Revenue
b) Incremental Revenue
c) Retained Revenue
d) Operational Efficiencies

Explanation: New Revenue is the revenue from a new opportunity after a project is completed.

592. Which of the following are the responsibilities of the scrum master?
a) Achieving initial and ongoing funding for the project
b) Teaching the product owner about the business or domain
c) Being responsible for developing the functionality of the product
d) Facilitating communication by removing the barrier between the team and the product owner

Explanation: The scrum master is responsible for removing impediments and facilitating communications between the team.

593. Which of the following is NOT a good way to split a large story into smaller stories?

a) Decomposing the story along operational boundaries
b) Making design, development, and testing as separate stories
c) Limiting the amount of data being processed
d) Separating the exceptions and cross-cutting concerns

Explanation: A story should deliver value to the customer. Just design or development without testing will not deliver value

594. Which one of the following phases of project management includes developing a capability and/or feature-based release plan for the product?
a) Envision
b) Speculate
c) Explore
d) Adapt

Explanation: In agile project management, the "Speculate" phase involves developing plans and strategies for delivering the project's features and capabilities. This phase includes creating a release plan based on the project's envisioned goals, focusing on how features will be developed and delivered in iterations.

595. Which of the following is a unit of measure to express the estimated difficulty or the effort of a user story that is based on an analogous estimation technique?
a) Days
b) Years
c) Story Points
d) Iterations

Explanation: A story point is a unit of measure to express the estimated difficulty or the effort of a user story and is arrived at by a comparison (analogy) with benchmark stories.

596. A team needs to be very responsive to changes and improve the product and process rapidly. How should the agile process be tailored for this team?
a) Hire more technically capable teams.
b) Spend more time in the design of the system to make it more flexible.
c) Clarify all the requirements upfront to limit the extent of change encountered.
d) Work in shorter increments and retrospect more often.

Explanation: Agile practice guide Table 3-1. The team should retrospect more often and work in shorter increments

597. A team is conducting the daily standup but the team member's updates don't conclude within 15 minutes. What should the scrum master do at this point?
a) Declare the meeting as closed and encourage the team to be more focused next time.
b) Limit the remaining updates to just half minute by each team member.

c) Provide the remaining updates in brief on behalf of the rest of the team.
d) Ask the team to provide the updates over email.

Explanation: If the scrum meetings extend beyond 15 minutes, the scrum master should declare the meeting as closed and encourage the team to be more focused next time.

598. In a project, all the user stories are estimated and the sum of those estimates is 200 story points. Based on past experience, it is known that the team velocity is 25 story points per two-week iteration. How many iterations does the project need to complete the user stories?
a) 9
b) 8
c) 5
d) 1

Explanation: To determine the number of iterations needed to complete the user stories, divide the total sum of story points (200) by the team's velocity per iteration (25). Therefore, the project would require 8 iterations to complete all the user stories, as 200 divided by 25 equals 8.

599. Who should participate in iteration review?
a) Team members and product owner
b) Team members and team facilitator
c) Team members, product owner and anybody who can give feedback about the product
d) Product specialists, customers, iteration manager, developers, and testers

Explanation: Iteration review is a good forum to collect feedback so anybody whose inputs are valuable may be invited to the meeting.

600. Who should participate in an iteration retrospective session?
a) Team, product owner, and managers of the team members
b) Team, customers, and team facilitator
c) Team and the team facilitator
d) Team, team facilitator, and subject matter experts

Explanation: Inviting external participants to the retrospective may affect the free exchange of ideas within the team.

601. Technical debt is the gap between a product's actual Cost of Change (CoC) and its optimal CoC. What is a good way to keep technical debt low?
a) Devoting every alternate iteration to "hardening" activities
b) Selecting more technically capable team members
c) Invoking strict penalties for not following good design practices
d) Performing frequent refactoring and following good development practices at all times

Explanation: Performing frequent refactoring and following good development practices at all times is the best way to keep technical debt low.

602. The team working on a product has become too large and the product demands are increasing. This is resulting in increasing the team size beyond manageable levels. What can be done?
    a) Split the teams into a development and test team to keep the size of each team manageable.
    b) Split the teams into feature teams, each taking care of a manageable chunk of user-visible product.
    c) Grow the team size to the desired level and follow a hybrid model combining agile and waterfall practices.
    d) Sacrifice some productivity in the interest of meeting the customer goals.

    Explanation: Feature teams are a good way for a team to scale to the requirements of the product.

603. Team 1 has been asked to complete work on Item 10's story and deliver it to Team 2 so that they can initiate work on Item 12's story, which is dependent on the completion of Item 10's story. The leader of Team 1 decides to add a buffer that protects the start date on Item 12's story against delays in the completion of Item 10's story. This buffer is known as:
    a) Feeding buffer
    b) Iteration buffer
    c) Story buffer
    d) Release buffer

    Explanation: The feeding buffer protects the start date on Item 12's story against delays in the completion of Item 10's story.

604. A meeting where a team reflects and looks back to adapt and learns to improve future projects is popularly known as:
    a) Retrospective
    b) Planning
    c) Review session
    d) Status meeting

    Explanation: Retrospective is the meeting where a team reflects and looks back to adapt and learn to do future projects better.

605. In the agile methodology, decisions are mostly taken by consensus. Consensus in this context means:
    a) Everybody is onboard and agrees with the decision.
    b) Everybody is able to live with the decision.
    c) The decision is taken by one or two people after consulting with the team.
    d) The decision is taken by majority of the team members present and voting.

Explanation: In agile methodology, decisions are taken in the spirit of "I can live with that". Whether it is estimates or technical approach or priorities - it is virtually impossible to reach 100% agreement

606. When should be the best time for the team to estimate a task?
    a) Throughout the iteration
    b) During iteration planning
    c) During the release planning and baselining the plan
    d) During the time of planning and whenever additional information is available

    Explanation: Project team will estimate a task at the time of planning and estimate whenever necessary

607. Which of the following charts show the number of tasks completed each day of the iteration to monitor the progress?
    a) Progression charts
    b) Task burnup chart
    c) Task check-off charts
    d) Task charts

    Explanation: Many agile practitioners use a task burnup chart, which shows the number of tasks completed each day of the iteration to monitor progress.

608. Self-organization implies a reliance on internal motivation of the team rather than external motivation. This is referred to as:
    a) Delegatory management
    b) Measurement-based management
    c) Democratic management
    d) Autocratic management

    Explanation: Self-organization implies a reliance on internal motivation of the team rather than external motivation (i.e., a measurement target). Author Austin calls this delegatory management as contrasted with measurement-based management. He goes on to say, "measurement-based management is in conflict with delegatory management. There is a negative interaction because of the implicit message of distrust that a measurement system conveys by the fact of its existence."

609. When is the definition of done first used by the team?
    a) During iteration reviews
    b) During retrospectives
    c) During iteration planning
    d) During feature estimation of user stories

    Explanation: The estimate must take into consideration all the work needed to complete the story - codified in the definition of tone

610. What is the Japanese term for a technique used to perform root cause analysis using a fishbone diagram?
   a) Lean
   b) Sashimi
   c) Ishikawa diagram
   d) Kaizen

   Explanation: Root cause analysis can be performed using Ishikawa diagrams or fishbone diagrams

611. How would risk be managed in an agile project?
   a) Decision tree
   b) Risk-adjusted backlog
   c) Risk assessment matrix
   d) Risk management plan

   Explanation: In an agile project, risk would be managed using the risk-adjusted backlog.

612. Which methodology deals with the movement of a product from development to operational environment?
   a) DevOps
   b) ITIL
   c) Kanban
   d) Scrum

   Explanation: DevOps is a methodology focused on streamlining the processes involved in software development and IT operations, with the goal of enabling the rapid and reliable deployment of software into production environments. It emphasizes collaboration, automation, and continuous integration and delivery to ensure efficient and smooth movement of products from development to operational environments. Therefore, DevOps is specifically designed to address the movement of products from development to operations.

613. Which of the following charts shows how much work remains in an iteration?
   a) Iteration burndown chart
   b) Burndown chart
   c) Daily burndown chart
   d) Burnup chart

   Explanation: The burndown chart shows how much work (usually measured in story points or tasks) remains to be completed over the course of an iteration or sprint.

614. A story that was planned as one story point, actually took two days to complete. How much does it contribute to the velocity when calculated at the end of the iteration?

a) **One Point**
b) Three Points
c) Two Points
d) Cannot be calculated

Explanation: Only the original planned effort of one story point is used to calculate the Velocity at the end of the iteration.

615. A good approach for small stories is to combine them into larger stories that represent from about a half-day to several days of work. The combined story is given a name and is scheduled and worked on just like any other story. Which of the following is an example of a small story?
a) Major database architecture changes
b) Updating over one hundred automated reports
c) **User interface changes**
d) Status updates

Explanation: Bug reports and user interface changes are common examples of stories that are too small for an individual story.

616. Who specifies acceptance tests in an agile project?
a) **Product owner**
b) Programmer
c) Developer
d) Scrum master

Explanation: As the software is being written to fulfill a vision held by the product owner, the acceptance tests need to be specified by the product owner. The product owner can work with a programmer or tester to actually create the tests, but the product owner needs to specify the tests that will be used to know when a story has been correctly developed.

617. What does the lean principle "amplify learning" mean?
a) Emphasize trainings to the team members.
b) Hire employees with good academic records.
c) Encourage team members to learn from their seniors by pairing.
d) **View every problem encountered as a learning opportunity to make the process better.**

Explanation: Lean is based on the principle of "amplify learning" where every incident or problem is a learning opportunity.

618. When can a plan be changed in agile management?
a) At the end of the Iteration
b) **When something new is learned**
c) At the daily meeting
d) Cannot be changed

Explanation: In agile, you can change the plan when something new is learned or when it is known that a mistake is to be avoided.

619. You have been assigned a new project that is large, complex, and uncertain. As a project manager, you will be using agile project management. Which of the following would you select while creating your team?
    a) A team that is expandable with the project
    b) A large team
    c) A functional manager who is responsible for the team
    d) A small cross-functional team

    Explanation: As per agile methodology, it is best to work with small teams staffed by cross-functional skills and increase the number of team members only when necessary.

620. Assuming one-week iterations and a team of four developers, how many iterations will it take for the team to complete a project with 27 story points if they have a velocity of 4?
    a) 4
    b) 6
    c) 7
    d) 24

    Explanation: With a velocity of 4 story points per iteration and a project size of 27 story points, the team would require approximately 6.75 iterations to complete the project. Since iterations are discrete units of time, the team would round up to the nearest whole number, resulting in 7 iterations needed to complete the project. This calculation ensures that the team adequately accounts for all the work required to finish the project within the given timeframe.

621. Which of the following defines a complex story?
    a) A user story that contains multiple stories
    b) A user story that users can mark as inactive
    c) A user story that is large by nature and cannot easily be dissected into smaller stories
    d) A user story that is not executable

    Explanation: A complex story is a user story that is large by nature and cannot easily be dissected into smaller stories.

622. Which of the following is NOT an attribute of a good story?
    a) Testable
    b) Large
    c) Independent
    d) Negotiable

Explanation: A good story has six attributes - Independent, Negotiable, Valuable to users or customers, Estimable, Small, and Testable.

623. When a team of people work together, the team goes through a specific sequence of group dynamics called the Tuckman model. Which of the following depicts the correct sequence of this process?
    a) Forming, Norming, Performing, Storming
    b) Forming, Storming, Norming, Performing
    c) Forming, Performing, Storming, Norming
    d) Storming, Norming, Performing, Forming

    Explanation: The Tuckman model has the following stages: Forming, Storming, Norming, Performing.

624. Name the XP technique where two people work: one person who codes is the driver and the other person who thinks is the navigator?
    a) Pair programming
    b) Planning poker
    c) Planning technique
    d) Coding technique

    Explanation: Pair programming is when two developers work on the same code together to improve efficiency.

625. An informative and collaborative workspace makes information easily communicated by just being in the room. Which of the following is least likely to be present in an informative workspace?
    a) Big visible charts
    b) Process improvement charts
    c) Hand-drawn charts
    d) Project charter

    Explanation: Big visible charts, process improvement charts, hand-drawn charts are all examples that might be found in an informative workspace. A project charter would be a project document that would be stored in the project archive and not visible to the team in a workspace.

626. There are different types of retrospectives: release retrospectives, project retrospectives, iteration retrospectives, and surprise retrospectives. What are surprise retrospectives?
    a) Retrospectives conducted when an unplanned event changes the situation
    b) Retrospectives conducted at the end of each iteration
    c) Retrospectives conducted half way of each project
    d) Retrospectives suggested by the scrum master

Explanation: Surprise retrospectives are conducted when an unplanned event changes the situation.

627. The red team finished 38 story points in their last two-week iteration. The blue team has half as many developers on a different project and completed 38 story points in their last two-week iteration. What can be determined from this information?
    a) The story points of one team are not comparable to the story points of any other team
    b) Team A is twice as productive as Team B.
    c) Team A should have more people in the team.
    d) Team B has the management support.

    Explanation: The story points of one team are not comparable to the story points of any other team.

628. Four developers are estimating a story. The estimates provided are two, three, six, and ten story points. Which estimate should be used?
    a) The one given by the expert
    b) The majority view
    c) The average of the estimates
    d) The team should decide by consensus

    Explanation: In Agile methodology, including Scrum, the team typically follows the principle of collective decision-making and collaboration. Consensus decision-making encourages open discussion and ensures that all team members contribute to the decision-making process. This fosters shared understanding and commitment to the estimate, leading to better alignment and accountability within the team.

629. A stand-up meeting is conducted in the morning every day. Each person describes the new information that the team should know. Which of the following is NOT discussed?
    a) What did I do yesterday?
    b) What are my tasks for the coming week?
    c) What do I plan to do today?
    d) What issues are preventing my progress?

    Explanation: In the daily stand-up meeting, the tasks for next week would not be discussed, only yesterday's activity, today's activity, and current issues are discussed.

630. Which of the following most appropriately describes a task board?
    a) Appears on a whiteboard
    b) Show the project trends
    c) Organizes work and communicates what work is left

d) Displays the velocity

Explanation: A task board is a visual representation of the work in progress within a project. It typically consists of columns representing different stages of work (e.g., to-do, in progress, done) and cards representing individual tasks or user stories. By organizing work and visually indicating what tasks are pending, in progress, or completed, the task board helps teams manage their workflow effectively and fosters transparency and collaboration.

631. Some stakeholders are interested in viewing the project progress, but are not interested in the details. Which of the following should you share with these stakeholders?
    a) Weekly demo
    b) Vision Statement
    c) Release plan
    d) Burnup chart

    Explanation: The burnup chart shows progress towards project completion and the expected completion date.

632. Which of the following is NOT included in the story cards information'?
    a) Estimated value points
    b) Story description
    c) Story dependencies
    d) Velocity

    Explanation: Story cards information includes story name, story description, estimated work effort, estimated value points, requirements uncertainty, story dependencies, and acceptance tests. Velocity is the number of user stories that are delivered in iteration.

633. Which of the following will NOT assist in critical problem solving during the project?
    a) Ask probing questions
    b) Use active listening
    c) Use reflective listening
    d) Injecting ideas

    Explanation: During critical problem solving, you can ask probing questions and use active and reflective listening to find an answer, but one should avoid injecting their ideas.

634. One of your stakeholders wants more details than the vision statement provides, but not the overwhelming details of the release and iteration plans. Which of the following progress report will you share with this stakeholder?
    a) Roadmap
    b) Status email

c) Iteration plan
d) Productivity chart

Explanation: A roadmap is the best report when stakeholders want more details than the vision statement provides. It summarizes planned releases and the significant features in each one.

635. With whom should a team share information about the progress and impediments?
a) All interested stakeholders
b) Team members, team facilitator and subject matter experts
c) Team members, team facilitator and product owner
d) Senior management and customer

Explanation: Transparency is a key principle of scrum. Therefore the information must be made transparent to all interested stakeholders

636. Which of the following is NOT true about the Iteration demos?
a) It demonstrates the team progress.
b) It realistically reports progress.
c) It is shown only to a select set of stakeholders.
d) It solicits feedback from customers.

Explanation: Iteration demos are not shown only to select set of stakeholders, but all stakeholders and are invited to provide feedback.

637. During which project management phase is the product backlog changed based on feedback received?
a) Envision
b) Speculate
c) Explore
d) Adapt

Explanation: During the Adapt phase of project management, teams adjust and refine their approach based on feedback received from stakeholders, customers, or team members. This includes updating the product backlog to reflect new priorities, insights, or requirements that emerge as the project progresses. By continuously adapting to changing circumstances, teams can ensure that their project remains aligned with stakeholders' needs and goals**.**

638. Which of the following is true about the throughput of a team?
a) Throughput = WIP * Cycle time
b) Throughput = 1 / Cycle Time
c) Throughput = WIP / Lead Time
d) Throughput = Lead time /Cycle time

Explanation: Throughput is the number of features the team can develop in a particular amount of time and is the inverse of the cycle time

639. Which of the following does NOT explain refactoring?
    a) Organizes the working code
    b) Improves code quality
    c) Makes the code more maintainable
    d) Reduces bugs dramatically

    Explanation: Refactoring is the reorganization of code to make it more organized, of higher quality, and more maintainable. It does not reduce bugs.

640. If your release is 12 iterations away, your velocity is 14 points, your risk exposure is 1 iteration and with a 10% chance, how many points will you have finished?
    a) 154
    b) 110
    c) 86
    d) 92

    Explanation: Using the following formula, risk adjusted points remaining = (iterations remaining - risk exposure) X velocity / risk multiplier --> points remaining = (12 - 1) X 14 = 154 points
    10 percent chance: 154 / 1 = 154 points.

641. The retrospective meeting on previous iteration should NOT focus on:
    a) What we did well in the previous iteration
    b) What should be planned for the next iteration
    c) What we did not do well in the previous iteration
    d) What we need to change as a team

    Explanation: The retrospective meeting on previous iteration should not get distracted by planning the next iteration.

642. What is wrong with the story - "As an administrator, I want to manage assignments so that I can control the workload"?
    a) The word manage does not provide enough detail about what exactly is needed.
    b) The story does not say who needs it.
    c) The story does not specify the reason.
    d) The story is too small.

    Explanation: The word manage does not provide details - it may potentially consist of multiple stories bunched together

643. Which of the following is NOT a principle from the agile Manifesto?
    a) Responding to change
    b) Customer collaboration

c) **Technical excellence**
d) Working software

Explanation: The four principles of the agile Manifesto are individuals and interactions, working software, customer collaboration, and responding to change.

644. In a design the box activity, the objective is to design the product box front and back. Which of the following would NOT be included in the activity?
a) Graphic design
b) Feature description
c) Operating requirements
d) Competitor's product name and features

Explanation: In a design-the-box activity, the objective is to design the product box front and back and would include the graphic design, feature description, and operating requirements.

645. In an iteration, the team committed 10 stories but completed only eight. What can be done about the remaining two stories?
a) They move back to the product backlog.
b) They move to the next iteration.
c) The iteration is extended till all stories are completed.
d) The team performs a short iteration to complete them.

Explanation: Incomplete stories move to the backlog and are prioritized against the rest of the items.

646. Which of the following factors will NOT assist in improving the project velocity?
a) Pay down technical debt.
b) Offload programmer duties.
c) Reduce customer involvement.
d) Provide needed resources.

Explanation: Improving customer involvement will assist in improving the project velocity (not reducing).

647. What financial return parameter will be best for comparing the values of two different cash flow streams?
a) Internal rate of return
b) Payback period
c) Discounted payback period
d) Compound interest

Explanation: Internal rate of return (IRR) is used to express the return on project in percentage when comparing the two different cash flow streams.

648. In the relative weighting prioritization technique, what is the relative contribution of feature X to the project if it has a value of 12 and the total value of all features is 35?
   a) 0.38
   b) 0.34
   c) 2.91
   d) 14.91

   Explanation: To calculate the relative contribution of each feature, divide its value by the sum of the Total Value of all features. Hence, relative contribution of feature X = 12/35 = 0.34.

649. Which of the following is NOT true about agile retrospectives?
   a) The team decides how they will work together to complete the project.
   b) The team reflects on past work and learns from the project history.
   c) The team decides how to drive the development process.
   d) The team decides who will be part of the team for the next project.

   Explanation: Retrospective is a meeting in which a team learns from their project history and discusses how they worked together and what they can learn from their progress on the project. This helps the team members to continue the development process and resolve issues when they arise.

650. After completing the calculation using the relative weights method, the numerical priority value for feature A was 0.92, B was 0.88, C was 0.97, and D was 0.41. Which of the features should be implemented first?
   a) A
   b) B
   c) C
   d) D

   Explanation: The order of the feature is determined by dividing priorities. Since C has highest priority value, it should be picked up first.

651. What is NOT true about participatory decision making?
   a) It is a technique that involves all the affected parties.
   b) It is a style of making decisions where timeliness of decision is important.
   c) It ensures that all interested parties agree with the decision.
   d) In it many concerned parties are consulted before making a decision.

   Explanation: Participatory decision making is a technique that involves all the affected parties. It does not mean all have to agree with the decision. They have to be able to "live with" the decision.

652. A team's average billing rate is $2,000 per iteration. They plan to work in thirteen two-week iterations during the project a team velocity of twenty story points per iteration. What will be the budget for the overall project?

a) 52,000
b) 13,000
c) 26,000
d) 520,000

Explanation: The budget for the overall project is calculated by multiplying the average billing rate per iteration ($2,000) by the number of iterations planned for the project (13). Therefore, 13 iterations * $2,000 per iteration equals $26,000. This represents the total budget allocated for the project, taking into account the team's billing rate and the planned duration of the project in iterations.

653. A team is wondering whether a given user story meets its intended scope. Which of the following is NOT true regarding scope in agile projects?
a) Scope in agile is primarily a function of rolling wave planning.
b) Scope creep doesn't exist in agile projects.
c) Scope is controlled using the product backlog.
d) Scope is verified by the scrum master.

Explanation: In agile projects, scope is primarily managed through the product backlog, where user stories are prioritized and detailed. While the scrum master facilitates the agile process, including sprint ceremonies, it's the responsibility of the product owner to verify and validate the scope of the product to ensure alignment with stakeholder requirements. The product owner acts as the primary authority on scope-related decisions, not the scrum master.

654. A team often starts the retrospective with each participant declaring allegiance to the prime directive. Why is this important?
a) To ensure that the retrospective does not degenerate into a blame game.
b) To ensure that the team achieves the goals set up during sprint planning.
c) To reaffirm the commitments made by the team during the planning.
d) To get the right feedback about the product from the customers and other stakeholders.

Explanation: The prime directive ensures that the retrospective is conducted without blaming each other for what happened. It ensures that the focus is instead on improvements.

655. A product owner is discussing prospective features that will be added in the near future with customers. According to the planning onion, the PO is describing the _____.
a) Strategy
b) Release plan
c) Roadmap
d) Iteration goals

Explanation: PO's often present product roadmaps to customers to give them a glimpse of the upcoming product features.

656. A product owner felt that the "performance" of the product in terms of response time needed to be improved. The team then discussed what modifications were necessary in the product to achieve this. They came up with a bunch of work items. According to the product backlog, these work items can be collectively referred to as a _____.
    a) Theme
    b) Epic
    c) Story
    d) Task

    Explanation: Theme is the main purpose behind a group of stories, an iteration, or a release.

657. A project uses a version control system to mediate changes, which includes updating codes, testing the code, and then checking-in the changes, which may occur several times in a day. This process is known as:
    a) Version control management
    b) Continuous integration
    c) Simple integration
    d) Configuration management

    Explanation: Continuous integration is the development practice of updating codes, testing the code, and then checking-in the changes, which may occur several times in a day.

658. When a team uses ideal days as a unit of estimation, which of the following is NOT true?
    a) There won't be any distractions.
    b) All dependencies are met.
    c) The team will focus only on the work.
    d) A relative size unit is used.

    Explanation: Ideal time is an absolute measure of size. Story points is a relative measure of size.

659. Agile teams may use Planning Poker for estimation. Which of the following is NOT true about Planning Poker?
    a) It combines expert opinion, analogy, and disaggregation to obtain reliable estimates.
    b) Participants include all the developers in the team.
    c) The team for Planning Poker should not exceed more than 10 participants.
    d) The highest estimate is considered the final estimate.

Explanation: The highest estimate is not necessarily considered the final estimate because the players with high and low estimates must explain their estimates and then a decision is taken by the players as to the final estimate value.

660. When the estimates of a team practicing Planning Poker estimation do not converge, how can a decision be made?
    a) Ask the team to come to a decision.
    b) Take the majority view.
    c) Take the highest of the estimates.
    d) Take an expert opinion.

    Explanation: The team has to decide how to arrive at a decision regarding estimates. This is true regardless of the technique used.

661. Which of the following chart indicates the scope changes made during an agile project?
    a) Burndown bar chart
    b) Sprint burndown chart
    c) Scrum board
    d) Parking lot chart

    Explanation: The burndown bar chart indicates the scope changes in addition to the work completed.

662. A team is wondering whether to use Planning Poker or affinity estimation to estimate a subset of the product backlog. Affinity estimation is preferred over Planning Poker when:
    a) A team-based approach is necessary.
    b) A large number of items need to be estimated.
    c) When the unit of estimation is ideal days instead of points.
    d) When the design or approach is unclear.

    Explanation: Affinity estimation is most valuable when there are a large number of items to be estimated.

663. The ideal time estimate for a given piece of work does not automatically translate into the elapsed time. A conversion factor is often necessary. Which of the following defines elapsed time?
    a) It is the same as ideal time.
    b) The actual amount of time that the team spent on developing a feature.
    c) The amount of time without any interruptions.
    d) The amount of time that the team spent on the iteration.

    Explanation: Elapsed time is the actual amount of time that the team spent on developing a feature.

664. A Kanban system for managing work is often used by teams. Which of the following is one of the essential steps of putting into place a Kanban system?
   a) Creating a value stream map
   b) Identifying the waste and removing it
   c) Visualizing the workflow
   d) Managing work in iterations

   Explanation: One of the essential steps in implementing a Kanban system is visualizing the workflow. This involves creating a visual representation, often using a Kanban board, of the work items moving through different stages of the process. By visualizing the workflow, teams gain insight into their work, identify bottlenecks, and improve overall efficiency and transparency in managing tasks.

665. The MoSCoW technique is used to create a prioritized requirements list and is popularly used in the DSDM agile technique. In the MoSCoW technique, 'M' stands for:
   a) Medium
   b) Must
   c) Must Not Have
   d) Minimum Marketable Features

   Explanation: The MoSCoW prioritization scheme derived its name from the first letters of its labels: must, should, could, won't.

666. Pair programming is a the widely used practice in the extreme programming method. This method allows two people to work on a task instead of one. The primary advantage of this method is that _____.
   a) It optimally utilizes the available development effort.
   b) It leads to knowledge sharing and collective code ownership.
   c) It generates self-testing code.
   d) It leads to greater commitment and ownership from the team.

   Explanation: The pair programming method leads to knowledge sharing across the team.

667. Many contract types are used on agile projects, including fixed price contracts. In a fixed price contract, the risk of cost escalation is borne mostly by the:
   a) Supplier
   b) Customer
   c) Product owner
   d) Project team

   Explanation: In a fixed price contract, the risk is mostly assumed by the supplier because if the estimate is incorrect the supplier must still honor the price fixed with the customer.

668. Teams utilizing unfamiliar technology may need time for training prior to a project's launch and time to establish a development environment. The best way to account for this is to include the training time ____.
   a) In iteration 0.
   b) As a new story.
   c) Outside the scope of the project as such.
   d) As a part of the functional manager's budget.

   Explanation: Iteration 0 is used to prepare the team for the project and need not deliver direct customer value.

669. A user story must be properly described so that the readers can understand what is the underlying requirement. Such a story is often called a closed story. Which of the following is NOT a closed story?
   a) A recruiter can review resumes from applicants to one of their ads.
   b) A recruiter can delete an application that is not a good match for a job.
   c) A recruiter can manage ads.
   d) A recruiter can change the expiry date of an ad.

   Explanation: A recruiter can manage the ads that they place is not a closed story because it is an ongoing activity.

670. Stories are prioritized based on many factors. In a release planning exercise, the primary task is assigning stories to iterations. This is done primarily on the basis of:
   a) Value and risk
   b) Risk and cost
   c) Customer direction
   d) Team capabilities

   Explanation: The primary task in release planning is assigning stories to iterations, chiefly on the basis of value and risk.

671. Agile methods are not uniformly applicable to all kinds of projects. Broadly speaking, one can say that agile is most suitable when:
   a) The team is largely inexperienced.
   b) The teams are distributed geographically.
   c) The requirements are changing and the technology is moderately complex.
   d) The customer prefers to work in a linear or incremental fashion.

   Explanation: Agile methods are most appropriate when the complexity of requirements and/or technology is moderate.

672. "Having problems does not indicate a problem in itself, so welcome problems because they bring with them the chance for the team to overcome, grow, and

become stronger together." As an agile coach, which of the following is NOT a perspective to analyze and detect the symptoms of a problem?

a) Process level
b) Quality and performance level
c) Team dynamics dimension
d) Management of team members

Explanation: The multiple perspectives to analyze include the process level, the quality and performance angle, and the team dynamics dimension, not the management of team members.

673. A newly appointed product owner is looking for a list of all the known features that are to be implemented in the future. Where can this information be found?

a) Prioritized sprint
b) Vision box
c) Product backlog
d) Sprint review meet

Explanation: The product backlog documents all the known features that are to be implemented throughout the project.

674. During the second level of conflict, disagreement, the team members distance themselves from one another, except in one-on-one conversations to obtain advice and support. What becomes as important as solving the problem?

a) Self-protection
b) Satisfying self-ego
c) Aiming to win
d) Aiming to grow

Explanation: During disagreement, self-protection becomes as important as solving the problem or the conflict situation.

675. John and Joseph are working in the same room. John is programming and Joseph is discussing with Kim. John overhears their conversation enough to know that Joseph has discussed an idea with Kim. What kind of communication is this referred to as?

a) Push communication
b) Osmotic communication
c) Pull communication
d) Active listening

Explanation: Osmotic communication occurs when people in the same environment hear and comprehend information without it being communicated to them directly.

676. Which of the following is NOT included on information radiators?
   a) Work breakdown
   b) Project status
   c) Automated build report
   d) Performance analysis of team members

   Explanation: Information radiators may display the work breakdown, project status, and automated build report, but not the performance analysis of individual team members.

677. Spike solutions are small experiments to answer a question and determine a path forward. Which of the following types of spikes will you use if you need to test an approach to solve a particular problem?
   a) Development spikes
   b) Design spikes
   c) Scheduling spikes
   d) Iteration spikes

   Explanation: A design spike is used to test the appropriateness of a particular solution or approach.

678. Which of the following statements about lean, Kanban and agile is not true?
   a) Lean thinking is a superset that encompasses both Kanban and agile approaches.
   b) Kanban approaches are less prescriptive than agile methods as it advocates "start where you are".
   c) Agile methods are a superset that include both lean and Kanban.
   d) Agile approaches encompass many approaches like scrum, DSDM, etc.

   Explanation: Lean thinking is the superset of agile and Kanban methods, not the other way around.

679. A team is experiencing conflict while solving a particular issue. It can be said that the level of conflict is 1 (a problem to solve). What is the appropriate response option at this level?
   a) Collaboration and consensus
   b) Accommodate
   c) Safety
   d) Root case analysis

   Explanation: At conflict level 1, where there's a problem to solve, the appropriate response options include collaboration and consensus-building among team members. By working together to address the issue and reach agreement on a solution, the team can effectively resolve the conflict and move forward constructively. This approach fosters teamwork, communication, and mutual understanding, leading to better outcomes for the project.

680. As per the Kano model of customer satisfaction, which features are most important for the product to be ready for the marketplace?
   a) Exciters
   b) Linear Features
   c) Delighters
   d) Threshold

   Explanation: In the Kano model, threshold features represent basic requirements that must be met for a product to be considered acceptable by customers. Without these features, the product is unlikely to be competitive in the marketplace, making them crucial for readiness.

681. A predictive life cycle uses the traditional approach to project management. All activities are performed in sequence and typically only once throughout the project. The predictive life cycle is suitable when:
   a) Requirements are highly dynamic.
   b) Small deliveries are frequently required.
   c) Requirements are more or less fixed.
   d) The solution is repeated until it is correct.

   Explanation: Predictive life cycles are best when requirements are more or less fixed.

682. A pharmaceutical product development process includes getting a mandatory FDA approval at the end. What would you recommend as an agile coach for the team?

   a) A hybrid process should be used in which the development is agile but the approval process is linear or predictive
   b) Agile should not be applied to projects when regulatory approvals are necessary
   c) Modify the process to get regulatory approvals in the middle of the life cycle
   d) Write a separate story for the regulatory requirements and schedule it throughout the life cycle

   Explanation: As an agile coach, you should recommend the use of a hybrid process in which the development is agile but the approval process is linear or predictive.

683. The primary difference between iteration-based agile and flow-based agile is that:
   a) Iteration-based agile may have different iteration durations for each feature
   b) Flow-based agile is based on time boxed iterations
   c) Flow-based agile may have different iteration durations for each feature
   d) Iteration-based agile relies more on feature teams

Explanation: Flow-based agile relies on teams picking up features for development and completing them based on their ability - rather than fixed set of time boxes.

684. A team facilitator in an agile team is an example of a servant leader. Which of the following is not a servant leader's responsibility?
    a) Enabling people to contribute
    b) Ensuring that the team delivers on commitments
    c) Removing impediments
    d) Facilitating collaboration

    Explanation: Servant leaders help, unblock, and enable, but they do not ensure that the team delivers on commitments.

685. To detect problems at the process level, the team uses a questionnaire that lists required elements and qualities that are the basic ingredients of agile. Using it, we can estimate the maturity of the agile process in the organization and how well the team follows the agile process. This questionnaire is referred to as:
    a) Health check
    b) Compliance requirement
    c) Audit
    d) Acceptance criteria

    Explanation: The questionnaire used to assess the maturity of the agile process in an organization and how well the team adheres to agile practices is commonly known as a "health check." This evaluation tool helps identify strengths, weaknesses, and areas for improvement within the agile framework, allowing teams to gauge their performance and make informed decisions to enhance their processes.

686. Extreme programming is a collection of practices that enable high quality products. Which of the following is NOT one of the XP Practices?
    a) Programmer welfare
    b) Shared understanding
    c) Fine-scale feedback
    d) MoSCoW prioritization

    Explanation: MoSCoW prioritization is a technique commonly associated with the Dynamic Systems Development Method (DSDM) rather than Extreme Programming (XP). XP practices primarily focus on aspects such as programmer welfare, shared understanding among team members, and fine-scale feedback mechanisms to ensure high-quality product development. MoSCoW prioritization, on the other hand, is used to prioritize requirements based on their importance to the project's success, and it is not specific to XP practices.

687. What is the primary output of the sprint planning meeting?

a) Sprint backlog
b) Product backlog
c) Working product
d) Refined product backlog

Explanation: The sprint planning meeting primarily results in the creation of the sprint backlog, which outlines the tasks and user stories that the team commits to completing during the upcoming sprint. This backlog is derived from the prioritized items in the product backlog and serves as a guide for the team's work during the sprint, ensuring alignment with the sprint goal and objectives.

688. When would you consider changing the team composition of an agile team?
a) When the team lacks a critical skill that is needed to deliver a product
b) When the functional manager wants to replace a team member
c) When the team wants to add or remove members
d) When the project is completed and a new project is about to start

Explanation: The team needs all the skills required to complete the product increment.

689. At a retrospective meeting, few issues are highlighted by the team. Who is responsible for solving the issues?
a) Scrum master
b) Product owner
c) Stakeholders
d) Team

Explanation: Agile teams are self-organizing and need to find solutions to their problems.

690. Which of the following is NOT done as part of planning in agile project management?
a) Risk management
b) Product vision
c) Buffer and slack management
d) Done-done

Explanation: The term done-done is used when the user story that the team is developing is completed. Done-done is part of releasing, not planning.

691. An important criterion for an agile team is to understand the market which is served by the product that they are working on. Which of the following best describes Horizontal Market Software?
a) Developed for many organizations
b) Developed and used across many industries
c) Developed for a dedicated customer
d) Developed for one organization

Explanation: Horizontal Market Software is intended to be used across a wide range of industries.

692. What is the best indicator of progress in a project?
    a) Completion of a sprint
    b) Delivery of working product increment
    c) Delivering all the committed features
    d) Releasing on schedule and within budget

    Explanation: Working software and delivery of working product increment is the primary measure of progress.

693. An agile method is both incremental and iterative. What is the primary value of an iterative life cycle?
    a) Allows faster delivery of value
    b) Allows the team to reflect on feedback and correct issues
    c) Makes it easier to develop
    d) Reduces overall cost of development

    Explanation: The iterative nature of the life cycle allows the team to inspect and adapt.

694. What is the process of adding more details as they become better defined during planning called?
    a) Burndown chart
    b) Release
    c) Rolling wave
    d) Roadmap

    Explanation: The process of planning in waves as project details become more defined is called rolling wave.

695. A team finds that the process of integrating a complex product is always problematic and time consuming. Which of the following approaches will help solve the problem?
    a) Delay the integration until all the components have been fully developed
    b) Move towards a continuous integration setup
    c) Appoint a separate team for integration
    d) Note the integration risk and make allowances in the plan

    Explanation: Continuous integration will help detect and solve integration issues early in the life cycle.

696. Which report should NOT be shared with your stakeholders?
    a) Defect reports
    b) Status reports
    c) Throughput reports

d) SLOC reports

Explanation: Source Lines of Code (SLOC) reports should not be shared with your stakeholders as they are flawed productivity metrics.

697. In a complex product consisting of multiple components and subcomponents, the vast majority of tests should be:
    a) Unit tests
    b) Acceptance tests
    c) Exploratory tests
    d) Usability tests

    Explanation: In a complex product with multiple components and subcomponents, the vast majority of tests should be unit tests. Unit tests focus on testing individual units or components of the software in isolation, verifying that each unit behaves as expected according to its specifications. This approach ensures thorough testing of the codebase's functionality at a granular level, facilitating early detection of bugs and promoting code reliability and maintainability.

698. A retrospective can be as short as 15 minutes or as long as 3 hours. The duration of the retrospective can be dependent on all of the following except:
    a) Length of the iteration
    b) Level of conflict or controversy
    c) Size of the team
    d) Participation of the functional managers

    Explanation: Functional managers ideally should not participate in the retrospectives as it may impede the free flow of ideas and feedback.

699. A team wants to increase the duration of the iteration. The coach of the team doesn't agree. What would be a scenario where increasing the iteration duration may be justified?
    a) When there is a recurring problem that causes the iteration goal to be compromised
    b) When the team has too many dependencies on others
    c) When the team lacks technical competence to complete the iteration on time
    d) When the iteration demos are critical delivery milestones

    Explanation: When the problem impacting the completion of the iteration is a recurring one, the team could consider increasing the duration of the iteration.

700. What is a valid scenario for a product owner to cancel a sprint or iteration?
    a) When the team is not on track to complete the deliverables
    b) When the iteration backlog is no longer valuable to the customer due to a major change

c) When the product owner wants a different feature instead of those that are planned
d) When the customer changes the priorities of the backlog

Explanation: When the entire iteration backlog becomes useless, it may be worth cancelling an iteration and starting a new one with revised priorities.

701. Activities are timeboxed processes that help the team move through the phases of a retrospective. Activities provide structure to help your team think together and have several advantages over freewheeling discussion. Activities do not have to be elaborate to be effective. Which of the following are NOT examples of activities that are useful in retrospectives?
   a) Brainstorming
   b) Voting with dots
   c) Pair interviews
   d) Synergy

   Explanation: Activities do not have to be elaborate to be effective. Examples of activities that are useful in retrospectives include brainstorming, voting with dots, doing check-ins, and performing pair interviews.

702. For a retrospective meeting, the seating arrangement should be such that there is enough space for people to move comfortably. Which of the following is a suitable seating arrangement for a retrospective meeting?
   a) Circle or semicircle of chairs
   b) Theatre-style arrangement
   c) Tables arranged in U-shape
   d) Meeting room

   Explanation: A circle or semicircle of chairs encourages participation because people can see each other. Classroom or theater-style arrangements stifle participation. Staring at the back of someone else head is not conducive to conversation. Tables can be a physical barrier that becomes a psychological barrier. Avoid rooms with an immovable conference table in the middle. That big table will inhibit creative collaboration. This is not a board meeting, after all. Tables arranged in a U-shape with a big gulf in the middle create distance and make it hard to move around.

703. Which of the following is most helpful in implementing collective code ownership in a team?
   a) Version control
   b) Pair programming
   c) Continuous integration
   d) Probability distribution

Explanation: Pair programming is helpful in implementing collective code ownership in a team.

704. How do you best describe the primary role of an agile leader?
    a) Manage WBS, schedule, and cost of project
    b) Manage stakeholders
    c) Manage work allocation to team members
    d) Manage teams without managing their tasks

    Explanation: The primary role of an agile leader is to empower and support the team, foster collaboration, and create an environment where team members can self-organize to accomplish their tasks. Agile leaders focus on removing obstacles, facilitating communication, and providing guidance and support, rather than micromanaging tasks. This approach encourages autonomy, ownership, and innovation within the team while aligning with agile principles of self-organization and collective responsibility.

705. Which of the following is NOT a critical skill for a product owner of a team?
    a) Business acumen
    b) Communication skills
    c) Ability to generate consensus
    d) Technical excellence

    Explanation: Technical excellence is not necessary for the PO as they can rely on the team for technical abilities.

706. Performing agile development in geographically distributed teams is a challenge in itself. Which of the following is an arrangement that will help them overcome communication challenges?
    a) Fishbowl window - with long live video conferences
    b) Formal communication through designated representatives only
    c) Restricting access to each other's repositories across locations
    d) Following more waterfall practices in distributed teams

    Explanation: Modern communication tools help geographically dispersed teams overcome challenges in agile teams.

707. Who performs value point estimating?
    a) Finance team
    b) Scrum team
    c) Product team
    d) Development team

    Explanation: Value point estimating is conducted by the product team, which includes product owners, managers, and stakeholders. This method aids in prioritizing features based on their perceived value to the product or business,

assisting the team in making informed decisions about which items to focus on during development.

708. In the absence of the product owner, who should assume the product owner's responsibilities at the sprint planning meeting?
    a) Scrum master
    b) Team
    c) Person selected by the product owner
    d) Person selected by the team

    Explanation: In Agile and Scrum practices, the Product Owner is responsible for the product backlog and prioritizing the work. If the Product Owner is absent, it is their responsibility to designate someone to assume their responsibilities during the sprint planning meeting. This ensures that the vision and priorities for the product are communicated effectively.

709. Who has the role of refocusing the team when they have drifted away from agile practices?
    a) The functional manager
    b) The team facilitator
    c) The product owner
    d) The cross functional project team

    Explanation: It is the job of the team facilitator to remind the team about agile practices.

710. The agile team has collective responsibility of the following, except _____.
    a) The quality of the product
    b) The code
    c) The product vision
    d) The speed at which stories are completed

    Explanation: The product vision is primarily the responsibility of the product owner, who ensures that the team aligns with the overall vision and goals of the product. While the agile team collectively holds responsibility for the quality of the product, the code, and the speed at which stories are completed, the product vision is typically defined and maintained by the product owner in collaboration with stakeholders.

711. After an iteration is planned, who manages the iteration backlog?
    a) The team
    b) The sponsor
    c) The product owner
    d) The scrum master

Explanation: The team manages the iteration backlog and the product owner manages the product backlog.

712. An important characteristic of any project (including agile projects) is progressive elaboration. Progressive elaboration is used to:
    a) Document all the requirements at once
    b) Detail progress on the project
    c) Gather characteristics of the product over time
    d) Test the requirements

    Explanation: Progressive elaboration is used to gather characteristics of the product over time.

713. Effective project management starts with the definition of requirements. In agile projects, requirements are expressed from the user perspective in the form of user stories. Which of the following is NOT an attribute of good user stories as per the INVEST model?
    a) Negotiable
    b) Acceptable
    c) Testable
    d) Estimable

    Explanation: INVEST stands for independent, negotiable, valuable, estimable, small, and testable.

714. An agile team is both cross functional and empowered. Which of the following is not one of the characteristics of an agile team?
    a) Decision-making manager
    b) Clear communication
    c) Customer involvement
    d) Shared vision

    Explanation: Empowered teams do not require a manager to make decisions. The teams cooperate with each other to make decisions.

715. Quality management in agile projects focuses on both verification and validation. Ensuring that the functionality meets specifications is _____.
    a) Review
    b) Ad-hoc testing
    c) Verification
    d) Validation

    Explanation: Verification in quality management ensures that the product or functionality conforms to specified requirements or specifications. It involves reviewing and checking whether the work products meet predefined criteria and standards. This process helps in identifying and correcting defects early in the

development lifecycle, contributing to overall product quality and customer satisfaction.

716. One of the important purposes of retrospective meetings is:
    a) Planning
    b) Doing work
    c) Checking compliance
    d) Continuous improvement

    Explanation: Retrospective meetings serve the purpose of facilitating continuous improvement within agile teams. These sessions provide a dedicated time for teams to reflect on their processes, identify areas for enhancement, and implement changes to improve their effectiveness and efficiency. By fostering open communication and collaboration, retrospectives help teams adapt and evolve over time, ultimately leading to better outcomes and higher performance.

717. Daily stand-up meetings are used by most agile teams. These meetings should be:
    a) Used for the manager to tell the team what to do
    b) Kept short and to the point
    c) Used to stand-up the software implementation
    d) Used to review weekly project status

    Explanation: Stand-ups should be kept short and to the point and only held when necessary.

718. The velocity of a team is an important metric for enabling reliability in planning. Velocity refers to _____.
    a) The number of story points that a team can complete in an iteration
    b) How fast the pair of programmers can switch off between programming and integration
    c) The number of fixes the team can make in an iteration
    d) The number of story points that a team can complete in a release

    Explanation: Velocity is the number of story points that a team can complete in an iteration.

719. Kanban is a popular system for managing work that originated from the Toyota Production System. The closest synonym for Kanban is _____.
    a) Process
    b) Signal
    c) Pull
    d) Box

    Explanation: Kanban is the Japanese word for signal.

720. Agile teams should focus on delivering value early. This means that they should:
    a) Keep options open to improve the ability to respond to change

b) Be empowered to make decisions
c) Deliver working features as quickly as possible
d) Refactor the code frequently

Explanation: Agile teams should focus on delivering value even in the earlier phases of a project. This can be done by delivering working features at every iteration.

721. A project has a firm delivery deadline. Which of the following will be most helpful in managing the project?
a) Timeboxing
b) Decomposition
c) Detailed planning upfront
d) Using ideal time

Explanation: Timeboxing is the practice of limiting the time available and delivering in short time windows.

722. A chicken stakeholder in an agile project is _____.
a) Someone who is committed
b) A stakeholder who refuses to get involved
c) Someone who is involved
d) Part of the core team

Explanation: A chicken is someone who is involved, but not committed. Chickens should not be on the core team but may give inputs.

723. In an agile project, which of the following is more important than processes and tools?
a) Individuals and interactions
b) Comprehensive documentation
c) Working product
d) Responding to change

Explanation: As per the agile manifesto, individuals and interactions are more important than processes and tools.

724. A team is developing a product using the agile methodology. The customer is very particular about quality and will not easily forgive any mistakes in development. How should the team proceed with development?
a) Spend a lot of time in planning up-front and establish delivery milestones
b) Plan frequent demos to get customer feedback
c) Get a detailed design approved by the customer before developing
d) Set up a continuous integration system before starting development

Explanation: "Inspect and adapt" is an important principle that advocates early direction validation.

725. A team member states that a high priority user story is too complex and it is difficult to provide an estimate for it or to complete it in one iteration. How should the team respond?
   a) Drop the story and replace it with a simpler one
   b) Give a couple of iterations to complete the story instead of one
   c) Break the story into a spike and implementation story
   d) Accept the story for the iteration as a "stretch goal"

   Explanation: Decomposing the story into a spike and implementation story is a sensible idea when a story is too complex.

726. One of the patterns of writing good code is DRY (Don't Repeat Yourself). What does this mean?
   a) Keep agile meetings focused
   b) Write a user story only once
   c) Keep common code in one place for easy maintenance
   d) Never assign the same name for more than one variable

   Explanation: DRY is an acronym for Don't Repeat Yourself. It is a reminder to maximize the work not done.

727. A very large story that may span several iterations is _____.
   a) Very difficult to implement
   b) An epic story
   c) Requires a large team
   d) Should be dissected before starting iterations

   Explanation: An epic story is a very large story that may span iterations.

728. Engaging stakeholders early and often is an important facet of an agile project. Which of the following is the best ways to involve external stakeholders?
   a) Invite them for the iteration demos
   b) Take their inputs for planning iterations
   c) Consulting them during iteration retrospectives
   d) Selectively calling them for the daily stand-up meetings

   Explanation: Stakeholder inputs are sought during iteration demos. The other forums are more appropriate for the team.

729. Incremental life cycles are often employed by teams working on product delivery. Which of the following is NOT a characteristic of incremental life cycles?
   a) Prioritization of functionality
   b) Planning everything up-front
   c) Delivering value quickly
   d) The customer working with the system before the project is complete

Explanation: Incremental life cycles prioritize delivering value quickly by breaking the project into increments and delivering them iteratively. This approach allows for early customer feedback and involvement, ensuring that the product meets their needs. However, planning everything upfront contradicts the iterative nature of incremental life cycles, as detailed planning typically occurs incrementally as the project progresses, rather than all at once at the beginning. Therefore, option (b) is not a characteristic of incremental life cycles.

730. What role does the sponsor play in an agile project?
    a) Provides project funding and resources
    b) Decides on product features and functions
    c) Approves a high-level project plan
    d) Helps define customer value

    Explanation: The role of a sponsor in an agile project is to provide funding and resources.

731. Each release in an agile project is made up of _____.
    a) One or more iterations
    b) A waterfall process
    c) Multiple approvals from stakeholders
    d) A detailed plan

    Explanation: A release, by definition, is made up of one or more iterations.

732. Information about stakeholders should be captured in a _____.
    a) Requirements document
    b) Project charter
    c) Stakeholder register
    d) Project plan

    Explanation: The stakeholder register is the place to store relevant information about stakeholders.

733. If the uncertainty about requirements and technology is low and detailed planning is desirable, which methodology is useful?
    a) Extreme
    b) Waterfall
    c) Scrum
    d) Lean

    Explanation: The waterfall methodology favors heavy up-front planning in order to remove project uncertainty.

734. At what stage in a project does an agile team produce a release plan?
    a) Envisioning
    b) Speculation

c) Exploration
d) A release plan is not produced in an agile project

Explanation: The available information is used to produce a release plan in the speculation stage.

735. Which value is NOT in the Agile Manifesto?
a) Comprehensive documentation
b) Process improvement
c) Contract negotiation
d) Processes and tools

Explanation: The Agile Manifesto values "Individuals and interactions over processes and tools," "Working software over comprehensive documentation," "Customer collaboration over contract negotiation," and "Responding to change over following a plan." While process improvement is important in Agile, it is not explicitly stated as one of the four core values of the Agile Manifesto.

736. A definition of done clarifies when a particular work item can be marked as "done". When does the definition of done need to be defined?
a) At the end of the project
b) Just prior to the testing phase of the project
c) At the beginning of the project
d) In the middle of the project

Explanation: The definition of done should be defined at the beginning of the project to ensure clarity and alignment among team members regarding the completion criteria for each work item. By establishing this definition early on, the team can maintain consistency and quality throughout the project, enabling smoother workflow and facilitating effective decision-making regarding the status of work items. Therefore, option (c) is the correct choice.

737. A team agrees that all project-related emails will be cc'd to the entire development team. This is an example of:
a) Team building
b) Organizational mandate
c) Compliance requirement
d) Ground rules

Explanation: The agreement to cc all project-related emails to the entire development team represents the establishment of ground rules within the team. Ground rules are guidelines or agreements that team members adhere to in order to foster effective communication, collaboration, and productivity. In this case, the agreement helps ensure transparency and inclusivity in communication, aligning with the team's collaborative principles and facilitating

the sharing of relevant information among team members. Therefore, option (d) is the correct choice.

738. The role of the product owner in an agile project is to _____.
   a) Run the business
   b) Choose the stakeholders to participate in the project
   c) Act as a single point of contact from the stakeholders to the agile project team
   d) Finance the agile project

   Explanation: The role of the product owner is to act as a single point of contact for the agile project team right from the stakeholders.

739. Extreme programming is a system of practices that lets you create high quality products. Which of the following is NOT a practice of XP?
   a) Planning game
   b) Planning Poker
   c) Refactoring
   d) Small releases

   Explanation: Planning Poker is not a practice of XP. It is an estimation techniques.

740. "See the whole" is a lean principle. The purpose of seeing is _____ .
   a) Focus on the big picture and optimize globally
   b) Involve the whole team in every problem
   c) Ensure that the team is continuously learning and improving
   d) Empower the team

   Explanation: Lean recommends focusing on the big picture and look for global optimization rather than local optimization

741. Creating focus while working is an important productivity enabler in an agile workspace. Reducing or eliminating distractions is an example of _____.
   a) The cone of silence
   b) Removal of problems
   c) Information Refrigerator
   d) Closed work area

   Explanation: In an agile workspace, reducing or eliminating distractions is often referred to as "the cone of silence." This concept emphasizes creating a focused environment where team members can concentrate on their work without interruptions. By minimizing distractions, such as noise, unnecessary meetings, or irrelevant communications, teams can enhance productivity and maintain their momentum towards achieving project goals. Therefore, option (a) is the correct choice.

742. Which of the following is NOT a practice of agile management?
   a) Empowerment
   b) Self-organizing teams
   c) Command and control
   d) Scrum

   Explanation: Command and control is a traditional principle where the decisions are made by individuals higher up in the organizational chart.

743. A spike is _____.
   a) A problem in the way of completing work
   b) An experiment designed to help the team decide the next step
   c) A team member who determines the course of action
   d) Growth period in the project

   Explanation: A spike is an experiment that is designed to help the team determine the next step and is a throwaway work.

744. Who is ultimately responsible for the team regularly refining the product backlog and has a clear set of requirements to work according to?
   a) The product owner
   b) The scrum master
   c) The development team
   d) The project manager

   Explanation: The product owner is responsible for making sure the backlog is in good shape and that backlog refinement is taking place regularly.

745. While planning a product release spanning across several iterations, which of the following should be a logical first step?
   a) Decompose the epics into smaller, more granular stories
   b) Define the conditions of satisfaction or goals of the release
   c) Estimate the stories
   d) Prioritize the targeted stories

   Explanation: In agile methodology, defining the conditions of satisfaction or goals of the release is a fundamental initial step during product release planning. This process involves clarifying the overarching objectives and expectations for the release, providing a clear direction for the subsequent activities such as decomposing epics, estimating stories, and prioritizing tasks. By establishing the release goals upfront, teams can ensure alignment with stakeholders and focus their efforts on delivering value in line with the project vision. Therefore, option (b) is the correct choice.

746. A team is working on a product on behalf of a supplier organization, which is in turn working on a contract with a customer. The team should share the burndown charts and other progress indicators with _____.

a) **All the relevant stakeholders**
b) The team, team facilitator, and product owner
c) The supplier organization
d) The customer organization discreetly

Explanation: The team should share burndown charts and progress indicators with all relevant stakeholders involved in the project. This ensures transparency and keeps everyone informed about the project's status, aligning expectations and facilitating collaboration among all parties. Option (a) emphasizes the importance of inclusive communication and transparency in agile project management practices.

747. A theme is a group of _____.
a) **Related stories**
b) Iterations
c) Releases
d) Features

Explanation: A theme in agile project management refers to a collection of related stories that collectively address a specific aspect or goal of the project. It provides a higher-level organizational structure for grouping user stories based on common objectives or functionality, helping teams prioritize and plan their work more effectively.

748. While agile is a good way of working for most teams, there are some formidable challenges in practicing agile. The most important organizational challenge facing an agile team is:
a) Complex technology
b) Lack of experienced team members
c) **Geographically dispersed teams**
d) Presence of an influential sponsor

Explanation: Geographically dispersed teams make life difficult for an agile team.

749. A charter gives the project manager authority to begin a project. What types of projects should begin with a charter?
a) Scrum projects
b) Waterfall projects
c) Agile projects
d) **All types of projects**

Explanation: A project charter is a fundamental document that formally authorizes the initiation of a project, irrespective of the project management methodology used. It outlines the project's purpose, objectives, scope,

stakeholders, and other key details essential for successful project initiation and execution.

750. After 3 iterations, an agile team found that the earned value was $1500 while the planned value was $1200. What is the cost variance?
   a) -$300
   b) Can't be said
   c) $300
   d) $0

   Explanation: To calculate the cost variance (CV), you need the actual cost (AC) of the work performed. The formula for cost variance is CV = EV - AC, where EV is the earned value and AC is the actual cost. Since the actual cost is not provided in the question, it is not possible to determine the cost variance.

751. A team performing the retrospective is trying to determine the root causes of a failed iteration. Which of the following techniques will be useful to them?
   a) Affinity diagrams
   b) Five whys
   c) Burn rate chart
   d) Skeleton diagram

   Explanation: The five whys technique and the cause and effect diagram are useful in root cause analysis.

752. In which year was the Agile Manifesto developed?
   a) 1989
   b) 2004
   c) 2001
   d) 2006

   Explanation: The Agile Manifesto was developed in 2001.

753. Story point, as a size measure, relies on the principle of relative sizing. This means that _____.
   a) Similar stories are grouped together
   b) A precise estimate is arrived at for each story
   c) An estimate is given only after performing a complete analysis of user stories
   d) One user story size is ranked in relation to another user story

   Explanation: In relative sizing, the size of a user story is estimated in relation to another user story.

754. What term or phrase means a team should make decisions as late as possible in order to preserve all options for as long as possible?
   a) Negotiation
   b) Backlog

c) Last responsible moment
d) Refactoring

Explanation: Last responsible moment means that a team should make decisions as late as possible in order to preserve all options for as long as possible.

755. During which meeting is the team likely to revisit its working agreement?
a) The review meeting
b) The retrospection meeting
c) The sprint planning meeting
d) The daily Meeting

Explanation: The team is likely to revisit its working agreement during the retrospection meeting.

756. One of the lean principles is to decide as late as possible. This means _____.
a) Keep options open for better ability to respond to change
b) Deliver functionality as quickly as possible
c) Empower the team to make decisions
d) Refactor the code

Explanation: Deciding as late as possible helps keep options open, which improves the ability to respond to change.

757. Swarming is the act of _____.
a) Deploying the entire team on a single user story
b) Buzzing around the solution
c) Grouping user stories together
d) Reacting to change

Explanation: Swarming refers to placing the entire team on a single user story.

758. Test-driven development relies on having an executable test before any code is written. What kind of testing is TDD most suitable for?
a) Integration testing
b) Regression testing
c) Unit testing
d) Ad-hoc testing

Explanation: Test-driven development (TDD) emphasizes writing tests before writing code, which promotes a focus on small units of code functionality. As such, TDD is most suitable for unit testing, where individual units or components of code are tested in isolation to ensure they function as intended. This approach helps enforce code modularity and improves code quality by catching defects early in the development process.

759. A team member estimated to complete a user story in four days with no interruptions. The team member is _____.

a) Timeboxing the user story
b) Giving an ideal time
c) Not following agile principles
d) Decomposing

Explanation: Ideal time is an estimate of time without interruptions or distractions.

760. Which of these do you need to implement agile projects?

a) Planning
b) A large team
c) Hierarchical managers and team
d) Cross-functional teams

Explanation: Planning is required when using the agile methodology. There is no excuse for chaos.

761. Which of the following is represented in the Y-axis of a burnup chart?

a) Time
b) Completed work
c) Resources
d) Quality

Explanation: The burnup chart shows story points on the Y-axis, which refers to completed work.

762. The main advantage of continuous integration is that:

a) Feedback is received quickly and code is integrated at all time
b) Knowledge is spread across the team
c) Good quality code is written
d) Deliveries are continuous

Explanation: Feedback is received quickly and code is integrated at all times. Continuous integration ensures that changes made by developers are integrated into the main codebase frequently, allowing for rapid feedback on the impact of these changes and enabling early detection of integration issues. This process helps maintain code stability and allows for faster identification and resolution of defects.

763. Which of the following estimation scales is most appropriate for a team performing story pointing?

a) 1, 2, 3, 5, 8, 13, 21, 34, 55, 89
b) 1, 3, 5,7, 9, 11, 13, 15
c) 2, 4, 6, 8, 10, 12
d) 1, 10, 100, 1000, 10000

Explanation: The Fibonacci sequence is frequently used as an estimation scale by agile teams.

764. Which of the following is an important prerequisite for continuous integration?
    a) A mature team
    b) Co-located teams
    c) A fast build process
    d) Error free code

    Explanation: A fast build process. Continuous integration relies on the ability to quickly build and test the codebase after every change. A fast build process ensures that developers receive rapid feedback on the integration status of their changes, enabling them to address any issues promptly. This helps maintain the momentum of development and ensures that the codebase remains stable.

765. Traditional managers view the plan as the goal, whereas agile leaders view _____ as the goal.
    a) Release
    b) Customer value
    c) Common sense
    d) Backlog

    Explanation: Agile leaders view customer value as the goal.

766. An information radiator is an artifact that radiates information and keeps everybody well-informed. The information radiator should do the following, except _____.
    a) Be kept in a closed space
    b) Be updated regularly
    c) Posted publicly
    d) Show important information

    Explanation: The information radiator should be visible to everyone and must be in the open.

767. A sign of a bad agile team space is:
    a) No partitions
    b) Lack of conversation
    c) Healthy food and drink in the pantry
    d) Private spaces for private work

    Explanation: In agile methodology, effective communication and collaboration are key. A lack of conversation within the team space can hinder collaboration and impede the agile process.

768. A team is making use of a flow-based agile practice that takes up a certain piece of work at a time and completes it - no matter how long it takes. Such a team is most likely to be:
   a) Cross functional team
   b) Component team
   c) Feature team
   d) Self-empowered team

   Explanation: A feature team makes use of a flow-based agile practice that takes up a certain piece of work at a time and completes it - no matter how long it takes.

769. One of the recommended XP practices is continuous refactoring. The purpose of refactoring is to _____.
   a) Create multiple solution factors
   b) Redesign the process
   c) Determine the root cause
   d) Make the code more maintainable

   Explanation: The purpose of refactoring is to reorganize the code and make it more maintainable.

770. You constructed a building valued at $750,000, which cost $500,000 to build. The net present value is _____.
   a) $1,250,000
   b) $0
   c) $250,000
   d) $750,000

   Explanation: Net present value (NPV) is calculated by subtracting the project's total costs from its total revenue. In this case, the building's value ($750,000) minus its cost to build ($500,000) equals a net present value of $250,000. This represents the overall profitability or value generated by the project.

771. Which one of the following options is NOT a primary role in scrum?
   a) Scrum master
   b) Project manager
   c) Product owner
   d) Team

   Explanation: The three primary roles of scrum are product owner, team, and scrum master.

772. What is the primary objective of amplify learning?
   a) Ensuring that the team is continuously learning and improving
   b) Creating a unified solution
   c) Aligning the code conceptually with the solution

d) Empowering the team

Explanation: Amplify learning makes sure that the team is continuously learning and improving.

773. What is the main objective of lean?
   a) See the whole
   b) Empower the team
   c) Amplify learning
   d) Eliminate waste

   Explanation: Lean methodology focuses on identifying and eliminating waste in processes to improve efficiency and value delivery. By reducing waste, organizations can optimize resources, enhance productivity, and streamline workflows, ultimately leading to better outcomes and customer satisfaction.

774. The discount rate at which the present value of benefits equals the present value of costs is known as the _____.
   a) NPV
   b) BCR
   c) IRR
   d) ROI

   Explanation: The IRR is the internal rate of return. It is the rate of discounting at which the present value of benefits is equal to the present value of costs.

775. An agile team is discussing the pros and cons of a system change during a retrospective meeting. To help them with this process, the coach should ask them to use _____.
   a) Cause and effect diagram
   b) Force field analysis
   c) Brainstorming
   d) Pareto diagram

   Explanation: The force field diagram is used to understand the forces for and against a system change.

776. Ensuring quality of a product also involves ensuring that the product does what it was intended to do. This is known as _____.
   a) Quality control
   b) Testing
   c) Verification
   d) Validation

   Explanation: Validation makes sure that the product does what it was intended to do.

777. Agile project management does NOT focus on _____.

a) Prioritizing the work so that the most important work gets done
b) Providing product direction
c) Setting up the team and ensuring they have the right environment to work in
d) Delivering value to the customer

Explanation: Agile project management uses a different approach that focuses more on enablement.

778. Wideband Delphi estimating has all of the following attributes, except _____.
a) Using large numbers of experts
b) Creating group thinking
c) Preparing estimates away from the group
d) Experts discussing ranges of estimates

Explanation: Wideband Delphi estimating is used to reduce group thinking by anonymously obtaining estimates.

779. A team found that they will not be able to complete all the stories they committed to in one of the iterations. What is the best thing to do?
a) Let the product owner know and request for reordering of the items planned
b) Work overtime to complete the goals
c) Extend the iteration to complete the required items
d) Accept fewer stories in the next iteration

Explanation: It is sensible to reassess priorities and complete at least the most important items.

780. The risk burndown chart is _____.
a) The number of reduced risks on a project
b) Where the risk to project success that is associated with each feature is displayed
c) Shows the number of resources that are used in the Agile project
d) Amount of mitigation needed to handle risks

Explanation: The risk burndown chart is where the risk to project success that is associated with each feature is displayed.

781. A one-year project executed using agile methods is in the last iteration when a major change was requested by the customer. What should the team do?
a) Add it to the backlog and figure out a way to accommodate the change
b) Include the request in the backlog for the next release
c) Request the customer to withdraw the change until the current project is completed
d) Add few sprints to the project to include the change

Explanation: Add it to the backlog and figure out a way to accommodate the change. In Agile, accommodating changes, even late in the project, is part of the iterative and flexible nature of the methodology. The team should add the requested change to the backlog, prioritize it accordingly, and find a way to incorporate it into the remaining iterations or releases while ensuring that the project objectives are still met.

782. A scrum-of-scrum is a forum for integrating the work of multiple scrum teams working on the same product. Which of the following is NOT included in a scrum-of-scrums?
    a) Progress discussion
    b) Coordination of multiple teams
    c) Attendance by the scrum master
    d) Product demos and feedback by stakeholders

    Explanation: Scrum-of-scrums includes discussions of progress and coordination of multiple teams and are attended by the scrum master. This technique is used to scale very large scrum projects.

783. The customer is the most important part of agile management. What is the most important role of the customer in an agile team?
    a) Write the requirements for the product
    b) Manage changes in the product backlog
    c) Provide timely feedback about the product
    d) Test the product to make sure it meets requirements

    Explanation: Customers must be closely involved in agile development because their feedback is useful in improving product functionality.

784. A sprint is a timebox. It is also called an iteration. Which of the following is the most important factor in determining the appropriate sprint duration for a team?
    a) Industry standard of sprint durations across different organizations
    b) Ease of getting feedback from the customer
    c) The average time taken to complete the largest of product features
    d) The velocity of the team

    Explanation: Ease of getting feedback from the stakeholders is an important factor in determining the sprint duration.

785. An organization is wondering whether there is any point in estimating the features of a project that has a fixed deadline. What is a prudent approach in this situation?
    a) Use the estimates that were provided during the bidding process as a guideline for making the project plan
    b) Renegotiate the deadline with the customer on the basis of detailed estimation
    c) Use relative weights technique to arrive at a reasonable plan

d) **Team needs to estimate the work to determine the feasibility, negotiate features, and decide the timeline.**

Explanation: Team needs to estimate the work to determine the feasibility, negotiate features, and decide the timeline. In Agile projects, estimation helps in understanding the scope of work, identifying potential risks, and negotiating with stakeholders. Even with a fixed deadline, estimation guides the team in prioritizing features, adjusting scope, and ensuring the project's overall feasibility within the given constraints.

786. A team completed 8 out of 10 stories committed during an iteration. What should be done about the remaining 2 stories?
   a) It should be moved to the next iteration
   b) **It should be moved back to the product backlog**
   c) The team should extend the iteration to complete them
   d) The team should plan a special short sprint to complete the two items

   Explanation: It should be moved back to the product backlog. In Agile methodology, unfinished work from an iteration is returned to the product backlog for reprioritization. This allows the team to reassess the remaining work and adjust priorities based on the current state of the project and stakeholder feedback.

787. A product owner is an important stakeholder in an agile team. Which of the common agile events or meetings is optional for the product owner to attend?
   a) Iteration planning meeting
   b) **Daily stand-up meeting**
   c) Iteration retrospective
   d) Release planning meeting

   Explanation: The daily stand-up meeting is optional for the product owner. However, if the team requests, the product owner should attend.

788. An agile project was completing the third (out of ten planned) iteration. After reviewing the progress during an iteration demo, the customer requested that they would like to start using the product as it is. What should the team do?
   a) **Make the product available to the customer with appropriate quality checks**
   b) Tell the customer that the product will be released only after ten iterations as planned
   c) Ask the customer to make a formal request and put it in front of the senior management
   d) Release the product but tell the customer that no liabilities will be accepted if it does not perform as expected

   Explanation: Every iteration should have a releasable product.

789. An agile project was completing the third (out of ten planned) iterations. After reviewing the progress during an iteration demo, the customer requested that they would like to start using the product as it is. Who is responsible for making a decision about this request?
   a) Team facilitator
   b) Team
   c) Product owner
   d) Senior management

   Explanation: In Agile methodology, the product owner is responsible for representing the customer or stakeholder interests and making decisions regarding the product. They prioritize the product backlog, determine when increments of the product are released, and ensure that the product meets the needs and expectations of the stakeholders. Therefore, the product owner is the appropriate individual to decide whether to release the product based on the customer's request.

790. A timeboxed period when the team prepares for a project and makes high-level technical choices and release plans is called:
   a) Hardening iteration
   b) Iteration zero
   c) Beta project release
   d) Sprint planning

   Explanation: Iteration zero is used to prepare for a long project.

791. Which of the following prioritization schemes explicitly specifies which work items will not be worked on?
   a) MoSCoW
   b) Kano model
   c) Relative weights
   d) Value-based prioritization

   Explanation: W in MoSCoW stands for Won't. It highlights the features that will not be worked on in a given timebox.

792. In a burndown bar chart, the movement in the bottom of the bars indicate:
   a) Work added or removed to a project
   b) Work added or removed to an iteration
   c) Work remaining in the project
   d) Work completed in an iteration

   Explanation: Movement in the bottom of the burndown bar chart indicates work that is added to or removed from a project.

793. A team found that the CPI was 0.9 and the SPI was 1.1 after 4 iterations. This means that the project is:

a) By and large on track
b) Ahead of schedule and under budget
c) Ahead of schedule and over budget
d) Behind schedule and over budget

Explanation: CPI < 1 means that project is over budget; SPI > 1 means that the project is ahead of schedule.

794. Managers, who participate with the team, are hands-on in projects, and get involved with the team every day, are _____.
a) Adaptive leaders
b) Servant leaders
c) Directive leaders
d) Active leaders

Explanation: A servant leader is someone, who participates with the team, is hands-on in projects, and gets involved with the team daily.

795. A user story is often written on a physical card called a story card. The main advantage of the story card is that:
a) It limits the space available and forces the author to be brief and to the point
b) It prevents the possibility of changing the story at a later date
c) It allows to discard stories that don't add value
d) It prevents an electronic audit trail that may be problematic in a compliance-heavy environment

Explanation: Story cards are physical and have limited space. Moreover, the act of writing by hand forces one to reflect deeply before writing a story.

796. A good user story follows the INVEST model and the 3 Cs. Which of the following is NOT included in a user story?
a) Clear and demonstrated value
b) Technical specifications
c) Scenarios
d) Customer value documentation

Explanation: Users think in terms of value, scenarios, features, and stories. Technical specifications are used by technical experts and are not in user stories.

797. The return on investment indicates the average savings or profits realized by making an investment. The main drawback of using RoI as a measure for making decisions is that:
a) It does not consider the time value of money
b) It does not specify who exactly gets the value
c) It is difficult to calculate
d) It does not say what happens after the savings are realized

Explanation: The return on investment (ROI) is the return made by investing in a project. The primary drawback of ROI is that it does not consider the time value of money.

798. Brainstorming is used to generate ideas and get inputs from many people in a short amount of time. Which of the following is NOT a good practice while brainstorming?
   a) Critiquing ideas when they are being collected
   b) Writing down the ideas immediately
   c) Allowing people to provide creative ideas
   d) Involving the entire team

   Explanation: Ideas should only be critiqued after they are collected and not during the brainstorming process.

799. Making decisions in complex situations is not easy. On agile projects, the decisions that should be made on a project should be based on _____.
   a) Providing many functions to the customers
   b) Giving the customer all the features that they requested
   c) Net present value
   d) Providing value to the customer

   Explanation: Decisions for a project should be made based on providing value to the customer.

800. After an iteration began, the product owner wanted to add an important item to the iteration backlog. How should the team respond?
   a) Accept the new item as agile believes in welcoming change
   b) Ask the product owner to wait until the next iteration
   c) Recommend cancellation of the current iteration
   d) Accept the new item as a stretch goal without making any commitment

   Explanation: Iteration backlog should be stable to let the team focus and complete it. The PO can always make changes to upcoming iterations by reordering the product backlog.

801. Items in product backlog tend to be:
   a) Smaller than the items in the sprint backlog
   b) Larger than the items in the sprint backlog
   c) Usually smaller than sprint backlog items, but depends on the context
   d) The same size as the items in the sprint backlog

   Explanation: Items in product backlog are usually larger than the items in the sprint backlog

802. Which of the following cannot be inferred from an agile story map?
   a) Order in which the stories are required in the actual user workflow

b) Detailed breakdown of the tasks in each story
c) Minimally marketable features and priorities
d) Suggested order of execution to complete a usable feature

Explanation: Detailed breakdown of the tasks in each story. Agile story maps typically focus on visualizing the flow of user stories to represent the product backlog in a spatial manner. While they show the order in which stories are required in the user workflow, minimally marketable features, and priorities, they do not provide a detailed breakdown of tasks within each story. Task breakdown usually occurs during sprint planning or as part of the team's daily activities, not explicitly within the story map itself.

803. After the process of refactoring, what step should be performed next?
a) Update documentation
b) Perform root cause analysis
c) Complete regression test
d) Fire the programmer who wrote the original code

Explanation: Refactoring is implemented to reorganize code that has already been written. After refactoring, a complete regression test is required to ensure that the updated code behaves as it should.

804. Scrum is among the most popular agile methods. What is the ideal team size of a scrum team?
a) Over 50 people
b) A single person
c) About 15 team members including product owner and scrum master
d) About seven people

Explanation: A Scrum team ideally consists of around three to nine members, as this size promotes effective communication, collaboration, and decision-making within the team.

805. Earned value management is a well known project management technique. Earned value _____.
a) Can only be used for traditional projects
b) Should never be used for agile projects
c) Can be calculated for agile projects
d) Is too difficult to calculate to be useful

Explanation: Earned value can be calculated for agile projects with some slight modifications.

806. A Kanban board can be used to show all of the following except _____.
a) Work in progress
b) Schedule and cost variance
c) High priority work items

d) Who is working on a work item

Explanation: A Kanban board is a WIP view that is posted publicly and shows important task information.

807. The product roadmap should be used to _____.
   a) Communicate the plan
   b) Articulate the product design
   c) Determine the path to meet the objectives
   d) Document the vision statement

   Explanation: The product roadmap is used to communicate the plan on a moderate level of detail. It communicates more than the vision statement and less than a release plan.

808. The average velocity of a team is 20, maximum is 25, and minimum is 12. If the release backlog is 125 points, how many iterations are required to complete the release?
   a) 7
   b) 5
   c) 6
   d) 11

   Explanation: The team's average velocity is used as a baseline for estimating the number of iterations needed to complete the release backlog. With an average velocity of 20 points per iteration and a release backlog of 125 points, dividing the backlog size by the average velocity gives 6.25 iterations, rounded up to 7. Therefore, the correct answer is a) 7 iterations.

809. One of the work items in an agile project is meant to prove compliance with a regulatory requirement. In terms of the Kano model, this requirement will be classified as:
   a) Exciter
   b) Linear
   c) Threshold
   d) Utility

   Explanation: Compliance on a project is not optional, but is mandatory to complete the project.

810. War rooms are co-located team spaces where teams get together to work on a project. The primary benefit of a war room setting is that it:
   a) Reduces the stress level of the team
   b) Promotes debates about features and functions
   c) Makes it easier to manage the team
   d) Promotes osmotic communication

Explanation: Osmotic communication is when information is picked up by merely being in a certain environment. The war room is a good place for this to happen.

811. An information radiator should do have all of the following characteristics except it's _____.
    a) **Plenty in number**
    b) Concise
    c) Displayed in a public space
    d) Transient - removed when irrelevant

    Explanation: Information radiators are essential tools in agile environments, providing visibility into project progress and status. While they should be concise, publicly displayed, and transient (removed when irrelevant), having "plenty in number" contradicts their purpose, as it could lead to confusion and information overload. Therefore, the correct answer is (a) Plenty in number.

812. An important metric used in Lean and Kanban teams is the cycle time. The cycle time is _____.
    a) The number of days to complete a project phase
    b) The average time between two agile deliveries
    c) **The amount of time for a feature to move from start to completion**
    d) The number of iterations to complete a project

    Explanation: Cycle time is a key metric in Lean and Kanban methodologies that measures the total time taken from when work on a feature begins to when it is completed. This metric helps teams understand their process efficiency, identify bottlenecks, and improve their workflow to deliver features faster.

813. Which of the following is not valued highly in the Agile Manifesto?
    a) **Processes and tools**
    b) Working software
    c) Customer collaboration
    d) Responding to change

    Explanation: Individuals and interactions, not processes and tools are part of the Agile Manifesto.

814. If the burndown chart indicates that the team is likely to finish the iteration goals earlier than planned, what should be done?
    a) The team should slow down till the trend is corrected
    b) The team should perform extra checks to improve product quality
    c) The team should plan some team building activities to effectively use the time
    d) **The team may either finish the iteration early or ask the product owner to add a few items to the iteration**

Explanation: If an iteration finishes early, it is acceptable for the team to pull in new items or declare an early completion.

815. The X-axis of a burnup chart represents which of the following?
    a) Time
    b) Functionality
    c) Resources
    d) Quality

    Explanation: In a burnup chart, the X-axis typically represents time, showing the progression of the project over a period. This allows stakeholders to track the project's progress and see how it evolves over time in terms of completed functionality or scope. Therefore, option (a) accurately identifies time as the representation on the X-axis of a burnup chart.

816. A product owner has a habit of detailing every story in the product backlog as soon as the details become available. The agile coach does not agree with this. The primary reason for this could be that the backlog should be:
    a) Emergent
    b) Detailed appropriately
    c) Prioritized
    d) Estimable

    Explanation: Details should be added to the backlog only when the items are near to being picked up for working

817. The project charter is _____.
    a) Still required in agile
    b) Not required in agile
    c) Written by the product owner
    d) A detailed project document

    Explanation: The project charter is still required in agile. The charter is needed to document and get consensus on the project.

818. Why is daily scrum held at the same time and place?
    a) Rooms are not normally available, hence they are booked in advance.
    b) The team demands it
    c) Consistency reduces complexity and overhead
    d) The product owner demands it

    Explanation: The daily stand-up meeting is held at the same place and time to reduce complexity.

819. Which of the following is not an appropriate discussion item at a daily stand-up meeting?
    a) What has been done since the previous meeting

b) What would be done before the next meeting
c) An approach to improve product performance
d) Blocks and challenges

Explanation: The approach to improve performance is a discussion that should take place outside of the stand-up.

820. In an agile team, the developer should focus on _____ and NOT _____.
a) Working software; comprehensive documentation
b) Architecture; lines of code
c) Customer; processes
d) Comprehensive documentation; working software

Explanation: The developer should focus on working software and not comprehensive documentation.

821. During an agile retrospective, a suggestion came up that would improve the productivity of the team. However, this requires implementation of a tool that requires a significant amount of effort. What should be done about this?
a) Document it, but don't work on it immediately
b) Drop the suggestion and work on simpler suggestions first
c) Discuss with the product owner if the work should be prioritized
d) Ask management to provide funding for the tool

Explanation: The work needs to be discussed with the product owner and prioritized.

822. The project team released a software to the customers. The escaped defects are _____.
a) Ignored by the core team
b) Errors that are discovered by the customer
c) Discovered by the developers
d) Not tracked

Explanation: Escaped defects are errors that are discovered by the customer and missed during testing and quality control.

823. Cumulative flow diagrams are designed to show _____.
a) The build-up toward completion as the project progresses
b) The sum of all the resources on a project
c) When to add more people to the project
d) Process flow of the system

Explanation: Cumulative flow diagrams show the build-up toward completion of the project.

824. On an agile team, each team member is _____.

a) **A generalizing specialist**
b) Responsible for their own part
c) A controller of their domain
d) A specialist

Explanation: In an agile team, each member is a "generalizing specialist," meaning they have a core specialization but also possess general skills in other areas. This promotes flexibility and effective collaboration, allowing the team to adapt quickly to changing requirements.

825. In a value stream mapping exercise, a step is marked either as a value adding step or waste based on:
    a) Whether the step is necessary
    b) Whether the step is optimized to the extent possible
    c) **Whether the step adds to customer value**
    d) Whether the step is on the critical path

    Explanation: Lean manufacturing uses the concept of Value Stream Mapping to eliminate waste.

826. The main job of the agile coach of a team should be _____.
    a) To bring about appropriate level of Agile tooling
    b) **To help the team follow the Agile processes**
    c) To maximize customer value
    d) To maximize the work not done

    Explanation: The coach is one who keeps the team focused on learning and the agile processes.

827. The product owner is not comfortable writing a particular user story as the content is too technical. In such a situation:
    a) A technical consultant may be hired to write specific technical stories
    b) The product owner may test the story but need not write it
    c) The product owner may write it and get it reviewed by the technical team
    d) **The technical team may write the story and the product owner can prioritize it**

    Explanation: It is not necessary that the PO should write all the stories but they must accept ownership for vetting and prioritizing them.

828. An agile team does not have enough bandwidth to serve the requirements for a large product and is being expanded. It was decided to form multiple agile teams to serve the same product. What would be the best way to divide responsibility among the teams
    a) Responsibility for each technical area (for example, databases, UI, reporting) with a specific team

b) Responsibility according to the team roles (for example, developers, testers, document writers) with different teams
c) Responsibility for a specific feature or module with each team
d) Co-located team members who preferably are part of the same team

Explanation: The new team members should be chosen based on interviews and selections conducted by existing team members.

829. A project manager is tasked with overseeing a software development project. Upon encountering a key stakeholder from the company in the hallway, the project manager took advantage of the opportunity to seek their feedback. The key stakeholder, impressed with the project's progress, urgently asked the project manager to add a new user story for marketing the product. How should the project manager respond to this request?
a) Refuse to create the user story since it's the product owner's responsibility to maintain and refine the product backlog
b) Add the user story to the product backlog and let the product owner decide its priority
c) Welcome the stakeholder's request and add the user story to the sprint backlog
d) Add the user story to the product backlog and schedule it for the next sprint to protect the team from disruptions

Explanation: The project manager should add the user story to the product backlog and let the product owner decide its priority. The product owner is responsible for maintaining and refining the product backlog, and they are the ones who should decide the priority of the user story. The project manager should not make decisions that are outside their scope of responsibility.

830. A project manager is leading an agile project for the first time. Considering their choice of the Scrum framework, which of the following describes the project team?
a) All of the team members are dependent on the Scrum master
b) All of the team members are I-shaped
c) All of the team members have a technical background
d) The team size ranges from three to nine members

Explanation: The right answer is D because the Scrum framework recommends a team size of three to nine members.

831. A senior business analyst has been appointed to lead the project selection committee within their organization. After shortlisting a list of projects, the committee will decide on the project to pursue by evaluating each project's benefit-to-cost ratio. Which project should the committee opt for?
a) Project A has a BCR of 6:3
b) Project B has a BCR of 6:2

c) **Project C has a BCR of 8:2**
d) Project D has a BCR of 3:1

Explanation: Project C has the highest BCR of 8:2, indicating that it will generate the most benefits for the least cost.

832. A project manager is overseeing an Agile web application project. In the first sprint, the team completed tasks with story points of 3, 5, 8, and 2, along with completing half of a 13-story points task. What is the team's velocity?
a) 24.5
b) 18
c) 31
d) 13

Explanation: Velocity is the total number of story points completed in a sprint. The team completed 3+5+8+2+(13/2) = 18 story points in the first sprint.

833. Faced with a quality concern, a project manager decides to utilize the _____, a fundamental quality management tool that applies the 80/20 Rule to pinpoint high-priority defects.
a) Fishbone Diagram
b) Pareto Chart
c) PERT Chart
d) Flowchart

Explanation: The Pareto Chart is a basic quality management tool that uses the 80/20 Rule to identify top-priority defects. It helps to identify the most frequent defects or issues, which can then be addressed to improve overall quality.

834. A project manager is submitting several potential projects to management for selection and approval. Given the organization's existing financial constraints, the project manager has been informed that each project's investment must not exceed $350,000, with a permissible variation of 5% on either side. The 5% leeway is a component of:
a) Threats
b) Risks
c) Enterprise Environmental Factors (EEFs)
d) Organizational Process Assets (OPAs)

Explanation: The 5% leeway is a factor that is external to the organization and is beyond the control of the project manager. Therefore, it is considered an Enterprise Environmental Factor (EEF).

835. A product owner initiated the day with a sprint planning meeting to discuss and define the goal and backlog for the forthcoming sprint. Which of the following accurately describes the sprint planning meeting?
   a) A discussion between the cross-functional team members only
   b) A discussion between the product owner and the scrum master
   c) A discussion between the scrum master and the cross-functional team members
   d) A discussion between the product owner, the scrum master, and the cross-functional team

   Explanation: The sprint planning meeting is a discussion between the product owner, the scrum master, and the cross-functional team to define the goal and backlog of the upcoming sprint.

836. A project manager is overseeing a hotel construction project in a foreign country where corruption is pervasive. Encountering difficulties in obtaining necessary construction permits from local authorities, a team member suggested offering a bribe to the local officials to expedite the process and ensure smooth project execution. How should the project manager react?
   a) Refuse to give a bribe
   b) Give the smallest amount of money possible
   c) Negotiate a non-monetary bribe
   d) Ask the team member to give the bribe on their behalf

   Explanation: Giving a bribe is illegal and unethical. Project managers should always act with integrity and follow the laws and regulations of the country they are working in. Refusing to give a bribe is the right thing to do in this situation.

837. A project manager was assigned to a park redevelopment project, encompassing the cleaning and enhancement of the park for the benefit of local families and residents. Due to the project's multiple phases and various tasks, it involves a lot of stakeholders. What should the project manager do?
   a) Disregard low-power stakeholders
   b) Only engage the high-interest stakeholders
   c) Analyze all stakeholders in order to prioritize engagement
   d) Report the issue to the project sponsor

   Explanation: The project manager should analyze all stakeholders in order to prioritize engagement because it is important to identify all stakeholders and their level of interest and power in the project. This will help the project manager prioritize engagement and ensure that all stakeholders are satisfied with the project outcome.

838. A product owner for a high-quality clothing project joins a meeting with the project team and other key stakeholders to assess a demonstration of a produced deliverable. This type of meeting is conducted in accordance with the _____ approach, which is iteration-based Agile.

a) At the beginning of every iteration
b) At the end of every iteration
c) At the end of the project
d) At the start of the project

Explanation: Since the project adopts an iteration-based Agile approach, this type of meeting is held at the end of every iteration to review the produced deliverables.

839. MomentumCo is a company operating in the Oil and Gas industry. They had signed a subcontracting agreement with GlobalTech Solutions to obtain experts, technical know-how, and mechanical spare parts. Since MomentumCo's projects involve constructing innovative high-tech machines, the specific details of the work are not clear. In this situation, what type of contract should they enter into with GlobalTech Solutions?

a) Fixed Price Incentive Fee (FPIF)
b) Cost Plus Award Fee (CPAF)
c) Time & Materials (T&M)
d) Firm Fixed Price (FFP)

Explanation: Since the exact description of the work involved is not clear, Time & Materials (T&M) contract should be signed with SPM Inc. This type of contract is used when the scope of work is not clear and the buyer wants to have flexibility in terms of the amount of work to be done and the resources required to complete the work.

840. Tasked with leading a significant project crucial to their organization, a project manager is under pressure to avoid any delays in completion. Upon obtaining the signed project charter, the project manager begins the process of identifying and categorizing stakeholders, employing various data representation tools. Which of the following tools can the project manager utilize? (Choose three)

a) Brainwriting
b) Stakeholder cube
c) Direction of influence
d) Salience model

Explanation: The project manager can use the stakeholder cube to categorize stakeholders based on their level of interest, power, and influence. The direction of influence tool can be used to identify the direction of influence between

stakeholders. The salience model can be used to prioritize stakeholders based on their power, urgency, and legitimacy.

841. A project manager decides to obtain his PMP certification within two months. He creates the following exam preparation plan: dedicating 4 weeks to studying, taking mock practice tests in weeks 5 and 6, and then taking the 7th week off before the exam in week 8. However, at the conclusion of week 5, the project manager gets into an accident, resulting in one week of hospitalization. To compensate for the lost time, the project manager decides to forgo his week off and instead focuses on mock tests during weeks 6 and 7, originally planned for weeks 5 and 6. This serves as an example of:
   a) Adaptability
   b) Risk mitigation
   c) Workaround
   d) Corrective action

   Explanation: This is an example of a workaround, which is a response to an unexpected event that is not part of the original plan. The project manager had to adjust her plan to make up for the lost time due to the accident, and the new plan involved working on mock tests during weeks 6 and 7 instead of weeks 5 and 6.

842. You are simultaneously overseeing six projects within the company. Two of these projects are of a similar type, while the remaining four are entirely different. You work as a _____.
   a) Portfolio manager
   b) Program manager
   c) Project manager
   d) Program coordinator

   Explanation: As a portfolio manager, you are responsible for managing multiple projects that are related to each other in some way. In this case, two of the projects are of a similar type, while the other four are entirely different. As a portfolio manager, you would oversee the projects and ensure that they align with the company's overall strategy and goals.

843. During the iteration review, the project team demonstrated new features to the Product Owner. The product owner maintained a relaxed and open posture, occasionally fiddling with his pen or coffee mug on the table without directly looking at the speaker. What types of communication is the product owner exhibiting? (Choose two)
   a) Paralingual communication
   b) Active listening

c) **Implicit message**
d) **Non-verbal communication**

Explanation: The product owner is using non-verbal communication and implicit messages. The relaxed and open posture with occasional fiddling with the pen or coffee mug without looking directly at the speaker indicates that the product owner is not fully engaged in the conversation but is still interested in the new features being demonstrated.

844. A project manager is implementing an internal design project and requires a driller with a rental cost of $25 per day, which she considers relatively high. Given the extended duration of use, a team member proposes purchasing a driller for $500. What is the minimum number of rental days required for it to be more cost-effective to buy a driller?
a) 2
b) 5
c) 10
d) **20**

Explanation: The minimum number of rental days is calculated by dividing the cost of the driller by the daily rental cost and rounding up to the nearest whole number. In this case, $500 divided by $25 per day is 20 days. Therefore, if the project manager needs the driller for more than 20 days, it is more advantageous to buy it.

845. An experienced project manager is assigned to lead a new hybrid development project. Thus, management enrolled the project manager in training to teach them how to combine agile principles with predictive techniques. Which aspect of the PMI talent triangle does this training target?
a) Business Acumen
b) Power Skills
c) **Working Methods**
d) Ways of thinking

Explanation: The training targets working methods because it aims to teach the project manager how to combine agile principles with predictive techniques, which are two different approaches to project development.

846. For forecasting purposes, the project manager can use the following methods to predict the project's future performance based on its current performance. (Choose three)
a) **Scenario building**
b) **Simulation**

c) Time series method
d) Variance analysis

Explanation: The project manager can use scenario building, simulation, and time series method to forecast the project's future performance based on its current performance.

847. An organization is overseeing multiple projects, each with its designated deadline. Resources need to be allocated concurrently to various projects. Concerned about potential delays in her project, a project manager consults with the functional manager to discuss additional resource allocation for her project. Which of the following is the most crucial skill that the project manager will need for this situation?
a) Planning
b) Negotiating
c) Facilitating
d) Documenting

Explanation: The project manager will need negotiating skills to convince the functional manager to allocate additional resources to her project.

848. An experienced project manager believes managers should embrace transparency in their projects because it has many advantages over its disadvantages. However, the project manager recognizes that, although it is beneficial to communicate many aspects with the project team, there are specific details that should not be disclosed to everyone. As a project manager, it is imperative to be transparent about:
a) Confidential information
b) Proprietary information
c) Unproven information and gossip
d) Decision-making processes

Explanation: As a project manager, it is important to be transparent about decision-making processes to ensure that the project team understands how decisions are made and can provide input when necessary. This helps to build trust and collaboration within the team, which can lead to better project outcomes.

849. A manufacturing company is implementing lean principles, conducting quarterly internal audits to ensure project adherence to these principles. In the most recent audit, auditors observed that the project manager is prioritizing fast delivery but deferring decisions until the last possible moment. Given this observation, is the company in alignment with lean principles?

a) **Yes, it is**
b) No, the project manager should both deliver and decide as late as possible
c) No, the project manager should both deliver and decide as fast as possible
d) No, the project manager should deliver as late as possible but decide as fast as possible

Explanation: The project manager is delivering as fast as possible, which is in line with the lean principle of delivering value quickly. However, the project manager is taking decisions as late as possible, which is also in line with the lean principle of delaying decisions until the last responsible moment to avoid waste. Therefore, the company is complying with lean principles.

850. A project manager is overseeing an international project with team members from diverse cultural backgrounds and nationalities. Despite the team members' fluent English proficiency, what actions should the project manager take?
a) Due to cultural differences, the project manager should create a separate code of conduct for each nationality.
b) The project manager may have to just accept that some team members may face some difficulties when working with colleagues of other nationalities.
c) **The project manager should take into consideration that spoken communication may give rise to certain misunderstandings that may not occur in written communications and that such misunderstandings are hard to identify.**
d) The project manager should keep in mind that certain groups won't mind late-night meetings and video conferences.

Explanation: The project manager should take into consideration that spoken communication may give rise to certain misunderstandings that may not occur in written communications and that such misunderstandings are hard to identify. This is because even though all team members speak English fluently, cultural differences can still lead to misunderstandings.

851. A project manager is overseeing an industrial project and requires a set of 1000 custom-made identical filters. Given their crucial significance, the project manager opts for a trustworthy and reputable supplier to manufacture these filters. Recognizing that a tested filter must be discarded as it cannot be reused, what course of action should the project manager take?
a) Skip inspection since he trusts that the supplier will deliver the filters according to his specifications
b) **Order more than 1000 filters to perform acceptance sampling on the batch surplus**
c) Require the seller to provide a conformance certificate of the filter's raw materials

d) Conduct a 100% inspection upon delivery, then order another batch of 1000 filters if he's satisfied with the quality

Explanation: Ordering more than 1000 filters to perform acceptance sampling on the batch surplus is the right answer because it allows the project manager to test a sample of the filters to ensure they meet the project's specifications. This is a common quality control technique that helps to ensure that the entire batch of filters is of acceptable quality.

852. While executing the project, the project manager discovered that tasks were not being carried out in the correct sequence and at the appropriate times, leading to rework and a bad in morale among the project team. Which project management tool should the project manager employ to handle this issue?
   a) Organization chart
   b) RACI matrix
   c) Communications management plan
   d) Work authorization system

   Explanation: The project manager should implement a work authorization system to ensure that tasks are performed in the right order at the right time. A work authorization system is a project management tool that ensures that work is done in the right sequence and at the right time. It is used to prevent rework and ensure that the project team is working efficiently.

853. A project manager has taken over the management of a climate-smart agriculture project, stepping in for the manager who resigned. In the initial stages of the execution phase, the project manager observes varying opinions among team members regarding project tasks, deliverables, and the overall complexity level. What steps should the project manager take in response to this situation?
   a) They should not intervene. Instead, they should allow the team enough time to get familiar with the project scope
   b) They should identify and assess potential risks caused by this divergence of opinion, then develop a plan to manage and respond to these risks
   c) **They should organize a meeting involving all team members to identify and resolve misunderstandings in order to avoid issues, disintegration, and rework.**
   d) They should talk to each team member separately to explain the project requirements

   Explanation: The project manager should organize a meeting involving all team members to identify and resolve misunderstandings in order to avoid issues, disintegration, and rework. This will help to ensure that everyone is on the same page and has a clear understanding of the project scope and deliverables.

854. A project manager at Data Innovate has recently completed her final project report and is preparing to meet with her manager. Over the past few weeks, she has conducted several lessons-learned sessions to gather feedback from project participants for the final report. What are the motivations behind the project manager organizing lessons learned sessions? (Choose three)

a) **Lessons learned databases are an important element of the organizational process assets**
b) Lessons learned meetings focus on identifying individuals accountable for failures and errors
c) **Lessons learned meetings include recommendations for future performance improvements**
d) **Phase-end lessons learned workshops represent a good team-building exercise for project members**

Explanation: Lessons learned sessions are held to identify what went well and what didn't go well in the project, and to document recommendations for future performance improvements. They also represent a good team-building exercise for project members.

855. After a long wait, a project manager receives a phone call from the project sponsor, confirming the allocation of the budget and signaling the commencement of the project. The sponsor also emphasizes the need for the project development approach to be defined promptly and insists on the kick-off meeting occurring within the next two weeks. In response, the project manager asserts that he must first obtain the project charter. What is the role of a project charter?

a) To document the project
b) **To formally authorize the project and document its initial requirements**
c) To be used as a reference during the project kick-off meeting
d) To set up the project code of conduct

Explanation: The project charter is a document that formally authorizes the project and documents its initial requirements. It is usually created by the project sponsor or initiator and provides the project manager with the authority to apply organizational resources to project activities. It also defines the project's high-level objectives, scope, and stakeholders, and outlines the project manager's authority level and responsibilities. Therefore, the project charter is essential for the project manager to start the project.

856. As a part of the project quality control process, the project manager at AYZ Electric Motorcycles chose to inspect only 10% of the produced motorbikes for environmental control. Which technique is the project manager employing?

a) Sample Selection
b) Control Charts
c) Statistical Sampling
d) Pareto Diagram

Explanation: The project manager is using statistical sampling, which involves selecting a sample of the population to test and make inferences about the entire population. In this case, the project manager is testing only 10% of the manufactured motorbikes for environmental control, which is a form of statistical sampling.

857. A local school plans to employ a screening system to choose suppliers of vegetables and fruits for their "NourishWell" organic meals initiative. Which of the following exemplifies a screening system?
a) You interview all the vendors and make your decision based on these interviews
b) You only consider vendors with more than $10,000 in revenue in the last financial year
c) You negotiate with all the vendors and engage the one who offers the best price
d) You engage the vendor who responds first to your announcement

Explanation: Lessons learned sessions are held to identify what went well and what didn't go well in the project, and to document recommendations for future performance improvements. They also represent a good team-building exercise for project members.

858. The business analyst for the project is conducting Earned Value Reporting. With a CPI of 0.9 and a budget at completion of $900, what is the estimated cost at completion for the project?
a) 819
b) 810
c) 900
d) 1000

Explanation: The project's estimated cost at completion is calculated by dividing the budget at completion by the CPI. In this case, the estimated cost at completion is $1000 (900/0.9).

859. A project manager is evaluating the potential of an industrial project. In order to ascertain whether the projected financial benefits surpass the current investment, the project manager employs the Net Present Value (NPV) as a valuable tool to gauge the project's profitability. For instance, if NPV > 0, it indicates:

a) The project will lose money
b) The project will break even
c) The project is profitable
d) We can't know until the ROI is calculated

Explanation: NPV > 0 means that the present value of the anticipated cash inflows is greater than the present value of the anticipated cash outflows, which indicates that the project is profitable.

860. In an attempt to enhance agile knowledge, a project manager has been teamed up with other Agile project managers to observe their team leadership practices. The project manager observes that the project team bears the responsibility for many project decisions, and project managers adopt a facilitative rather than authoritative role. They cultivate a shared vision and enable the team to concentrate on their tasks. What leadership approach does this exemplify?
a) Participative leadership
b) Autocratic leadership
c) Transformational leadership
d) Servant leadership

Explanation: This depicts servant leadership as the project manager is facilitating the team and sharing a common vision, allowing the team to focus on their work.

861. A project manager leads a project with an exceptionally skilled team. The team members possess varied skills and expertise, prompting the project manager to foster an environment where they can leverage their knowledge to make optimal decisions on project issues. Which leadership style is most appropriate for the project manager's objective?
a) Democratic
b) Laissez-faire
c) Autocratic
d) Directive

Explanation: The laissez-faire leadership style is characterized by a hands-off approach where the leader delegates decision-making power to the team members. This style is suitable for highly skilled and experienced teams who can make optimal decisions concerning project issues. The project manager wants to encourage the team members to use their knowledge to make optimal decisions concerning project issues, hence the most suitable leadership style is laissez-faire.

862. Over the past few years, an organization has expanded into a corporate group by acquiring several companies in the same industry. This has notably elevated the number of its projects, programs, and portfolios. How many portfolios should an organization manage simultaneously?
   a) One portfolio at a time
   b) Based on the size of human resources allocated to portfolio management
   c) Each portfolio manager should handle one portfolio at a time
   d) **As many as the organization can handle**

   Explanation: An organization can manage as many portfolios as it can handle. There is no specific limit to the number of portfolios that can be managed concurrently by an organization. The number of portfolios that can be managed depends on the organization's capacity and capability to manage them effectively.

863. A project manager at Clinica Labs, a biopharmaceutical corporation, intends to talk to her manager about obtaining additional resources for complex activities that her team is unable to perform. These resources will roll off the project as soon as the activities are completed. Which of the following skills does the project manager need the most in this situation?
   a) Planning skills to identify resource requirements
   b) Interpersonal skills to convince her manager of her need for additional resources
   c) Interviewing skills to hire the required resources
   d) Technical skills to respond to the risk associated with adding more resources

   Explanation: The project manager needs strong interpersonal skills to effectively communicate and persuade her manager about the necessity of obtaining additional resources for complex activities in order to ensure project success.

864. A Project manager ensures that the project team documents all assumptions. The team records assumptions on an ongoing basis and then validates these assumptions. Which of the following helps the team validate a project assumption?
   a) Organizational policies
   b) Historical data
   c) Constraints
   d) Team members recollection

   Explanation: Historical data helps validate project assumptions by providing past evidence and trends, allowing the team to assess the reliability and accuracy of their current assumptions based on previous experiences.

865. A project manager has just joined a new organization. A majority of the organization's projects are conducted by experienced project managers. What should be the primary focus of the new project manager in order to ensure his efficiency?
   a) Acquire a skillful team
   b) Learn about the organization's culture
   c) Get to know the organization's executive managers
   d) Get to know the organization's project managers

   Explanation: Understanding the organization's culture is crucial for the new project manager to navigate existing workflows, communication styles, and stakeholder expectations, ensuring efficient integration into the experienced project management environment.

866. A project manager is facilitating a meeting to define the user stories of the next iteration. Two team members, Charlotte and Sam, get too vocal during the meeting. Charlotte believes that user story #2 is clear enough, while Sam thinks it's ambiguous and needs further elaboration. The team moves on to the next user story, ignoring Sam's point of view. Sam accepts the team's decision and remains quiet for the rest of the meeting. At the end of the meeting, the project manager tells Sam that user story #2 could be reviewed during the iteration's first week when more information is available. Which of the following conflict-resolution techniques did Sam use in this scenario?
   a) Smoothing
   b) Compromising
   c) Forcing
   d) Withdrawal

   Explanation: Sam used the conflict-resolution technique of Withdrawal by accepting the team's decision, remaining quiet, and agreeing to review the user story later.

867. A project manager chose for their new project a management framework that is a hybrid of two Agile approaches. The work will be organized in sprints, and the team will use a board to display and monitor work progress. Which of the following frameworks is the project manager using?
   a) Scrumfall
   b) eXtreme Programming
   c) Dynamic Systems Development Method
   d) Scrum ban

Explanation: The project manager is using Scrumban, a hybrid management framework that combines elements of Scrum and Kanban, incorporating sprints and a board for work visualization and monitoring.

868. A project manager and their Agile team are demonstrating a potentially shippable product increment to the project stakeholders. What type of Agile meetings is the project manager conducting?
   a) Review meeting
   b) Standup meeting
   c) Retrospective meeting
   d) Deliverables meeting

   Explanation: The project manager is conducting a Review meeting in Agile, showcasing a potentially shippable product increment to project stakeholders for feedback and validation.

869. A project manager is leading a project that was supposed to last 6 months. The project is currently in its second year. Which of the following statements best describes the situation?
   a) The project scope was not clearly defined
   b) The project should have been divided into subprojects
   c) The project team didn't protect the scope from changes
   d) The project should have been halted after 6 months

   Explanation: The project's extended timeline suggests inadequate scope definition, leading to unforeseen challenges and delays.

870. While performing beta testing with a small number of users, the project manager noticed that the web application had defects due to the following: User interface and responsiveness issues (20%) User experience issues (25%) Bugs in functionalities (40%) Missing important features (10%) Others (5%). To illustrate these problems, the project manager can use a:
   a) Histogram
   b) Quality checklists
   c) Scatter diagram
   d) Flowchart

   Explanation: The project manager can use a histogram to visually represent the distribution of defects in the web application, categorizing them based on their respective percentages in user interface and responsiveness issues, user experience issues, bugs in functionalities, missing important features, and others.

871. After facing endless issues with one of the project suppliers, the project manager decided to put an end to the disputes by concluding their professional collaboration. Which document should the project manager refer to in order to examine the termination clause and the alternative dispute resolution mechanism?
   a) Source selection criteria
   b) Supplier bid
   c) **Agreement**
   d) Project charter

   Explanation: The project manager should refer to the Agreement document to examine the termination clause and alternative dispute resolution mechanism for concluding the professional collaboration with the problematic supplier.

872. A project manager engaged Outsource, an offshore software development company, to develop an e-learning platform using the adaptive approach. During the second iteration, the project manager wasn't satisfied with the company's work and decided to immediately terminate the contract. The project manager only had to pay the incurred costs with no additional fees. What type of Agile contract did the project manager sign with the supplier?
   a) Incremental delivery contract
   b) Time and materials contract
   c) Target cost contract
   d) Early termination contract

   Explanation: The project manager signed a Time and Materials contract, allowing termination at any phase with payment based on incurred costs without additional fees.

873. A Project Manager is leading a software project using a hybrid approach. During a lessons-learned meeting, one of the team members complained that every time she submits a new piece of code, she finds many errors caused by the code changes made by her fellow developers. The Project Manager suggested testing, updating, and integrating new software code more frequently in order to reduce such errors. Which of the following techniques did the Project Manager suggest?
   a) Constant Integration
   b) Consecutive Integration
   c) Consistent Integration
   d) Continuous Integration

   Explanation: The Project Manager suggested Continuous Integration, a technique involving frequent testing, updating, and integration of new code to minimize errors caused by code changes made by different developers.

874. A Project Manager is assigned to a packaging project. The organization purchased a custom-built laser printing machine for $130,000 for the project, which was later found to be useless. The $130,000 amount represents a(n) _____ cost.
   a) Fixed
   b) Sunk
   c) Indirect
   d) Opportunity

   Explanation: The $130,000 cost for the custom-built laser printing machine is a sunk cost because it has already been incurred and cannot be recovered, regardless of the project's outcome.

875. During an iteration review meeting, the product owner rejected one of the features demonstrated by the development team. What will happen next to the rejected user story?
   a) It will be automatically moved to the next sprint backlog
   b) It will be deleted from the product backlog and from the project
   c) It will be updated to address the reasons why it was rejected
   d) It will be moved back to the product backlog for reprioritization

   Explanation: The rejected user story will undergo updates to address the reasons for rejection in response to the product owner's feedback.

876. In order to assess the project performance, a Project Manager calculates both its CPI and SPI. Knowing that it has a high CPI and a low SPI, what is the project status?
   a) Behind schedule and under budget
   b) Ahead of schedule and under budget
   c) Behind schedule and over budget
   d) Ahead of schedule and over budget

   Explanation: The project is progressing behind schedule but is under budget, indicated by a high Cost Performance Index (CPI) and a low Schedule Performance Index (SPI).

877. A project manager is running a multinational organization. They want to learn the difference between high and low-context cultures in order to ensure effective communication and avoid doing anything offensive or embarrassing. What is a typical characteristic of communication in high-context cultures?
   a) The use of technical means of communication that focus on transferring spoken and written language.

b) Communication is preferred to avoid missing out on a great part of additional information.
c) A conveyed message has little meaning without a full understanding of the surrounding context.
d) There is no need for history or personal opinions to understand a message.

Explanation: In high-context cultures, messages rely heavily on contextual cues, and understanding the background and relationships is crucial for accurate interpretation.

878. During the risk identification phase, the project manager and their team identified more than 100 risks. The project manager believes that the quantitative evaluation of each of these risks will take a lot of time, whereas not all of the identified risks are important enough to justify such a measure. What should the project manager's next step be?
a) Identifying risk triggers, then performing a quantitative analysis of risks that have no triggers
b) Using qualitative risk analysis to qualify and prioritize risks for the quantitative risk analysis
c) Conducting a qualitative assessment of each risk's probability, then perform a quantitative analysis of risks with a high probability
d) Conducting a qualitative assessment of the potential impact of each risk, then perform a quantitative analysis of risks with a high impact

Explanation: The project manager should prioritize and qualify risks through qualitative analysis before conducting quantitative risk analysis to focus on significant risks efficiently.

879. A Project Manager is leading a 3-year project that involves upgrading and improving an electric engine for industrial use. Since the majority of the project activities are hazardous, the project team will receive extensive safety training. The training will ensure the safety of workers and prevent delays caused by unsatisfactory work. The cost of the conducted safety training falls under:
a) Cost of regulations
b) Cost of quality
c) Cost of conformance
d) Cost of nonconformance

Explanation: The cost of the conducted safety training for the hazardous project activities falls under the Cost of conformance, as it is an investment to ensure adherence to safety standards and prevent unsatisfactory work delays.

880. While leading agile release planning with a newly assembled team, a project manager is dedicated to achieving outstanding results. However, due to a lack of technical knowledge, the client requests a product with minimal requirements. What steps should the project manager take to ensure the client's satisfaction with the final product?
   a) Add more features to the final product
   b) Have the client provide their own technical engineer
   c) Engage the client in every iteration
   d) Set the requirements in a signed contract

   Explanation: Engaging the client in every iteration is a key principle of agile methodology. This ensures that the client is satisfied with the final product and that their requirements are met throughout the project.

881. As a project manager in a construction project, you encounter disagreements among the project team regarding their assigned tasks, leading to a decline in overall performance. Which tool or technique would you employ to address the team's issues?
   a) Recognition and rewards
   b) Conflict management
   c) Training
   d) Networking

   Explanation: Conflict management is the right answer because it helps to resolve disagreements and conflicts between team members, which can lead to a reduction in performance across the project.

882. In an agile team assigned to develop a robot, the project manager aims to ensure early feedback on requirements and adaptive design before constructing the actual robot. Which approach should the project manager employ to achieve this objective?
   a) Business case
   b) Prototyping
   c) Design review
   d) Checklists

   Explanation: Prototyping is an agile technique that allows the team to create a working model of the robot early in the development process. This model can be used to gather feedback from stakeholders and users and to make changes to the design before the actual robot is built.

883. A buyer has engaged a vendor to create a new device. The vendor emphasized a comprehensive contract capturing device specifications, while the buyer lacks

clear upfront requirements and prefers an Agile approach. Despite delivering per the contract, the buyer is dissatisfied. Which Agile Manifesto value was likely not adequately applied?

a) Pair programming over continuous integration
b) Customer collaboration over contract negotiations
c) Individuals and interactions over processes and tools
d) Business people and developers must work together

Explanation: The Agile Manifesto value of customer collaboration over contract negotiations was not appropriately applied in this scenario because the buyer insisted on a contract that thoroughly captured the device specifications, rather than collaborating with the vendor to develop the device using agile.

884. A project team accustomed to the waterfall project management approach is transitioning to their initial agile project, retaining some waterfall elements for a smoother shift. In the iteration review, a misunderstanding among the development team resulted in duplicated efforts. How should the project manager resolve this situation?

a) Do nothing since the team members are transitioning from waterfall to agile.
b) Submit a change request to update the communications management plan.
c) Bring up the topic during the upcoming iteration retrospective.
d) Fallback to a waterfall project management framework going forward.

Explanation: The project manager should address the issue during the upcoming iteration retrospective to identify the root cause of the misunderstanding and prevent it from happening again in the future. This is a key aspect of the agile approach, where continuous improvement is encouraged through regular retrospectives.

885. While overseeing a complex web development project, upon scrutinizing the project's statistics, you calculate that the team consistently completes an average of 50 story points per iteration. After assessing the backlog, you find approximately 500 points yet to be addressed. With each iteration time-boxed at two weeks, when do you anticipate completing the project?

a) Agile teams cannot predict project completion
b) In about 10 weeks from now
c) In about 20 weeks from now
d) In about 75 weeks from now

Explanation: The project has 500 story points remaining, and the team completes 50 story points on average per iteration. Therefore, it will take approximately 10 iterations (20 weeks) to complete the remaining work.

886. Which method is employed to present a concise, high-level timeline spanning 3-6 months, derived from the product roadmap and vision, during the Develop Schedule process?
   a) Root cause analysis
   b) Data analysis
   c) Agile release planning
   d) Project management information system

   Explanation: Agile release planning is a technique used in the development schedule process to provide a high-level summary timeline of the 3-6 month schedule based on the product roadmap and the product vision for the product evolution.

887. In this complex project scenario, after a scope modification necessitates additional expertise, you've requested consultants from your company. However, these experts are dispersed in various locations. How do you plan to gather their input?

   a) Ask management to bring them to you
   b) Recruit new members with the same skills
   c) Create a virtual team
   d) Take no action

   Explanation: Creating a virtual team is the best option in this situation as it allows the experts to work remotely and still provide their input to the project. This saves time and money compared to bringing them to the project location.

888. As the project manager overseeing a critical organizational project that's lagging behind schedule and exceeding the budget, the project sponsor proposes the idea of team members working on weekends. How can you effectively encourage and persuade your team members to dedicate their weekends to the project?
   a) Ignore the sponsor
   b) Ask the management what should you do
   c) Recognition and rewards
   d) Update the issue log

   Explanation: Recognition and rewards are a good way to persuade team members to work during their weekends. This will motivate them to work harder and achieve the project goals.

889. The project manager aims to document details concerning roles and responsibilities, release plans, training requirements, resource availability, recognition, and rewards. Which document would be the most suitable for capturing this information?

a) Project Management Plan
b) Resource Management Plan
c) Project Team Assignments
d) Issue Log

Explanation: The Resource Management Plan includes information related to roles and responsibilities, resource calendar, and training needs. It also includes information on recognition and rewards for the project team.

890. Within your project team, you observe heightened emotions due to the competition among team members. Instead of understanding each other's needs, there's a tendency for quick reactions, hindering collaboration toward project goals. Disturbed by this, you consult your sponsor, who recommends a specific training. What type of training is likely being suggested?
a) Team building
b) Emotional intelligence
c) Goal setting and results management
d) Active listening

Explanation: Emotional intelligence training can help team members understand and manage their emotions, as well as recognize and respond appropriately to the emotions of others. This can lead to better communication, collaboration, and conflict resolution within the team.

891. A project manager overseeing the development of a new retail banking product faces schedule delays due to a lack of motivation among team members. What strategies could the project manager employ to boost motivation within the project team?
a) Offer recognition and rewards
b) Use coaching and monitoring skills
c) Delegates the responsibilities
d) Apply creative problem solving

Explanation: Offering recognition and rewards is a good way to motivate the project team. It can help to increase their morale and encourage them to work harder to achieve the project goals.

892. The project manager incorrectly categorized a stakeholder as a supporter, leading to reluctance. What steps should the project manager take to avoid such misclassification in future projects?
a) Identify stakeholders
b) Develop interpersonal skills to manage stakeholder expectations
c) Create the stakeholders engagement assessment matrix

d) Conduct a meeting with senior management

Explanation: Creating a stakeholders engagement assessment matrix helps in identifying the stakeholders and their level of support or opposition towards the project. This will help the project manager to avoid misclassifying stakeholders in future projects.

893. A key stakeholder in your construction project has recently stepped down from their role due to personal reasons. What is the appropriate action to take in terms of stakeholder management?
   a) Contact the sponsor
   b) Remove all changes suggested by the stakeholder
   c) Update stakeholder register
   d) Initiate an investigation into his resignation

   Explanation: Updating the stakeholder register is the right answer because it is a key output of the identify stakeholders process, and it helps to keep track of the stakeholders' information and engagement levels throughout the project lifecycle.

894. Throughout an organization-wide project, certain stakeholders seem to have diminished expectations for the product. Consequently, their performance is not meeting the anticipated standards, posing a risk to project objectives. What steps should the project manager take?
   a) Speak with the problematic stakeholders to increase their performance
   b) Follow the stakeholder engagement plan
   c) Ask the project sponsor to intervene and motivate the underperforming stakeholders
   d) Submit a change request

   Explanation: The stakeholder engagement plan is a key component of the project management plan and should be followed to ensure that all stakeholders are engaged and their expectations are managed. By following the plan, the project manager can identify the stakeholders' needs and expectations and develop strategies to address them. This will help to ensure that all stakeholders are performing as expected and that project objectives are not at risk.

895. A company traditionally employs a predictive project management approach, but the project management office intends to integrate agile best practices for quicker revenue realization. The project manager is set to initiate planning for the upcoming project and is worried about the team's readiness for a hybrid environment. What should be the project manager's optimal approach?
   a) Empower the team to self-organize and study agile best practices

b) Check the lessons learned repository for agile best practices
c) Request the training from the Project Management Office (PMO)
d) Add the risk of the team's inadequate agile skills to the risk register

Explanation: The project manager's best course of action is to request training from the Project Management Office (PMO) to equip the team with the necessary skills to work in a hybrid environment.

896. In recent iterations, an agile team dedicated additional hours to produce product documentation and scrutinize contracts with the customer. How could the team have utilized their time more effectively?
a) Developing functionality as requested by the project sponsor
b) Creating the product increment and working with the product owner
c) Reporting the extra hours in the company's time and attendance tools
d) Building features according to the initial project management plan

Explanation: The team's time would have been better spent creating the product increment and working with the product owner because this is the core of agile development, and the team's main focus should be on delivering a working product increment at the end of each iteration.

897. A few days before the conclusion of the last iteration, the team notifies the project manager that they are unable to complete two work packages, A and B, due to insufficient refinement in the WBS. The team believes they can potentially finish package A on schedule but requires an additional week for package B, which holds higher priority. What should the project manager do in this situation?
a) Instruct the team to complete package A since it can be done on time
b) Instruct the team to complete package B since it has a higher priority
c) Consult with the product owner and make a decision accordingly
d) Extend the iteration length in order to finish both packages A and B

Explanation: Consulting with the product owner is the best course of action as they are the ones who can make the final decision on the priority of the work packages. The project manager should not make this decision alone.

898. A project involves a senior stakeholder whose input holds significance, and staying informed about the project's status is crucial. Instead of receiving updates directly from the project manager, this stakeholder prefers to contact team members directly. The stakeholder has personal connections with these team members unrelated to the project and believes it's a means to obtain authentic information about the project's actual status. What steps should the project manager take in this situation?

a) Leave the connection in place since the information from the team member is unofficial and will not create problems.
b) Tell the team member not to accept calls from the stakeholder because these calls adversely affect the team member's productivity.
c) Keep the team member properly informed about the project status so that the stakeholder receives the same information from both sources.
d) Ask the stakeholder how the additional information that they feel is missing from current communications may be provided.

Explanation: Option 4 is the right answer because it addresses the root cause of the stakeholder's behavior and seeks to find a solution that benefits both parties. By asking the stakeholder how they would like to receive additional information, the project manager can work to establish a communication plan that meets the stakeholder's needs while also ensuring that all project information is communicated through official channels.

899. A project manager oversees a substantial construction project valued at 4 billion USD. Multiple teams from diverse countries are engaged in the project, requiring seamless collaboration for success. The responsibilities of individual team members are interdependent. What factor is crucial for ensuring the overall success of project integration?
a) Detailed definition of the work packages
b) Adherence to professional code of conduct
c) Communication management plan
d) Change management

Explanation: The communication management plan is important for the overall success of project integration because it ensures that all stakeholders are informed about the project status, issues, and risks. It also helps to ensure that all team members are aware of their roles and responsibilities and that they are working towards the same goals.

900. A project manager collaborating with another company in a joint venture needs to disclose confidential information related to intellectual property rights. To determine the individual authorized to release this confidential information, she should consult which project document?
a) Project Charter
b) Stakeholder Management Plan
c) Organizational Breakdown Structure
d) Project Communication Management Plan

Explanation: The Project Communication Management Plan outlines the procedures for sharing project information, including confidential information. It

identifies the person responsible for authorizing the release of confidential information.

901. In the middle of project execution, multiple stakeholders expressed concerns about team performance and delivery. Despite the project manager's belief that the project is progressing according to the approved scope, budget, and schedule, what should the project manager do?
    a) Email the stakeholders to assure them of action.
    b) Ask stakeholders to escalate their concerns to the CEO.
    c) Ask the project sponsor to interview.
    d) Consult the communications management plan to manage stakeholders' expectations.

    Explanation: The project manager should consult the communications management plan to manage stakeholders' expectations. This will help the project manager to understand the stakeholders' concerns and communicate the project status effectively.

902. Alice is overseeing a project. The new project sponsor often meets with functional managers to discuss the project's status due to missed weekly status meetings. Despite regularly sending meeting minutes, it seems they go unread. What should Alice do next?
    a) Direct the functional managers to stop discussing project status.
    b) Resend the weekly meeting minutes to the sponsor.
    c) Meet with the sponsor individually.
    d) Send the communication management plan to the sponsor.

    Explanation: Alice should meet with the sponsor individually to discuss the issue and find out why the minutes are not being read. This will help her understand the sponsor's concerns and expectations and address any issues that may be causing the sponsor to miss the weekly status meetings.

903. The client has entered into a contract with your organization for the development of a product upgrade. The current product is no longer effective for their operations due to changes in the business environment. In a scheduled meeting, you discover that they are struggling to articulate their revised requirements clearly. What approach will you use to gather the requirements?
    a) Observation/job shadowing
    b) Interviews
    c) Context diagram
    d) Facilitation workshop

Explanation: Observation/job shadowing is the right approach to gathering requirements when the customer is not able to articulate their revised requirements clearly. This approach involves observing the customer's work environment and processes to understand their needs and requirements.

904. As a project manager, you have several team members who are new to the project. Although they are collaborating effectively as a team, you are worried about whether they are adhering to the procedures correctly. What tool or technique could assist you in identifying any potential issues?
    a) Ishikawa diagrams
    b) Histogram
    c) Quality audit
    d) Inspection

    Explanation: A quality audit is a structured review of the quality management activities that help identify lessons learned that could improve performance on current or future projects. It is a tool used to determine if the project is following the procedures correctly and to identify areas for improvement.

905. The project manager wants to determine whether the variance relative to the scope baseline has been improving or deteriorating as the project progresses. What should the project manager do next?
    a) Conduct product analysis
    b) Perform variance analysis
    c) Perform trend analysis
    d) Complete stakeholder analysis

    Explanation: Trend analysis is used to determine if the variance has been improving or deteriorating over time. It involves comparing the actual project performance to the planned performance over time to identify trends.

906. In the fiercely competitive technology market, delivering a product swiftly is crucial for the success of a tech start-up. An agile team opts to release the product with minimal features and intends to gather insights from early adopters' feedback. What agile tool or technique is the team employing in this scenario?
    a) WBS
    b) Product Analysis
    c) Kano Analysis
    d) Minimum Viable Product (MVP)

    Explanation: The team is embracing the Minimum Viable Product (MVP) approach, which is a key agile tool or technique. This approach involves

launching a product with minimal features and then learning from the early adopters' feedback to improve the product.

907. While enhancing the layout of a screen in a mobile application, a developer within an agile team proposes an idea to improve the screen, enhancing usability and potentially increasing customer satisfaction. This improvement introduces a new requirement. The developer's initial motivation is to impress the client. What is the appropriate next step for the developer?
    a) Implement this requirement to show the product owner the improvement in terms of the application's usability
    b) Do nothing since this requirement is out of scope and it should not be added to the product
    c) Collect the supporting information for this requirement and present this information to the product owner
    d) Suggest this requirement to the product owner for possible inclusion in the next iteration

    Explanation: The developer should suggest this requirement to the product owner for possible inclusion in the next iteration. This is the right answer because the developer should not make unilateral decisions to add new requirements to the product without consulting the product owner. The product owner is responsible for prioritizing requirements and deciding which ones should be included in the product. The developer should present the idea to the product owner and let them decide whether to include it in the next iteration or not.

908. In a daily stand-up meeting, a programmer mentions having additional time and incorporating functionality not part of the original design. She believes the customer will value the added features. What is the most appropriate course of action?
    a) Update the cost baseline since the project will run over budget due to the cost of adding new functionality
    b) Accept the added functionality, but ask the programmer to build only what has been designed going forward
    c) Remove the added functionality and ensure that the project team builds the software as designed
    d) Thank the programmer for the extra effort and notify the customer of the new functionality that has been added

    Explanation: The added functionality was not included in the original design, and therefore, it is considered scope creep. The best course of action is to remove the added functionality and ensure that the project team builds the software as designed.

909. Following three iterations, the agile team achieves an average velocity of 50 story points. With 300 more story points remaining for the project, how many additional iterations will be required for the team to complete the project?
   a) 5 iterations
   b) 6 iterations
   c) 7 iterations
   d) 10 iterations

   Explanation: The team's average velocity is 50 story points per iteration. Therefore, the team can complete 150 story points in three iterations. To complete the remaining 300 story points, the team will need 6 more iterations (300/50=6).

910. The finance group collaborates with the project manager on large construction projects for a logistics company, establishing timelines for expenses and exploring options to even out spending, preventing concentrated expenditures in one quarter and none in the next. This is an example of:
   a) Funding Limit Reconciliation
   b) Rescheduling
   c) Resource leveling
   d) Crashing

   Explanation: Funding Limit Reconciliation is the process of determining when expenses will be incurred in a given project and whether there are ways to smooth out or level the spending to avoid a single large expenditure in one quarter and none in the next. This is exactly what is described in the question.

911. Commencing a project with a hybrid approach, the project manager encounters a major design change approved by the customer. However, the software team leader, operating under an agile approach, hesitates to commit to final delivery dates. Given the interdependence with the hardware team, the project manager must now navigate the situation. What steps should be taken?
   a) Escalate this to higher management and ask for help to resolve the issue.
   b) Ask the software team leader to use a predictive approach and commit to the delivery date.
   c) Update the risk register with this risk and monitor it according to the risk management plan.
   d) Impose the delivery dates on the software team leader and notify the functional manager about the situation.

   Explanation: Considering the context of a hybrid approach, the project manager needs to find a balance between agile flexibility and the predictability required

by the hardware team. Therefore, the best option is likely to involve both addressing the immediate coordination need and managing the risk.

912. In the retrospective ceremony, the agile team discusses unmet expectations from the product owner. What actions should the project manager have taken at the project's outset?
   a) Communicated project vision and sprint goals clearly to the team
   b) Defined the expected agile team contribution in the creation of the backlog
   c) Defined the team ground rules and shared project vision
   d) Assessed the capabilities of the agile team and planned for the required training

   Explanation: Communicating project vision and sprint goals clearly to the team at the beginning of the project helps ensure that the product owner's expectations are met. This allows the team to understand what is expected of them and what they need to deliver.

913. A recently assigned junior staff member expresses confusion to the project manager, stating a lack of familiarity with project expectations. How should the project manager assist the new team member?
   a) Advise the new team member to enroll in project management training.
   b) Review the resource management plan and provide mentoring to the new team member if necessary.
   c) Instruct the new team member to review the responsible, accountable. consult. and inform RACI matrix before asking for help.
   d) Advise the new team member to review the project charter and stakeholder engagement plan.

   Explanation: Reviewing the resource management plan and providing mentoring gives the new team member a clear understanding of their role and project expectations while offering personalized support. This approach builds confidence and facilitates smooth integration into the team, ensuring they can contribute effectively.

914. The sponsor has approved the schedule management plan for an integrated project involving both hardware and software. The project team has assessed the functionality feasibility of the product and expedited testing. What actions should the project manager take in their role as a servant leader?
   a) Submit a change request to the project sponsor for accelerated testing.
   b) Develop the product as per the schedule to perform integration and testing.
   c) Amend the project budget to include minimum viable product (MVP)
   d) Support the team as necessary to find the minimum viable product (MVP)

Explanation: As a servant leader, the project manager should support the team as necessary to find the minimum viable product (MVP). This approach aligns with the Agile methodology, which emphasizes collaboration and flexibility to deliver value to the customer. By supporting the team in finding the MVP, the project manager can ensure that the project stays on track and meets the sponsor's approved schedule management plan.

915. In the implementation phase of a project, a new regulation impacts the project, necessitating a crucial status meeting to discuss scope changes. The project manager realizes that a key stakeholder is unable to attend the planned meeting. How should the project manager proceed?
    a) Meet with the stakeholder prior to the meeting to obtain their opinion
    b) Meet with the project sponsor to discuss how to address the situation
    c) Call for a change control board (CCB)
    d) Update the communications management plan and implement the change.

    Explanation: Meeting with the stakeholders prior to the meeting to obtain their opinion is the right answer because it allows the project manager to gather the stakeholder's input and perspective before the critical status meeting. This will help ensure that the stakeholder's concerns are addressed and that the meeting is productive.

916. A client approaches a design team member, asking to halt a current task and develop a new concept, which would substantially alter the design and affect the project budget and schedule. The team member seeks guidance from the project manager on the next steps. What should the project manager do?
    a) Ask the client to refrain from directly contacting team members
    b) Accept the request and make the changes
    c) Refer to the perform integrated change control process
    d) Hold a team meeting to discuss the request

    Explanation: The integrated change control process is the appropriate process to follow when a change request is received. This process ensures that all changes are properly reviewed, evaluated, and approved or rejected before they are implemented. The project manager should refer to this process to determine the impact of the change request on the project budget and schedule.

917. A specific stakeholder consistently requests changes to the project scope that deviate from the original project requirements. One of these changes was proposed just one month before the project launch date, and it is anticipated to result in significant project delays. What course of action should the project manager take?
    a) Inform the project steering committee about the project delay risk.

b) Update the change management document.
c) Submit a change request to the change control board (CCB)
d) Continue with the launch date as per the approved project management plan.

Explanation: Submitting a change request to the change control board (CCB) is the right thing to do because the change is not consistent with the original project requirements and is projected to cause significant project delays. The CCB will evaluate the change request and determine whether to approve or reject it based on its impact on the project's scope, schedule, and budget.

918. Some team members are not cooperating and lack mutual support, often turning to you for decisions on minor issues. How can you improve this situation?
a) Involve them in project management work to increase their motivation
b) Initiate team building to increase trust among team members
c) Involve them in your decision making activities
d) Check whether there are any issues due to diverse cultures

Explanation: Initiating team-building activities can help increase trust among team members and improve their cooperation. This can lead to better decision-making and problem-solving within the team.

919. In the executing phase of a project, you, as the project manager, observe conflicts within the team arising from the inclusion of subject matter experts working remotely. What will be your initial step to enhance the situation?
a) Exchange the locations of the other geography resources with the local resources
b) Improve the communications plan for the project
c) Bring the other geography resources to the local office for a day every month to meet the local members face to face
d) Hold frequent meetings with the other geography resources

Explanation: Improving the communications plan for the project is the best option to address the conflicts in the project team due to remote work. This will help to ensure that all team members are on the same page and have a clear understanding of their roles and responsibilities, as well as the project goals and objectives.

920. Having conducted surveys, interviews, assessments, and focus groups within your project team, you've compiled a team performance assessment. This report includes observations on skills, competencies, turnover rate, and cohesiveness. How do you intend to utilize this report moving forward?
a) Make changes to the team members who are performing poorly

b) **Identify specific training, coaching, mentoring or other changes required to improve the team's performance**
c) Estimate the cost of improving the team's performance
d) Estimate the revised project schedule based on the team's performance

Explanation: The team performance assessment report will help identify specific training, coaching, mentoring or other changes required to improve the team's performance. This will help the project manager to take corrective actions to improve the team's performance.

921. A customer has been consistently tardy with approvals on design documentation. Despite John, the project manager, addressing the issue previously, the habit of delayed approval persists. John has been meticulously documenting these delays. Which conflict resolution technique is he employing?
a) Compromising
b) Collaborating
c) Smoothing
d) Withdrawal

Explanation: John is following the withdrawal conflict resolution technique by avoiding the conflict altogether and documenting the delays instead of confronting the customer.

922. While executing a project, the project manager identifies that a team member in one branch is exerting counterproductive influence on tasks. What should the project manager employ to ensure positive working relationships among team members?
a) Ground rules
b) Recognition and rewards
c) Interpersonal skills
d) Team-building activities

Explanation: Team-building activities can help improve communication and collaboration among team members, which can help resolve conflicts and improve working relationships. In this case, the project manager can use team-building activities to address the issue with the team member who is causing problems.

923. You're the project manager in charge during the execution phase, and one of your team members left the site early on a Friday at 3:00 pm, displaying inappropriate behavior towards colleagues. To prevent such incidents in the future, you should establish clear expectations by creating:
a) Management skills

b) Team-building activities
c) The team charter
d) Interpersonal skills

Explanation: The team charter is a document that outlines the expectations and responsibilities of the project team. It includes guidelines for acceptable behavior and can be used to address situations like this.

924. On a sizable project, a lead business analyst consistently works late to finalize requirements for the upcoming sprint. During the weekly team lead meeting, the project manager discovers that several other business analysts on the team have available time. To ensure an equitable workload, what should the project manager do?
a) Facilitate a discussion within the team to redistribute the work within the project team.
b) Delay the start date of the next sprint to allow more time for gathering requirements.
c) Direct the business analysts with free cycles to take more ownership of the project.
d) Review the list of open requirements and reassign them to team members with free cycles.

Explanation: Facilitating a discussion within the team to redistribute the work within the project team is the right answer because it ensures an even workload and makes use of the available resources.

925. You're overseeing a software upgrade project for your company. A crucial stakeholder holds significant decision-making authority but is resistant to changes resulting from the project. Continuous support from this stakeholder is vital for project success. Which stakeholder management strategy should you employ with this influential participant?
a) Keep the key stakeholder satisfied
b) Manage the key stakeholder closely
c) Keep the key stakeholder informed
d) Monitor the key stakeholder actions

Explanation: Managing the key stakeholder closely is the right answer because this stakeholder has a great deal of authority and is resistant to changes. Therefore, it is important to keep him engaged and informed throughout the project to ensure his continuous support.

926. Certain stakeholders in your project are not reacting as expected to your strategies. To address this, you're intending to explore the underlying reasons. What would be the most effective approach for this investigation?
   a) Mind mapping
   b) Root cause analysis
   c) Stakeholder prioritization
   d) Funding limit reconciliation

   Explanation: Root cause analysis is the best methodology for investigating the reasons for stakeholders not responding appropriately to strategies. It helps to identify the underlying causes of the problem and develop effective solutions.

927. Six months into its anticipated nine-month duration, the project manager reports a delay of at least three months due to key stakeholders resisting changes resulting from the project. What could have been adjusted in the project planning phase to prevent this scenario?
   a) Creation of a comprehensive stakeholder register
   b) Establishment of a more robust communications management plan
   c) Better execution of the Manage Stakeholder Engagement process
   d) Development of a stakeholder engagement assessment matrix

   Explanation: The development of a stakeholder engagement assessment matrix would have helped identify the key stakeholders and their level of influence and interest in the project. This would have allowed the project manager to develop a more effective stakeholder management plan and address any resistance to change early on in the project.

928. The government sponsors a project for the construction of a new highway. The project manager informs stakeholders that the project charter has been approved. While working on the project management plan, the project manager receives an email from an environmental organization inquiring about the project's objectives and start date. What steps should the project manager take to handle the situation?
   a) Conduct a stakeholder analysis and communicate to stakeholders accordingly.
   b) Update the communication management plan accordingly.
   c) Conduct a brainstorming session with the stakeholders and the team.
   d) Respond to the email with the requested information.

   Explanation: The project manager should conduct a stakeholder analysis to identify the stakeholders and their needs, interests, and expectations. This will help the project manager to communicate with the stakeholders effectively and efficiently. The project manager should then communicate the objectives and

start date of the project to the environmental organization and any other relevant stakeholders.

929. Project team members are collaborating on a document outlining key values crucial to the team's success, including sharing best practices, maintaining open-minded listening, and fostering respectful communication during meetings. What term characterizes this document?
   a) Resource calendars
   b) Team charter
   c) Communication management plan
   d) Resource management plan

   Explanation: The document described in the question is a team charter, which outlines the team's values, goals, and roles. It is an important tool for establishing a shared understanding among team members and promoting collaboration and accountability.

930. A project manager, emphasizing systems and structure to achieve project goals, experiences strained relations with the project team. Despite the current project aligning with its objectives, a high team turnover jeopardizes success. To enhance personal skills, what type of skills should the project manager prioritize for development?
   a) Communication skills
   b) Management skills
   c) Operational and problem-solving skills
   d) Leadership skills

   Explanation: The project manager should spend time developing leadership skills to improve their relationship with the project team and reduce turnover. This will help them to better motivate and inspire the team to achieve project goals.

931. While overseeing a software development project that demands proficiency in various programming languages, a developer is reassigned by her functional manager, and a replacement is provided. What initial steps should you take in response?
   a) Introduce the new developer to the project sponsor
   b) Provide the new developer with an ability test
   c) Assign the developer the next user story in the backlog
   d) Schedule training for the new developer

   Explanation: Providing the new developer with an ability test is the first thing to do because it will help you assess the new developer's skills and knowledge of

the programming languages required for the project. This will help you determine if the new developer is a good fit for the project and if additional training is required.

932. Assigned as the leader of an agile project, your aim is to facilitate the swift identification of misalignments, dependencies, or issues related to the dynamic requirements. What would be the most effective approach to ensure the sharing of project information within and across the organization?
   a) By promoting aggressive transparency
   b) By inviting stakeholders to daily standups
   c) By distributing the project report weekly
   d) By sharing comprehensive documentation

   Explanation: Agile promotes aggressive transparency, which means that all information is shared within and across the organization. This helps to surface any misalignments, dependencies, or other issues related to the frequently changing requirements as quickly as possible.

933. In an agile project led by a scrum master, the development team has fallen short of the sprint goal in the last three iterations. Following the sprint retrospective, the team has chosen to implement various changes to enhance velocity. What actions can the scrum master take to confirm any improvement in performance?
   a) Study the Gantt chart
   b) Review the burnup chart
   c) Calculate the project's TCPI
   d) Create Pareto chart

   Explanation: The burnup chart is a tool used in agile project management to track progress and forecast completion. It shows the amount of work completed over time and compares it to the total amount of work planned for the project. By reviewing the burnup chart, the scrum master can verify any performance improvement resulting from the changes implemented by the development team.

934. With years of collaboration, the project manager and team have cultivated a strong sense of community. Each member is proficient and continually enhances their skills with each project. In this context, which leadership style does the project manager employ to enhance these connections effectively?
   a) Communications leadership
   b) Servant leadership
   c) Transformational leadership
   d) Charismatic leadership

Explanation: Servant leadership is a leadership style that emphasizes collaboration, trust, empathy, and the well-being of the team members. It is particularly effective in deepening connections and building a sense of community among team members.

935. Your organization's PMO routinely requests project artifacts across all projects to ensure adherence to corporate project management standards. Additionally, they readily provide templates and guidelines upon request. What type of PMO is being described in this scenario?
    a) Controlling PMO
    b) Governing PMO
    c) Supporting PMO
    d) Directing PMO

    Explanation: The PMO described in the question is a Controlling PMO because it is focused on ensuring that corporate standards are being followed and periodically checking project artifacts to ensure compliance.

936. The Project Manager, aiming to construct a concrete waterfall at a theme park, prefers thorough upfront planning and follows the waterfall method. However, the customer desires flexibility, anticipating potential changes during the project and seeking assurance that the Project Manager can accommodate such adjustments. What type of project life cycle should the Project Manager employ?

    a) Explicit
    b) Predictive
    c) Adaptive
    d) Tacit

    Explanation: The Project Manager should use an adaptive project life cycle, which allows for changes and adjustments to be made throughout the project. This will enable the customer to change her mind in certain areas without causing significant delays or issues in the project.

937. You've recently assumed the role of Marketing Manager in an organization. Informed about a new product requiring swift market entry within the next month, involving various promotional channels and a significant launch event, you seek guidance from a project manager. However, you discover that the organization lacks a dedicated Project Manager, only having a part-time Project Coordinator. What is the organizational structure of the organization?
    a) Balanced matrix
    b) Project-oriented
    c) Strong matrix

d) PMO

Explanation: The organizational structure of the organization is a balanced matrix because it has a part-time Project Coordinator who will be responsible for coordinating the project, but the Marketing Manager will be responsible for the project's success.

938. A project manager is overseeing the design and development of a state-of-the-art computer system, currently in the second phase of a five-phase project. Facing delays and substantial budget overruns due to a force majeure event, the project is being conducted internally to retain control over proprietary technology, offering a competitive edge. The project has now reached a phase gate. What is the purpose of conducting the phase gate?
   a) Determine how many resources are required to complete the project according to the project baseline
   b) Obtain customer acceptance of project deliverables
   c) The project's performance is compared to the business documents to determine if the project should be Continue to the next phase, End the project. or Repeat the phase
   d) Adjust the schedule and cost baselines based on past performance

   Explanation: The purpose of performing the phase gate is to compare the project's performance to the business documents to determine if the project should continue to the next phase, end, or repeat the phase. This helps to ensure that the project is still aligned with the business objectives and that it is still feasible to continue with the project.

939. A project manager has recently assumed control of a project aiming to create a product through a hybrid life cycle. In this scenario, the hardware component has well-defined requirements, while the software requirements are still undergoing refinement. How should the project manager implement the hybrid life cycle in this project?
   a) Develop the hardware with a predictive life cycle and use an adaptive life cycle for the software
   b) Both the hardware and software should have their scope defined during initial project planning
   c) Develop the software with a predictive life cycle and use an adaptive life cycle for the hardware
   d) Both the hardware and software should have their scope defined iteratively

   Explanation: The hardware has fixed requirements, so it can be developed using a predictive life cycle. The software requirements are still evolving, so an adaptive life cycle should be used for software development. Therefore, the

project manager should apply a hybrid life cycle by developing the hardware with a predictive life cycle and using an adaptive life cycle for the software.

940. In your project, the supplier provides equipment that does not meet customer requirements. Upon investigation, it becomes clear that the supplier misunderstood the requirements and made incorrect assumptions without seeking clarification from the customer. As a project manager, what communication methods could you have employed to prevent this?
   a) Pull communication
   b) Push communication
   c) Interactive communication
   d) Manual communication

   Explanation: Interactive communication is a two-way communication method that allows for clarification and feedback. By using interactive communication, the project manager could have ensured that the supplier understood the customer's requirements and had no incorrect assumptions.

941. You are overseeing an agile software project after previously managing non-agile projects. What type of meetings would you identify as distinct in your current project compared to your previous ones?
   a) Sprint meetings
   b) Design meetings
   c) Vendor performance review meetings
   d) Quality review meetings

   Explanation: Sprint meetings are unique to Agile projects and are not found in traditional project management methodologies.

942. In overseeing a multinational project with team members spanning different countries and time zones, you aim to select a communication medium that best aligns with project needs. What would be the most suitable approach to tackle this?
   a) Discuss the communication requirements and the available options with stakeholders to seek their inputs
   b) Use a combination of communication technology as is done in your peer's project
   c) Explore communication tools available in the market. and ask for quotation exclusively for your project instead of using existing tool in the organization
   d) Use the most cost effective technique, and involve everyone to explain its usage

Explanation: Choosing the communication media that is most efficient for project requirements is a critical aspect of project management. The best way to address this is to discuss the communication requirements and the available options with stakeholders to seek their input. This will help in identifying the most appropriate communication media that meets the project requirements.

943. In the course of a new product development project, the project manager recognizes that the requirements are conceptual, making it challenging to establish a shared understanding among all stakeholders. What actions can the project manager take to foster a shared understanding of the project's requirements?
   a) Build a prototype
   b) Hold brainstorming sessions with the stakeholders
   c) Conduct a requirements walkthrough
   d) Benchmark with the other organizations
   Explanation: Building a prototype can help stakeholders visualize the requirements and provide feedback, leading to a shared understanding of the project's requirements.

944. On your project, there is a diverse range of stakeholders representing various functional areas. As you gather requirements from each of them, you aim to delineate cross-functional requirements and reconcile any differences among stakeholders. What requirement collection techniques would be most effective in accomplishing this objective?
   a) Interviews
   b) Focus Groups
   c) Facilitation
   d) Brainstorming

   Explanation: Facilitation is the best technique to use when there are multiple stakeholders with different perspectives and requirements. It helps to bring everyone together and reconcile differences.

945. As a Project Manager, you're utilizing the triangular distribution formula to estimate the duration of a project activity. For a specific activity, your Optimistic estimate is 14 days, the Pessimistic estimate is 49 days, and the Most Likely estimate is 27 days. What would be the anticipated duration of this activity?
   a) 25
   b) 30
   c) 32
   d) 50

Explanation: The expected duration of an activity using a triangular distribution formula is calculated as (Optimistic + 4 x Most Likely + Pessimistic) / 6. Substituting the values, we get (14 + 4 x 27 + 49) / 6 = 30. Therefore, the expected duration of the activity is 30 days.

946. As a project manager overseeing a complex construction project, you seek a five-year schedule for the endeavor. Despite conducting numerous brainstorming sessions with your project team, you encounter challenges in furnishing specific details for the project management plan. This results in a delay in securing approvals for the project management plan. What steps could you have taken to prevent this situation?
   a) Used the rolling wave planning technique for activity planning
   b) Used decomposition techniques
   c) Created a high level Gantt chart to represent the milestone only
   d) Used subject matter expert to develop the detailed schedule

   Explanation: Using the rolling wave planning technique for activity, planning would have allowed the project manager to develop a high-level plan for the entire project and a detailed plan for the near-term work. This would have allowed the project manager to obtain the project management plan approvals on time.

947. A software development project has been assigned to a selected vendor. Your company and the vendor have reached an agreement to execute the project using the Waterfall Methodology. Upon inquiring about progress and the start of development, the vendor communicates that the actual development can only commence after the completion of the design. What type of relationship most accurately characterizes the dependency between design and actual development?
   a) Finish-to-Start
   b) Start-to-Start
   c) Start-to-Finish
   d) Finish-to-Finish

   Explanation: The dependency between design and actual development is a finish-to-start relationship because the actual development can only start after the completion of the design.

948. The central emphasis of a project is maintaining adherence to the schedule. In order to oversee this aspect, the project manager solicits monthly progress reports from the contractor. What specific details should the project manager request the contractor to incorporate into these reports?
   a) Gantt chart

b) Schedule attributes
c) Work breakdown structure (WBS)
d) Network chart

Explanation: A Gantt chart is a visual representation of a project schedule, which shows the start and end dates of each task, as well as dependencies between tasks. By including a Gantt chart in the monthly progress reports, the project manager can easily monitor the project's progress and identify any delays or potential issues.

949. As a project manager in a manufacturing company overseeing the production of piston rings on a specific line, you aim to assess the output. Your goal is to differentiate between variations inherent to the process and those indicating a need for adjustments. In this scenario, which tool would be the most beneficial?
a) Pareto charts
b) Statistical sampling
c) Control charts
d) Trend analysis

Explanation: Control charts are used to determine whether the variations in the jobs are inherent to the process or if the process is in need of adjustment. They are used to monitor the stability of a process over time and to identify any unusual variations that may indicate a problem.

950. The goal of a sprint should be to:
a) Deliver and deploy the product to production
b) Complete as many features as possible
c) Produce a product increment of a releasable quality
d) Fix all the defects in the product

Explanation: A sprint is supposed to produce a product increment with a releasable quality

951. Continuous integration is:
a) When the development happens in real time
b) When code is integrated after the feature is updated
c) When all code changes are checked and tested every day
d) When the team works on a single set of code

Explanation: Continuous integration is executed when code changes are checked and tested daily.

952. A definition of done clarifies when a particular work item can be marked as done. When is the definition of done first used by the team?

a) While estimating the work
b) While marking a work item as complete
c) During the sprint to ascertain what all needs to be done
d) By the tester while checking the stories

Explanation: While estimating the work the team should know what needs to be done, so that they can make informed commitments.

953. How frequently should the test-code-test cycle repeat itself in test-driven development?

a) At least once a day
b) At the end of an iteration
c) Every few minutes
d) For each line of code

Explanation: Test-driven development means the code should follow the tests and the cycle should repeat frequently

954. A team assigned 3 story points to a story. The manager wanted to know how this can corelate to the time required to complete it. How should the team respond?

a) A story point is roughly one hour of work
b) A story point is roughly one day of work
c) Story points are not directly correlated to time, rather through velocity, which specifies how many points a team can complete in an iteration
d) A team will establish a standard ratio between points and hours, which is constant for a specific team

Explanation: A story point is a unit of measure to express the estimated effort of the user story.

955. In order to implement Lean, the first step should be to:

a) Eliminate waste
b) Limit work-in-progress
c) Draw a value stream map
d) Continuously improve the process

Explanation: Value stream mapping is a way of analyzing a chain of processes and is the first step to achieve elimination of waste.

956. A team facilitator (aka scrum master) facilitates the daily stand-up meeting for a team. The facilitator is going on a leave for 2 weeks. What should be done about the daily stand-up meetings in this period?
   a) The product owner should facilitate for that period
   b) The stand-up meetings should be deferred until the scrum master returns
   c) The stand-up meetings should happen regardless of whether the facilitator is present or not
   d) The team should inform each other about the status through emails or other means of communication

   Explanation: Daily stand-up meeting is important to the flow of communication and for early detection of any issues. This should not be skipped even if the scrum master or facilitator is absent.

957. The process of test-driven development has the following steps:
   a) Test, Code, Deliver, Refractor
   b) Code, Refactor, Test, Deliver
   c) Code, Test, Refactor, Deliver
   d) Test, Code, Refactor, Deliver

   Explanation: The steps for the process of test-driven development are test, code, refactor, and deliver.

958. Agile approaches are becoming very popular in every industry today because of their flexibility to adapt to change. This is important because:
   a) Reducing the extent of change is important for success
   b) Planning is greatly hampered if there is excessive change
   c) Changes bring about a competitive advantage to the customer
   d) Teams need to gear up to handle change to improve their technical abilities

   Explanation: Changing requirements brings about a competitive advantage to the customer.

959. Agile focuses on iterative, incremental development, whereas Kanban focuses on:
   a) Minimizing waste
   b) Maximizing WIP
   c) Maximizing Lead Time
   d) Flow of work

   Explanation: Agile and Kanban share common goals such as delivering value quickly and adapting to change, but they differ in their approaches. While Agile emphasizes iterative and incremental development through methods like Scrum, Kanban focuses on visualizing and managing the flow of work. Kanban aims to

optimize efficiency by minimizing bottlenecks, reducing work in progress (WIP), and ensuring a smooth flow of tasks through the system. Therefore, option (d) accurately identifies Kanban's focus on the flow of work.

960. A PMO (Project Management Office) can be a key enabler for an agile transformation. What could be an area that the PMO can help an agile team with?
    a) How to deliver the work
    b) Guidelines to ensure compliance with established frameworks
    c) Standards and benchmarks for team velocity
    d) Technical expertise required to deliver

    Explanation: The PMO can help the team stay on the right side of required compliance needs

961. The developers in a team felt that all the stories were completed. A tester in the team disagreed with this. What is the best way to resolve this issue?
    a) Redefine the stories after each iteration
    b) Approach the customer
    c) Don't change the stories. The team agreed on the definition in advance
    d) Rewrite ground rules

    Explanation: Done must be explicitly defined and agreed upon by the entire team.

962. A team is debating about the effectiveness of two different design approaches. How to arrive at a conclusion?
    a) Approach the customer to clarify requirements
    b) Conduct small experiments to help the team determine a course of action
    c) Add a story to finalize the design
    d) Set aside one iteration for design

    Explanation: Spike is a quick experiment used to help the team answer a question and determine a path forward.

963. Which of the following is the best description of the 5 Whys technique?
    a) Ask why while selecting the team, scrum master, and product owner
    b) Ask why questions while understanding and analyzing user stories
    c) Asking the why question at least 5 or as many times as necessary to understand the deeper root causes
    d) Five people asking the why question to see if the answers are consistent

Explanation: Toyota used the 5 Whys in their manufacturing processes to detect and correct deeper root causes of issues

964. A team is estimating the time needed to create a software package by posting stories on the wall in the order from of least to greatest. This is an example of:
    a) Relative sizing
    b) Top down estimating
    c) Wireframes
    d) Game player

    Explanation: Relative sizing is estimating the size of a function or user story based on the size of another one.

965. The most important job of a scrum master is:
    a) To keep the team focused on agile principles
    b) To manage the projects
    c) To be a liaison with the customer
    d) To report status to executives

    Explanation: The scrum master is responsible for helping the team to follow the scrum process.

966. Senior management would like to keep tabs on progress of a project. What is the best demonstration of progress that the team can come up with?
    a) Burn down chart
    b) Information radiators
    c) Value points delivered
    d) Working product

    Explanation: Working software is the primary measure of progress as per the Agile Manifesto

967. Extreme programming believes in programmers working in pairs. Which of the following is an exception to this rule?
    a) When the team works on a SPIKE
    b) When there is an emergency
    c) When programmers are fixing defects
    d) When programmers implement minor features

    Explanation: Extreme Programming (XP) emphasizes pair programming as a core practice to enhance code quality and knowledge sharing. However, there are exceptions to this rule. When the team is working on a SPIKE, which involves exploration or research to address uncertainties, it's acceptable for

programmers to work individually as the focus is on learning and experimentation rather than production code. Therefore, option (a) correctly identifies an exception to the pair programming rule in XP.

968. Distributed agile teams often make use of live video conferences to stay in touch with the remote team members. This is important because these methods enable:
   a) Information radiation
   b) Face-to-face communication
   c) Brainstorming
   d) Conflict resolution

   Explanation: Face-to-face communication is the most effective type of communication and therefore colocation is the best arrangement for the team.

969. Who is primarily accountable for delivery of the work in an agile team?
   a) Development team
   b) Product owner
   c) Scrum master
   d) Functional managers

   Explanation: In agile teams, the collective accountability for delivering the work lies primarily with the development team. While the product owner prioritizes the backlog and the scrum master facilitates the agile process, it's the development team that is responsible for executing the work and delivering the agreed-upon increments of functionality. Therefore, option (a) accurately identifies the primary accountability for delivery within an agile team.

970. In a given iteration, a team completed 5 stories of sizes 1, 3, 5, 8, and 13 respectively. Two other stories of size 5 each were almost complete but narrowly missed the definition of done. What is the velocity of the team in the iteration?
   a) 20
   b) 30
   c) 35
   d) 40

   Explanation: Velocity only accounts for completed stories as per the definition of done

971. Iteration retrospective is used to:
   a) Shape product roadmap
   b) Improve processes in future iterations
   c) Resolve team conflicts
   d) Allow the opportunity to re-organize the team

Explanation: The purpose of an iteration retrospective in agile is to improve processes for future iterations. It provides a dedicated time for the team to reflect on what went well, what didn't, and how they can enhance their practices to be more effective and efficient in subsequent iterations. This continual process of reflection and adaptation is fundamental to agile's principle of continuous improvement, making option (b) the correct choice.

972. Project A has an NPV of $15,000 and IRR of 15%. Project B has an NPV of $20,000 and IRR of 13%. Project C has an NPV of $18,000 and IRR of 12.5%. Project D has an NPV of $25,000 and IRR of 14%. Which project should be selected if only one can be funded?
    a) A
    b) B
    c) C
    d) D

    Explanation: Project D should be selected because it has the highest Net Present Value (NPV) of $25,000, which indicates the greatest potential increase in the firm's value. NPV is the primary metric for decision-making in capital budgeting, making Project D the best choice despite its IRR being lower than Project A.

973. Which attributes should be attached to user stories?
    a) Validated, Economical
    b) Vital, Large
    c) Valuable, Estimable
    d) Verifiable, Editable
    Explanation: User stories should add value to the user and be small in effort and therefore, estimable.

974. The scrum master of a team is working on a critical task when another team member approached him for help. The scrum master should:
    a) Complete the critical task and then help the team member
    b) First help the team member and then return to his task
    c) Ask another team member to help
    d) None of the above

    Explanation: Scrum master duties always come first - before any other task that the scrum master is performing

975. What is the appropriate level to measure earned value management in an agile project?
    a) Feature

b) Release
c) Iteration
d) Continuous

Explanation: Earned value management in an agile project should be measured at the iteration level, because this is the level where velocity is measured and resources are applied.

976. During the iteration review, a customer representative pointed out that the addition of a few features would add a lot of value to the product. Who should take this into consideration first?
a) The sponsor
b) The product owner
c) The scrum master
d) The team

Explanation: The product owner is primarily responsible for managing the product backlog and making decisions about its contents based on the input from various stakeholders, including customer representatives. Therefore, the product owner should be the one to consider the suggested features first and evaluate their potential value for inclusion in the product backlog.

977. In a scrum planning meeting, the team had a few questions about the features with the highest priority. This should be resolved by:
a) The project manager
b) The customer
c) The product owner
d) The scrum master

Explanation: The product owner leads the first part of each iteration planning meeting where the features are discussed.

978. You see a burndown chart, burnup chart, feature backlog, and a velocity chart on the wall. What are they?
a) Enterprise Environmental Factors
b) Organizational process assets
c) Agile status
d) Information radiator

Explanation: The information radiator is used to communicate project status to the team and other stakeholders.

979. The charter for an agile project is a bit different from the charter for a traditional project in that:
   a) It is not mandatory for an agile project to have a charter
   b) Agile charter may not require a business case
   c) Agile charter is more lightweight, recognizing the inevitability of changes
   d) Traditional charter is more high level, agile charter is revised per iteration

   Explanation: The charter is still important for the success of an agile project but is relatively lightweight as compared to traditional charters.

980. Simplicity - the art of maximizing the work not done - is essential. What is the most appropriate interpretation of this agile principle?
   a) Working in short iterations automatically leads to simplicity.
   b) Look for simpler solutions to avoid overengineering and prevent unnecessary work.
   c) Simplicity is more important than avoiding maximum work.
   d) None of the above

   Explanation: The most appropriate interpretation of the agile principle of "Simplicity - the art of maximizing the work not done" is option b) Look for simpler solutions to avoid overengineering and prevent unnecessary work. This principle emphasizes the importance of finding the most straightforward solutions that meet the requirements, avoiding complexity and unnecessary effort, which ultimately enhances efficiency and effectiveness in product development.

981. At a stakeholder meeting, all stakeholders are given fake money to purchase the features they like and need. This is an example of:
   a) An innovation game
   b) Product mapping
   c) Economic analysis
   d) Market analysis

   Explanation: An innovation game is used to frame the process of requirements gathering in a creative way.

982. What are the questions asked during daily stand-up meetings?
   a) What did you do today?
   b) What are the impediments to getting work completed?
   c) What are you planning to do tomorrow?
   d) All of the above

Explanation: The questions asked during the daily stand-up meetings are what did you accomplish yesterday, what do you plan to do today, and are you encountering obstacles?

983. The expected time for planning a four-week iteration is:
   a) 1-2 hours
   b) 2-4 hours
   c) 4-8 hours
   d) 12-24 hours

   Explanation: Typically an iteration planning should take 1-2 hours for each week of sprint duration (5-10% of effort)

984. The iteration review or demo is an important meeting for the team to bring the iteration to a logical conclusion. Which of the following should NOT be part of this meeting?
   a) A demonstration of the working product
   b) A detailed overview of the challenges encountered during the iteration and how they were overcome
   c) Feedback by the participants about the product
   d) Discussion about the potential release of the product

   Explanation: Discussion about challenges is probably more appropriate for the retrospective.

985. Which of the following is NOT very consistent with agile principles?
   a) Large teams
   b) Incremental delivery
   c) Small releases
   d) Progressive elaboration

   Explanation: Agile methodologies prioritize small, self-organizing teams to foster effective communication, collaboration, and adaptability, which may be compromised in larger teams. Therefore, "Large teams" (option a) is less consistent with agile principles compared to the other options provided.

986. During an iteration, an end user keeps calling the team members requesting to include more work items to the iteration. What should be done?
   a) The team should talk to the end user asking them not to be disturbed.
   b) The coach should talk to the end user and explain how an agile project is planned.
   c) The end user should be asked to join the team.
   d) Senior management should be brought in to solve the problem.

Explanation: The coach is the person who keeps the team and end users focused on learning and using agile processes.

987. A developer has not created a detailed documentation for a code that is only understood by him. Why is this a problem?
   a) Code should be collectively owned by the team.
   b) Conflict should be handled immediately.
   c) Documentation should be written immediately.
   d) The team should demand the coder be fired.

   Explanation: In an agile project, the code is collectively owned by team and each team member should understand the code.

988. Extreme programming believes in deeper involvement of the onsite customers in the team. Which of the following is NOT a customer role in extreme programming?
   a) Interaction designer
   b) Business analyst
   c) Product manager
   d) Programmer

   Explanation: In Extreme Programming (XP), customer roles involve individuals who define and refine product features, such as interaction designers, business analysts, and product managers. However, programmers primarily focus on implementation rather than representing the customer's perspective. Hence, "Programmer" is not a customer role in XP.

989. Mary is the scrum master for a new project and is working with the product owner and team on selecting the user stories for the first sprint. The team has committed to 30 story points for the sprint, but the product owner wants the team to deliver more. The team is uncomfortable committing more as it hasn't understood the stories well. What should Mary do?
   a) Explain to the product owner that the team determines their commitment and she needs to accept what they have identified.
   b) Encourage the product owner to review the stories and add more details and acceptance criteria to help the team better determine if they can deliver more stories
   c) Ask the team to identify two additional stories for the sprint as a stretch goal.
   d) Add a story for the next sprint and add more details to the stories.

   Explanation: The product owner needs to provide the details that the team requires to make their commitments.

990. Team workspace should be arranged so the team members can:

a) Form smaller groups
b) Face the information radiator
c) Face each other
d) Have privacy

Explanation: Team workspace should be arranged to maximize the interaction between team members for the best results.

991. Refactoring should produce software that is:
a) Better organized
b) Regression tested
c) Finished
d) Just passing acceptance tests

Explanation: Refactoring is the reorganization of working code to bring the organization in line with its functionality and maintain it better.

992. During an iteration, there was a major reorganization and the team is in a flux about future direction for the product. They feel that the iteration should be put on hold or cancelled until clarity emerges. Who is responsible for making this decision?
a) Scrum master
b) Product owner
c) Team facilitator
d) Team
Explanation: Product owner decides whether it is appropriate to cancel an iteration.

993. A team is analyzing factors that drive change and restrict change. This is called _____.
a) Change management
b) Product mapping
c) Force field analysis
d) Root cause analysis

Explanation: Force field analysis is a technique for analyzing the forces that are encouraging and resisting potential or real change and the strength of these forces.

994. After the team has removed the bottlenecks in a Kanban system and established WIP limits, what should they focus on next?
a) Introduce iterative development
b) Prioritize the requirements

c) Hand over to operations
d) Continuous improvement

Explanation: Kaizen means continuous improvement and prevents inertia from becoming a constraint.

995. Pruning the Product Tree is a technique for:
a) Release planning
b) Iteration planning
c) Requirement gathering
d) Integration testing

Explanation: The act of pruning the product tree is a technique for requirement gathering.

996. A team completed 10 work items in a five-day week and each item took about five days to complete. What is the cycle time?
a) 5 days
b) 0.5 day
c) 1 day
d) 2 per day

Explanation: Cycle time is the average time between the delivery of completed work items. 10 items in 5 days means half day per item.

997. The number of story points a team can complete in a single iteration is known as _____.
a) Velocity
b) Time to value
c) Value points delivered
d) Cycle time

Explanation: Velocity is the number of features or user stories that a team delivers in a fixed iteration.

998. All of the following are common ways to enable osmotic communication in distributed agile teams except:
a) Appointing a spokesperson for each location
b) Looping the entire team in all project-related communications
c) Conducting live video conferences
d) Traveling frequently between locations and meetings

Explanation: Appointing a spokesperson restricts the channels of communication available.

999. Who is responsible for updating the Kanban board?

a) The product owner
b) The scrum master
c) The tester
d) The team

Explanation: The Kanban board is an artifact that shows work in progress (WIP) that can only be updated by the team doing the work.

1000. Which technique helps in analyzing how different groups interact with the product?

a) Stakeholder analysis
b) Creating personas
c) Wireframes
d) Risk analysis

Explanation: A persona is an imaginary person or identity created by the team, to model interactions with the system to gather requirements. Creating personas helps in analyzing how different groups interact with the product.